

claude-windows / 9f6357d8-0e35-491a-a830-052e094c7f37

Metadata

```
- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\9f6357d8-0e35-491a-a830-052e094c7f37.jsonl`  
- SHA-256: `03bb544d42e3e4f71bd34de0c2d56ea8e5e3907908c19b6258c0cc4036052607`  
- Source modified: `2026-06-21T14:47:22+00:00`  
- Imported at: `2026-07-05T16:48:18+00:00`  
- project: `C--Users-avidu-Projects-badminton-highlight-indexer`  
- session_id: `9f6357d8-0e35-491a-a830-052e094c7f37`
```

Transcript

1. user (2026-06-21T09:12:55.834Z)

? Picking up KhelSutra after the big localization + dry-run session ? the product now speaks Kannada and Hindi end to end, and we proved the whole loop live: a bucket video ? the engine ? a downloaded reel, driven from the UI, in both languages.

Clean slate to build on: both repos on master, 0 open PRs, no worktrees, no GCP VMs (khelsutra 2421557 · engine 2afd54f). Proof lives in ~/Downloads/khelsutra-dryrun/ + gs://khelsutra-rally-corporus/promo/khelsutra-i18n-screencast.mp4.

My mission this session: make the bucket?reel CUJ fully UI-driven ? Path B.

Read first (the "use the docs" rule): khelsutra/docs/DRY_RUN_CUJ.md (the run + the gap + the two paths) and FRIENDLY_AND_MULTILINGUAL.md §7. ? The one thing to verify first: today /api/process is local-path-only, BUT _execute_processing (engine main.py:140) already calls storage.acquire_local_input(req.video_path, ?) ? so a gcs:// video_path might already fetch. Confirm that before building ? B-P1 could be a one-line acceptance + a test, not a feature.

First moves:

B-P1 ? /api/process accepts a gcs:// URI ? fetch ? process (engine PR + a real-subprocess test, like the #281 guard).

B-P2 ? the SPA ingest + progress screen (enter a bucket URI ? poll /api/jobs ? load rallies; the back-half already works), localized en/kn/hi.

Droplet (opt 3) ? deploy the FIXED engine to 168.144.126.176 (un-provisioned, 2-core/3.8GB) ? re-run the CUJ on Linux.

Don't redo: the full en/kn/hi SPA, the offline guard, the flyers, the dry run, the #281 fix. ? Still open for the localization DoD: B3 (the site/ marketing pages are still English-only) + a native-speaker review of the AI-drafted kn/hi.

Ultracode's on ? fan out research with a workflow, verify against the docs, keep tight one-item PRs, merge the good ones on green CI. Let's make the appliance point-and-go. ?

2. assistant (2026-06-21T09:13:07.215Z)

I'll start by ramping up: pull both repos, read the key docs, and verify the critical engine behavior before building anything. Let me scout inline first.

3. user (2026-06-21T09:13:12.663Z)

Already up to date.

---ENGINE STATUS---

feat/serving-m4-device-context...origin/feat/serving-m4-device-context

?? artifacts/

?? docs/KHELSUTRA_PRODUCT_OVERVIEW.md

---ENGINE HEAD---

44c9f01 docs(serving): mark \$7 tracker M4 ? + changelog (PR #283)

4. user (2026-06-21T09:13:14.678Z)

Already up to date.

---KHELSUTRA STATUS---

master...origin/master

?? .claude/

?? site/access.db

---KHELSUTRA HEAD---

2421557 docs(dry-run): record Path-A local result (en+hi reel downloaded) + the engine fix it surfaced (#54)

5. user (2026-06-21T09:13:14.738Z)

```
100         receiver: Optional[str] = None
101         duration_min: Optional[float] = None
102         event_type: Optional[str] = None
103
104     class CompileRequest(BaseModel):
105         video_id: str
106         segment_ids: List[int]
107         output_filename: str
108
109     def _finalize_reingest(video_id: str, segments, play_wm: int, cand_wm: int) -> None:
110         """Commit a (re)segmentation with destroy-AFTER-confirm semantics.
111
112         On a non-empty run, drop the PRIOR rows (id <= the pre-run watermark) so only the
113         fresh segments remain. On an empty run, drop the NEW partial rows (id > watermark)
114         and
115         keep the prior good results intact ? a failed/empty re-run never destroys prior
116         data.
117         """
118         if segments:
119             db_instance.prune_segments(video_id, play_upto=play_wm, cand_upto=cand_wm)
120         else:
121             db_instance.prune_segments(video_id, play_after=play_wm, cand_after=cand_wm)
122
123     # --- API Endpoints ---
124     @app.get("/api/config")
125     def get_config():
126         return global_config
127
128     @app.get("/api/health")
129     def health():
130         """Cheap liveness/readiness probe (no auth) for the SPA's offline banner
131         (HOSTING_PLAN.md:42). Reports the configured sport + default segmenter so the UI can
132         confirm the engine is reachable and which detector is wired. Never raises."""
133         sport = getattr(global_config, "sport", None)
134         indexing = getattr(global_config, "indexing", None)
135         return {
136             "status": "ok",
137             "sport": sport,
138             "default_segmenter": getattr(indexing, "default_segmenter", None),
139         }
140
141     def _execute_processing(req: ProcessRequest, progress=None) -> Dict[str, Any]:
142         """The full hash ? validate ? segment ? report pipeline for one video.
143
144         This is the former synchronous /api/process body, unchanged in behavior, now run on
145         the job worker thread (DEFERRED_HARDENING_PLAN #16). Returns the same response dict
146         the sync endpoint used to return (UI compatibility); the per-video observability
147         report is still emitted in `finally:` and grafted as response["reports"].
```

```

147
148     `progress` is an optional callable(stage: str) used to surface coarse job progress.
149     Raises on unexpected internal errors ? the caller records those as a job failure
150     (after this function has already restored the pre-run segments, see below).
151     """
152     notify = progress or (lambda stage: None)
153
154     notify("hashing")
155     # M2: resolve the input to a LOCAL path. A bare/local path is returned unchanged
156     (identity ? # byte-identical to today); a bucket/Drive URI is fetched to scratch first. The
original
157     # req.video_path (the source URI) is kept for the DB record + report; only the file-
reading
158     # consumers (hash / validators / segmenter) use the resolved local path.
159     local_video = storage.acquire_local_input(req.video_path, getattr(global_config,
"storage", None))
160     video_id = compute_video_id(local_video)
161     was_reingest = db_instance.get_video(video_id) is not None
162
163     # 1. Run pluggable validators (with-context variant surfaces ffprobe_meta for the
report)
164     notify("validating")
165     logger.info(f"Running validations for {req.video_path}...")
166     val_results, val_context = registry.run_all_with_context(local_video, global_config)
167     failures = [r for r in val_results if not r.passed]
168
169     # typed AppConfig: attribute access for the default segmenter (with safe fallback)
170     default_seg = getattr(getattr(global_config, "indexing", None), "default_segmenter",
"motion")
171     seg_name = req.segmenter or default_seg
172     segmenter_actual = None
173     segmentation_ran = False
174     response: Optional[Dict[str, Any]] = None
175     try:
176         if failures:
177             # add_video is an UPSERT (no cascade) ? a failed re-validation only updates
178             # status; it no longer destroys the video's prior good segments.
179             db_instance.add_video(video_id, req.video_path, "failed", [r.model_dump()
for r in val_results])
180             response = {
181                 "video_id": video_id,
182                 "status": "failed",
183                 "message": "Video failed validation checks.",
184                 "results": [r.model_dump() for r in val_results],
185             }
186             return response
187
188     # 2. Run segmenter & indexer
189     logger.info("[API] Video passed checks. Segmenting and indexing...")
190     db_instance.add_video(video_id, req.video_path, "processed", [r.model_dump() for
r in val_results])
191
192     segmenter_cls = segmenter_registry.get(seg_name)
193     if not segmenter_cls:
194         # Submit-time validation already 400s unknown names; this is the defensive
195         # in-worker path (e.g. config default pointing at an unregistered
segmenter).
196         available = list(segmenter_registry.list_available().keys())
197         response = {
198             "video_id": video_id,
199             "status": "failed",
200             "message": f"Segmenter '{seg_name}' not found. Available segmenters:
{available}",
201             "results": [r.model_dump() for r in val_results],

```

```

202         "play_segments": [],
203     }
204     return response
205
206     logger.info(f"[API] Using segmenter: {seg_name}")
207     notify(f"segmenting ({seg_name})")
208     segmenter_actual = seg_name
209     # Destroy-AFTER-confirm: remember the pre-run rows, then keep new + drop old
only
210     # if the run yields segments; otherwise drop the new partial rows and keep the
old.
211     play_wm, cand_wm = db_instance.segment_watermarks(video_id)
212     segmenter = segmenter_cls(db_instance, global_config)
213     try:
214         result = segmenter.process_video(video_id, local_video, req.player_pool,
req.max_frames)
215     except Exception:
216         # A raising segmenter must not leave half-written rows: restore the pre-run
217         # state (same destroy-after-confirm contract as the Failure branch), then
let
218         # the job worker record the failure.
219         db_instance.prune_segments(video_id, play_after=play_wm, cand_after=cand_wm)

```

6. user (2026-06-21T09:13:14.757Z)

```

1     # Dry-run CUJ ? UI drives the tool on a bucket video ? reel downloaded local
(multilingual)
2
3     > **Status:** `DRAFT` ? owner-gated . **Owner:** Avi Dullu . **Created:** 2026-06-21 .
**Goal for this session.**
4     > **Umbrella:** a milestone of `docs/LOCAL_APPLIANCE_ARCHITECTURE.md` (the appliance) +
`docs/PER_RALLY_SELECTION.md` (the shipped SPA back-half).
5     > **Honesty note:** grounded in a file:line integration-surface map (this session);
claims tagged `[verified]` / `[gap]`.
6
7     ## 0. The goal (owner's words)
8     Operate the **web UI** ? point at a **video on the GCS bucket** ? the engine (tool)
processes it ? **review/select rallies in the UI** ? **compile** ? **download the reel to the
local machine**. Then **repeat with the UI in Hindi** to prove the multilingual workflow.
9
10    ## 1. What already works vs the gap `[verified]`
11    **? Back half (shipped ? PER_RALLY_SELECTION M0):** the SPA loads `?video=<id>` ? `POST
/api/query` (rallies, default all-selected) ? deterministic query/select ? `POST /api/compile` ?
`<video>` preview + **`<a download>` saves the MP4 locally**. Reel download-to-local **works
today** (`GET /api/highlights/{video_id}/{filename}` streams it; `/api/compile` also returns
`download_url`). Fully **en/kn/hi** (the switcher / `?lang=`).
12
13    **? Front half (the gap):**
14    - The SPA has **no ingest/process UI** ? it can't point at a video (bucket or local),
can't call `/api/process`, can't poll `/api/jobs`. It only gets a `video_id` from `?video=` in
the URL.
15    - **No HTTP bucket-ingest:** `POST /api/process` accepts only a **local** `video_path`.
The worker CLI (`python -m backend.worker infer --video-uri gcs://`) *can* fetch+process a
bucket video (GCS is a real `StorageBackend`), **but it writes `intervals.csv`/`report.json` to
the bucket ? NOT to the API's SQLite `videos`/`play_segments`**. So a worker run alone does not
give the SPA a queryable `video_id`. **Bridging "gcs://" ? a queryable video_id in the API DB" is
net-new** (appliance T12/T14).
16
17    ## 2. Two paths to the goal
18    **Path A ? Cheapest dry run NOW (~zero new code).** Pre-process the bucket video out-of-
band to mint a `video_id` in the engine DB, then drive the existing SPA back-half. Proves the
**multilingual UI-driven workflow** end-to-end. The only non-UI step is "point at the bucket" (a
one-line CLI pre-process).
19
20    **Path B ? Full "point at a bucket video IN the UI" (bigger, multi-session).** Build the

```

in-UI ingest: a net-new ****HTTP bucket-ingest endpoint**** (`gcs://` ? process ? DB `video\_id`) + the SPA ****ingest + progress**** screens (appliance S1/S2). The proper appliance front-half.

21

22 **## 3. Path A runbook (the cheapest dry run) `[verified-steps]`**

23 1. ****Engine host + deps:**** run the engine FastAPI (`python -m backend.main --server`, binds `127.0.0.1:8000`) on a box with `ffmpeg` + OpenCV. Gemini/motion paths are ****CPU-only, no WSL**** ? local Windows works; the CPU droplet works too. (GPU/WSL only needed for native WASB.)

24 2. ****Get the bucket video local + into the API DB:****

25 `gsutil cp gs://<bucket>/<match>.mp4 ./incoming/` (or the worker fetch), then `POST /api/process {"video\_path":"<abs local path>","segmenter":"gemini"}` ? poll `GET /api/jobs/{job\_id}` until `result.video\_id`. *(`gemini` needs `GEMINI\_API\_KEY`; `motion` is key-free but hides fabricated metadata.)*

26 3. ****Open the SPA against the engine:**** `npm run dev` (Vite proxies `/api` ? `:8000`, same-origin) ? `http://localhost:5173/?video=<video\_id>` ? rallies auto-load.

27 4. ****Select ? Build ? download:**** pick rallies ? Build ? `<video>` preview ? click the download link ? MP4 saved local.

28 5. ****Repeat in Hindi:**** click ********* (or `?lang=hi`) ? redo select?compile?download ? confirm the whole flow works with the UI in Hindi.

29

30 ****Small unblocked SPA polish that makes Path A more "via the UI" (independent of A/B, useful in both):** a *****Load a video by ID***** input (paste the `video\_id` ? load, instead of editing the URL) + consume the engine's `download\_url` (instead of hand-building it). One tight PR.

31

32 **## 4. Open decisions ? OWNER (these gate the dry run)**

33 1. ****Scope:**** Path A (cheapest dry run now) or Path B (build the full in-UI bucket ingest)? ***Recommend A first*** (proves the multilingual workflow fast), then B as the appliance build.

34 2. ****Compute / detector:**** ****Gemini**** (real rallies; needs your `GEMINI\_API\_KEY`; uploads downsized chunks to Google = paid + cloud) . ****motion**** (key-free, runs anywhere, but lower-quality boundaries + fabricated server/shot metadata stays hidden) . ****GPU WASB**** (best quality, but needs a GPU host + is owner-parity-gated, not production-verified). ***Recommend Gemini for "real-ish, cheapest" ? pending your key + OK (paid + uploads footage).***

35 3. ****Engine host:**** local Windows (simplest, Gemini/motion) . the CPU droplet `168.144.126.176` (shared testbed ? careful) . a GPU VM (only for WASB).

36 4. ****Which bucket video:**** e.g. `gs://khelsutra-rally-corpus/videos/mahadevpura\_smoke\_180s.mp4` (already staged, known-good) or another clean ?720p clip (must pass the engine validators).

37

38 **## 5. Honest caveats**

39 - ****Gemini = paid + uploads footage to Google**** (an external/paid action ? needs your explicit OK + a clean clip). ****Rotate the chat-shared `GEMINI\_API\_KEY`.**

40 - ****motion**** fabricates `server`/`receiver`/`shot\_count` (the SPA correctly hides them) ? boundaries are real but metadata isn't.

41 - ****GPU/native WASB**** is partially wired + owner-byte-parity-gated ? ****not production-verified****.

42 - Path A's "point at the bucket" is a ****CLI pre-step****, not in-UI ? full in-UI ingest is Path B.

43 - Edge auth: none in the engine (localhost dry-run is fine; required before any non-localhost exposure).

44

45 **## 6. Definition of done (the dry run)**

46 - [x] A bucket video ? (engine processes) ? the ****SPA loads its rallies**** ? select ? compile ? ****reel downloaded to the local machine****, all driven from the UI (Path A allows a one-line CLI pre-process). ? ? ****Path A done (local, 2026-06-21)****, see §7.

47 - [x] The ****same flow works with the UI in Hindi**** ? multilingual workflow confirmed. ? (Kannada also works via the same switcher.)

48 - [] ***(Path B, if chosen)*** the bucket video is pointed-at ****from the UI**** (no CLI pre-step).

49 - [] ***(option 3)*** the same CUJ confirmed on the ****CPU droplet**** (Linux deploy target) ? droplet reachable but currently ****un-provisioned**** (no engine checkout; ~3.8GB RAM) ? needs a from-scratch engine install or a proper deploy of the fixed engine.

50

51 **## 7. ? Run result ? Path A, local (2026-06-21)**

52 Ran the real CUJ end to end on the local Windows box, isolated engine off

```

`origin/master` (fresh DB/output worktree), Gemini detector:
53 1. `gs://khelsutra-rally-corpus/videos/mahadevpura_smoke_180s.mp4` (1080p/30fps) ?
`gsutil cp` local ? `POST /api/process {segmenter: gemini}`.
54 2. Engine processed via Gemini (3 chunks) ? 12 rallies (16 found, 4 too-short
rejected), `video_id 05e0e577?`.
55 3. SPA (`npm run dev`, Vite proxies `/api/?`:8000) opened at `?video=05e0e577?` ?
loaded the 12 real rallies ("Loaded 12 rallies from this session").
56 4. Build ? `/api/compile` ? reel (1920x1080, 76.5s) ? downloaded to
`~/Downloads/khelsutra-dryrun/highlights_local_en.mp4` (50.9 MB).
57 5. Repeated with the UI in Hindi (`?lang=hi`): the page + rally cards + "???" +
"???" all in Hindi; reel compiled + downloaded (`highlights_local_hi.mp4`).
? multilingual workflow confirmed.
58
59 Screencast/proof: per-locale screenshots; both reels on the local machine. Screencast of
the i18n SPA is at `gs://khelsutra-rally-corpus/promo/khelsutra-i18n-screencast.mp4`.
60
61 Fix the dry run surfaced (engine) ? MERGED-PENDING
62 The async-job server hung every job in `queued` when run the documented way (`python
-m backend.main`): main.py runs as `__main__` (sets `__main__.db_instance`), but `job_worker`
reads `import backend.main` ? a separate module whose `db_instance` is `None` ? the drain
crashes silently. Latent because the unit tests monkeypatch the executor inline and the UI was
verified vs a stub engine. Fix + a real-subprocess regression test ? engine
[#281](https://github.com/Khelsutra/badminton-highlight-indexer/pull/281) (dispatch through
`backend.main`; FAILS-without / PASSES-with). Unblocked the run via the single-module
invocation; the fix applies to the droplet too.
63

```

7. assistant (2026-06-21T09:13:23.079Z)

Confirmed the critical pivot: `\_execute\_processing` at
backend/main.py:159 already calls
`storage.acquire\_local\_input(req.video\_path, ?)` which per the M2 comment fetches a bucket URI
to scratch. Now I need to find what actually **gates** `/api/process` to local-only today. Let me
look at the request model, the endpoint, and the storage resolver.

8. user (2026-06-21T09:13:24.519Z)

```

60 return "<h3>Sports Highlight Indexer Server is Running. Please create index.html in
backend/api/static/</h3>"
61
62 # Global variables initialized at startup
63 db_instance: Optional[Database] = None
64 global_config: Optional[AppConfig] = None # now typed (Pydantic model)
65
66 # Async job worker ? drain/wake loop (DEFERRED_HARDENING_PLAN #16, EXECUTION_POLICY.md
P3)
67 # now lives in backend.api.job_worker. Re-exported here so the names stay addressable on
68 # `backend.main` (`JobWaiting`, `_drain_due_jobs`, `_arm_wake_timer`, `_get_executor`,
69 # `_run_claimed_job`, `_MAX_DRAIN_WAKE_SEC`, ...). The worker resolves these seams
THROUGH
70 # this module object at call-time, so `monkeypatch.setattr(backend.main, ...)` still
71 # intercepts the inter-function calls (a parked job arms the PATCHED `_arm_wake_timer`,
72 # so no real daemon timer leaks in tests). Behavior is unchanged by the extraction.
73 from backend.api.job_worker import ( # noqa: F401 (re-exports: names stay addressable
on backend.main)
74     JobWaiting,
75     _arm_wake_timer,
76     _drain_due_jobs,
77     _get_executor,
78     _job_to_request,
79     _MAX_DRAIN_WAKE_SEC,
80     _run_claimed_job,
81     _schedule_next_drain_if_waiting,
82 )
83

```

```

84     # Legacy thin wrapper for any remaining direct calls during transition.
85     # New code should import load_app_config / AppConfig from backend.config directly.
86     def load_config(config_path: str = "config.json") -> Dict[str, Any]:
87         cfg = load_app_config(config_path)
88         return cfg.model_dump(mode="json")
89
90     # --- API Models ---
91     class ProcessRequest(BaseModel):
92         video_path: str
93         player_pool: Optional[List[str]] = None
94         max_frames: Optional[int] = None
95         segmenter: Optional[str] = None
96
97     class QueryRequest(BaseModel):
98         video_id: str
99         server: Optional[str] = None
100        receiver: Optional[str] = None
101        duration_min: Optional[float] = None

```

9. user (2026-06-21T09:13:25.514Z)

```

230         "results": [r.model_dump() for r in val_results],
231         "play_segments": [],
232     }
233     return response
234
235     segments = result.value if hasattr(result, "value") else result
236     notify("finalizing")
237     _finalize_reingest(video_id, segments, play_wm, cand_wm)
238
239     response = {
240         "video_id": video_id,
241         "status": "success",
242         "message": f"Successfully indexed {len(segments)} play segments using
243 '{seg_name}' segmenter.",
244         "results": [r.model_dump() for r in val_results],
245         "play_segments": segments,
246     }
247     return response
248 finally:
249     # Always emit the per-video observability report (next to the video, or to
250     # report.output_dir). Never raises. Surface the paths in the response.
251     paths = emit_indexer_report(
252         db=db_instance, video_id=video_id, video_path=req.video_path,
253         config=global_config,
254         val_results=val_results, val_context=val_context,
255         segmenter_requested=req.segmenter, segmenter_actual=segmenter_actual,
256         was_reingest=was_reingest, segmentation_ran=segmentation_ran,
257     )
258     if paths and isinstance(response, dict):
259         response["reports"] = paths
260
261 @app.post("/api/process", status_code=202)
262 def process_video(req: ProcessRequest):
263     """Submit a processing job (202 + job_id). Poll GET /api/jobs/{job_id} for state.
264
265     Fast-fail checks (bad path, missing file, unknown segmenter) happen here so the
266     client gets an immediate 4xx; everything slow ? including MD5 hashing of the file ?
267     runs on the worker (DEFERRED_HARDENING_PLAN #16). The row is created 'queued'; the
268     worker drains it via `claim_due_job` (EXECUTION_POLICY.md P3 self-scheduling loop).
269     """
270     allowed_root = getattr(getattr(global_config, "security", None),
271 "allowed_video_root", None)
272     try:

```

```

271         validate_input_path(req.video_path, allowed_root=allowed_root)
272         # M2: resolve the input once (path or storage URI) ? raises ValueError for an
unsupported
273         # scheme, which we map to a clean 400 (an unknown scheme must never 500).
274         input_ref = storage.resolve_input_ref(req.video_path, getattr(global_config,
"storage", None))
275         except ValueError as e:
276             raise HTTPException(status_code=400, detail=str(e)) from e
277         # The local-existence fast-fail applies only to a LOCAL input, and stats the
RESOLVED local path
278         # (input_ref.key ? e.g. local://? stripped to the bare path), not the raw URI
string. A
279         # bucket/Drive URI can't be cheaply os.path.exists()'d ? its existence is verified
when the worker
280         # fetches it (a fetch miss fails the job loudly). validate_input_path above still
runs for every
281         # input: its null-byte guard always applies, and a configured allowed_video_root
(hosted mode)
282         # rejects a non-local URI as out-of-root.
283         if input_ref.backend == "local" and not os.path.exists(input_ref.key):
284             raise HTTPException(status_code=404, detail=f"Video file not found at
'{req.video_path}'")
285         if req.segmenter and not segmenter_registry.get(req.segmenter):
286             available = list(segmenter_registry.list_available().keys())
287             raise HTTPException(
288                 status_code=400,
289                 detail=f"Segmenter '{req.segmenter}' not found. Available segmenters:
{available}")
290         )
291
292         job_id = uuid.uuid4().hex
293         if not db_instance.create_job(job_id, req.video_path, req.segmenter,
294                                     req.player_pool, req.max_frames):
295             raise HTTPException(status_code=500, detail="Failed to persist job record.")
296         _get_executor().submit(_drain_due_jobs)
297         return JSONResponse(
298             status_code=202,
299             content={"job_id": job_id, "status": "queued", "poll": f"/api/jobs/{job_id}"},
300         )
301
302
303 @app.get("/api/jobs/{job_id}")
304 def get_job(job_id: str):
305     """Job state: {status, progress, result, reports}. `status` = execution state
306     (queued|running|done|failed); the pipeline verdict is result["status"]."""
307     job = db_instance.get_job(job_id)
308     if not job:
309         raise HTTPException(status_code=404, detail=f"Unknown job_id '{job_id}'")
310     result = job.get("result")
311     return {
312         "job_id": job["id"],
313         "status": job["status"],
314         "progress": job.get("progress"),
315         "video_id": job.get("video_id"),
316         "error": job.get("error"),
317         "result": result,
318         "reports": (result or {}).get("reports"),
319         "created_at": job.get("created_at"),
320         "started_at": job.get("started_at"),
321         "finished_at": job.get("finished_at"),
322     }
323
324 # --- Chunked upload (B2, PR-2) -----
325 # The chunked-upload island (in-process registry + helpers + the 4 endpoints) now lives
in

```



```

326 # backend.api.uploads as a self-contained APIRouter. It resolves the live startup config
via
327 # backend.main.global_config, so behavior ? route paths, status codes, body-size limits,
328 # sequential-part logic, and abort cleanup ? is unchanged.
329 from backend.api import uploads as uploads_api
330
331 app.include_router(uploads_api.router)
332
333
334 # --- Highlight download (B3, PR-2) -----
335
336 @app.get("/api/highlights/{video_id}/{filename}")
337 def download_highlight(video_id: str, filename: str):
338     """Stream a compiled highlight reel ? `filename` is the bare name given to
/api/compile
339     (which only writes into output_dir; the browser previously got back an unreachable
340     server-local path)."""
341     try:
342         name = safe_output_filename(filename)
343     except ValueError as e:
344         raise HTTPException(status_code=400, detail=str(e)) from e
345     if not db_instance.get_video(video_id):
346         raise HTTPException(status_code=404, detail="Indexed video record not found.")
347     output_dir = getattr(getattr(global_config, "output", None), "output_dir", "output")
or "output"
348     path = os.path.join(output_dir, name)
349     try:
350         path = resolve_under_root(path, output_dir)
351     except ValueError as e:
352         raise HTTPException(status_code=400, detail=str(e)) from e
353     if not os.path.isfile(path):
354         raise HTTPException(status_code=404,
355                             detail=f"No compiled file named '{name}'. Compile first via
/api/compile.")
356     media = "video/quicktime" if name.lower().endswith(".mov") else "video/mp4"
357     return FileResponse(path, media_type=media, filename=name)
358
359
360 @app.post("/api/query")
361 def query_play_segments(req: QueryRequest):
362     filters = {
363         "server": req.server,
364         "receiver": req.receiver,
365         "duration_min": req.duration_min,
366         "event_type": req.event_type
367     }
368     segments = db_instance.query_play_segments(req.video_id, filters)
369     return {"play_segments": segments}
370
371 @app.post("/api/compile")
372 def compile_highlights(req: CompileRequest):
373     video = db_instance.get_video(req.video_id)
374     if not video:
375         raise HTTPException(status_code=404, detail="Indexed video record not found.")
376
377     # Retrieve matching play segments from db
378     all_segments = db_instance.query_play_segments(req.video_id, {})
379     matched_segments = [s for s in all_segments if s["id"] in req.segment_ids]
380
381     if not matched_segments:
382         raise HTTPException(status_code=400, detail="No matching play segments found to
stitch.")
383
384     # Reject path traversal / absolute output names (os.path.join would otherwise let an
385     # absolute or '..'-laden output_filename write anywhere on disk).

```

```

386         try:
387             out_name = safe_output_filename(req.output_filename)
388         except ValueError as e:
389             raise HTTPException(status_code=400, detail=str(e)) from e
390
391     # The configured shot-count floor (stitching.min_shot_count) applies on the API
392     # path too ? parity with scripts/run_process_and_stitch.py. Segments WITHOUT
shot_count
393     # metadata are exempt: the floor filters known counts only, so explicitly
394     # requested manual/legacy segments can't silently vanish under the default floor.
395     stitching = getattr(global_config, "stitching", None) or StitchingConfig()
396     kept = []
397     for s in matched_segments:
398         meta = s.get("metadata") or {}
399         sc = meta.get("shot_count") if isinstance(meta, dict) else None

```

10. user (2026-06-21T09:13:26.223Z)

No files found

11. assistant (2026-06-21T09:13:44.995Z)

Major finding: ``/api/process`` at backend/main.py:260 is **already URI-aware** (M2/#280) ? line 274 calls ``storage.resolve_input_ref``, and line 283 explicitly skips the local-existence check for non-local backends, deferring fetch to the worker. So B-P1 may already be done in code. But the ``grep`` for ``def acquire_local_input`` found nothing ? I need to trace the actual storage module to confirm ``gcs://`` is really handled end-to-end. Let me dig in.

12. user (2026-06-21T09:13:46.057Z)

No files found

13. user (2026-06-21T09:13:47.348Z)

```

1:import os
2:import sys
3:import argparse
4:import logging
5:import uuid
6:from urllib.parse import quote
7:import uvicorn
8:from dotenv import load_dotenv
19:from fastapi import FastAPI, HTTPException
20:from fastapi.middleware.cors import CORSMiddleware
21:from pydantic import BaseModel
22:from typing import List, Dict, Any, Optional
24:from fastapi.staticfiles import StaticFiles
25:from fastapi.responses import FileResponse, HTMLResponse, JSONResponse
26:from backend.pipeline.validators.base import registry
27:from backend.infrastructure.database import Database
28:from backend.pipeline.segmenters.base import segmenter_registry
29:from backend.pipeline.stitcher.compiler import VideoCompiler
30:from backend.utils.hashing import compute_video_id # canonical file-identity hashing
31:from backend.utils.paths import resolve_under_root, safe_output_filename, validate_input_path
32:from backend.config import load_config as load_app_config, AppConfig, StitchingConfig # new
typed config
33:from backend.results import Failure # standardized error/result contract
34:from backend.reporting import emit_indexer_report
35:from backend import storage # M2: URI-aware input acquisition (local = identity passthrough)
73:from backend.api.job_worker import ( # noqa: F401 (re-exports: names stay addressable on
backend.main)
159:    local_video = storage.acquire_local_input(req.video_path, getattr(global_config,
"storage", None))
272:    # M2: resolve the input once (path or storage URI) ? raises ValueError for an

```

```

unsupported
274:         input_ref = storage.resolve_input_ref(req.video_path, getattr(global_config,
"storage", None))
329:from backend.api import uploads as uploads_api
598:         storage_cfg = getattr(global_config, "storage", None)
600:         input_ref = storage.resolve_input_ref(args.video_path, storage_cfg)
607:         local_video = storage.acquire_local_input(args.video_path, storage_cfg)

```

14. assistant (2026-06-21T09:13:51.376Z)

The `storage` module is `backend/storage`. Let me read it to trace the `gcs://` path through `resolve\_input\_ref` and `acquire\_local\_input`, and find where GCS fetch actually happens.

15. user (2026-06-21T09:13:52.398Z)

No files found

16. user (2026-06-21T09:13:53.110Z)

```

1      # Friendly & Multilingual ? non-tech-savvy UX + Hindi/Kannada localization
2
3      > **Status:** `DRAFT` ? owner-gated (implementation starts only on explicit approval) .
**Owner:** Avi Dullu . **Created:** 2026-06-21 . **Last updated:** 2026-06-21
4      > **Lifecycle:** `DRAFT` ? IN PROGRESS ? DONE ? archived` (move to `docs/archives/` when
DONE).
5      > **Tracking anchors:** $7 progress tracker is the source of truth; indexed in
`docs/README.md`; thread in `NEXT_STEPS.md`; pointer in engine `docs/DOC_STATUS.md`; memory
`current-state` + `next-session-ux-localization`.
6      > **Relation to existing docs:** this is the **implementation project** for two engine
design docs ? it *adapts & supersedes-in-practice* `badminton-highlight-
indexer/docs/I18N_PLAN.md` (localization) and the UX half of `?/docs/UI_PROPOSAL.md` (non-tech-
savvy console). Those stay the canonical *design*; this doc carries the *tracked execution*
against today's surfaces.
7      > **Honesty note:** each claim tagged `[verified]` (checked against code this session),
`[researched]` (needs sign-off), or `[design]` (proposed). Not legal advice ? the OFL font
question (D-FONT) is owner/counsel-gated.
8
9      ---
10
11     ## 0. TL;DR
12     A coach or parent in Bangalore should land on KhelSutra, feel at home, set up their
camera with confidence, and get highlights ? **in Kannada or Hindi**, with **zero hand-
holding**. This project isolates that goal into one tracked workstream-pair: **(A)** make the UI
super friendly for non-tech-savvy users, and **(B)** localize to Hindi + Kannada. Both surfaces
(`web/` React SPA + `site/` vanilla marketing) have **zero i18n today** `[verified]`, so the
foundation ? shared `locales/{en,kn,hi}` catalogs + a negotiation contract + CI parity ? is the
real deliverable; translations follow. **The single most important caveat:** translations are
**human-reviewed** (Gemini drafts are *inference only, never training* ? the standing
guardrail), and Kannada/Hindi rendering is **blocked on the OFL font decision (D-FONT)**. The
cheapest unblocked high-impact first slice is the **SPA i18n scaffold + language switcher (en-
only)** ? it locks the shared catalog/negotiation contract and adds the render-test safety net
the SPA currently lacks.
13
14     ## 1. Problem & goal
15     **Why now.** The public site (kheLSutra.guru) is live, clean, honest, and tested. The
PMF beachhead is **Bangalore badminton academies** ; many coaches, parents, and players are far
more comfortable in **Kannada** than English, and most are **not highly tech-savvy** `[verified
? design intent]` (`UI_PROPOSAL.md:25,28`, `I18N_PLAN.md:10-13`, `PMF_MARKET_RESEARCH.md:116`,
`KHELSUTRA_PRODUCT_OVERVIEW.md:13/50/61`). An English-only, knob-heavy product adds friction at
the exact point of first contact.
16
17     **Concrete outcome ("good" looks like):** a Kannada/Hindi-comfortable, phone-first, non-
technical coach/parent can ? without help ? (1) understand what the product does and how to
record reliably, (2) request access / use the console, (3) pick rallies and get a reel, and (4)

```

share it on WhatsApp, ****entirely in their language****.

18

19 ****Read to get current:**** this doc's §3§4; the engine design `I18N\_PLAN.md` (canonical i18n design) + `UI\_PROPOSAL.md` §2 (personas, the 6 alpha screens, "teach recording ***before*** upload"); `VIDEO\_RECORDING\_GUIDELINES.md`; this repo's `PER\_RALLY\_SELECTION.md` (the honest SPA scope) + `REQUEST\_ACCESS\_FORM.md`.

20

21 **## 2. Decisions locked (structural ? owner directive 2026-06-21)**

22 | # | Decision | Source / date | Implication |

23 |---|-----|-----|-----|

24 | D1 | ****Isolate as ONE tracked project**** spanning both workstreams (A: UX, B: kn/hi l10n) | owner, 2026-06-21 | This doc; §7 = source of truth; archived on DoD |

25 | D2 | ****Home = `khelsutra/docs/`**** (the implementation repo) | owner-gated tracked-doc convention | `web/` + `site/` both live here; §7 tracks khelsutra PRs; engine P2 backend-l10n is a cross-repo deferred row |

26 | D3 | ****Adapt `I18N\_PLAN.md` to today's surfaces:**** P1 (vanilla `t()` shim) ? `site/`; P3 (react-i18next) ? `web/`; ****drop the dead `backend/api/static` retrofit**** | ****[verified]**** the doc targets `backend/api/static` (`I18N\_PLAN.md:25,30,156,191`) which exists only in the engine repo, not here | The doc's framework-agnostic ***foundation*** (catalog format + negotiation + ICU plurals + CI) survives unchanged; only the wiring re-points |

27 | D4 | ****Engine P2 backend-localization stays owner-gated + cross-repo**** (not in this DoD) | `CLAUDE.md` guardrails | Dynamic engine error/report text localizes later; tracked as a deferred row |

28 | D5 | ****Archive trigger = en+kn+hi live on BOTH surfaces**** (+ the non-tech UX baseline), then run the engine `DOC\_STATUS.md` §2 archival checklist | owner, 2026-06-21 | See §9 |

29 | D6 | ****D-LOCALES: `en` + `kn` for v1, `hi` next; always-fallback to `en`**** | owner, 2026-06-21 | Kannada-first beachhead; `hi` = catalog-only (both 2-form CLDR) |

30 | D7 | ****D-FONT: OFL-for-media carve-out ? self-host Noto Sans Kannada/Devanagari**** (counsel may still confirm) | owner, 2026-06-21 | Unblocks B4/B5/A3 + the kn/hi reel watermark; an explicit exception to the MIT/Apache/BSD code guardrail (fonts = media) |

31 | D8 | ****D-SURFACE-FIRST: lead with the `web/` SPA scaffold; SKIP the dead static-UI retrofit**** | owner, 2026-06-21 (via B1 go-ahead) | B1 is greenlit; site recording-education localized early |

32 | D9 | ****D-CATALOG-HOME: co-locate `locales/` + CI in `khelsutra`**** (shared, consumed by SPA + site) | owner, 2026-06-21 (via B1 go-ahead) | No cross-repo coupling |

33

34 ***Only ****D-BACKEND-DEPTH**** (§8.3) remains open ? and it's a deferred, cross-repo engine concern (B7), not a v1 blocker.***

35

36 **## 3. Foundation ? research / prior art**

37 ****Current-state facts **[verified]** this session**** (adversarially checked against the working tree):

38 - ****`web/` SPA: zero i18n.**** `web/package.json` deps are only `react`/`react-dom`; no `i18next`/`react-intl`/`formatjs`; no `locales/` dir; no `t()`/`useTranslation`/`data-i18n` anywhere. User-facing English is hardcoded in `.tsx` (e.g. `QueryBar.tsx:46-47,56,72,76,78`; `App.tsx:115,211,224`) ? ~50 literals + ~16 derived labels.

39 - ****`site/` marketing: vanilla, zero i18n.**** Three `` files (`public/{index,capture,own-model}.html`) with inline CSS, no framework; the only scripts are a CF analytics beacon (`index.html:595`) + one inline form/reel IIFE (`index.html:597-633`). `server.js` hardcodes English responses; no locale handling.

40 - ****Recording-education is ***not*** greenfield.**** The homepage "Record it right" gist (`index.html:324-364`) + the `/capture` checklist (`capture.html`, routed `server.js:51`, guarded `assets.test.js:94`) already ship the user-facing best-practices surface. ***Unshipped:** the empirical-threshold framing and the programmatic pre-flight validator (a code change, not a page).

41 - ****Persona is design intent, not built.**** kn/hi + zero-knobs are ****proposed/owner-gated****; the app is English-only today (`KHELSUTRA\_PRODUCT\_OVERVIEW.md:61`). It's Kannada-****first****, Hindi-****next**** ? not Kannada-only.

42

43 ****Localization prior art **[researched]**:**

44 - ****CLDR plural categories:**** Hindi = `{one, other}`; Kannada = `{one, other}`. Both are 2-form (same shape as English) ? once ICU MessageFormat is in the catalog, adding `hi`/`kn` plurals is ****catalog-only****, no code. (ICU still required because English `count>1`'s:`:` hand-rolling ? present today in the SPA ? doesn't generalize.)

```

45     - **react-i18next** (web/): `i18next` + `react-i18next` + `i18next-browser-
    languagedetector` + an ICU/plural post-processor; lazy-load per-locale JSON; persist via
    `localStorage`.
46     - **Shared catalog + tiny shim** (site/): the SAME i18next-flavored JSON consumed by a
    ~30-line `t(key, vars)` shim that walks `data-i18n` attributes ? no build step, no framework,
    mirrors i18next interpolation/plural rules for the 2-form languages.
47     - **Noto Sans Kannada / Noto Sans Devanagari** are **OFL-1.1**; OFL permits commercial
    bundling/redistribution (fonts = media, not code/weights). Self-host (e.g. `@fontsource`) with
    `font-display: swap` + subsetting. **Gated on D-FONT.**
48     - **Non-tech-savvy / HCI4D patterns:** icon **+ text** (never icon-only), non-color
    selection indicators, large touch targets, mobile-first, WhatsApp as the share channel, a
    prominent persistent language switcher.
49
50     ## 4. Design / architecture
51     **Two surfaces, one shared source of truth.**
52
53     ```
54     locales/                                # shared catalogs ? co-located in khelsutra (D-CATALOG-
    HOME, see §8)
55     en/  common.json                        # source of truth ? devs author keys here
56     kn/  common.json                        # Kannada   (Gemini-draft ? human-review)
57     hi/  common.json                        # Hindi     (next ? the extensibility proof)
58
59     web/  ? react-i18next                    # I18N_PLAN P3: provider in main.tsx; useTranslation in
    components
60     site/ ? t(key,vars) shim                 # I18N_PLAN P1 retargeted: data-i18n attrs over the
    static HTML
61     ```
62
63     - **Negotiation contract (one order, both surfaces):** `persisted pref (localStorage) ?
    switcher / ?lang=<code> ? navigator/Accept-Language ? "en"`. Supported locales in a single
    module; anything else falls back to `en`; **a missing key degrades to English, never to a raw
    key.**
64     - **Keys, never prose** ? `t("compile.button")`, with ICU for interpolation/plurals:
    `"{count, plural, one {# rally} other {# rallies}}"` . This replaces the SPA's current hand-
    rolled `Segment${count>1?'s':''}`.
65     - **Translation pipeline (guardrail-clean):** dev authors `en` keys ?
    `scripts/i18n_draft.py`-style Gemini **draft** (inference only, never training; flags every
    value for review) ? **owner/native-speaker review** ? CI key-parity ? merge. Gemini never
    touches `en`, never invents keys.
66     - **CI enforcement (keeps it "by design"):** (1) **key-parity** ? every locale has
    exactly the `en` key set (drop a stub `ta.json` ? CI lists every gap = the extensibility proof);
    (2) **no-bare-strings** ? flags user-facing literals not routed through `t()` (curated allowlist
    to start).
67     - **Fonts:** bundle Noto Sans Kannada + Devanagari **iff D-FONT = OFL carve-out**; this
    also unblocks a Kannada/Hindi **reel watermark** (engine `WatermarkConfig.font_path` already
    takes a path ? config, not code).
68
69     **Non-tech-savvy UX levers** (ranked impact × effort, grounded in personas; surface-
    tagged):
70
71     | Lever | Surface | Impact | Effort | Note |
72     |---|---|---|---|---|
73     | Plain-language copy ? strip engine jargon, keep honest caveats | both | high | small | Banners + footer carry backend vocabulary today |
74     | Localized **WhatsApp share** for the downloaded reel | both | high | small | WhatsApp
    is the channel for this audience |
75     | **Language switcher** persisted via a shared localStorage key | both | high | small |
    site promises kn/hi but ships no switcher |
76     | Localize **recording-education** (capture + gist) kn?hi | site | high | medium | gates
    input quality; the highest-leverage kn content |
77     | Mobile-first touch targets; tooltip-text ? visible | web | high | medium | tiny
    targets + desktop layout + invisible tooltips on phones |
78     | Icon **+ text** + non-color selection indicator | web | medium | small | never icon-
    only |

```

```

79      | Friendly engine-error mapping over raw FastAPI `detail` | web | medium | medium |
`CompileResult` passes engine text untranslated |
80      | Plain-language framing for the Request-Access GPU/storage Qs | site | medium | small |
"GPU"/"S3/GCS" are intimidating |
81      | Background-processing reassurance + status as icon chips | web | medium | small |
honest expectations on slow uplinks |
82      | Fix English-source recording gaps (a diagram) before translating | site | medium |
medium | the diagram is reusable across all 3 languages |
83      | Tap-only filter controls alongside the English query bar | web | medium | large | the
`parseQuery()` parser is English-grammar only |
84      | Keep the honest *today-vs-roadmap* split through translation | both | medium | small |
never soften "experimental"/"not available yet" |
85
86      **Reuse map (what's new vs reused):** *Reused* ? the i18next-flavored JSON + negotiation
contract + ICU plurals + key-parity/no-bare-strings CI + draft?review pipeline (all from
`I18N_PLAN.md`); the shipped recording pages; the SPA's `lib/format.ts`/`lib/query.ts`;
`site/src/assets.test.js` + the SPA's vitest harness. *New* ? `locales/` catalogs; the SPA
react-i18next provider + switcher; the site `t()` shim + `data-i18n` wiring; Noto font bundle
(gated); a jsdom render-test env for the SPA.
87
88      ## 5. Privacy / scope note
89      i18n touches **presentation only** ? no new personal-data collection, no biometrics,
minors-excluded posture unchanged. WhatsApp share opens the user's own client with a pre-filled
message; it sends nothing server-side. Gemini-draft translation is inference, not training (no
user data in the loop).
90
91      ## 6. Honest limits ? what this does NOT do
92      - A green CI / DONE status proves **key-parity + render + no-bare-strings**, **not**
translation *quality* ? that's human-gated and owner-paced.
93      - Kannada/Hindi will **not render** until the Noto fonts land (D-FONT); until then kn/hi
catalogs exist but display tofu/fallback.
94      - The SPA is **job-tethered** ; localizing **dynamic engine errors/report text** needs
engine P2 (owner-gated, cross-repo, D-BACKEND-DEPTH) ? out of this DoD.
95      - The query bar stays **English-grammar parsing** ; kn/hi users rely on tap-only controls
(a later, larger slice), not free-text Kannada queries.
96      - **Mobile-*perfect*** layout is an alpha non-goal (usable-on-phone is in scope; pixel-
perfect is not).
97
98      ## 6.5 Testing strategy ? automatic language negotiation (rigorous, owner-required)
99      **The bar (owner, #41):** a user whose **computer/browser language is Kannada or Hindi
must get the respective version automatically** ? that is the core promise and it is **proven by
tests at every layer, never assumed.** Auto-selection IS the negotiation order (persisted pref ?
`?lang=` ? browser/`Accept-Language` ? `en`); each step gets a test.
100
101      **Negotiation matrix ? every row is an automated test:**
102
103      | # | Signal under test | Setup | Expected | Proven in |
104      |---|---|---|---|---|
105      | N1 | **Browser/OS = Kannada** | no persisted pref, no `?lang`,
`navigator.languages=["kn-IN", ?]` | resolves to **kn** ? Kannada content (once the kn catalog
ships) | **B1** detection . **B5** content |
106      | N2 | **Browser/OS = Hindi** | `navigator.languages=["hi-IN", ?]` | resolves to **hi**
? Hindi content | **B6** (hi added to the supported set + catalog) |
107      | N3 | **`?lang=` override** | `?lang=kn`, browser = en | kn | B1 |
108      | N4 | **Persisted pref wins** | `localStorage=kn`, browser = en, no `?lang` | kn (a
returning user keeps their choice) | B1 |
109      | N5 | **Region subtag normalizes** | `navigator.language="kn-IN"` | base **kn** (not a
"kn-IN" miss) | B1 |
110      | N6 | **Unsupported language** | `navigator="ta-IN"` (Tamil, not yet supported) |
**en** fallback ? never a locale we can't serve | B1 |
111      | N7 | **Missing key in active locale** | kn active, a key absent from `kn/common.json`
| **en** fallback for that key (no raw key shown) | B1/B5 |
112      | N8 | **Static site (vanilla shim)** | `Accept-Language: kn` / `?lang=kn` on a `site/`
page | kn-rendered page (same order as the SPA) | B3 |
113      | N9 | **Backend (engine P2)** | `Accept-Language: kn` to the API | kn customer/report

```

```

text | B7 (owner-gated) |
114
115     **Levels & tooling:**
116     - **SPA (`web/`):** jsdom tests stub `navigator.language`/`navigator.languages`,
`localStorage`, and the `?lang` query ? assert `i18n.language`/`resolvedLanguage` + rendered
text per the matrix. *(B1 lands N1/N3?N7 at the **detection** level now ? `kn` is supported; the
kn/hi **content** assertions ride the catalogs in B5/B6.)*
117     - **Static site (B3):** the shared `t()` shim's negotiation is unit-tested the same way;
plus a server test that `Accept-Language: kn`/`?lang=kn` returns the kn-rendered page.
118     - **CI key-parity:** guarantees every locale has every key, so a correctly-negotiated
locale can never fall through to a raw key.
119     - **End-to-end (manual, owner machine ? per I18N_PLAN $6):** set the real OS/browser to
Kannada, then Hindi ? load both surfaces ? confirm the right language renders **and the Noto
fonts display** (the cloud CI container can't render fonts; the owner eyeball is the final
gate).
120     - **? Query parsing / semantic understanding (forward-looking, owner directive
2026-06-21) ? i18n testing is MANDATORY there.** The query bar is **English-grammar-only** today
(a documented limit, tracker A9): the user types English queries (`over 10s`, `top 15 longest`)
and a localized **sample-query prompt** shows them what to type in **every** locale (kn/hi
included). The **moment** the parser gains **semantic understanding** or accepts **Kannada/Hindi
query input** (e.g. typing a Kannada/Hindi phrase, or a conversational/NL builder ? I18N_PLAN
P4), i18n tests become a **merge gate**: a kn/hi query (typed or spoken) must parse to the
**correct selection**, the localized **parse-feedback** ("parsed-pill") must render in the
active locale, and the unsupported/empty-result messages must localize. **No query-NL /
semantic-parsing work merges without those tests.**
121
122     A locale that's reachable by the switcher but **not auto-selected from the browser
language is a bug, not a "later". ** This matrix is a DoD gate ($9).
123
124     ## 7. Deliverables & progress tracker ? **source of truth**
125     Legend: ? Todo · ? In progress · ? Done · ? Blocked/gated. **One small PR per row**,
branched from `origin/master`, no stacking.
126
127     | ID | Deliverable | Workstream | Depends on | Gated? | Status | PR |
128     |----|-----|-----|-----|-----|-----|----|
129     | P0 | This tracked project doc + inbound pointers | ? | ? | No | ? |
[#41](https://github.com/avidullu/khelsutra/pull/41) · engine
[#275](https://github.com/Khelsutra/badminton-highlight-indexer/pull/275) |
130     | B1 | **SPA i18n scaffold + language switcher, en-only** (react-i18next + detector +
`en` catalog + switcher + persistence + **jsdom render tests** + **browser-language negotiation
tests** $6.5 N1/N3?N7) | B | P0 | No | ? | [#42](https://github.com/avidullu/khelsutra/pull/42)
|
131     | B2 | Extract SPA strings ? `en` catalog (~50 literals + ~16 derived; ICU plural keys);
add **key-parity + no-bare-strings CI** | B | B1 | No | ? |
[#45](https://github.com/avidullu/khelsutra/pull/45) |
132     | A1 | Plain-language copy pass on the `en` catalog (de-jargon; icon+text; non-color
selection) | A | B2 | No | ? | ? |
133     | B3 | Tiny shared `t()` shim + `data-i18n` plumbing for `site/`, en-only (proves shared
catalogs across surfaces) | B | B2 | D-CATALOG-HOME | ? | ? |
134     | A2 | Localized **WhatsApp share** for the reel (SPA + site) | A | B1/B3 | No | ? | ? |
135     | A6 | Plain-language framing for Request-Access GPU/storage questions | A | B3 | No | ?
| ? |
136     | A7 | Fix English-source recording gaps (add a placement diagram) before localizing | A
| ? | No | ? | ? |
137     | B4 | **Noto Sans Kannada + Devanagari** bundle (OFL carve-out) ? **SPA done; reel
watermark `font_path` = engine follow-up* | B | ? | D-FONT ? | ? |
[#46](https://github.com/avidullu/khelsutra/pull/46) |
138     | B5 | **Kannada (`kn`) catalog** ? render ? **SPA done (AI-draft, native review
pending); `site/` rides B3* | B | B2/B4 | D-LOCALES ? | ? |
[#47](https://github.com/avidullu/khelsutra/pull/47) |
139     | A3 | Localize **recording-education** (capture + homepage gist) to kn | A | B3/B5 |
D-LOCALES+D-FONT | ? | ? |
140     | A4 | Mobile-first touch targets + tooltip-text ? visible (SPA) | A | B1 | No | ? | ? |
141     | A5 | Friendly engine-error mapping over raw FastAPI `detail` (SPA, client-side map) |
A | B2 | No | ? | ? |

```

142 | B6 | ****Hindi (`hi`) catalog**** ? render (the extensibility proof ? just a catalog) ?
 SPA done (AI-draft, native review pending) | B | B5 | D-LOCALES ? | ? |
 [#48](https://github.com/avidullu/khelsutra/pull/48) |

143 | A8 | Background-processing reassurance + verdict/status as icon chips (SPA) | A | B2 |
 No | ? | ? |

144 | A9 | Tap-only filter controls alongside the English query bar (SPA) | A | B2 | No | ?
 | ? |

145 | B7 | ***(deferred, cross-repo)* Engine **P2 backend customer-facing localization****
 (report `customer\_` + rejection reasons) | B | engine | ****D-BACKEND-DEPTH, owner-gated**** | ? |
 ? |

146

147 ***** MILESTONE ? SPA fully trilingual (2026-06-21):**** the `web/` Studio SPA now speaks
****en + kn + hi**** end to end (B1?B2?B4?B5?B6, all merged). A ****Kannada-default browser auto-**
renders Kannada**, and the switcher walks **kn ? hi ? en ?** ****proven by a Playwright screencast****
 with `locale: kn-IN` (`khelsutra-i18n-screencast.mp4`; frames captured per locale + `
 lang>` tracked). This meets the owner's success criterion ****on the SPA surface****. ****Still open**
 for the full §9 DoD:**** B3**** (the `site/` marketing surface via the shared `t()` shim ? the
 homepage/capture pages are still English-only), ****A3**** (recording-education in kn/hi), the
****non-tech UX A-items****, and a ****native-speaker review**** of the AI-drafted kn/hi catalogs before
 a wide public launch. Do ****not**** archive until both surfaces are trilingual.

148

149 **## 8. Open questions ? owner (blocking gates) `[design]`**

150 ***Status (2026-06-21): 4 of 5 **LOCKED/ADOPTED** by the owner ? see §2 (D6?D9). Only**
D-BACKEND-DEPTH remains open (and it's deferred, not a v1 blocker).*

151 1. ****D-FONT ? ? LOCKED ? (a) OFL-for-media carve-out, self-host Noto Sans**
Kannada/Devanagari** (counsel may still confirm). An explicit exception to the MIT/Apache/BSD
 code guardrail (fonts = media). Unblocks B4/B5/A3 + the kn/hi reel watermark.

152 2. ****D-LOCALES ? ? LOCKED ? `en` + `kn` for v1, `hi` next, always-fallback to `en`****
 (both 2-form CLDR ? `hi` is catalog-only).

153 3. ****D-BACKEND-DEPTH ? ? OPEN (deferred, not a v1 blocker).**** Recommend ****customer**
 verdict/report `customer\_` + validation-rejection reasons first; technical/operator text stays
 English; phase two****** (engine repo, owner-gated ? tracker row B7).

154 4. ****D-SURFACE-FIRST ? ? ADOPTED ? lead with the SPA scaffold (B1), localize site**
 recording-education early, SKIP the dead static-UI retrofit.******

155 5. ****D-CATALOG-HOME ? ? ADOPTED ? co-locate `locales/` + CI in `khelsutra`**** (a shared
 i18n dir consumed by both the SPA and the site). (The engine `I18N\_PLAN` assumed the indexer
 repo; this is the adaptation.)

156

157 **## 9. Definition of done**

158 - [] ****Automatic language negotiation (§6.5) is green**** ? a ****Kannada browser gets**
 Kannada and a Hindi browser gets Hindi****** (N1/N2); persisted-pref / `?lang` / region-subtag /
 unsupported-fallback / missing-key-fallback all proven (N3?N7) on the SPA, and the static-site
 shim honors `Accept-Language` + `?lang` (N8).

159 - [] ****Manual cross-check (owner machine):**** OS/browser set to Kannada and to Hindi
 each render the respective language on BOTH surfaces, with Noto fonts displaying.

160 - [] **`en + kn + hi` catalogs live and ****rendering**** on BOTH `web/` SPA and `site/`**
 marketing (switcher + persistence + `en` fallback; no raw keys shown).

161 - [] ****Noto Sans Kannada + Devanagari render correctly**** ? owner eyeball on a real
 machine.

162 - [] CI ****key-parity + no-bare-strings**** green; a stub locale flags every gap.

163 - [] ****Recording-education**** (capture + homepage gist) available in `kn` + `hi`.

164 - [] ****Non-tech UX baseline:**** plain-language copy (no raw engine jargon in user-facing
 strings), mobile-usable touch targets, icon+text, WhatsApp share, honest today-vs-roadmap
 preserved through translation.

165 - [] §7 all rows ? (or explicitly deferred with a reason) ? flip ****Status ? DONE**** +
 completion note ? run engine `DOC\_STATUS.md §2` archival checklist ? `git mv` to
 `docs/archives/`.

166

167 **## 10. References**

168 ****Internal code anchors `[verified]`:**** `web/package.json`,
 `web/src/{App.tsx,main.tsx,components/QueryBar.tsx,components/CompileResult.tsx,lib/query.ts}`;
 `site/public/{index,capture,own-model}.html`, `site/server.js`, `site/src/assets.test.js`.
****This repo's docs:**** `docs/PER\_RALLY\_SELECTION.md`, `docs/REQUEST\_ACCESS\_FORM.md`,
 `docs/README.md`, `NEXT\_STEPS.md`. ****Engine design docs (canonical):**** `badminton-highlight-inde`
 `xer/docs/{I18N\_PLAN.md,UI\_PROPOSAL.md,VIDEO\_RECORDING\_GUIDELINES.md,PMF\_MARKET\_RESEARCH.md,PMF\_F`


```

IELD_SIGNALS.md,KHELSUTRA_PRODUCT_OVERVIEW.md,DOC_STATUS.md}`. **External `[researched]`:**
Unicode CLDR plural rules (kn/hi = one/other); react-i18next + i18next-icu; SIL Open Font
License 1.1; Google Noto Sans Kannada/Devanagari.
169
170    ---
171
172    ### Changelog
173    - `2026-06-21` ? created. Isolated the UX + Hindi/Kannada work into one tracked project
(owner directive). Grounded in a multi-agent research+verification pass (web/+site/ string maps,
Indic-l10n prior art, 5 adversarial verdicts). Decisions D1?D5 locked; 5 product/licensing
decisions surfaced as open; $7 tracker seeded with B1 as the recommended first slice.
174    - `2026-06-21` ? owner locked **D6 (en+kn?hi)** + **D7 (OFL Noto carve-out)** and
adopted **D8 (SPA-first)** + **D9 (catalogs in khelsutra)**; only D-BACKEND-DEPTH remains open.
**B1 greenlit ? ? in progress.**
175    - `2026-06-21` ? **B1 opened ? [#42](https://github.com/avidullu/khelsutra/pull/42)**:
SPA i18n scaffold (react-i18next + ICU + detector) + `en` catalog + language switcher + the
SPA's first jsdom render tests. Verified typecheck/79-tests/build green + live in-browser
(switcher persists, en-fallback). Doc/pointer PRs opened: khelsutra
[#41](https://github.com/avidullu/khelsutra/pull/41), engine
[#275](https://github.com/Khelsutra/badminton-highlight-indexer/pull/275).
176    - `2026-06-21` ? **$6.5 Testing strategy added (owner ask, #41):** a rigorous
**automatic-negotiation matrix** (N1?N9) so a Kannada/Hindi **browser** auto-gets the respective
version ? proven at the detection level now (B1), at the content level with the catalogs
(B5/B6), on the static shim (B3), and by a manual font-render eyeball. Wired into the DoD ($9).
B1 (#42) gains the detection tests (N1/N3?N7).
177    - `2026-06-21` ? **? SPA MILESTONE: fully trilingual (owner directive "take it to
completion + screencast").** B2 (#45, full string extraction + key-parity/no-bare-strings CI) ?
B4 (#46, self-host OFL Noto) ? B5 (#47, Kannada) ? B6 (#48, Hindi ? the extensibility proof) all
merged. 99 web tests green incl. N1/N2 browser-detection + B5/B6 render + key-parity.
**Playwright screencast** (`locale: kn-IN`): a Kannada-default browser auto-renders Kannada ?
switch hi ? switch en, each with `` tracked (`khelsutra-i18n-screencast.mp4` + per-
locale frames). kn/hi are **AI-drafted ? native review pending**. **Remaining for full $9 DoD:**
B3 (the `site/` marketing surface), A3 + the UX A-items, native-speaker translation review.
178

```

17. assistant (2026-06-21T09:14:01.500Z)

The storage import is `from backend import storage`. The Glob for `backend/storage/\*\*` found nothing, so it's likely `backend/storage.py`. Let me find and read it to trace the `gcs://` resolution + fetch path.

18. user (2026-06-21T09:14:02.174Z)

No files found

19. user (2026-06-21T09:14:02.300Z)

No files found

20. user (2026-06-21T09:14:08.345Z)

```

backend/storage
---
C:\Users\avidu\Projects\badminton-highlight-indexer\backend\storage\__init__.py
---grep resolve_input_ref def---
backend/storage/__init__.py

```

21. assistant (2026-06-21T09:14:10.706Z)

The Glob tool wasn't matching the directory. Let me read the storage package and find the GCS backend.

22. user (2026-06-21T09:14:11.184Z)

```

1      """Pluggable storage layer: local (default) + S3/GCS first-class buckets + Google Drive.
2
3      URI-driven (see ``ref.parse_uri``). The package import is cheap ? cloud SDKs are
imported
4      lazily, only when a non-local bucket URI is actually used (install the ``storage-s3`` /
5      ``storage-gcs`` extras). The default ``local`` backend needs no third-party deps and
preserves
6      today's behaviour byte-for-byte; ``gdrive://`` likewise needs none ? it reads a locally
MOUNTED
7      Google Drive (Google Drive for Desktop) as plain files.
8
9      Worker/dispatcher entry points speak URIs via the thin helpers ``fetch`` / ``put`` /
10     ``list_uris`` / ``get_storage``; ``local://`` (and bare paths) make ``fetch`` an
identity
11     passthrough. See ``docs/DECOUPLED_COMPUTE/``.
12     """
13     from __future__ import annotations
14
15     import os
16     import tempfile
17     from typing import Any, Dict, Iterator, Optional
18
19     from backend.storage.base import StorageBackend
20     from backend.storage.ref import StorageRef, StorageStat, parse_uri, resolve_input_uri
21
22     __all__ = [
23         "StorageBackend", "StorageRef", "StorageStat", "parse_uri", "resolve_input_uri",
24         "get_storage", "fetch", "put", "list_uris",
25         "acquire_local_input", "input_is_local", "resolve_input_ref",
26     ]
27
28
29     def _storage_dict(storage_config: Any) -> Dict[str, Any]:
30         """Coerce a StorageConfig (Pydantic) / dict / None into a plain settings dict."""
31         if storage_config is None:
32             return {}
33         if hasattr(storage_config, "model_dump"):
34             return storage_config.model_dump() # type: ignore[no-any-return]
35         return dict(storage_config)
36
37
38     def resolve_input_ref(raw: str, storage_config: Any = None) -> StorageRef:
39         """Resolve a user-supplied input (a path or storage URI) into a ``StorageRef`` under
the input
40         config (``input_backend``/``input_root``). Raises ``ValueError`` for an unsupported
URI scheme ?
41         callers at the API/CLI boundary catch that and return a clean client error (never a
500).
42
43         Use ``ref.backend`` to branch local-vs-remote and ``ref.key`` for the resolved LOCAL
path (e.g.
44         ``local://?`` is stripped to the bare path), rather than stat-ing the raw URI
string."""
45         sc = _storage_dict(storage_config)
46         uri = resolve_input_uri(raw, input_backend=sc.get("input_backend", "local"),
47                                input_root=sc.get("input_root", ""))
48         return parse_uri(uri)
49
50
51     def input_is_local(raw: str, storage_config: Any = None) -> bool:
52         """True if ``raw`` resolves (under the input config) to a LOCAL filesystem path ?
i.e. a
53         plain on-disk file the caller can stat/validate directly. False for a bucket / Drive
URI
54         that must be fetched first. Raises ``ValueError`` for an unsupported scheme (same as

```

```

55     ``resolve_input_ref``)."""
56     return resolve_input_ref(raw, storage_config).backend == "local"
57
58
59     def acquire_local_input(raw: str, storage_config: Any = None, *,
60                             scratch_parent: "str | None" = None) -> str:
61         """Resolve ``raw`` (a path or storage URI) to a LOCAL file path ready for
processing.
62
63         Local / bare paths are returned UNCHANGED (identity passthrough, no copy ? today's
behaviour
64         byte-for-byte). A bucket / Drive source (``gs://``/``s3://``/``gdrive://``, or a
bare name under a
65         non-local ``input_root``/``input_backend``) is downloaded into a fresh scratch dir
and the local
66         path returned. The scratch copy is left under the system temp dir (a pure
intermediate,
67         regenerable from the source) and relies on OS temp reclamation ? the same pull
pattern the
68         worker's ``cmd_infer`` already ships."""
69         sc = _storage_dict(storage_config)
70         ref = resolve_input_ref(raw, sc)
71         if ref.backend == "local":
72             return ref.key # identity ? no temp, no copy, byte-identical to passing the
path directly
73         scratch = tempfile.mkdtemp(prefix="rally-input-", dir=scratch_parent)
74         return fetch(ref.uri, os.path.join(scratch, ref.name or "in.mp4"), config=sc)
75
76
77     def _backend_class(name: str):
78         """Lazy-resolve a backend class so cloud SDKs aren't imported until needed."""
79         if name == "local":
80             from backend.storage.local import LocalStorage
81             return LocalStorage
82         if name == "s3":
83             from backend.storage.s3 import S3Storage
84             return S3Storage
85         if name == "gcs":
86             from backend.storage.gcs import GCSStorage
87             return GCSStorage
88         if name == "gdrive":
89             from backend.storage.gdrive import GDriveStorage
90             return GDriveStorage
91         raise ValueError(f"unknown storage backend: {name!r} (expected
local|s3|gcs|gdrive)")
92
93
94     def get_storage(backend: Optional[str] = None, *,
95                     config: Optional[Dict[str, Any]] = None, **overrides) -> StorageBackend:
96         """Construct a backend. ``backend`` (or ``config['backend']``, default 'local')
selects
97         the implementation; its per-backend settings come from ``config[backend]`` merged
with
98         ``overrides``. Settings carry NO secrets ? only endpoints/regions/paths/source-
selectors."""
99         cfg = config or {}
100         name = backend or cfg.get("backend", "local")
101         settings: Dict[str, Any] = dict(cfg.get(name, {}) or {})
102         settings.update(overrides)
103         return _backend_class(name)(**settings)
104
105
106     def fetch(uri: str, dst: "str | os.PathLike[str] | None" = None, *,
107              config: Optional[Dict[str, Any]] = None, overwrite: bool = False) -> str:
108         """Resolve ``uri`` to a local file path. For ``local`` this is an identity

```

```

passthrough
109         (no copy); for s3/gcs it downloads to ``dst``."""
110         ref = parse_uri(uri)
111         return get_storage(ref.backend, config=config).fetch_to_local(ref, dst,
overwrite=overwrite)
112
113
114     def put(local_path: "str | os.PathLike[str]", uri: str, *,
115            config: Optional[Dict[str, Any]] = None, content_type: Optional[str] = None) ->
str:
116         """Upload/copy ``local_path`` to ``uri``; return ``uri``."""
117         ref = parse_uri(uri)
118         get_storage(ref.backend, config=config).put(local_path, ref,
content_type=content_type)
119         return uri
120
121
122     def list_uris(prefix_uri: str, *, config: Optional[Dict[str, Any]] = None) ->
Iterator[str]:
123         """Yield the URI of every object under ``prefix_uri``."""
124         ref = parse_uri(prefix_uri)
125         for r in get_storage(ref.backend, config=config).list(ref):
126             yield r.uri
127

```

23. user (2026-06-21T09:14:12.061Z)

```

__init__.py
__pycache__
base.py
gcs.py
gdrive.py
local.py
ref.py
s3.py

```

24. assistant (2026-06-21T09:14:54.401Z)

The storage layer is fully built ? `acquire\_local\_input` fetches `gs://` URIs via `GCSStorage`, and `/api/process` already calls `resolve\_input\_ref` at submit and `acquire\_local\_input` during execution. This strongly suggests **B-P1** is already implemented in code, exactly as the session prompt suspected. Let me confirm the two crux pieces (URI parsing + path validation) and scout the test surface before committing.

25. user (2026-06-21T09:14:55.356Z)

```

1         """Storage value objects + URI parsing for the pluggable storage layer.
2
3         A ``StorageRef`` is a backend-agnostic pointer to one object. URIs:
4
5         <bare path>                -> local    (today's behaviour; e.g. C:/v/x.mp4,
/data/x.mp4)
6         local://<path>             -> local    (local:///abs or local://rel/p)
7         s3://<bucket>/<key>        -> s3       (AWS + R2 + DO Spaces + MinIO via
endpoint_url)
8         gcs://<bucket>/<key>      -> gcs      (gs:// accepted, normalized to gcs://)
9         gdrive://<path>           -> gdrive   (path under your locally-MOUNTED Google
Drive "My Drive")
10
11         No cloud SDKs are imported here ? this module is pure stdlib so ``import
backend.storage``
12         stays cheap.
13         """
14         from __future__ import annotations

```

```

15
16 import re
17 from dataclasses import dataclass
18 from datetime import datetime
19 from typing import Optional
20
21 _SCHEME_RE = re.compile(r"^([A-Za-z][A-Za-z0-9+.\-]*)://")
22 _KNOWN = ("local", "s3", "gcs", "gdrive")
23
24
25 @dataclass(frozen=True)
26 class StorageStat:
27     """Lightweight object metadata (uniform across backends)."""
28     size: int
29     modified: Optional[datetime] = None
30     etag: Optional[str] = None
31     content_type: Optional[str] = None
32
33
34 @dataclass(frozen=True)
35 class StorageRef:
36     """A backend + a location. ``bucket`` is the S3/GCS bucket; for ``local`` and
37     ``gdrive`` it is
38     empty and ``key`` carries the path (a filesystem path, or a path under the mounted
39     Drive)."""
40     backend: str
41     bucket: str
42     key: str
43     uri: str
44
45     @property
46     def name(self) -> str:
47         return self.key.rstrip("/").rsplit("/", 1)[-1]
48
49 def parse_uri(uri: str) -> StorageRef:
50     """Parse a storage URI (or a bare filesystem path) into a ``StorageRef``.
51
52     A string with no ``scheme://`` prefix (including Windows ``C:\\...`` paths, which
53     have no ``//``) is treated as a local filesystem path ? preserving today's
54     behaviour.
55     """
56     s = str(uri)
57     m = _SCHEME_RE.match(s)
58     if not m:
59         return StorageRef("local", "", s, f"local://{s}")
60     scheme = m.group(1).lower()
61     rest = s[m.end():]
62     if scheme == "gs":
63         scheme = "gcs"
64     if scheme in ("local", "gdrive"):
65         # Path-keyed, no bucket: local FS, or a path under the mounted Google Drive "My
66         Drive".
67         return StorageRef(scheme, "", rest, s)
68     if scheme in _KNOWN:
69         bucket, _, key = rest.partition("/")
70         return StorageRef(scheme, bucket, key, s)
71     raise ValueError(f"unsupported storage URI scheme: {scheme}://{rest} (in {s})")
72
73 def resolve_input_uri(raw: str, *, input_backend: str = "local", input_root: str = "")
74 -> str:
75     """Resolve a user-supplied input reference into a storage URI
76     (COMPUTE_DECOUPLED_SERVING M2).
77

```

```

74     Precedence (first match wins):
75     1. An explicit ``scheme://`` on ``raw`` (``gs://``, ``s3://``, ``local://``,
76     ``gdrive://``) ? as-is.
77     2. An ABSOLUTE local path (``/data/x.mp4``, ``C:\\?``, ``C:/?``) ? as-is (always
78     local).
79     3. A bare/relative name + a non-empty ``input_root`` ? joined under it
80     (``gs://bucket/videos`` + ``x.mp4`` ? ``gs://bucket/videos/x.mp4``).
81     4. A bare/relative name + ``input_backend != "local"`` ? prefixed (``gcs`` +
82     ``bucket/x`` ?
83     ``gcs://bucket/x``).
84     5. Otherwise ? ``raw`` unchanged (a relative/bare LOCAL path ? today's behaviour).
85
86     So the default (``input_backend="local"`` , ``input_root=""``) is an identity
87     passthrough for every
88     path used today, preserving byte-for-byte behaviour.
89     """
90     s = str(raw)
91     if _SCHEME_RE.match(s):          # 1. explicit scheme wins
92         return s
93     # 2. an anchored LOCAL path is always local, never joined. Detect this OS-
94     INDEPENDENTLY ?
95     # os.path.isabs is platform- AND Python-version-dependent (on Windows + py3.13 a
96     POSIX-style
97     # "/data/x" is NOT absolute). Anchored == a leading "/" or "\\", or a Windows
98     drive prefix "X:".
99     if s.startswith(("/", "\\")) or (len(s) >= 2 and s[1] == ":" and s[0].isalpha()):
100         return s
101     if input_root:                   # 3. join a relative name under the configured
102         root
103         return input_root.rstrip("/") + "/" + s.lstrip("/")
104     if input_backend and input_backend != "local": # 4. prefix a bucket/key with the
105         declared backend
106         return f"{input_backend}:{s}"
107     return s                         # 5. bare/relative local path ? unchanged
108

```

26. user (2026-06-21T09:14:56.629Z)

```

53: def validate_input_path(path: str, allowed_root: Optional[str] = None) -> None:
54-     """Reject null bytes and (when an allowed_root is configured) out-of-root inputs.
55-
56-     With no allowed_root (default, single-user local mode) only the null-byte check applies,
57-     so the local 'point it at any of my videos' workflow is preserved. Setting an
58-     allowed_root (a future hosted/tenant deployment) enforces containment.
59-     """
60-     if "\x00" in path:
61-         raise ValueError("video_path must not contain a null byte")
62-     if allowed_root:
63-         resolve_under_root(path, allowed_root)

```

27. assistant (2026-06-21T09:15:22.805Z)

Crux confirmed with file:line evidence ? ****B-P1 is already implemented in code****:

- `parse_uri("gs://?")` normalizes `gs`?`gcs`` backend
([ref.py:60-67](backend/storage/ref.py:60))
- `resolve_input_uri` returns an explicit-scheme URI as-is even with default
`input_backend="local"` ([ref.py:87-88](backend/storage/ref.py:87))
- `/api/process` skips the existence check for non-local backends
([main.py:283](backend/main.py:283)), and `validate_input_path` only does a null-byte check when
`allowed_root` is unset ([paths.py:60](backend/utils/paths.py:60))
- the worker then calls `acquire_local_input` ? `fetch` ? `GCSStorage`` and writes to the API DB
([__init__.py:70-74](backend/storage/__init__.py:70))

So the DRY_RUN \$1 claim ("`/api/process`` accepts only a local `video_path``" / "bridging
`gcs://?API-DB video_id` is net-new") is ****stale****. The real B-P1 deliverable is a regression test

locking this + an honest doc correction. Let me scout the test surface to feed the verification workflow.

28. user (2026-06-21T09:15:26.647Z)

```
=== tests referencing gcs/gs:// or process+uri ===
tests/test_api_endpoints.py
tests/test_input_source.py
tests/test_provider_gemini.py
tests/test_storage.py
tests/test_training_gen0.py
tests/test_worker.py
tests/test_workspace.py
tests/__pycache__/test_api_endpoints.cpython-313-pytest-9.0.3.pyc
tests/__pycache__/test_gemini_api.cpython-311-pytest-9.0.3.pyc
tests/__pycache__/test_gemini_api.cpython-313-pytest-9.0.3.pyc
tests/__pycache__/test_input_source.cpython-313-pytest-9.0.3.pyc
tests/__pycache__/test_provider_gemini.cpython-313-pytest-9.0.3.pyc
tests/__pycache__/test_provider_gemini.cpython-313.pyc
tests/__pycache__/test_storage.cpython-313-pytest-9.0.3.pyc
tests/__pycache__/test_training_gen0.cpython-313-pytest-9.0.3.pyc
tests/__pycache__/test_worker.cpython-313-pytest-9.0.3.pyc
tests/__pycache__/test_workspace.cpython-313-pytest-9.0.3.pyc
```

```
=== #281 / double-module / subprocess server tests ===
```

```
tests/test_api_endpoints.py
tests/test_async_jobs.py
tests/test_base.py
tests/test_compiler.py
tests/test_import_smoke.py
tests/test_job_self_scheduling.py
tests/test_job_worker_loop.py
tests/test_job_worker_module_seam.py
tests/test_native_wasb_runner.py
tests/test_run_telemetry.py
```

```
=== /api/process tests ===
```

```
tests/test_api_endpoints.py
tests/test_async_jobs.py
tests/test_base.py
tests/test_fusion_hybrid.py
tests/test_gemini.py
tests/test_generate_highlights.py
tests/test_job_worker_loop.py
tests/test_motion_segmenter.py
tests/test_output_base.py
tests/test_server_entrypoint.py
```

29. user (2026-06-21T09:15:26.665Z)

```
1      """Google Cloud Storage backend ? first-class object buckets.
2
3      Uses ``google-cloud-storage``, imported lazily so ``import backend.storage`` never pulls
it;
4      install with the ``storage-gcs`` extra. Credentials come from Application Default
Credentials
5      (``GOOGLE_APPLICATION_CREDENTIALS`` pointing at a service-account JSON, or workload
identity) ?
6      never from config. A ``client`` may be injected for tests.
7      """
8      from __future__ import annotations
9
10     from datetime import timedelta
11     from typing import Any, Iterator, Optional
12
```

```

13     from backend.storage.base import StorageBackend
14     from backend.storage.ref import StorageRef, StorageStat
15
16
17     class GCSStorage(StorageBackend):
18         def __init__(self, *, client: Any = None, project: Optional[str] = None,
19                     signed_url_ttl: int = 3600, **_ignored) -> None:
20             self._ttl = signed_url_ttl
21             if client is not None:
22                 self._client = client
23             return
24             from google.cloud import storage # type: ignore # lazy: only when a gcs:// URI
is used
25             self._client = storage.Client(project=project)
26
27         def _blob(self, ref: StorageRef):
28             return self._client.bucket(ref.bucket).blob(ref.key)
29
30         def fetch_to_local(self, ref, dst=None, *, overwrite=False) -> str:
31             if dst is None:
32                 raise ValueError("GCSStorage.fetch_to_local requires a dst path")
33             import os
34             dst = str(dst)
35             if os.path.exists(dst) and not overwrite:
36                 return dst
37             os.makedirs(os.path.dirname(dst) or ".", exist_ok=True)
38             self._blob(ref).download_to_filename(dst)
39             return dst
40
41         def put(self, local_path, ref, *, content_type=None) -> StorageRef:
42             self._blob(ref).upload_from_filename(str(local_path), content_type=content_type)
43             return ref
44
45         def exists(self, ref) -> bool:
46             return bool(self._blob(ref).exists())
47
48         def stat(self, ref) -> StorageStat:
49             b = self._blob(ref)
50             b.reload()
51             return StorageStat(size=int(b.size or 0), modified=b.updated, etag=b.etag,
52                               content_type=b.content_type)
53
54         def list(self, prefix) -> Iterator[StorageRef]:
55             for blob in self._client.list_blobs(prefix.bucket, prefix=prefix.key):
56                 yield StorageRef("gcs", prefix.bucket, blob.name,
f"gcs://{prefix.bucket}/{blob.name}")
57
58         def delete(self, ref) -> None:
59             self._blob(ref).delete()
60
61         def signed_url(self, ref, *, ttl=3600, method="GET") -> str:
62             return self._blob(ref).generate_signed_url(
63                 version="v4", expiration=timedelta(seconds=ttl or self._ttl), method=method)
64

```

30. assistant (2026-06-21T09:15:58.404Z)

The full chain is confirmed and the GCS backend is clean (injectable `client` for tests). I have enough to ground a rigorous verification. Per ultracode, I'll fan out a workflow that (1) independently re-verifies the "B-P1 already works" claim, (2) maps the test pattern to mirror (the #281 guard), (3) sketches the B-P2 SPA, then ****adversarially tries to refute**** the claim before I write a line of code.

31. user (2026-06-21T09:17:04.552Z)

Workflow launched in background. Task ID: wiwhk59cq

Summary: Verify whether /api/process already accepts gcs:// ? API-DB video_id (B-P1), map the test pattern + SPA front-half, adversarially refute the claim, then synthesize the plan
Transcript dir: C:\Users\avidu\claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\9f6357d8-0e35-491a-a830-052e094c7f37\subagents\workflows\wf_e3cc36d8-eca
Script file: C:\Users\avidu\claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\9f6357d8-0e35-491a-a830-052e094c7f37\workflows\scripts\bp1-verify-and-plan-wf_e3cc36d8-eca.js

(Edit this file with Write/Edit and re-invoke Workflow with {scriptPath: "C:\Users\avidu\claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\9f6357d8-0e35-491a-a830-052e094c7f37\workflows\scripts\bp1-verify-and-plan-wf_e3cc36d8-eca.js"} to iterate without resending the script.)

Run ID: wf_e3cc36d8-eca

To resume after editing the script: Workflow({scriptPath: "C:\Users\avidu\claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\9f6357d8-0e35-491a-a830-052e094c7f37\workflows\scripts\bp1-verify-and-plan-wf_e3cc36d8-eca.js", resumeFromRunId: "wf_e3cc36d8-eca"}) ? completed agents return cached results.

You will be notified when it completes. Use /workflows to watch live progress.

32. assistant (2026-06-21T09:17:15.967Z)

The workflow is running. While it works, let me reconcile the engine's git state ? HEAD is on `feat/serving-m4-device-context` (44c9f01) but the session said the engine clean-slate is master@2afd54f. I need to know exactly where `origin/master` is before I cut a B-P1 branch (branching-safety protocol).

33. user (2026-06-21T09:17:20.488Z)

```
=== HEAD vs origin/master ===
cee4baa feat(serving): M4 ? DeviceContext + native WASB CPU-fallback (device="auto", cuda stays strict) (#283)
2afd54f feat(serving): M3 ? configurable output/scratch base (de-hardcode output/, default-OFF) (#282)
41d45fd fix(server): drain async jobs under `python -m backend.main` (double-module hang) (#281)
--- is feat/serving-m4-device-context merged into origin/master? ---
--- commits on HEAD not in origin/master ---
44c9f01 docs(serving): mark $7 tracker M4 ? + changelog (PR #283)
6932d4a feat(serving): M4 ? DeviceContext + native WASB CPU-fallback (device="auto", cuda stays strict)
=== untracked of interest ===
?? artifacts/
?? docs/KHELSUTRA_PRODUCT_OVERVIEW.md
```

34. assistant (2026-06-21T09:17:36.311Z)

Git state reconciled: `origin/master` is at `cee4baa` (#283 M4, squash-merged) ? my local `feat/serving-m4-device-context` branch holds the *pre-squash* commits of that same work, and the #281 async-job fix (41d45fd) is already in master. So B-P1 branches from `origin/master`. The untracked `artifacts/` + `KHELSUTRA\_PRODUCT\_OVERVIEW.md` aren't mine ? I'll stage explicit paths only.

While the workflow runs, let me read the #281 test pattern myself (I'll be writing the mirror) and the conftest fixtures.

35. user (2026-06-21T09:17:37.080Z)

```
1      """"Async job model (DEFERRED_HARDENING_PLAN #16).
2
3      Contract under test:
4      - POST /api/process ? 202 + job_id (fast-fail 4xx still happens at submit time)
5      - GET /api/jobs/{id} ? {status, progress, result, reports, ...}
6      - job `status` = EXECUTION state (queued|running|done|failed); the pipeline's own
```

```

7         verdict (e.g. failed validation) lives in result["status"].
8     - worker crash restores pre-run segments (destroy-AFTER-confirm) and marks the job
failed
9     - startup sweep (fail_stale_jobs) terminalizes jobs orphaned by a restart
10    - migration: fresh DB lands at user_version 2 with a jobs table; v1 DBs upgrade in place
11    """
12    import sqlite3
13    import time
14    from concurrent.futures import ThreadPoolExecutor
15
16    import pytest
17
18    pytest.importorskip("httpx") # TestClient needs httpx; skip cleanly where absent
19    from fastapi.testclient import TestClient
20
21    import backend.main as m
22    from backend.infrastructure.database import Database
23    from backend.config.models import AppConfig
24    from backend.pipeline.validators.base import ValidationResult
25
26
27    class _InlineExecutor:
28        """submit() runs the fn synchronously ? deterministic tests, no threads."""
29
30        def submit(self, fn, *args, **kwargs):
31            fn(*args, **kwargs)
32
33
34    @pytest.fixture
35    def env(tmp_path, monkeypatch):
36        db = Database(str(tmp_path / "t.db"))
37        cfg = AppConfig()
38        monkeypatch.setattr(m, "db_instance", db)
39        monkeypatch.setattr(m, "global_config", cfg)
40        monkeypatch.setattr(m, "_get_executor", lambda: _InlineExecutor())
41        client = TestClient(m.app)
42        return client, db, tmp_path, monkeypatch
43
44
45    def _video(tmp_path, name="m.mp4"):
46        p = tmp_path / name
47        p.write_bytes(b"not-a-real-video-but-hashable")
48        return p
49
50
51    def _pass():
52        return [ValidationResult(validator_name="x", passed=True, message="ok", details={})]
53
54
55    def _fail():
56        return [ValidationResult(validator_name="blur", passed=False, message="blurry",
details={})]
57
58
59    class _FakeSeg:
60        def __init__(self, db, cfg):
61            self.db = db
62
63        def process_video(self, video_id, video_path, *a, **k):
64            sid = self.db.add_play_segment(video_id, 0.0, 5.0)
65            return [{"id": sid, "start_time": 0.0, "end_time": 5.0, "duration": 5.0,
"metadata": {}}]
66
67
68    class _RaisingSeg:

```

```

69     def __init__(self, db, cfg):
70         self.db = db
71
72     def process_video(self, video_id, video_path, *a, **k):
73         self.db.add_play_segment(video_id, 9.0, 10.0) # half-written row pre-crash
74         raise RuntimeError("boom")
75
76
77     # --- migration -----
78
79     def test_migration_fresh_db_is_v3_with_jobs_table(tmp_path):
80         db = Database(str(tmp_path / "fresh.db"))
81         with db._get_connection() as conn:
82             # v5 = per-user model store (PERSONALIZATION_PLAN.md §2); the jobs table arrived
83             assert conn.execute("PRAGMA user_version").fetchone()[0] == 5
84             cols = {r["name"] for r in conn.execute("PRAGMA table_info(jobs)").fetchall()}
85             assert {"id", "video_path", "video_id", "status", "progress", "error", "result"} <=
86             cols
87             assert {"policy", "next_poll_at"} <= cols # v3 self-scheduling columns
88
89     def test_migration_upgrades_v1_db_in_place(tmp_path):
90         path = str(tmp_path / "v1.db")
91         conn = sqlite3.connect(path)
92         conn.execute(
93             "CREATE TABLE videos (id TEXT PRIMARY KEY, filepath TEXT NOT NULL, "
94             "processed_at TIMESTAMP, status TEXT NOT NULL, validation_results TEXT)")
95         # A genuine v1 DB also has candidate_segments (created in the version<1 block);
96         include it # so the v4 personalization ALTER (which adds user_id to that table) has a table to
97         alter.
98         conn.execute(
99             "CREATE TABLE candidate_segments (id INTEGER PRIMARY KEY AUTOINCREMENT, "
100             "video_id TEXT NOT NULL, detector TEXT NOT NULL, start_time REAL NOT NULL, "
101             "end_time REAL NOT NULL, duration REAL NOT NULL, stats TEXT)")
102         conn.execute("PRAGMA user_version = 1")
103         conn.commit()
104         conn.close()
105
106         db = Database(path) # ladder should take it 1 ? 5
107         with db._get_connection() as c:
108             assert c.execute("PRAGMA user_version").fetchone()[0] == 5
109             assert c.execute(
110                 "SELECT name FROM sqlite_master WHERE type='table' AND
111                 name='jobs'").fetchone()
112             # Restore v5 personalization coverage (user_models table + user_id on
113             candidate_segments).
114             # These were dropped in prior edit; add (do not replace) per review.
115             tables = {r[0] for r in c.execute(
116                 "SELECT name FROM sqlite_master WHERE type='table'").fetchall()}
117             assert "user_models" in tables
118             cand_cols = {r["name"] for r in c.execute(
119                 "PRAGMA table_info(candidate_segments)").fetchall()}
120             assert "user_id" in cand_cols
121
122     def test_parity_job_model_with_direct_processing(env, tmp_path, monkeypatch):
123         """Item 5 positive: direct CLI-style path (m._execute_processing) vs job-worker path
124         (submit + _InlineExecutor leading to _run_claimed_job / drain, plus explicit
125         _run_claimed_job)
126         produce equivalent outcomes for the same ProcessRequest shape. Exercises success,
127         Failure, and empty-run branches. Real DB queries using vid (no self-compares or
128         tautologies).
129         """

```

```

126     client, db, tpath, mp = env
127     vid_file = _video(tpath)
128     vid = m.compute_video_id(str(vid_file))
129     mp.setattr(m.registry, "run_all_with_context", lambda *a, **k: (_pass(), {}))
130     mp.setattr(m.segmenter_registry, "get", lambda name: _FakeSeg)
131
132     req = m.ProcessRequest(video_path=str(vid_file), segmenter="x")
133
134     # Direct CLI-style path
135     direct_out = m._execute_processing(req)
136     segs = db.query_play_segments(vid, {})
137     assert len(segs) >= 1, f"direct path must affect segments for vid {vid}"
138     assert direct_out.get("result", {}).get("status") in (None, "success", "processed")
139     or "play_segments" in str(direct_out)
140
141     # Job-worker path via public API submit (exercises job creation + inline executor +
142     internal _run_claimed_job)
143     rj = client.post("/api/process", json={"video_path": str(vid_file), "segmenter":
144     "x"})
145     job_out = client.get(f"/api/jobs/{rj.json()['job_id']}").json()
146     segs_j = db.query_play_segments(vid, {})
147     assert job_out["result"]["status"] == "success"
148     assert len(segs_j) >= 1
149
150     # Explicit job-worker via _run_claimed_job (create + mark running + drive the
151     claimed runner)
152     vid_file2 = _video(tpath, name="m2.mp4")
153     vid2 = m.compute_video_id(str(vid_file2))
154     jid = f"parity-{vid2[:8]}"
155     db.create_job(jid, str(vid_file2))
156     db.mark_job_running(jid)
157     job_row = db.get_job(jid)
158     m._run_claimed_job(job_row)
159     segs2 = db.query_play_segments(vid2, {})
160     assert len(segs2) >= 1, f"_run_claimed_job path must produce segments for vid
161     {vid2}"
162
163     # Failure branch (direct + job submit)
164     mp.setattr(m.registry, "run_all_with_context", lambda *a, **k: (_fail(), {}))
165     req_f = m.ProcessRequest(video_path=str(vid_file))
166     direct_f = m._execute_processing(req_f)
167     rf = client.post("/api/process", json={"video_path": str(vid_file)})
168     jobf = client.get(f"/api/jobs/{rf.json()['job_id']}").json()
169     assert direct_f.get("result", {}).get("status") == "failed" or
170     jobf["result"]["status"] == "failed"
171
172     # Empty-run branch (success verdict but zero play_segments from this execution)
173     class _EmptySeg:
174         def __init__(self, db, cfg):
175             self.db = db
176         def process_video(self, video_id, video_path, *a, **k):
177             return []
178     mp.setattr(m.segmenter_registry, "get", lambda name: _EmptySeg)
179     mp.setattr(m.registry, "run_all_with_context", lambda *a, **k: (_pass(), {}))
180     req_e = m.ProcessRequest(video_path=str(vid_file))
181     de = m._execute_processing(req_e)
182     re = client.post("/api/process", json={"video_path": str(vid_file)})
183     je = client.get(f"/api/jobs/{re.json()['job_id']}").json()
184     assert de.get("result", {}).get("status") in (None, "success")
185     assert je["result"]["status"] == "success"
186     # (reingest safety for prior on empty/fail covered in dedicated tests)
187
188     # --- submit-time fast-fail -----

```

```

185     def test_submit_returns_202_with_job_id(env):
186         client, db, tmp_path, mp = env
187         vid_file = _video(tmp_path)
188         mp.setattr(m.registry, "run_all_with_context", lambda *a, **k: (_pass(), {}))
189         mp.setattr(m.segmenter_registry, "get", lambda name: _FakeSeg)
190         r = client.post("/api/process", json={"video_path": str(vid_file), "segmenter":
"x"})
191         assert r.status_code == 202
192         body = r.json()
193         assert body["job_id"]
194         assert body["status"] == "queued"
195         assert body["poll"].endswith(body["job_id"])
196
197     def test_submit_rejects_null_byte_path(env):
198         client = env[0]
199         r = client.post("/api/process", json={"video_path": "a\x00b.mp4"})
200         assert r.status_code == 400
201
202
203
204
205     def test_submit_missing_file_is_404(env):
206         client, db, tmp_path, mp = env
207         r = client.post("/api/process", json={"video_path": str(tmp_path / "nope.mp4")})
208         assert r.status_code == 404
209
210
211     def test_submit_unknown_segmenter_400_and_no_job_row(env):
212         client, db, tmp_path, mp = env
213         vid_file = _video(tmp_path)
214         mp.setattr(m.segmenter_registry, "get", lambda name: None)
215         mp.setattr(m.segmenter_registry, "list_available", lambda: {"motion": "..."})
216         r = client.post("/api/process", json={"video_path": str(vid_file), "segmenter":
"nope"})
217         assert r.status_code == 400
218         assert "Available segmenters" in r.json()["detail"]
219         with db._get_connection() as c:
220             assert c.execute("SELECT COUNT(*) FROM jobs").fetchone()[0] == 0
221
222
223     # --- end-to-end job lifecycle (inline executor = deterministic) -----
224
225     def test_job_done_with_segments_and_video_id(env):
226         client, db, tmp_path, mp = env
227         vid_file = _video(tmp_path)
228         mp.setattr(m.registry, "run_all_with_context", lambda *a, **k: (_pass(), {}))
229         mp.setattr(m.segmenter_registry, "get", lambda name: _FakeSeg)
230
231         job_id = client.post(
232             "/api/process", json={"video_path": str(vid_file), "segmenter":
"x"}).json()["job_id"]
233         r = client.get(f"/api/jobs/{job_id}")
234         assert r.status_code == 200
235         body = r.json()
236         assert body["status"] == "done"
237         assert body["progress"] == "finished"
238         assert body["error"] is None
239         assert body["result"]["status"] == "success"
240         assert len(body["result"]["play_segments"]) == 1
241         assert "reports" in body # None without obsreport; key must exist per the contract
242
243         vid = m.compute_video_id(str(vid_file))
244         assert body["video_id"] == vid
245         assert len(db.query_play_segments(vid, {})) == 1
246         assert db.get_video(vid)["status"] == "processed"

```

```

247
248
249 def test_job_done_but_pipeline_verdict_failed_preserves_prior(env):
250     client, db, tmp_path, mp = env
251     vid_file = _video(tmp_path)
252     vid = m.compute_video_id(str(vid_file))
253     db.add_video(vid, str(vid_file), "processed", [])
254     db.add_play_segment(vid, 1.0, 2.0) # prior good result
255     mp.setattr(m.registry, "run_all_with_context", lambda *a, **k: (_fail(), {}))
256
257     job_id = client.post("/api/process", json={"video_path":
str(vid_file)}).json()["job_id"]
258     body = client.get(f"/api/jobs/{job_id}").json()
259     # Worker finished fine ? the PIPELINE verdict is the failure.
260     assert body["status"] == "done"
261     assert body["result"]["status"] == "failed"
262     # PR-C semantics survive the async move: prior segments intact.
263     assert len(db.query_play_segments(vid, {})) == 1
264
265
266 def test_worker_crash_marks_failed_and_restores_prior_segments(env):
267     client, db, tmp_path, mp = env
268     vid_file = _video(tmp_path)
269     vid = m.compute_video_id(str(vid_file))
270     db.add_video(vid, str(vid_file), "processed", [])
271     db.add_play_segment(vid, 1.0, 2.0) # prior good result
272     mp.setattr(m.registry, "run_all_with_context", lambda *a, **k: (_pass(), {}))
273     mp.setattr(m.segmenter_registry, "get", lambda name: _RaisingSeg)
274
275     job_id = client.post(
276         "/api/process", json={"video_path": str(vid_file), "segmenter":
"x"}).json()["job_id"]
277     body = client.get(f"/api/jobs/{job_id}").json()
278     assert body["status"] == "failed"
279     assert "RuntimeError" in body["error"]
280     assert body["result"] is None
281     # The half-written 9.0?10.0 row was pruned; the prior 1.0?2.0 row survives.
282     segs = db.query_play_segments(vid, {})
283     assert len(segs) == 1 and segs[0]["start_time"] == 1.0
284
285
286 def test_progress_stages_are_recorded(env):
287     client, db, tmp_path, mp = env
288     vid_file = _video(tmp_path)
289     mp.setattr(m.registry, "run_all_with_context", lambda *a, **k: (_pass(), {}))
290     mp.setattr(m.segmenter_registry, "get", lambda name: _FakeSeg)
291
292     stages = []
293     real = db.set_job_progress
294
295     def spy(job_id, progress):
296         stages.append(progress)
297         return real(job_id, progress)
298
299     mp.setattr(db, "set_job_progress", spy)
300     client.post("/api/process", json={"video_path": str(vid_file), "segmenter": "x"})
301     assert "hashing" in stages
302     assert any(s.startswith("segmenting") for s in stages)
303
304
305 # --- polling endpoint -----
306
307 def test_get_unknown_job_is_404(env):
308     client = env[0]
309     assert client.get("/api/jobs/deadbeef").status_code == 404

```

```

310
311
312 # --- crash recovery -----
313
314 def test_fail_stale_jobs_terminalizes_queued_and_running_only(env):
315     client, db, tmp_path, mp = env
316     db.create_job("q1", "/v.mp4")
317     db.create_job("r1", "/v.mp4")
318     db.mark_job_running("r1")
319     db.create_job("d1", "/v.mp4")
320     db.mark_job_done("d1", {"status": "success"})
321
322     swept = db.fail_stale_jobs("interrupted by server restart")
323     assert swept == 2
324     assert db.get_job("q1")["status"] == "failed"
325     assert db.get_job("r1")["status"] == "failed"
326     assert "restart" in db.get_job("q1")["error"]
327     assert db.get_job("d1")["status"] == "done" # terminal states untouched
328
329
330 # --- real executor (cross-thread SQLite/WAL sanity) -----
331
332 def test_real_executor_completes_job(env):
333     client, db, tmp_path, mp = env
334     vid_file = _video(tmp_path)
335     mp.setattr(m.registry, "run_all_with_context", lambda *a, **k: (_pass(), {}))
336     mp.setattr(m.segmenter_registry, "get", lambda name: _FakeSeg)
337     pool = ThreadPoolExecutor(max_workers=1)
338     mp.setattr(m, "_get_executor", lambda: pool)
339     try:
340         job_id = client.post(
341             "/api/process", json={"video_path": str(vid_file), "segmenter":
"x"}).json()["job_id"]
342         deadline = time.time() + 10.0
343         body = None
344         while time.time() < deadline:
345             body = client.get(f"/api/jobs/{job_id}").json()
346             if body["status"] in ("done", "failed"):
347                 break
348             time.sleep(0.05)
349         assert body is not None and body["status"] == "done"
350         assert body["result"]["status"] == "success"
351     finally:
352         pool.shutdown(wait=True)
353

```

36. assistant (2026-06-21T09:18:15.320Z)

`test\_async\_jobs.py` is the ideal pattern to mirror ? the `env` fixture (fresh DB + `\_InlineExecutor` + `TestClient`), `\_FakeSeg`, and `test\_job\_done\_with\_segments\_and\_video\_id` which already drives `/api/process` ? `/api/jobs/{id}` ? asserts `video\_id` + DB segments. For B-P1 I'll add a `gs://` variant that fakes only the network download (`backend.storage.fetch`) while keeping the real routing.

Let me kick off a baseline test run in the background (the suite is offline/mocked) so any later failure is clearly mine, while the workflow finishes.

37. user (2026-06-21T09:18:17.474Z)

```

run_all_tests.py
FOUND run_all_tests.py
=== quick: any existing gs:// process test? ===
tests/test_api_endpoints.py:376:      ""A gs:// input is NOT 404'd at submit (can't
os.path.exists a bucket object); the worker
tests/test_api_endpoints.py:389:      r = client.post("/api/process", json={"video_path":

```

```

"gs://bucket/videos/x.mp4", "segmenter": "x"})
tests/test_api_endpoints.py:394:     assert not _CaptureSeg.last_path.startswith("gs://")      # a
# fetched LOCAL path, not the URI
tests/test_api_endpoints.py:398:     assert rec is not None and rec["filepath"] ==
"gs://bucket/videos/x.mp4"
tests/test_api_endpoints.py:439:     for bad in ("gs://b/x.mp4", "s3://b/x.mp4",
"gddrive://Sharing/x.mp4"):
tests/test_api_endpoints.py:471:     monkeypatch.setattr(sys, "argv", ["prog", "--video-path",
"gs://bucket/videos/cli.mp4",
tests/test_api_endpoints.py:475:     assert not _CaptureSeg.last_path.startswith("gs://")
# fetched LOCAL path
tests/test_api_endpoints.py:479:     assert rec is not None and rec["filepath"] ==
"gs://bucket/videos/cli.mp4" # source URI kept
tests/test_input_source.py:25:     ("gs://b/x.mp4", "local", "", "gs://b/x.mp4"),
tests/test_input_source.py:26:     ("s3://b/x.mp4", "gcs", "gs://r", "s3://b/x.mp4"),
tests/test_input_source.py:27:     ("local://abs/x.mp4", "gcs", "gs://r", "local://abs/x.mp4"),
tests/test_input_source.py:31:     ("/data/x.mp4", "gcs", "gs://r", "/data/x.mp4"),
tests/test_input_source.py:32:     ("\\\\server\\share\\x.mp4", "gcs", "gs://r",
"C:\\v\\x.mp4"),
tests/test_input_source.py:33:     ("C:\\v\\x.mp4", "gcs", "gs://r", "C:\\v\\x.mp4"),
tests/test_input_source.py:34:     ("C:/v/x.mp4", "gcs", "gs://r", "C:/v/x.mp4"),
tests/test_input_source.py:36:     ("x.mp4", "local", "gs://bucket/videos",
"gs://bucket/videos/x.mp4"),
tests/test_input_source.py:37:     ("sub/x.mp4", "local", "gs://bucket/videos/",
"gs://bucket/videos/sub/x.mp4"), # root trailing slash
tests/test_input_source.py:39:     ("bucket/x.mp4", "gcs", "", "gs://bucket/x.mp4"),
tests/test_input_source.py:51:     for p in ("/data/clip.mp4", "rel/clip.mp4", "clip.mp4",
"gs://b/k", "local://x"):
tests/test_input_source.py:63:     assert input_is_local("gs://b/x.mp4") is False

```

38. user (2026-06-21T09:18:18.355Z)

Command running in background with ID: bah8ybkhr. Output is being written to:
C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-
indexer\9f6357d8-0e35-491a-a830-052e094c7f37\tasks\bah8ybkhr.output. You will be notified when
it completes. To check interim output, use Read on that file path.

39. assistant (2026-06-21T09:18:30.675Z)

****Major discovery**** ? `tests/test\_api\_endpoints.py:376` already tests the gs:// ? `/api/process`
? API-DB path. B-P1 isn't just implemented, it's already **tested**. Let me read that test in full
to see exactly what it proves vs. what the SPA's contract needs (a **queryable** video_id).

40. user (2026-06-21T09:18:31.407Z)

```

355     """Default (local) input must reach the segmenter as the EXACT same path ? identity,
no copy
356     (the M2 zero-behaviour-change guarantee for today's local workflow)."""
357     client, db, tmp_path, mp = env
358     vid = _video(tmp_path)
359     mp.setattr(m.registry, "run_all_with_context", lambda *a, **k: (_pass(), {}))
360     mp.setattr(m.segmenter_registry, "get", lambda name: _CaptureSeg)
361     _CaptureSeg.last_path = None
362     r = client.post("/api/process", json={"video_path": str(vid), "segmenter": "x"})
363     assert r.status_code == 202
364     assert _CaptureSeg.last_path == str(vid)
365
366
367     def test_process_local_missing_file_still_404(env):
368         """A LOCAL input that doesn't exist still fast-fails 404 ? the existence gate is
preserved
369         for local paths (only non-local URIs skip it)."""
370         client = env[0]
371         r = client.post("/api/process", json={"video_path": "/no/such/local/file.mp4"})

```



```

372         assert r.status_code == 404
373
374
375     def test_process_nonlocal_uri_skips_exists_and_fetches(env):
376         """A gs:// input is NOT 404'd at submit (can't os.path.exists a bucket object); the
worker
377         fetches it to a local scratch file and the segmenter sees that LOCAL path, not the
URI."""
378         client, db, tmp_path, mp = env
379
380         def fake_fetch(uri, dst=None, *, config=None, overwrite=False):
381             with open(dst, "wb") as fh:
382                 fh.write(b"fetches-video-bytes")
383             return dst
384
385         mp.setattr(m.storage, "fetch", fake_fetch)
386         mp.setattr(m.registry, "run_all_with_context", lambda *a, **k: (_pass(), {}))
387         mp.setattr(m.segmenter_registry, "get", lambda name: _CaptureSeg)
388         _CaptureSeg.last_path = None
389         r = client.post("/api/process", json={"video_path": "gs://bucket/videos/x.mp4",
"segmenter": "x"})
390         assert r.status_code == 202                                # NOT 404 ? non-local skips
the exists gate
391         job = client.get(f"/api/jobs/{r.json()['job_id']}").json()
392         assert job["status"] == "done"
393         assert _CaptureSeg.last_path is not None
394         assert not _CaptureSeg.last_path.startswith("gs://")      # a fetched LOCAL path, not
the URI
395         assert os.path.basename(_CaptureSeg.last_path) == "x.mp4"
396         # Provenance: the DB record keeps the SOURCE URI (not the ephemeral scratch path).
397         rec = db.get_video(job["video_id"])
398         assert rec is not None and rec["filepath"] == "gs://bucket/videos/x.mp4"
399
400
401     def test_process_unknown_scheme_returns_400(env):
402         """An unsupported scheme (https/ftp/file) is a clean 400, never a 500 ? parse_uri's
ValueError
403         must not escape the request handler."""
404         client = env[0]
405         for bad in ("https://h/x.mp4", "ftp://h/x.mp4", "file:///etc/passwd"):
406             r = client.post("/api/process", json={"video_path": bad})
407             assert r.status_code == 400, f"{bad} -> {r.status_code}"
408
409
410     def test_process_local_scheme_uri_accepted(env):
411         """An explicit local://<existing file> resolves to the bare path: the existence gate
stats the
412         RESOLVED key (not the raw URI) so it is accepted, and the segmenter sees the
stripped path."""
413         client, db, tmp_path, mp = env
414         vid = _video(tmp_path, "ls.mp4")
415         mp.setattr(m.registry, "run_all_with_context", lambda *a, **k: (_pass(), {}))
416         mp.setattr(m.segmenter_registry, "get", lambda name: _CaptureSeg)
417         _CaptureSeg.last_path = None
418         # Forward-slash local:// URI (parse strips the scheme to the bare path).
419         uri = "local://" + str(vid).replace("\\", "/")
420         r = client.post("/api/process", json={"video_path": uri, "segmenter": "x"})
421         assert r.status_code == 202
422         assert _CaptureSeg.last_path == str(vid).replace("\\", "/")
423
424
425     def test_process_local_scheme_uri_missing_still_404(env):
426         """local://<missing> still 404s ? the existence gate runs on the resolved local
key."""
427         client = env[0]

```

```

428         r = client.post("/api/process", json={"video_path": "local:///no/such/ls-
missing.mp4"})
429         assert r.status_code == 404
430
431
432     def test_process_hosted_mode_rejects_nonlocal_uri(env):
433         """Hosted mode (allowed_video_root set) must reject a bucket/Drive URI ? a tenant
can't point
434         the engine at an arbitrary bucket. Pins the security contract instead of an
incidental side effect."""
435         client, db, tmp_path, mp = env
436         root = tmp_path / "tenant"
437         root.mkdir()
438         mp.setattr(m.global_config.security, "allowed_video_root", str(root))
439         for bad in ("gs://b/x.mp4", "s3://b/x.mp4", "gdrive://Sharing/x.mp4"):
440             r = client.post("/api/process", json={"video_path": bad})
441             assert r.status_code == 400, f"{bad} -> {r.status_code}"
442         # A local file UNDER the root is still accepted (passes validation; processing
mocked elsewhere).
443         under = root / "ok.mp4"
444         under.write_bytes(b"hashable")
445         mp.setattr(m.registry, "run_all_with_context", lambda *a, **k: (_pass(), {}))
446         mp.setattr(m.segmenter_registry, "get", lambda name: _FakeSeg)
447         r = client.post("/api/process", json={"video_path": str(under), "segmenter": "x"})
448         assert r.status_code == 202
449
450
451     def test_cli_nonlocal_input_fetches_and_keeps_uri(tmp_path, monkeypatch):
452         """The CLI --video-path block (a second M2 call-site) routes a bucket URI through
fetch: the
453         segmenter sees a LOCAL path and the DB keeps the source URI."""
454         db = Database(str(tmp_path / "cli.db"))
455         cfg = AppConfig()
456         cfg.output.output_dir = str(tmp_path)
457         monkeypatch.setattr(m, "load_app_config", lambda *a, **k: cfg)
458         monkeypatch.setattr(m, "Database", lambda *a, **k: db)
459         monkeypatch.setattr(m, "global_config", cfg)
460         monkeypatch.setattr(m, "db_instance", db)
461         monkeypatch.setattr(m.registry, "run_all_with_context", lambda *a, **k: (_pass(),
{}))
462         monkeypatch.setattr(m.segmenter_registry, "get", lambda name: _CaptureSeg)
463
464         def fake_fetch(uri, dst=None, *, config=None, overwrite=False):
465             with open(dst, "wb") as fh:
466                 fh.write(b"cli-video-bytes")
467             return dst
468
469         monkeypatch.setattr(m.storage, "fetch", fake_fetch)
470         _CaptureSeg.last_path = None
471         monkeypatch.setattr(sys, "argv", ["prog", "--video-path",
"gs://bucket/videos/cli.mp4",
472                                     "--output-dir", str(tmp_path)])
473         m.main()
474         assert _CaptureSeg.last_path is not None
475         assert not _CaptureSeg.last_path.startswith("gs://") # fetched LOCAL path
476         assert os.path.basename(_CaptureSeg.last_path) == "cli.mp4"
477         vid = m.compute_video_id(_CaptureSeg.last_path)
478         rec = db.get_video(vid)
479         assert rec is not None and rec["filepath"] == "gs://bucket/videos/cli.mp4" # source
URI kept
480
481
482     def test_cli_missing_local_returns_without_segmenting(tmp_path, monkeypatch):
483         """A missing LOCAL --video-path fast-fails (prints + returns) before ever touching
the segmenter."""

```

484 db = Database(str(tmp_path / "cli2.db"))

41. assistant (2026-06-21T09:18:59.705Z)

Conclusive: the existing test ``test_process_nonlocal_uri_skips_exists_and_fetches`` ([test_api_endpoints.py:375](tests/test_api_endpoints.py:375)) already proves the whole B-P1 chain ? gs:// accepted (202, not 404), job runs to ``done``, segmenter gets a fetched local path, and ``db.get_video(video_id)`` returns a row with the source URI preserved. Plus ``test_process_hosted_mode_rejects_nonlocal_uri``, ``test_process_unknown_scheme_returns_400``, and the CLI variant. ****B-P1 is implemented AND tested (shipped in M2/#280).****

So the DRY_RUN doc's "net-new" claim is factually wrong. Let me check the test helpers to see whether the SPA's exact ``/api/query`` contract is covered, then reconcile with the workflow.

42. user (2026-06-21T09:19:00.894Z)

```
1      """TestClient coverage for the FastAPI seam (previously zero ? the compiler SyntaxError
2      that broke `import backend.main` shipped green because nothing imported it)."""
3      import os
4      import subprocess
5      import sys
6      from unittest.mock import patch, MagicMock
7
8      import pytest
9
10     pytest.importorskip("httpx") # TestClient needs httpx; skip cleanly where absent
11     from fastapi.testclient import TestClient
12
13     import backend.main as m
14     from backend.infrastructure.database import Database
15     from backend.config.models import AppConfig
16     from backend.pipeline.validators.base import ValidationResult
17
18
19     class _InlineExecutor:
20         """submit() runs the fn synchronously ? async jobs complete before the response
21         is inspected (full lifecycle coverage lives in test_async_jobs.py)."""
22
23         def submit(self, fn, *args, **kwargs):
24             fn(*args, **kwargs)
25
26
27     @pytest.fixture
28     def env(tmp_path, monkeypatch):
29         db = Database(str(tmp_path / "t.db"))
30         cfg = AppConfig()
31         monkeypatch.setattr(m, "db_instance", db)
32         monkeypatch.setattr(m, "global_config", cfg)
33         monkeypatch.setattr(m, "_get_executor", lambda: _InlineExecutor())
34         client = TestClient(m.app)
35         return client, db, tmp_path, monkeypatch
36
37
38     def _video(tmp_path, name="m.mp4"):
39         p = tmp_path / name
40         p.write_bytes(b"not-a-real-video-but-hashable")
41         return p
42
43
44     def _pass():
45         return [ValidationResult(validator_name="x", passed=True, message="ok", details={})]
46
47
48     def _fail():
49         return [ValidationResult(validator_name="blur", passed=False, message="blurry",
```

```

details={}]]
50
51
52     class _FakeSeg:
53         def __init__(self, db, cfg):
54             self.db = db
55
56         def process_video(self, video_id, video_path, *a, **k):
57             sid = self.db.add_play_segment(video_id, 0.0, 5.0)
58             return [{"id": sid, "start_time": 0.0, "end_time": 5.0, "duration": 5.0,
"metadata": {}}]
59
60
61     def test_api_config_ok(env):
62         client = env[0]
63         r = client.get("/api/config")
64         assert r.status_code == 200
65         assert r.json()["sport"] == "badminton"
66
67
68     def test_process_success_with_mocked_segmenter(env):
69         """#16: submit returns 202 + job_id; the (inline) job carries the old response."""
70         client, db, tmp_path, mp = env
71         vid_file = _video(tmp_path)
72         mp.setattr(m.registry, "run_all_with_context", lambda *a, **k: (_pass(), {}))
73         mp.setattr(m.segmenter_registry, "get", lambda name: _FakeSeg)
74         r = client.post("/api/process", json={"video_path": str(vid_file), "segmenter":
"x"})
75         assert r.status_code == 202

```

43. user (2026-06-21T09:19:05.754Z)

```

341: class _CaptureSeg:
342-     """Records the video_path the segmenter actually received (M2: must be a LOCAL path)."""
343-     last_path = None
344-
345-     def __init__(self, db, cfg):
346-         self.db = db
347-
348-     def process_video(self, video_id, video_path, *a, **k):
349-         _CaptureSeg.last_path = video_path
350-         sid = self.db.add_play_segment(video_id, 0.0, 5.0)
351-         return [{"id": sid, "start_time": 0.0, "end_time": 5.0, "duration": 5.0, "metadata":
{}}]
352-
353-

```

44. assistant (2026-06-21T09:19:34.992Z)

`\_CaptureSeg` does persist a segment, so the gs:// job's video *is* queryable ? but the existing test only asserts `db.get\_video(...)` is non-null; it never asserts the SPA's actual loading contract (`POST /api/query {video\_id}` ? rallies). That's the one genuinely additive coverage gap tying the engine test to the dry-run's "SPA loads a bucket-ingested video" promise.

Let me check the baseline test run's progress while the verification workflow finishes.

45. user (2026-06-21T09:19:36.624Z)

<system-reminder>Warning: the file exists but is shorter than the provided offset (1). The file has 1 lines.</system-reminder>

46. assistant (2026-06-21T09:19:48.366Z)

Baseline tests still running. While both background tasks finish, let me complete my ramp-up by

reading the live memory checkpoint and handoff (non-overlapping with the workflow's code/SPA mapping) ? these carry open decisions and recent landings the docs don't.

47. user (2026-06-21T09:19:49.425Z)

<system-reminder>[Truncated: PARTIAL view ? showing lines 1-54 of 106 total (41193 tokens, cap 25000). Call Read with offset=55 limit=54 for the next page, or Grep to find a specific section. Do NOT answer from this page alone if the answer may be further in the file.]</system-reminder>

```
1      ---
2      name: current-state
3      description: "Live micro-checkpoint state for Claude ? updated every checkpoint (not
4      metadata:
5          node_type: memory
6          type: project
7          originSessionId: b13673cc-f087-43b1-9527-35dc0b67b301
8      ---
9
10     # Current state ? updated continuously (see [[working-agreement-checkpoints]])
11
12     > **Scope of this file = LIVE only:** the current handoff + *today's* checkpoints + open
13     PRs/decisions. Keep it read-first-short (?1 page). Older checkpoints are archived in [[current-
14     state-archive]] ? when today's checkpoints age out (next clean-slate day), move them there,
15     don't let them pile up here.
16
17     ***? NEXT SESSION FOCUS (2026-06-21, after the big localization+dry-run session):** SPA
18     is fully **en+kn+hi (DONE)**; a real **UI?engine?reel?download CUJ ran (en+hi)**; a **critical
19     engine bug fixed** (#281). ***? NEXT BUILD (owner-picked, in order): (1) Path B ? point at a
20     bucket video IN THE UI** ? **B-P1 = engine `gcs://` ingest endpoint** (extend `/api/process` for
21     a `video_uri` ? fetch via `backend/storage` ? process; today local-path-only). ? Scouted:
22     `_execute_processing` (main.py:140) ALREADY calls
23     `storage.acquire_local_input(req.video_path,?)` (:159) ? a `gcs://` `video_path` may already
24     fetch ? **verify that first**, then SPA ingest/progress screen (B-P2). **(2) Droplet deploy (opt
25     3):** deploy the FIXED engine to `168.144.126.176` (un-provisioned, 2-core/3.8GB) ? re-run CUJ
26     on Linux. **STILL OPEN (localization DoD):** B3 (`site/` marketing still English-only) + native-
27     speaker review of the AI-drafted kn/hi. **Ramp-up:** `khelsutra/docs/DRY_RUN_CUJ.md` +
28     `FRIENDLY_AND_MULTILINGUAL.md` §7 + the dry-run checkpoint below. **Clean slate:** both repos on
29     master, **0 open PRs**, no worktrees/VMs. khelsutra master `2421557` . engine master `2afd54f`.
30
31
32     **Checkpoint:** 2026-06-21 (NEW SESSION ? UX+l10n focus STARTED) ? ramped up via a
33     research+verification **workflow** (web/+site/ string maps, Indic-l10n prior art, **5
34     adversarial verdicts:** both surfaces **ZERO i18n today**; recording-edu **already shipped** on
35     the homepage gist + `/capture`; `I18N_PLAN` targets the **dead** `backend/api/static` so its
36     P1/P3 wiring retargets to site/+web/). ***? Isolated the work into ONE umbrella tracked project ?
37     `khelsutra/docs/FRIENDLY_AND_MULTILINGUAL.md`** (A: non-tech UX + B: kn/hi l10n; §7 P-tracker =
38     source of truth; **archive when en+kn+hi render on BOTH surfaces**). Pointers wired: khelsutra
39     `docs/README`+`NEXT_STEPS`; engine `I18N_PLAN`+`UI_PROPOSAL` banners + `DOC_STATUS` §3a/§3b rows
40     (engine edits in an **isolated worktree off origin/master** ? engine checkout is a concurrent
41     session's branch; their findings-#1/#2 PR **#274 merged mid-session** ? master `03fa951`).
42     **Owner (AskUserQuestion) locked D6 (en+kn?hi) . D7 (OFL Noto self-host) . D8 (SPA-first) . D9
43     (catalogs in khelsutra); only D-BACKEND-DEPTH deferred. 3 PRs OPEN:** khelsutra docs
44     [#41](https://github.com/avidullu/khelsutra/pull/41); engine pointers
45     [#275](https://github.com/Khelsutra/badminton-highlight-indexer/pull/275); **B1 SPA i18n
46     scaffold + language switcher (en-only) ? [#42](https://github.com/avidullu/khelsutra/pull/42) ?
47     CI GREEN (web+site pass), verified LIVE in-browser** (switcher EN|???? persists; content falls
48     back to en with no kn catalog; active-highlight follows the *chosen* locale ? a bug caught by
49     browser verification + fixed). B1 = react-i18next + ICU + detector + `en` catalog + the SPA's
50     FIRST jsdom render tests (79 tests green); deps MIT/BSD-clean. Engine pointer PR built in an
51     isolated worktree (now removed); engine checkout kept flipping branches (concurrent session
52     active) ? never disturbed. ***? NEXT slices (tracker §7): B2** (full ~50-string extraction + key-
53     parity/no-bare-strings CI) ? **B3** (site `t()` shim) ? **B4/B5** (Noto fonts + kn catalog via
54     Gemini-draft?review). **Also (owner ask, 2026-06-21): the en/kn/hi academy flyers already
55     existed** (a sibling merged PR **#40** ? master `37053b9`; flyer.html / flyer-kn.html / flyer-
56     hi.html served at /flyer{-kn,-hi}) but were UNLINKED ? **linked all three from the homepage
```

footer + added VM-host clean routes ? [#43](https://github.com/avidullu/khelsutra/pull/43)
(MERGED ? master `f121c7f`).* ** Then **owner asked for MORE flyer placement ?
[#44](https://github.com/avidullu/khelsutra/pull/44): "Flyer" in the desktop top-nav + the
3-language links on own-model's foot** (44 site tests green; verified live). (Flyers use Google-
Fonts Noto ? separate from the app's D7 self-host; not refactored.) **Owner #41 comment ? added
a rigorous \$6.5 "automatic language-negotiation" testing strategy** (matrix N1?N9: a kn/hi
BROWSER auto-gets the respective version; DoD gate) *** concrete browser-language detection
tests in B1 [#42]** (N1/N3?N7; 84 web tests green). #41/#42/#275/#44 all merged. ***? THEN
owner: "take it to completion until you can run a screencast with a kannada-default browser ?
right page renders ? switch hi ? en; merge PRs as you go." ? ? DONE for the `web/` SPA.** Merged
in sequence (I merged each on green CI, owner-authorized): **B2 [#45]** (full ~50-string
extraction + key-parity/no-bare-strings CI) . **B4 [#46]** (self-host OFL Noto
Kannada/Devanagari, @fontsource-variable) . **B5 [#47]** (Kannada catalog) . **B6 [#48]** (Hindi
catalog ? the extensibility proof: just a catalog + supported-list entry). **SPA now fully
en+kn+hi; 99 web tests green** (incl. N1/N2 browser-detection, B5/B6 render, key-parity
kn==hi==en, `<html lang>` sync). *** SCREENCAST PRODUCED** (Playwright `locale: kn-IN` ? Kannada
auto-renders ? switch hi ? en; `C:\Users\avidu\Projects\khelsutra-
screencast\khelsutra-i18n-screencast.mp4` (9s) + per-locale frames). Tracker milestone ? [#49].
? **kn/hi are AI-DRAFTED (me) ? native-speaker review pending** before wide public launch. ***
REMAINING for full \$9 DoD (NOT archive-ready):** **B3** (the `site/` marketing surface ?
homepage/capture still English-only ? via the shared `t()` shim) . **A3** recording-education
kn/hi . the **non-tech UX A-items** (A1/A2/A4/A6...) . native review. Playwright installed in
isolated `C:\Users\avidu\Projects\khelsutra-screencast\` (kept out of web/ deps). **THEN owner
polish batch (2026-06-21) ? all merged:** (1) **OFFLINE/self-contained verified + guarded ?
[#50]:** the SPA is air-gap-renderable (Noto = local woff2 via @fontsource, `@font-face`
url?local `/\_assets`, no Google-Fonts/CDN, `web/index.html` no external link); CI guard added so
it can't regress. *(The marketing `site/` still uses Google-Fonts ? hosted/online surface, not
the offline tool; self-host = optional follow-up.)* (2) **SCREENCAST uploaded ? `gs://khelsutra-
rally-corpus/promo/khelsutra-i18n-screencast.mp4`** (+ local). (3) **Localized SAMPLE-QUERY
PROMPT (en/kn/hi) ? [#51]:** `query.hint` = instructive "type `over 10s` for rallies >10s ? or
tap a sample" + `query.try` labels pills "Sample queries"; verified live kn. (4) **\$6.5 testing
rule:** i18n testing is now a MANDATORY merge-gate for any future query-parsing/semantic-
understanding/NL work (parser is English-grammar-only today, A9). **103 web tests green.** NO
open PRs. [[next-session-ux-localization]] [[doc-status-register]] [[working-agreement-tracked-
project-docs]]

17

18 **Checkpoint:** 2026-06-21 (DRY-RUN CUJ + a CRITICAL ENGINE FIX) ? owner: "stitch the UI
to the tool; dry-run a **GCS-bucket video** via the UI ? **reel downloaded local**; repeat in
Hindi. Run locally (opt 1) then droplet (opt 3); report both; identify fixes + merge." Chose
A-now-then-B, **Gemini**. **Tracked goal ? `khelsutra/docs/DRY\_RUN\_CUJ.md`** ([#53]/[#54]).
***? PATH-A LOCAL RUN DONE (en + hi):** isolated engine off origin/master (fresh DB/output
worktree) + `.env` Gemini key ? `gsutil cp gs://khelsutra-rally-
corpus/videos/mahadevpura_smoke_180s.mp4` ? `POST /api/process {gemini}` ? **12 rallies**
(`video\_id 05e0e577?`) ? SPA `npm run dev` (Vite proxies /api?:8000) `video=05e0e577?` ? loaded
12 real rallies ? **Build ? /api/compile ? reel (1080p, 76.5s) DOWNLOADED to
`~/Downloads/khelsutra-dryrun/highlights\_local\_{en,hi}.mp4`** (50.9MB). **Hindi repeat **
(page/cards/"??? ?"/"??? ?" all Hindi; reel downloaded). ***? CRITICAL BUG FOUND
+ FIXED ? engine [#281](https://github.com/Khelsutra/badminton-highlight-indexer/pull/281)
MERGED:** running the documented `python -m backend.main --server` runs main.py as `\_\_main\_\_`
(sets `\_\_main\_\_.db\_instance`), but `job\_worker`/`uploads` read `import backend.main` ? a
SEPARATE module whose `db\_instance` stays None ? the async-job drain crashes silently on
`None.claim\_due\_job()` ? **every /api/process job hangs in `queued` forever**. Latent: unit
tests monkeypatch the executor inline; the UI was verified vs a STUB engine. Fix = dispatch
`\_\_main\_\_`?`backend.main`; + `tests/test\_server\_entrypoint.py` (real subprocess guard, FAILS-
without/PASSES-with). Full engine suite green (2m23s). **Unblocked the run via the single-module
invocation** (`python -c "...; from backend import main as m; m.main()"`). ? **OPT-3 DROPLET
(168.144.126.176) NOT YET RUN:** reachable (Linux, py3.12, ffmpeg, 2-core/3.8GB) but **un-
provisioned** (no `~/rally` engine checkout) ? needs a from-scratch engine install OR a proper
deploy of the FIXED engine (the fix applies there too). ***? NEXT:** opt-3 droplet run (deploy
fixed engine ? re-run CUJ); Path B (in-UI bucket ingest ? net-new HTTP gcs:// ingest +
ingest/progress screens; `api/process` is local-path-only today + the worker writes CSV to the
bucket not the API DB). [[monetization-plan]]

19

20 ***? HANDOFF (2026-06-20) ? KHELSTRA.GURU brand site: shipped, live, portable. Resume
the product threads below.** Separate track from the indexer rally-detection handoff that

follows. Full durable record + ramp-up: [[khelsutra-domain-launch]]. Repo =
****`avidullu/khelsutra`**** (NOT this indexer repo).

21 - ****? START HERE (2026-06-20, latest ? big session, ALL MERGED, NO open PRs).**** Per-rally ****MVP/M0 COMPLETE**** + a full product/promo wave. ****khelsutra master `a7214ad`**** . ****engine (Khelsutra/badminton-highlight-indexer) master has M1 [#268]****.

22 - ****Per-rally MVP (khelsutra `web` SPA):**** P3 `parseQuery()` query bar [#28] (length/order/count/shot_count; greys shot-type/player/score "not available yet" ? never faked; ****13 adversarial-review findings folded in****) . P4 `SelectionFooter`+honest `min\_shot\_count` warning [#30] (`web/src/lib/floor.ts` mirrors `main.py:362-383`; no-shot_count exempt; all-below ? Build DISABLED) . P5 real load `?video=<id>` + Build?`compileReel`?`<video>` preview+download [#31]. Full obs (`query.\*`/`build.\*`/one cid). ****Then: lead-with-LENGTH, shot-count flagged EXPERIMENTAL**** [#33] (cards show `m:ss long`; amber "experimental" chip; duration-led examples; badge `P5`?`alpha`; parseQuery still accepts shot-count, de-featured). Suite ****75 green****.

23 - ****Engine M1 [#268] (GCP/org repo, built in an ISOLATED git worktree off origin/master so the concurrent session's checkout was untouched):**** `GET /api/health` ({status,sport,default_segmenter}) + `POST /api/compile` now returns ****`download\_url`**** (additive; `filepath` unchanged). Full engine suite ****1019 passed/7 skipped****, ruff+mypy clean, CI green (coverage floor incl. torch). ****SPA can adopt `download\_url`/offline-banner as a follow-up.)****

24 - ****GCS round-trip (reuse artifacts, CPU/no-GPU):**** pulled `mahadevpura\_smoke\_180s.mp4` + intervals from the bucket ? ****stitched a curated top-10-longest highlight**** (ffmpeg) ? ****put it back**** at `gs://khelsutra-rally-corpus/highlights/curated\_top10\_longest.mp4` (61s).

25 - ****3 screencasts (Playwright?webm?ffmpeg):**** `promo\_cuj.mp4` (old UI, real footage) . ****`full\_cuj\_real.mp4`**** (NEW duration-first UI + REAL curated reel ? the definitive full-CUJ) ? both in `gs://khelsutra-rally-corpus/promo/` (real footage ? bucket, NOT public repo) . ****`site/public/video/tool-cuj.mp4`**** (NEW UI, ****footage-free**** branded-placeholder reel ? safe to self-host). Local copies in `badminton-highlight-indexer/artifacts/promo/` (gitignored).

26 - ****Site (excite-to-join) [#34/#35]**** after sign-up the reveal now shows ****1)** the tool in action****** (self-hosted footage-free `tool-cuj.mp4`) ****+ (2) a ~30s reel preview**** (`REEL\_SECS` 90?30, gated nocookie YouTube ? real faces stay in the unlisted reel, not the repo); `revealAfterSignup`; coherent CTA/teaser copy; `server.js` serves `\*.mp4` as `video/mp4`; `\*.mp4`/`\*.gif` marked binary. Site suite ****34 green****; ****droplet re-synced + parity GREEN**** (index+mp4 byte-identical CF?VM:8787).

27 - ****? GPU note:**** could NOT spin a fresh GPU VM ? `GPUS\_ALL\_REGIONS=1` slot held by the concurrent session's `rally-gpu` M2 training (did NOT disturb). Validated CUJ against the first-L4-run REAL artifacts instead. A fresh live `api/process` GPU inference stays ****deferred**** until the GPU frees (low-risk existing path). ****Owner observation ? [[machine-pooling-idea]]**** (pool GPUs vs spin/kill per task).

28 - ****Owner decision flags (PUBLIC promo):**** `full\_cuj\_real.mp4`/`promo\_cuj.mp4` show real court footage (faces) ? for public use beyond the gated site, confirm. The footage-free `tool-cuj.mp4` is safe.

29 - ****? COLD-START RUN (2026-06-21, owner-approved; GPU freed ? the other session deleted `rally-gpu`):**** provisioned a FRESH ****L4**** `rally-coldstart` (T4 stocked-out asia-south1-a ? L4 -c), built engine+native-WASB env in ****~7 min**** (`bootstrap --gpu`+`gpu\_setup.sh`+weights; main torch 2.6+cu124 & wasb-env 1.11+cu113 both cuda True), `worker infer` GCS-in?GCS-out, ****VM DELETED (\$0, ~\$0.40 total)****. ****? Cold-start MECHANISM fully validated end-to-end.**** ****? BLOCKER: native WASB returned 0 rallies**** on mahadevpura-1 (full-res), GX030094_proxy, AND the known-good ****smoke clip (gave 22 in the first L4 run)**** ? a ****fresh-build CONFIG/wiring issue, NOT the CUJ****: my minimal reconstructed `config.json` (`indexing.detector\_impl=native`, `default\_segmenter=wasb\_hybrid`, no extra fusion/gating tuning) doesn't reproduce the engine session's TUNED `worker infer detector\_impl=native` invocation. ****? THE missing piece for a repeatable cold-start = the engine session's EXACT native-WASB config/cmd**** (the one that gave 22 rallies). Side finding: `mahadevpura-1.MP4` is full 1080p/~25Mbps; the working clips are downscaled `\*\_proxy` ? feed proxies (or add a downscale step). ? The diagnostic infer ****clobbered**** `outputs/05e0e577?/{intervals.csv,report.json}` ? ****RESTORED to 22 segments**** from local `artifacts/promo/intervals.csv` (reel untouched; junk 0-seg dirs removed). ****? ROOT-CAUSE OF "no logs" FIXED ? engine [#271](https://github.com/Khelsutra/badminton-highlight-indexer/pull/271) MERGED**** (isolated worktree, full gate 1027 green): `backend/worker/\_\_main\_\_.py` ****never configured logging**** ? all pipeline diagnostics (detector choice, "Running native WASB inference", availability failures) were silently dropped at default WARNING. Now: `\_setup\_logging()` at the CLI entry (INFO; `force=True` so it owns the level) + a ****`-v/--verbose`** DEBUG flag (native cmd, frame counts); and `cmd\_infer` emits a ****LOUD WARNING** on a 0-segment success****** (video_id + segmenter + detector_impl + likely causes + "re-run -v"). So ****next cold-**

start, a 0-rally run is fully analysable from the log** ? that's the diagnostic that was missing this time.

30 - **? VIDEO?HIGHLIGHTS REVIEW TABLE (owner asked):** in `gs://khelsutra-rally-corpus/`
only `mahadevpura\_smoke\_180s.mp4` is fed (id 05e0e577?; **22 rallies, 94.2s play, ~48% dead-time cut**) ? reels `outputs/05e0e577?/highlight\_smoke\_l4.mp4` (full ~95s) +
`highlights/curated\_top10\_longest.mp4` (top-10, 61s). **`mahadevpura-1.MP4` (GT 10) +
`GX030094\_proxy.mp4` (GT 41) staged, NOT yet fed** (cold-start blocked above). Local review
copies: `badminton-highlight-indexer/artifacts/review/` (source + both reels). 7 GT
`\*.rallies.csv` in `labels/`.

31 - **OPEN owner-gated (unchanged):** appliance \$10 decisions; move khelsutra site repo
? Khelsutra org (needs Cloudflare GitHub-App reconnect). **Next options:** (1) **get the engine
session's exact tuned native-WASB `worker infer` config/cmd** ? a cold-start that actually
yields rallies+reel (the one missing piece); (2) SPA adopt `download\_url`+offline-banner (uses
M1); (3) engine P7 (`api/videos/{id}/rallies` alias); (4) polished public promo cut. [[machine-
pooling-idea]]

32 - **? SESSION WRAP (2026-06-20 late) ? site fully LIVE on Cloudflare + the DO droplet
twin (`/srv/khelsutra-site`), byte-in-sync (parity green); PRs #1?#26 merged, NO open PRs,
master `9f0942b`.** This session shipped: **per-rally MVP P1+P2** (`web/` SPA:
Vite/React/TS/Tailwind + typed engine client + `RallyGrid` + `useRallySelection`; 16 vitest;
P3 now done ? see START HERE above) . **3 tracked design docs merged**
(`PER\_RALLY\_SELECTION.md` IN-PROGRESS, `LOCAL\_APPLIANCE\_ARCHITECTURE.md`,
`ANNOTATE\_AND\_TRAIN.md`) . live **`/own-model`** page (real annotation-tool screenshots [no
people] in `site/public/img/`, sample reel **gated** behind request-access, "default works out
of the box / BYO-training **optional**" framing, **? KhelSutra home** back-link) . **provider-
portable** `site/scripts/provision-vm.sh`+`teardown-vm.sh` (LIVE-tested: fresh DO VM ?
operational ? state-export ? destroy) + the **`git -c core.autocrlf=false archive`** CRLF fix .
homepage sample reel gated post-signup (random-start `youtube-nocookie` + 90s auto-close) .
`rally-annotator` TRANSFERRED ? `Khelsutra` org (links repointed). **Site CI = 32 green**
(`npm test`) + scheduled live-parity workflow. **GPU deferred to the owner's Windows+WSL box**
(DO has no GPU SKUs + GCP quota=0; no Gemini e2e per owner). **OPEN owner threads:** per-rally
P3+ ; appliance **\$10 decisions** (edge-auth / retention / per-video cost cap / GPU host / P5
bucket-worker bridge); **move sister repos ? `Khelsutra` org as PRIVATE** (verified feasible ?
free org allows 10000 private repos + collaborators, owner stays admin; ENGINE-repo move needs
coordinating w/ active sessions ? see `khelsutra/NEXT\_STEPS.md` "Org & repos"); optional VLC-CSV
annotate walkthrough screenshots. ? Droplet STILL a shared testbed (only `/srv/khelsutra-site`
touched; never the engine `~/rally/MinIO/GCS`). **Ramp-up:** this block ? [[khelsutra-domain-
launch]] ? `khelsutra/NEXT\_STEPS.md` ? the 4 design docs (`PER\_RALLY\_SELECTION` /
`LOCAL\_APPLIANCE\_ARCHITECTURE` / `ANNOTATE\_AND\_TRAIN` / `PORTABILITY`). Droplet SSH:
`~/ssh/khelsutra\_droplet` ? `root@168.144.126.176`; sync recipe in `PORTABILITY.md`.

33 - **? ORG CONSOLIDATION DONE (2026-06-20 late):** transferred **`badminton-highlight-
indexer` (engine), `sports-data-collector`, `sports-obsreport` ? the `Khelsutra` GitHub org**
(all stay **private** ; verified `Khelsutra/<repo>` private). Safe: no open PRs, no forks,
already-private (no visibility change); GitHub **redirects git ops** so active quality-iteration
sessions' push/pull/CI keep working (local files untouched). **Local remotes repointed** to
`https://github.com/Khelsutra/<repo>.git` (engine ls-remote verified). **`rally-annotator` stays
PUBLIC** (OSS MIT plugin the site links to ? do NOT make private). **`khelsutra` (site repo) NOT
moved** ? a transfer breaks the **Cloudflare Workers-Builds git connection** (auto-deploy) until
the CF GitHub App is installed on the `Khelsutra` org + the Worker reconnected (owner dashboard
step) ? **pending owner** . Engine harness/memory path UNCHANGED (keyed to the local dir name
`badminton-highlight-indexer`, not the GitHub URL). ? Could not read org Actions policy (gh
token lacks `admin:org`); engine CI history intact at the new path (last run = success) + free
orgs default Actions-on ? verify CI on the next engine PR; enable in org settings if needed.

34 - **DONE + LIVE this session (PRs #1?#8, all merged, repo clean):** owner bought
khelsutra.guru ? built & launched end-to-end. On **Cloudflare** (Worker `hidden-king-khelsutra`
+ D1 `khelsutra-access` ; **git-connected ? merge `master` auto-deploys** ; CI runs `npm test` =
20 green): landing page + `capture` checklist + OG image + a **segmented Request-Access form**
(academy/individual + GPU + storage Qs + free text) ? persists to D1 **and emails a summary to
avi.dullu@gmail.com per registration** (Email Routing send binding ? VERIFIED live in inbox).
Valid SSL; `hello@` Email Routing verified.

35 - **PORTABLE + proven:** identical form runs on a dependency-free **Node + SQLite** host
(`site/server.js`), **live + browsable on the owner's DO droplet ? http://168.144.126.176:8787**
(systemd `khelsutra-site`, isolated `/srv/khelsutra-site`; SSH key `~/ssh/khelsutra\_droplet`).
Both-hosts compat ENFORCED in CI (`site/src/server.test.js` boots the real server). ? Droplet is
a SHARED storage testbed ? another session runs MinIO 9000/9001 + GCS emu 9023 +
`/root/do\_gcs\_`; STAY CLEAR. Docs: `docs/REQUEST\_ACCESS\_FORM.md`, `docs/PORTABILITY.md`,

`NEXT\_STEPS.md`.

36 - **?? PENDING / next threads (owner ideas this session):** (1) **Marketing video** ? use Grok/Gemini image-gen ? a short video themed for **serious enthusiasts** showing the whole record?reel process ? embed on the site. (2) **Interactive per-rally selection** (big product feature) ? the engine detects rallies ? present them in a UI so the user **picks** which rallies go into the highlight (dynamic, NOT a fixed setting). ? **Design doc MERGED** ? [khelsutra#11](https://github.com/avidullu/khelsutra/pull/11); owner APPROVED ? decisions D8?D11 locked (status IN PROGRESS). ? CI regression-guards MERGED ? [khelsutra#12](https://github.com/avidullu/khelsutra/pull/12): `site/src/deploy-config.test.js` (pins wrangler.toml deploy invariants ? the #5 name-drift class), `site/src/assets.test.js` (static-site content contract), `.github/workflows/parity.yml` (scheduled+dispatch live-parity, NON-blocking); `npm test`?`node --test` discovery; suite 20?28 green; live-parity dispatched ? BOTH hosts byte-in-sync. ? **P1 MERGED** ? [khelsutra#14](https://github.com/avidullu/khelsutra/pull/14) (squash `aac7e0c`): `web` SPA scaffold (Vite6+React19+TS+Tailwind4) + **typed engine API client** (`listRallies`/`compileReel`/`highlightUrl`/`getConfig`+`minShotCount`, grounded+verified; honest types flag `server/receiver/events` as motion fabrications) + **correlation-id** structured logging (`web/src/lib/log.ts`, observability seed, loud failures) + **8 vitest green** + new **web` CI job** (npm ci?typecheck?test?build, gated lockfile); live-verified renders in-browser with ZERO console errors. Isolated (no README/NEXT\_STEPS edits ? no #13 conflict). ? **P2 DONE** ? [khelsutra#21](https://github.com/avidullu/khelsutra/pull/21): **RallyGrid** (honest fields only) + `useRallySelection` hook over pure `lib/selection.ts`+`lib/format.ts` (default ALL-on per D8); 16 vitest green; live-verified (deselect ? footer recompute 6/1:22?5/0:53). **NEXT = P3** `parseQuery()` ? P4 footer+`min_shot_count` warning ? P5 build?compile?download. All on existing endpoints. **Built via a multi-agent design Workflow** (research real engine contract ? 3 approaches ? synthesize) with **every load-bearing claim personally re-verified vs engine source**. Key finding: **MVP needs ZERO new engine endpoints** ? rides existing `POST /api/query` ? select `segment\_ids` ? `POST /api/compile` ? `GET /api/highlights`. Recommended hybrid: selectable rally **grid** (workhorse) + **deterministic `parseQuery()`** (no LLM; length/count/order/shot_count ONLY) writing into ONE shared selection set + sticky compile footer. Honest scope: shot/player/score = roadmap (no column/detector); `server`/`receiver`/`events` = `motion`-only fabrications ? hidden; default `gemini` writes real-ish `shot\_count`; per-rally **preview** blocked until a source/clip stream endpoint lands (deferred milestone = biggest UX-vs-effort lift). Owner decisions RESOLVED (D8?D11, 2026-06-20): (a) default selection **ON** (curate-by-removal); (b) preview-before-select **NOT** a ship-blocker (text-metadata MVP; preview = P9); (c) **job-tethered** MVP (no `GET /api/videos`); (d) `min\_shot\_count` = **warn, don't override** (override deferred ? Launch Report). Doc = `khelsutra/docs/PER\_RALLY\_SELECTION.md`. Pairs with the "ask for a reel" conversational builder already teased on the site. (3) productionize droplet (TLS + `vm.khelsutra.guru`), VM-path SMTP email (currently noop), Always-Use-HTTPS/HSTS, Web Analytics token, `/capture.html`?`/capture` footer link, delete any orphan `khelsutra-site` Worker, clean CF D1 test rows (`deploy-test@`/`email-test@`). Revoke the droplet SSH key when done. [[paper-coauthor-aim]]

[[monetization-plan]]

37 - **?? APPLIANCE DESIGN ? doc SHIPPED for review ?**

[khelsutra#13](https://github.com/avidullu/khelsutra/pull/13) (DRAFT, owner-gated; CI green) ? "locally-installed, UI-powered rally indexer + reel generator" (KheISutra Studio). **Owner target (HEADLINE CUJ):** a **hosted website** + a **NEW GPU-enabled machine** (provider-agnostic: DO GPU droplet / GCE GPU VM / Cloud Run+GPU) **running the decoupled e2e**; from the UI the user picks an input ? **Google Drive / bucket URI (S3/GCS/DO Spaces) / local upload?bucket** ? then **generates + watches + downloads** highlights **completely from the UI**. Framed as a **deployment PROFILE** of the decoupled-compute arch (engine `backend/storage`/`StorageBackend` + `python -m backend.worker {infer|train}` bucket pipeline; PRs #246?#254 on engine master, 936 green; **the CPU droplet 168.144.126.176** already hosts BOTH the site `srv/khelsutra-site` AND the engine `~/rally`, GPU MOCKED via stub). ? **OWNER PRIORITY (2026-06-20): MOVE TO A GPU-ENABLED MACHINE ASAP** ? provider-agnostic: DO GPU droplet / GCE GPU VM / Cloud Run+GPU (storage stays the S3/GCS bucket; GCS = storage, GCE/Cloud Run = compute ? do NOT conflate). Critical-path deps that are NOT mine to do unilaterally: (owner) stand up the GPU compute target + provide the cloud API token/creds ? DO runbook = engine `docs/SETUP\_NEW\_MACHINE.md` \$2/\$9 + `deploy/digitalocean`; **Cloud Run / GCE** need the `backend.worker` CONTAINERIZED (likely a GAP; WASB's torch-1.11 conda env makes the image non-trivial) ? **verify**. Cloud Run+GPU (scale-to-zero) is the strong fit for the per-video-billing economics (engine DECOUPLED_COMPUTE #246). (engine session) build **NativeWasbRunner** (M3.5) for real (non-mock) WASB. The **UI/appliance work runs in PARALLEL** (rides the existing FastAPI: chunked upload, `api/process`?`api/jobs`, `api/query`, `api/compile`, `api/highlights`). ? Engine API has **NO auth (CORS ` `)** ? **EDGE AUTH** required before any public exposure. ? Google **Drive**

ingest may be a `gdrive` STUB** in `backend/storage/` (#249) ? verify before promising. Doc ?
`khelsutra/docs/LOCAL\_APPLIANCE\_ARCHITECTURE.md` (PR pending). The merged per-rally selection
(#11) = ONE screen of this operator console. CUJ/live-test plan is a first-class doc section
(owner asked re: live CUJ replay). **RECOMMENDATION (B+C hybrid):** reverse-proxied **UNIFIED
ORIGIN** ? `web/` SPA behind edge auth, `/api/` ? engine `127.0.0.1:8000` (CORS `\*` moot; gate
= tenancy boundary; bind verified `main.py:638`) ? driving the **EXISTING** FastAPI loop
(upload/process?jobs?query?compile?highlights) with `storage=local`; **compute-as-a-setting**,
bucket/remote-GPU worker = reserved scale path. **Verified-honesty refinements:** native WASB
PARTIALLY wired (`fusion\_hybrid` builds `NativeWasbRunner` :86; `trajectory\_hybrid` refuses
:57-63) + owner-byte-parity-gated; `ai\_max\_workers` default=**5** not 1; chunked-upload
endpoints EXIST (`uploads.py:64/89/123`) but `app.js` never calls them (client-wiring only).
NET-NEW engine endpoints to COORDINATE with the decoupled sessions: `/api/capabilities`,
`/api/videos`, `/api/reels`, `/api/health`. **\$10 owner decisions PENDING:** edge-auth (Caddy
basicauth vs cloudflared+Access), retention defaults, per-video cost ceiling, GPU host choice,
P5 bridge ownership. **Observability = first-class \$5** (correlation-id e2e, structured logs per
path, loud failures, health/metrics) per [[feedback-launch-quality-bar]]. Doc =
`khelsutra/docs/LOCAL\_APPLIANCE\_ARCHITECTURE.md`. **DELIVERY LOCKED (D10, 2026-06-20):** self-
hosted **webserver + UI the user spins up** (NOT a desktop app; supersedes
`DESKTOP\_APP\_PLAN.md`); user points at local storage+GPU/CPU OR cloud, UI drives the tool.
PROVISIONING ACCESS (2026-06-20): `doctl` **INSTALLED v1.162.0** (winget
`DigitalOcean.Doctl`; at `%LOCALAPPDATA%\Microsoft\WinGet\Links\doctl.exe`, on PATH) ? **doctl
AUTHED** (team "Khelsutra Guru", droplet-limit 3; 1 used = `claude-do-rally-test` blr1 CPU
@168.144.126.176). ? **GPU BLOCKED ON BOTH PROVIDERS (verified 2026-06-20):** DO catalog has
NO GPU sizes (17 CPU-only sizes; DO GPU is access-gated + region-limited to NYC/TOR/ATL, not
blr1) AND **GCP `GPUS\_ALL\_REGIONS` quota = 0** (Compute API enabled). Both need an **owner-
initiated, provider-approved access/quota request** ? NOT CLI-doable (GCP quota-increase =
fastest realistic path since gcloud is authed as avi.dullu@gmail.com, project `gen-lang-
client-0306026826`). ? \*\*DO NOT DELETE the old droplet\*\* (`claude-do-rally-test`) ? it's the
SHARED decoupled-session testbed (`~/rally` engine, MinIO 9000/9001, GCS emu 9023, validated DO
Spaces e2e); deleting destroys their env. Site-migration + droplet-deletion is MOOT until a GPU
VM exists anyway. ? **Real product e2e (upload?index?reel) is testable NOW via Gemini (no GPU
needed)** ? GPU is only for the owned/native-WASB model + the train-your-own-model USP (also
pending the engine session's `NativeWasbRunner` M3.5). ? **Annotate+train USP design SHIPPED ?
[khelsutra#15](https://github.com/avidullu/khelsutra/pull/15)** (DRAFT, owner-gated; CI green).
Plan: ONE marketing page `site/public/own-model.html` (Annotate/Train/Setup sections, clones
`capture.html`, links MIT rally-annotator) + a design-tracked owner-gated SPA correction screen
(NO "Train now" button ? training is offline `backend/worker train` CLI). VERIFIED: annotator
CSV (`rally\_number,start\_time,end\_time,ending\_reason,sport,shots\_count`) ingests with **NO
transform** (synth `DictReader row["start\_time"]`, `stub\_runner.py:139` + `gt\_loader`); ONE
prereq = **MD5+stem video?label join** (never re-encode after annotating). Honest: annotator
real+shipping (v1.6.4 MIT VLC plugin); train half = open CLI + roadmap (GPU-free=CPU **mock**
only; real needs GPU + in-flight native WASB; ship-gate `gate\_verdict` `harness.py:98` refuses
commercial ship; `label\_source=human` + `no\_paid\_llm\_training\_labels` attested :178/:189-192);
annotator repo has **ZERO images ? all screenshots TO-CAPTURE** (consented, no minors, NEVER
fabricate); USP = ownership/licensing/privacy/consent/minors-excluded, **NOT accuracy** (no
benchmark vs paid LLM). **GPU DECISION (owner 2026-06-20):** use the **Windows+WSL GPU box** for
real testing; **DEFER the cloud GPU VM**; **NO Gemini e2e now**; **NO droplet
migration/deletion**. Real WASB e2e ? owner's box when run deliberately. **#13 (appliance) + #15
(annotate/train) MERGED** (owner-approved). ? **PROVIDER-PORTABILITY now first-class + LIVE-
TESTED ? [khelsutra#16](https://github.com/avidullu/khelsutra/pull/16) MERGED:**
`site/scripts/provision-vm.sh` (fresh Ubuntu VM on ANY provider ? `git archive` push [private-
repo-safe, no secret on VM] ? Node24 + systemd + firewall + verify) + `teardown-vm.sh` (EXPORT
SQLite state to local backup BEFORE destroy ? clean resume via `RESTORE\_DB`) + PORTABILITY
runbook + CI `bash -n` parse-check. **PROVEN on a throwaway DO droplet (then destroyed):** bare
? operational ? public HTTP 200 ? state-export ? re-provision-with-RESTORE_DB preserved the row
(`rows:2`) ? droplet+temp-key removed. So destroy/migrate-across-providers is now safe +
reproducible (owner principle: "deleting is fine IF we can resume cleanly via scripts"). ? Still
DO NOT delete the shared `claude-do-rally-test` droplet without coordinating the decoupled
session (their `~/rally` engine env reproduces via the ENGINE repo's `SETUP\_NEW\_MACHINE.md`, not
mine). ? **UI PROPOSAL rendered** (visualize mockup): an **"Own your model" section for the
EXISTING landing page ? Annotate-in-VLC (live) / Train-your-own (in-build, honest) /
Yours+private ? links the MIT annotator + a `own-model` setup guide; ? **own-model page LIVE ?
[khelsutra#22](https://github.com/avidullu/khelsutra/pull/22)** (`site/public/own-model.html`:
Annotate=live / Train=in-build / Setup; honest framing, GitHub-linked, **placeholder**
screenshots; `own-model` alias + nav/footer cross-links; deployed to BOTH hosts, parity green;

real screenshots = owner-gated ****P5****). ? ****own-model** reel REUSED as click-to-play ?
[khelsutra#24](https://github.com/avidullu/khelsutra/pull/24):**** owner** asked to reuse the existing reel rather than capture screenshots ? the video is a REEL (engine output), NOT a shot of the annotator UI, so it can't replace a tool screenshot, but it's embedded as a ****branded click-to-play**** (ink poster + play button, NO footage/faces until tap ? `youtube-nocookie` IFrame-API player, ****random start ?90s before end****). Footage hidden until tap (privacy-consistent w/ homepage); verified live both hosts (`facade=4`, static-iframe=0, thumbnail=0, parity green md5 `90de6a9e`). ? ****SUPERSEDED** by
[khelsutra#25](https://github.com/avidullu/khelsutra/pull/25) ? more coherent layout (owner call):**** the reel** (shows players) is now GATED in ONE place only (homepage request-access); the own-model page no longer embeds it. Instead own-model shows the ****real annotation-TOOL screenshots**** (owner provided 2, NO people: `site/public/img/annotate-enable.png` = VLC View?Rally Annotator, `annotate-tool.png` = the Mark START/END/Save dialog) + a one-line ****"request access to see a sample output" pointer + a "you don't have to train anything ? a ready-made default works out of the box; BYO-training is OPTIONAL" note** (keeps non-tech users engaged). Tests assert screenshots-present + no-reel-embed + default/optional framing. Verified live BOTH hosts: images HTTP 200 image/png, md5 `b6350339` identical local/droplet/CF (binary integrity through git-archive), own-model `iframe=0/youtube=0`, parity green. (Reel still lives gated on the homepage: random-start nocookie + 90s auto-close after signup.) ? ****rally-annotator ? Khelsutra org:**** owner created the `Khelsutra` GitHub org; transfer is FEASIBLE (owner = org admin, no name conflict) ? ? ****DONE: TRANSFERRED**** to `Khelsutra/rally-annotator` (org owns it; old `avidullu/?` URL auto-redirects). Site/doc links repointed ?
[khelsutra#23](https://github.com/avidullu/khelsutra/pull/23); deployed to BOTH hosts, `own-model` parity green (md5 `c0af70fb`). Repo `git remote` for any local rally-annotator checkout should be updated to the Khelsutra URL. GPU still deferred to the Windows+WSL box (no cloud GPU; no Gemini e2e). ? ****DEMO REEL ? SIGN-UP-GATED + on BOTH hosts (#17 open ? #18 gated ? #19 fix)**** The owner's UNLISTED reel `ajTCZ7pJW\_o` is NOT shown openly; instead a "Sign up to see it in action" teaser + on a successful request-access submit it reveals a `youtube-nocookie` autoplay embed that ****AUTO-CLOSES in 90s**** (`#reel-reveal` + live countdown;
[khelsutra#18](https://github.com/avidullu/khelsutra/pull/18) merged; verified live submit?reveal on the Node host; `assets.test.js` asserts GATED-not-open; suite 29 green). ****Deployed to BOTH:**** Cloudflare auto-deployed; ****droplet `/srv/khelsutra-site` twin re-synced to master ? PARITY GREEN**** (`OK / 231dd90` . `OK /capture 2ada2cf` ; access.db + self-contained Node 26 preserved; only `srv/khelsutra-site` touched). ? ****CRLF-drift bug found live + fixed ?**
[khelsutra#19](https://github.com/avidullu/khelsutra/pull/19):**** Windows `core.autocrlf=true` overrides `.gitattributes eol=lf` in `git archive` ? CRLF on the VM ? byte parity drift vs CF's LF; `provision-vm.sh` + `parity-check.sh` now pin `git -c core.autocrlf=false archive` (LF cross-platform; no-op on Linux). Privacy now stronger: the clip only plays after signup (not for every visitor/crawler); YT stays owner-controlled. ? ****#20 (random start + join FAQ):**** the reel now starts at a ****RANDOM point ?90s before the end**** (YT IFrame Player API: `getDuration()` ? random `[0,dur-90]` ? `seekTo` ; falls back to 0 if dur unknown; verified live = 35s then 20s across two sign-ups) so people don't all see the same chunk; the ****"How do I join?" FAQ**** now points to the request-access form (`#join` , no `mailto`) ? consistent with the actual signup. Both hosts re-synced LF-safe ? ****parity GREEN**** (`848312?`); site suite ****30 green****.
[[feedback-launch-quality-bar]] ****gcloud` IS authed\*\*** here (avi.dullu@gmail.com, project `gen-lang-client-0306026826` ? likely needs GPU quota + billing verification before GCE/Cloud Run). Owner prefers ****DO first**** for GPU. Creating paid GPU infra = confirm-first every time; I propose exact cmd + hourly cost + teardown, owner OKs. [[monetization-plan]] [[khelsutra-domain-launch]]**

38
39 ---
40
41 ****? HANDOFF (2026-06-20) ? DECOUPLED_COMPUTE: M3.5 native runner MERGED + FIRST REAL GPU RUN DONE on GCP L4 (M1? M3.5? M2?). VM deleted (clean slate). Resume/repro from `deploy/gcp/README.md` + `docs/DECOUPLED\_COMPUTE/DEPLOY\_REPORT\_GCP\_2026-06-20.md`.**** Resumes the engine-side dependency the appliance block above was waiting on.
42 - ? ****M3.5 MERGED ? [#256](https://github.com/avidullu/badminton-highlight-indexer/pull/256):**** `NativeWasbRunner` (`backend/pipeline/detectors/native\_wasb\_runner.py`) runs `wasb\_infer.py` NATIVELY on a Linux GPU box ? no WSL/conda-activate/mnt; `detector\_impl="native"` seam; `device` cuda/mps/cpu (cuda default = byte-parity with WSL); weights/repo/python/scratch env-first
(`WASB\_WEIGHTS` / `WASB\_REPO\_DIR` / `WASB\_PYTHON` / `WASB\_SCRATCH`). `wasb\_infer.py` got `--device` + device-keyed cache (back-compat). ****worker train --trajectory-source detector**** does REAL per-video extraction (fps PROBED via ffprobe + skip-on-miss, video-ext filter, graceful native-misconfig). Suite ****998 green****, ruff+mypy clean, cov 83.7%. Adversarial multi-agent review run;

fixes folded in. ****Parity ? VALIDATED 2026-06-20 on the owner's RTX 2070 (via WSL ? a Linux+CUDA proxy): native==WSL 1194/1195 frames, two native runs byte-identical (deterministic); gate = ~1e-3px TOLERANCE (cuDNN sub-pixel cross-GPU), NOT byte-exact. Native runner now DEFAULTS to streaming (#259). See the validation checkpoint below.****

43 - ? ****DO?GCP PIVOT (ADR-010) + runbook ? [#257](https://github.com/avidullu/badminton-highlight-indexer/pull/257) (docs):** DO has NO Singapore/India GPU (NA-only droplets) ? GPU+bucket move to ****GCP asia-south1 (Mumbai)****. ``deploy/gcp/{README,provision.sh,teardown.sh}`` = full runbook + live-resource inventory + switch-off. ****PROVISIONED LIVE**** (project ``gen-lang-client-0306026826``, billing ****Khelsutra Billing**** ``01420D-419EE0-53D83C``): all APIs on ``bucket`` ****gs://khelsutra-rally-corpus`** (asia-south1) ``SA`` ****`rally-worker`** + key at ``~/.gcp/rally-worker-sa.json`` (local only, gitignored) ``$100`` budget ``rally-100usd-guard`` (50/90/100%).**

44 - ? ****FIRST REAL GPU RUN DONE on GCP (2026-06-20) ? report [#261](https://github.com/avidullu/badminton-highlight-indexer/pull/261) `docs/DECOUPLED\_COMPUTE/DEPLOY\_REPORT\_GCP\_2026-06-20.md`.** GPU quota GRANTED (``GPUS_ALL_REGIONS``?1, via Cloud Quotas REST API ? ****Spot does NOT bypass it, tested****). Spun an ****L4**** (``g2-standard-4``, asia-south1-c; T4 was stocked-out in -a; image ``common-cu129-ubuntu-2204-nvidia-580``), ported via ``bootstrap --gpu`` + the FIXED ``gpu_setup.sh``. ****M1 ?**** ``worker infer detector_impl=native`` on the smoke clip ? ****22 rallies**** (gcs in?out); ****highlight reel**** stitched ? ``gs://khelsutra-rally-corpus/outputs/05e0e577/?highlight_smoke_l4.mp4``. ****M3.5 ?**** native runner on a real cloud GPU; ****cross-GPU parity = sub-pixel (684/1195 frames differ, cuDNN L4-vs-2070) ? gate at the RALLY level, not CSV bytes**** (same-GPU is byte-exact). ****M2 ?**** real-extract train validated IN-FLIGHT (16k+/21k windows extracted on the L4) ? full weights = next session. ****VM DELETED**** (clean slate, \$0; bucket persists). ****Ops Agent fixed**** (root cause: project's ****logging+monitoring APIs were OFF**** ? now enabled + SA monitoring/logging roles + agent installed).

45 - ****BUCKET state**** ``gs://khelsutra-rally-corpus``: ``models/wasb_badminton_best.pth.tar`` ``labels/*.rallies.csv`` (7) ``videos/`` (smoke + mahadevpura-1 + GX030094_proxy) ``outputs/05e0e577/{intervals.csv,report.json,highlight_smoke_l4.mp4}`` ``outputs/parity/native_smoke_l4.csv``.

46 - ****NEXT ITEMS (next session picks up):**** (1) ****Finalize M2**** ? re-port one-shot via the fixed ``gpu_setup.sh`` (repro: ``deploy/gcp/README.md``), run ``worker train --trajectory-source detector`` with ****`indexing.wasb.stream\_video=false`** (disk extraction keeps the GPU fed on LONG videos ? streaming starved it; finding #6) ? weights ``weights/0.1.0-l4/``. (2) ****Full corpus**** for real training: gdown the 4 big videos + held-out Video 1 from the Drive folder ``1sHGL7iacnmM80ChT-p2cKy638gK504T`` ? bucket ``videos/`` (raw GoPro at ``F:[Khelsutra] GoPro Hero Black 12``). (3) ****DO GPU is BLOCKED at the provider**** ? token valid but ****0 GPU Droplet sizes on the account****; enable GPU Droplets in the DO console to use DO credits for training (else training stays on GCP). (4) DO cleanup still owner-side: copy ``/srv/khelsutra-site`` off droplet ``579001481`` ? destroy it ? rotate the exposed Spaces key. (5) Rally-level cross-GPU parity check + the streaming-vs-disk train default. PRs this session: ****#256?#261**** all merged. ``[[monetization-plan]]`` ``[[working-agreement-merging]]`` ``[[paper-coauthor-aim]]``

47

48 ****Checkpoint:**** 2026-06-20 late (NEW SESSION ? DO still BLOCKED ? M2 finalize on a GCP L4; ****ADR-012**** GPU/Cloud-Run RATIFIED; ops-agent fixed+made-permanent; per-video workspace kicked off.) Repo is now under the ****Khelsutra**** org (``github.com/Khelsutra/badminton-highlight-indexer``).

49

50 ****Checkpoint:**** 2026-06-21 ? ****manifest git_sha provenance fix ? [#269](https://github.com/Khelsutra/badminton-highlight-indexer/pull/269) MERGED (squash `c8e60ee` on master; all 7 CI checks green; branch pruned).**** Root cause: a ``git archive`` tarball has no ``.git``, so ``training/gen0/harness.py::git_sha()`` stamped ``git_sha:"unknown"`` on the remote box ? that's why ``gs://khelsutra-rally-corpus/weights/0.1.0-l4/manifest.json`` has no real provenance. Fix = resolution chain ``git rev-parse`` ? ``$RALLY_GIT_SHA`` env ? baked ``GIT_SHA`` file ? ``"unknown"`` (each hex-validated; local git-present path byte-identical/default-OFF-safe). ``deploy/gcp/README.md`` \$4 now bakes the SHA into the archive (``git archive --add-file``); ``/GIT_SHA`` gitignored; +6 unit tests. Suite ****1032 green****, ruff+mypy clean, harness cov 93%. ? Forward-looking only ? the already-shipped ``0.1.0-l4`` manifest still reads ``"unknown"`` (re-stamp needs an owner-gated re-run). ? Branched off ``origin/master`` in a SHARED checkout ? a sibling session had staged ``docs/DECOUPLED_COMPUTE`` ? ``docs/archives/past_projects/?`` renames in the index; committed via explicit pathspec so only my 4 files landed (their staged renames left untouched). ``[[gotcha-shared-checkout-branch-collisions]]`` ``[[working-agreement-merging]]``

51 - ****DO GPU STILL BLOCKED at the provider:**** fresh token present (User env ``DIGITALOCEAN_ACCESS_TOKEN``) + account active, but DO ``/v2/sizes`` = ****0 GPU sizes**** (16 CPU-only ``s-``). Owner must ****ENABLE GPU Droplets in the DO console**** (access-gated, NA-only

tor1/nyc2/atl1) ? NOT API-doable. Balance API can't see the \$200 credit (billing console only).
 ? owner chose the ****GCP fallback**** to finalize M2.

52 - ****? M2 FINALIZED on a GCP L4 ? `asia-southeast1` (Singapore), NOT Mumbai ? report PR [#267](https://github.com/Khelsutra/badminton-highlight-indexer/pull/267).**** asia-south1 was ****fully GPU-stocked-out**** (L4+T4, all 3 zones, $\pm$ local-SSD); quota fine everywhere ? pure capacity ? landed an ****L4 g2-standard-4 in asia-southeast1-a**** (200GB pd-ssd `/scratch`). One-shot port via fixed ``gpu_setup.sh`` (wasb torch1.11+cu113, ****CUDA True on L4**** via PTX-JIT).
``worker train --trajectory-source detector --version 0.1.0-l4``,
``stream_video=false`+`timeout_sec=10800``, on the 2 staged pairs (GX030094_proxy + mahadevpura-1)
 ? ****pushed `gs://?/weights/0.1.0-l4/{weights.json,manifest.json}` (rc=0).**** Held-out ****LOVO n=2 mean F1 0.045**** (AUC 0.782/0.582) ? an ****honest plumbing result**** (real-detector train e2e on cloud GPU + versioned bundle), NOT a quality model; full-corpus train = real quality (next).
``label_source=human``, ``gate.ship=False`` (no floor/uncalibrated/WASB-C3). ****Decode-bound**** (~12 fps cv2 extract on 4 vCPU; ~1.5 hr ~\$1.5). ****VM DELETED**** (not stopped); audit = ****0 instances + 0 disks ? \$0****. ? ``git_sha=unknown`` in the bundle (tarball drops ``git``) ? record the SHA next time. Scripts: ``scratch/m2_deploy/``.

53 - ****ADR-012 RATIFIED ? PR [#263](https://github.com/Khelsutra/badminton-highlight-indexer/pull/263):**** ****GPU-native (L4/T4); TPU DEFERRED**** (benchmark-gated, only a large owned transformer at sustained batch scale); ****Cloud Run + GPU (scale-to-zero) = the supported SERVING model**** (concrete form of ADR-011's deferred M3/M4). Lever = ****NVDEC GPU-decode**** (decode, not GPU math, is the bottleneck). Owner ratified: "this model we should support (Cloud Run)". Doc-only; ****MERGED****.

54 - ****? ADR-013 RATIFIED ? PR [#266](https://github.com/Khelsutra/badminton-highlight-indexer/pull/266):** DROP DigitalOcean ? GCP is the SOLE compute stack (training + serving).****** Owner: "drop DO and document it? record it and bake it for all future work." Researched: DO GPU not enabled (0 sizes) + NA-only + DO charges newer customers' cards for GPU (not credit-funded); Paperspace free GPU subscription-gated (\$39/mo) + bills separately. Baked: ADR-013 + ADR-011 annotated superseded; ``deploy/digitalocean/README`` DEPRECATED (scripts kept = provider-agnostic, reused by GCP); ``deploy/gcp/README`` + ``SETUP_NEW_MACHINE`` + ``HOSTING_PLAN`` + ``docs/README`` + ****`CLAUDE.md` guardrail**** updated. \$200 DO credits ? non-compute only; DO Spaces stays a possible S3 *storage* target. Memory: `[[compute-stack-gcp-only]]`. Shared DO droplet + DO-token retirement = owner-coordinated.

48. assistant (2026-06-21T09:19:55.266Z)

Baseline suite is green (exit 0). Let me confirm the count, then await the workflow's adversarial verdict.

49. user (2026-06-21T09:19:55.987Z)

```

..... [ 38%]
..... [ 44%]
..... [ 51%]
..... [ 57%]
..... [ 63%]
..... [ 70%]
..... [ 76%]
..... [ 83%]
..... [ 89%]
..... [ 95%]
..... [100%]
1126 passed in 79.04s (0:01:19)
=====
RESULT: PASS - all tests green
=====
```

50. assistant (2026-06-21T09:20:29.297Z)

Baseline is ****1126 passed**** on the current tree ? a clean reference point. Here's the key interim finding before the workflow's adversarial pass returns:

? B-P1 is already done ? and already tested

The session's "verify first" hunch was right, and stronger than expected. ``/api/process`` is

****not**** local-path-only:

- ****Submit**** ([main.py:274-284](backend/main.py:274)) resolves the input via ``storage.resolve_input_ref``, and ****explicitly skips**** the existence check for non-local backends ? a ``gs://`` URI returns ****202**, not 404.
- ****gs://` ? `gs` backend**** normalization is real ([ref.py:60](backend/storage/ref.py:60)), and an explicit scheme wins even under default ``input_backend="local"`` ([ref.py:87](backend/storage/ref.py:87)).
- ****The worker**** then calls ``acquire_local_input` ? `fetch` ? `GCSStorage.fetch_to_local``, processes through the normal pipeline, and ****writes to the API DB**** (``add_video` + `play_segments``).
- ****Coverage already ships:****
[``test_process_nonlocal_uri_skips_exists_and_fetches``](tests/test_api_endpoints.py:375) proves `gs://` ? 202` ? job `done` ? segmenter sees a fetched *local* path ? `db.get_video(video_id)`` returns a row with the source URI preserved. Plus hosted-mode rejection, unknown-scheme 400, and the CLI variant.

So the ``DRY_RUN_CUJ.md`` \$1/\$2 claim that ``"/api/process`` accepts only a local ``video_path`` and "bridging `gs://`?a queryable video_id` is ****net-new**** (T12/T14)" is ****factually stale**** ? it predates the M2/#280 URI-aware work that routes through ``/api/process``. ****B-P1** needs zero new engine feature code.** The honest deliverables shift to: (1) correct the doc, (2) optionally pin the one un-covered contract (the SPA's ``/api/query`` load over a bucket-ingested video), and (3) the real Path-B work is ****B-P2** (the SPA ingest/progress screen).

I'm letting the verification workflow finish its 3-skeptic adversarial refutation of this claim + its SPA map for B-P2 before I write anything ? that's the decision-rigor gate. Standing by for it.

51. user (2026-06-21T09:21:24.292Z)

please go ahead and fix all the stale doc you find and ensure the next sessions dont spiral in such territories again only to find the staleness.

52. assistant (2026-06-21T09:22:00.785Z)

Understood ? I'll fix the staleness everywhere it lives and add a durable guard so the next session doesn't re-discover it. Let me first map the ***full*** staleness surface across both repos and the memory steering files (the workflow's SPA map + adversarial verdict will fold in when it lands).

53. user (2026-06-21T09:22:03.254Z)

C:\Users\avidu\Projects\kheIsutra\docs\LOCAL_APPLIANCE_ARCHITECTURE.md:49:- ****Absent (net-new):**** ``GET /api/capabilities``, ``GET /api/videos`` (only single-row ``get_video(id)``), ``GET /api/reels/{video_id}``, any source/proxy streaming endpoint, any ****Dockerfile****, any per-video ****cost ledger**** ``[verified ? grep returns none]``.

C:\Users\avidu\Projects\kheIsutra\docs\LOCAL_APPLIANCE_ARCHITECTURE.md:87:| ****Bucket URI**** (``s3://``, ``gs://``) | the worker already fetches ``--video-uri` ? local ? pipeline`` ``[verified worker/_main_.py:98]``. From the UI, a small ****ingest-to-process**** step (fetch the ref ? ``/api/process``, or enqueue the worker) is ****net-new**** | ****design**** (storage layer ready) |

C:\Users\avidu\Projects\kheIsutra\docs\LOCAL_APPLIANCE_ARCHITECTURE.md:98:| ****Remote GPU worker via bucket**** (reserved) | console CPU + ``storage=s3``; separate GPU host runs ``backend.worker infer`` | bucket creds; a GPU host; a ****net-new orchestration endpoint**** | REAL (remote) | FastAPI loop and worker CLI are ****separate**** today; "remote GPU as a setting" is ****design**** ? selector shown ****disabled**** in MVP |

C:\Users\avidu\Projects\kheIsutra\docs\LOCAL_APPLIANCE_ARCHITECTURE.md:169:| T12 | Ingest from ****bucket URI**** (``s3://`/`gs://``) wiring | T4 | coord | ? | ? |

C:\Users\avidu\Projects\kheIsutra\docs\LOCAL_APPLIANCE_ARCHITECTURE.md:171:| T14 | ***(reserved)*** Bucket-worker bridge ? remote-GPU "compute as a setting" (P5) | T8,T12 | ? owner | ? | ? |

C:\Users\avidu\Projects\kheIsutra\docs\README.md:6: the integration gap (no in-UI bucket ingest yet) + a cheapest-path runbook + open decisions. ``DRAFT``

C:\Users\avidu\Projects\kheIsutra\docs\DRY_RUN_CUJ.md:15:[Omitted long matching line]

C:\Users\avidu\Projects\kheIsutra\docs\DRY_RUN_CUJ.md:20:****Path B ? Full "point at a bucket video IN the UI" (bigger, multi-session).**** Build the in-UI ingest: a net-new ****HTTP bucket-**

ingest endpoint** (`gcs://` ? process ? DB `video\_id`) + the SPA **ingest + progress** screens (appliance S1/S2). The proper appliance front-half.
C:\Users\avidu\Projects\kheIsutra\docs\DRY_RUN_CUJ.md:33:1. **Scope:** Path A (cheapest dry run now) or Path B (build the full in-UI bucket ingest)? *Recommend A first* (proves the multilingual workflow fast), then B as the appliance build.
C:\Users\avidu\Projects\kheIsutra\docs\PER_RALLY_SELECTION.md:56:**Reuse map (existing ? `[verified]`, no change for MVP):** `POST /api/query` (list) ? `POST /api/compile` (stitch) ? `GET /api/highlights/{video\_id}/{filename}` (download); `GET /api/config` (floor) ; `GET /api/jobs/{id}` (carry `result.video\_id`). **All net-new code is the SPA** in `kheIsutra` `web/` (Vite + React + TS + Tailwind, per `web/README.md`), plus a small set of *optional* real-gap engine enablers (§4.3) ? none required for MVP.
C:\Users\avidu\Projects\kheIsutra\docs\PER_RALLY_SELECTION.md:58:**4.1 Components (net-new SPA)**

54. user (2026-06-21T09:22:04.665Z)

docs\archives\past_projects\DECOUPLED_COMPUTE\04-BUILD-AND-TEST-PLAN.md:103:*Accept:* `POST {job\_type:"infer", video\_uri:"?"}` with a valid API key returns a `job\_id`; polling yields

55. assistant (2026-06-21T09:22:46.798Z)

Clear staleness map emerging. The stale claims cluster in **kheIsutra** (DRY_RUN_CUJ.md, LOCAL_APPLIANCE_ARCHITECTURE.md, README.md) plus the **memory steering** that seeds the next session. Note `PER\_RALLY\_SELECTION.md:56` is actually *correct* ("all net-new code is the SPA") ? it already agrees with my finding. Let me read the exact sections to edit, verify the appliance's "absent" endpoint list is still true, and check the other memory files ? then cut a clean branch.

56. user (2026-06-21T09:22:48.310Z)

```
40
41     ## 3. Foundation ? what the engine gives us today  `[verified this session unless
noted]`
42     - **FastAPI loop:** `POST /api/process` (202 + `job_id`) ? poll `GET /api/jobs/{id}` ?
`POST /api/query` ? `POST /api/compile` ? `GET /api/highlights/{video_id}/{filename}`
(FileResponse) `[main.py]`. **Chunked upload exists** ? `POST /api/upload/init` . `PUT
/api/upload/{id}/part/{i}` . `POST /api/upload/{id}/complete` . `DELETE /api/upload/{id}`
`[verified uploads.py:64,89,123,147]` ? but the bundled `app.js` **never calls them** (it only
hits `/api/process` + `/api/compile` `[verified app.js:88,373]`). So browser upload is **client-
wiring only**, no engine change.
43     - **In-process worker:** one module-global `ThreadPoolExecutor(max_workers=1)`
`[verified job_worker.py:37]` ? strictly **one job at a time** ; the DB `jobs` table is the
clock; `fail_stale_jobs` sweeps queued/running on boot. Gemini chunk concurrency is a
**separate** knob `indexing.ai_max_workers` (**default 5** `[verified models.py:183]`; set to 1
to serialize) ? it is **not** already 1.
44     - **Storage:** `storage.backend` default = "local" `[verified models.py:298]`;
`LocalStorage.fetch_to_local` is **identity** `[verified local.py:33]`, `put` no-ops on
`src==dst` ? so on a single box the worker's pull?pipeline?push **collapses to read-local/write-
local**. URI scheme `parse_uri` `[verified ref.py]`: bare/`local://` ? local; `s3://`
(AWS/R2/**DO Spaces**/MinIO, by endpoint) ; `gcs://`/`gs://` ; `gdrive://` ? **follow-up stub
(emulated only)**. Credentials never in config ? boto3 default chain / `~/aws` `[verified
s3.py]`.
45     - **Compute seam:** `indexing.detector_impl` ? `{auto=WSL, native=Linux-native,
`stub`=CPU mock}; default "auto" `[verified models.py:129]`, resolved in
`trajectory_hybrid._resolve_runner` `[verified :76-85]`. **Native WASB is in-flight on branch
`feat/native-wasb-runner`:** `native_wasb_runner.py` **exists** `[verified]`;
**`fusion_hybrid._build_native_runner` returns `NativeWasbRunner`** `[verified
fusion_hybrid.py:86]` while **`trajectory_hybrid._build_native_runner` still raises** `[verified
:57-63]`. Acceptance gate = **owner byte-parity** (WSL CSV == native CSV) on a real GPU box ?
not exercisable on the droplet or in the offline suite. Treat native as **not production-
verified**.
46     - **Worker CLI:** `python -m backend.worker {infer|train}` pulls a `--video-uri` ?
pipeline ? pushes `outputs/<video_id>/{intervals.csv,report.json}` `[verified
worker/__main__.py:88,149]`. The validated GPU-free/key-free combo: `detector_impl=stub` +
```

```

--set indexing.skip_ai_handoff=true` + `storage.s3`.
47 - **Gating:** cloudflared Tunnel (outbound-only, no inbound ports) + Cloudflare Access
email-OTP; backend trusts `Cf-Access-Authenticated-User-Email` `[HOSTING_PLAN.md]`. `GET
/api/health` is **expected but does not exist** `[HOSTING_PLAN.md:42]`.
48 - **Droplet reality:** `168.144.126.176`, **CPU-only, NO GPU**, region `blr1`; co-hosts
the marketing site (`/srv/khelsutra-site`, node `:8787`) **and** the engine (`~/rally`); ran the
full train+infer e2e against DO Spaces with the **GPU mocked** `[SETUP_NEW_MACHINE.md,
DECOUPLED_COMPUTE/HANDOFF.md]`. Shared testbed: MinIO `9000/9001` + GCS emulator `9023` also run
there.
49 - **Absent (net-new):** `GET /api/capabilities`, `GET /api/videos` (only single-row
`get_video(id)`), `GET /api/reels/{video_id}`, any source/proxy streaming endpoint, any
**Dockerfile**, any per-video **cost ledger** `[verified ? grep returns none]`.
50
51 ## 4. Design / architecture
52
53 ### 4.1 Topology & request path
54 ONE box: **edge gate** (cloudflared+Access or Caddy basicauth) ? **reverse proxy /
static host** serving the `web/` SPA bundle and proxying `/api/*` ? engine **FastAPI
`127.0.0.1:8000`** ? **LocalStorage** ? **detector seam** (Gemini/stub today; native WASB on a
GPU host). The firewall blocks inbound `8000` + the test ports `9000/9001/9023`. The marketing
site stays a **separate** systemd unit on a **separate** subdomain.
55
56 ```
57 PUBLIC INTERNET
58 ? (1) EDGE GATE ? the ONLY ingress; auth lives HERE, not in the app
59 ? cloudflared Tunnel + Cloudflare Access (no inbound ports) OR Caddy :443 +
basicauth
60 ?
61 ?????????????????????????????????????????????????????????????????????????????????
62 ? ONE BOX firewall BLOCKS inbound 8000 + 9000/9001/9023 ?
63 ? (2) reverse proxy / static host (Caddy/cloudflared) ?
64 ? /, /assets/* ? web/ React SPA /api/*, /api/highlights/* ? :8000 ?
65 ? ? ONE ORIGIN to the browser ? engine CORS '*' is irrelevant ?
66 ? ? ?
67 ? (3) ENGINE FastAPI (host=127.0.0.1:8000, HARD-bound [main.py:638]) ?
68 ? control: jobs table + ThreadPoolExecutor(max_workers=1) drain ?
69 ? data: uploads.py chunked upload · /api/compile · /api/highlights ?
70 ? ? ?
71 ? (4) STORAGE = LocalStorage (default) fetch=identity · put=no-op(src==dst) ?
72 ? ? ?
73 ? (5) COMPUTE seam: gemini (cloud, key) · stub (CPU mock) [droplet] ?
74 ? native WASB (GPU host) · auto (owner Win+WSL) ?
75 ?????????????????????????????????????????????????????????????????????????????????
76 RESERVED SCALE PATH (out of MVP): console "compute target" selector ? storage=s3 (DO
Spaces)
77 ? SEPARATE GPU host runs `python -m backend.worker infer --video-uri s3:///` ?
outputs/<vid>/ back
78 (needs a NET-NEW orchestration endpoint reflecting outputs into the jobs/poll
contract)
79 ```
80
81 ### 4.2 Ingest sources (the headline) ? one menu, one normalized video object
82 All sources normalize to **"a video the pipeline can read"** ? a local path (MVP) or a
`StorageRef` (scale):
83
84 | Source | Path today | Status |
85 |---|---|---|
86 | **Local upload** | SPA chunks the file ? `max_part_mb` via the existing
`/api/upload/init\|part\|complete` `[verified uploads.py]` ? server path ? `POST /api/process`.
(Scale: also `StorageBackend.put` to `s3:///`/`gcs:///`) | **wire client only** (endpoints exist)
|
87 | **Bucket URI** (`s3:///`, `gs:///`) | the worker already fetches `--video-uri` ? local ?
pipeline `[verified worker/___main__.py:98]`. From the UI, a small **ingest-to-process** step
(fetch the ref ? `/api/process`, or enqueue the worker) is **net-new** | **design** (storage
layer ready) |

```



```

88      | **Google Drive** (`gdrive://`) | `parse_uri` recognizes it, but the gdrive backend is
an **emulated stub** `[verified ref.py, DECOUPLED_COMPUTE research]` | **stub ? roadmap;** MVP
requires a Drive?bucket pre-step; do **not** present Drive as live until the backend is real |
89
90      ### 4.3 Compute placement matrix (one UI, surfaced from `/api/capabilities`)
91      | Placement | Runs | Needs | Real/mock | Honest caveat |
92      |---|---|---|---|---|
93      | **CPU droplet ? gemini** (MVP target) | `segmenter=gemini`, in-proc worker |
`GEMINI_API_KEY`; network; **no GPU** | **REAL** | Cloud brain: uploads ?1280px chunks to Google
? **must be in consent copy**; **no per-video cost ceiling** today |
94      | **CPU droplet ? stub** | trajectory hybrid + `detector_impl=stub`+`skip_ai_handoff` |
nothing | **MOCK** | not real tracking ? UI must label "stub (CPU mock ? not real detection)";
demo/regression only |
95      | **CPU droplet ? motion** | `segmenter=motion` | nothing | REAL, low-quality | only no-
key/no-GPU real segmenter; never surface `server/receiver/events` (fabricated) |
96      | **GPU host ? native WASB** (in-flight) | `wasb`+`detector_impl=native` ?
`NativeWasbRunner` (in-proc torch) | a **GPU host ? DO GPU droplet / GCE GPU VM / Cloud
Run+GPU**; `WASB_WEIGHTS` | **REAL**, on-device | partially wired (fusion builds it, trajectory
refuses); **owner byte-parity gate**; **Cloud Run/GCE need a containerized worker ? no
Dockerfile exists, and WASB's torch-1.11 conda env makes the image non-trivial** |
97      | **Owner Win+WSL ? auto** | `wasb`+`detector_impl=auto` (WSL2+conda) | Windows+WSL2+GPU
| REAL (today) | Windows-only transport; the working real-GPU path for owner verification |
98      | **Remote GPU worker via bucket** (reserved) | console CPU + `storage=s3`; separate GPU
host runs `backend.worker infer` | bucket creds; a GPU host; a **net-new orchestration
endpoint** | REAL (remote) | FastAPI loop and worker CLI are **separate** today; "remote GPU as
a setting" is **design** ? selector shown **disabled** in MVP |
99
100     **Provider note:** **Cloud Run + GPU** (scale-to-zero, pay-per-job) is the strongest fit
for the engine's own **per-video-billing** economics (`DECOUPLED_COMPUTE` #246) ? gated on the
**containerization feasibility check** above.
101

```

57. user (2026-06-21T09:22:49.227Z)

```

158     | T1 | Mock engine + capability fixtures + browser-replay CUJ scripts (L1) | T0 | no | ?
| ? |
159     | T2 | Appliance shell: Caddyfile + systemd units; firewall; **edge auth**; rotate key
(P1) | ? | owner | ? | ? |
160     | T3 | L3 live posture test (raw `:8000`/`:900x` refused from outside) | T2 | no | ? | ?
|
161     | T4 | S1 ingest (local upload wired) + segmenter picker | T1 | no | ? | ? |
162     | T5 | S2 progress (poll + one-job + restart-failure) | T4 | no | ? | ? |
163     | T6 | S3 embed **RallyPicker** + compile + download | T4 | no | ? | ? |
164     | T7 | *(engine)* `GET /api/capabilities` + `GET /api/health` + banner | T1 | coord | ?
| ? |
165     | T8 | *(engine)* `GET /api/videos` + `GET /api/reels/{video_id}` (S4 reload-survival) |
? | coord | ? | ? |
166     | T9 | S5 Settings (config+capabilities; disabled compute/storage target) | T7,T8 | no |
? | ? |
167     | T10 | Observability pass: correlation id end-to-end + structured logs + metrics ($5) |
T4 | no | ? | ? |
168     | T11 | Retention sweep for `output/`+`uploads/` | ? | owner | ? | ? |
169     | T12 | Ingest from **bucket URI** (`s3://`/`gs://`) wiring | T4 | coord | ? | ? |
170     | T13 | **GPU host stand-up** (provider-agnostic) + wire `detector_impl=native`; owner
byte-parity (P4) | branch | ? owner | ? | ? |
171     | T14 | *(reserved)* Bucket-worker bridge ? remote-GPU "compute as a setting" (P5) |
T8,T12 | ? owner | ? | ? |
172
173     **Owner priority (2026-06-20): T13 (GPU host) is elevated to run in parallel from the
start ? owner provisions the host + token; the engine session lands native WASB; the UI MVP
does **not** block on it (ships on Gemini).
174
175     ## 10. Open questions ? owner / external

```

58. user (2026-06-21T09:22:50.102Z)

```

1      # khelsutra docs index
2
3      ## Tracked work (this repo)
4      - **`DRY_RUN_CUJ.md`** ? **? active goal (owner directive 2026-06-21):** the UI drives
the tool on a
5          **GCS-bucket video** ? review/select ? compile ? **reel downloaded local**, verified
in Hindi. Grounds
6          the integration gap (no in-UI bucket ingest yet) + a cheapest-path runbook + open
decisions. `DRAFT`
7          (owner-gated). Milestone of `LOCAL_APPLIANCE_ARCHITECTURE.md`.
8      - **`FRIENDLY_AND_MULTILINGUAL.md`** ? **? active focus (owner directive 2026-06-21):**
make the
9          product super user-friendly for non-tech-savvy coaches/parents **and** localize it in
Hindi +
10         Kannada, across both surfaces (`web/` SPA ? react-i18next, `site/` ? a shared `t()`
shim). The
11         implementation project for the engine's `I18N_PLAN.md` (l10n) + the UX half of
`UI_PROPOSAL.md`.
12         `DRAFT` (owner-gated). Archived when en+kn+hi render on both surfaces.
13     - **`LOCAL_APPLIANCE_ARCHITECTURE.md`** ? the single-box, UI-powered rally indexer +
reel
14         generator (KhelSutra Studio): the `web/` SPA drives the full ingest ? index ? reel
loop on one

```

59. user (2026-06-21T09:22:51.046Z)

No files found

60. assistant (2026-06-21T09:23:28.069Z)

Confirmed: `/api/videos`, `/api/reels`, `/api/capabilities` are genuinely still absent (grep returns none) ? so that list stays. But I found **another stale claim**: LOCAL_APPLIANCE line 47 says `GET /api/health` "does not exist" ? it **was** added in M1 #268 ([main.py:127](backend/main.py:127)). And line 87 / T12 frame bucket-URI ingest as a net-new **endpoint**, which is wrong.

Let me read the remaining memory steering files and cut the khelsutra branch from `origin/master`.

61. user (2026-06-21T09:23:29.041Z)

```

1      ---
2      name: next-session-ux-localization
3      description: Consolidated handoff/plan for the NEXT session ? focus SOLELY on (A) making
the UI super user-friendly for non-tech-savvy users and (B) localizing in Hindi + Kannada.
Grounded in existing repo docs (owner directive 2026-06-21).
4      metadata:
5          node_type: memory
6          type: project
7          originSessionId: 577fd4df-ed3f-43d3-854d-d63d732e2e07
8      ---
9
10     **Owner directive (2026-06-21, session wrap):** the next session focuses SOLELY on
making the product super user-friendly (non-tech-savvy) AND localized (Hindi + Kannada)**.
*Research first using the existing repo docs, THEN craft the plan* (don't reinvent ? there's
prior design). The end goal: **a non-tech-savvy coach/parent reads the page, confidently sets up
recording, runs the tool, and gets highlights ? in their own language.**
11
12     ## ? NOW ISOLATED INTO A TRACKED PROJECT (2026-06-21 ? done)
13     The work is isolated as ONE umbrella tracked project:
**`khelsutra/docs/FRIENDLY_AND_MULTILINGUAL.md`**
14     (workstream A = non-tech UX, B = kn+hi l10n; §7 P-tracker = source of truth;
**DoD/archive trigger = en+kn+hi
15     render on BOTH `web/` SPA + `site/` marketing**). Inbound refs point at it: khelsutra

```

```

`docs/README.md` +
16     `NEXT_STEPS.md`; engine `I18N_PLAN.md` + `UI_PROPOSAL.md` banners + `DOC_STATUS.md`
$3a/$3b rows. **Branches
17     OPEN: khelsutra docs [#41], engine pointers [#275], B1 scaffold [#42]. ** Owner locked D6
(en+kn?hi) · D7
18     (OFL Noto self-host) · D8 (SPA-first) · D9 (catalogs in khelsutra); only D-BACKEND-DEPTH
deferred.
19     **B1 DONE + CI GREEN + verified live** (react-i18next + ICU + detector + `en` catalog +
language switcher +
20     the SPA's first jsdom render tests; 79 tests green). **? NEXT (work the doc's $7, don't
re-plan): B2** (full
21     ~50-string extraction + key-parity/no-bare-strings CI) ? **B3** (site `t()` shim) ?
**B4/B5** (Noto fonts +
22     kn catalog via Gemini-draft?human-review).
23
24     ## Clean slate at handoff (2026-06-21)
25     Both repos on `master`, **0 open PRs**, no orphaned worktrees, **no GCP VMs** (no
billing). khelsutra master `5de981f`; engine (`Khelsutra/badminton-highlight-indexer`) master
`4134721` (DECOUPLED_COMPUTE archived as delivered). Engine local checkout is the **concurrent
session's** branch ? don't disturb it; use an isolated `git worktree` off `origin/master` for
any engine work (the #116/#117 collision rule).
26
27     ## ? Ramp-up kit (READ THESE before crafting the plan ? the owner's "use the docs" ask)
28     **Localization (the pillar):** **`engine docs/I18N_PLAN.md`** ? Kannada-first?Hindi-
next, **by design** (no hardcoded strings; `t("key")` + ICU MessageFormat plurals; one `en`
source catalog + per-locale siblings; locale-negotiation order = persisted pref ?
switcher/?`lang=` ? `Accept-Language` ? `en` fallback; **Gemini-DRAFT?human-REVIEW** [inference,
never training ? guardrail-OK]; CI **key-parity** + **no-bare-strings** lint; **Noto Sans
Kannada/Devanagari**, OFL font decision). Phases P0?(doc)?P1(static-UI retrofit)?P2(backend
text)?P3(**SPA react-i18next**)?P4(conversational/coach-mode).
29     **UX + personas (the pillar):** **`engine docs/UI_PROPOSAL.md`** $2 personas
(Coach/academy = primary B2B India, Kannada-comfortable, records on static cameras; Enthusiast
B2C) + the "teach recording BEFORE upload / zero hand-holding" principle + the 6 alpha screens
(A1 Matches · A2 Upload+recording-checklist · A3 Job progress · A4 Rally review ? · A5
Highlights · A6 Report).
30     **Non-tech-savvy reality:** **`engine docs/PMF_FIELD_SIGNALS.md`** +
**`docs/PRODUCT_FIELD_ASSOCIATE_FLYER.md`** (field associate works in **Kannada/Hindi/English**;
coaches/players/**parents**; "simple summary"; **WhatsApp-native**; possible B2C-to-parents
pivot). **`docs/PERSONALIZATION_PLAN.md`** (coach overlay).
**`docs/KHELSUTRA_PRODUCT_OVERVIEW.md`** (who it's for).
31     **Current product surfaces (what to localize + simplify):** khelsutra **`web/` SPA**
(per-rally operator console ? `docs/PER_RALLY_SELECTION.md`) + **`site/`** marketing (homepage,
`/capture` checklist, `/own-model`, request-access form, post-signup reveal). `web/README.md`.
32
33     ## ? THE adaptation the next session must make
34     `I18N_PLAN.md` was written for the **OLD static UI** (`backend/api/static/`). The
product is now the khelsutra **`web/` SPA** (Vite/React/TS/Tailwind) + the static marketing
**`site/`** (vanilla HTML). So adapt the design's framework-agnostic foundation to:
35     - **`web/` SPA ? react-i18next** (the doc's P3) consuming `locales/{en,kn,hi}/*.json`
catalogs (ICU plurals); a language switcher + persistence; localize the per-rally UI (query
examples/pills, rally cards, footer, build/compile/download, errors). Honest scope still applies
(duration-first; shot-count experimental).
36     - **Marketing `site/` (vanilla)** ? a tiny `t(key,vars)` shim + `data-i18n` over
`index.html`/`capture.html`/`own-model.html` (no framework), sharing the SAME catalog format.
The form + reveal copy + the recording gist localize too.
37     - **Backend (engine)** ? P2 (errors/report `customer_` as keys+params) is owner-gated +
a separate worktree PR.
38
39     ## Two threads to plan (one tracked project doc each, per `[[working-agreement-tracked-
project-docs]]`)
40     ** (A) Non-tech-savvy UX** ? research ? plan. Likely levers (validate against the docs +
personas): plain-language/low-jargon copy, fewer steps + bigger touch targets, **mobile-first**
(coaches use phones), visual/iconographic over text, **recording-education-first** (? already
shipped the homepage "Record it right" gist + `/capture`), WhatsApp-friendly share, a guided
"upload ? wait ? here's your reel" happy path with zero knobs. The per-rally query bar already

```

leads with **duration** and flags shot-count "experimental" ? keep simplifying.

```

41      (B) Hindi + Kannada localization ? adapt `I18N_PLAN.md` to `web/` + `site/`; stand
up the catalog + negotiation + a first `kn` (then `hi`) draft via `scripts/i18n_draft.py`-style
Gemini-draft?owner-review; CI key-parity; Noto fonts.
42
43      ## Open OWNER decisions to resolve first (from I18N_PLAN §4)
44      1. Indic font licensing ? recommend (a) OFL exception for bundled fonts, self-
host Noto Sans Kannada/Devanagari (treats fonts as media, not code/weights ? also unblocks a
Kannada/Hindi reel watermark via the existing `WatermarkConfig.font_path`).
45      2. v1 locale set ? recommend `en` + kn for v1, hi next; always-fallback
to `en`.
46      3. Backend-localization depth ? recommend customer verdict / report `customer_` +
validation-rejection reasons first; operator/technical text stays English.
47      4. UX scope ? which surface first: the marketing `site/` (first-contact friction)
and/or the `web/` SPA (the operator console).
48
49      ## Already shipped this session (DON'T redo ? build on it)
50      Per-rally MVP P1?P5 (query bar/grid/footer/build?compile?download, honest duration-
first, shot-count parked experimental); engine M1 (`/api/health` + compile `download_url`);
marketing site ? post-signup reveal (tool screencast + 30s reel, watchable: manual close +
holds-while-playing), submit "Sending?" feedback, "Record it right" homepage gist (from
`VIDEO_RECORDING_GUIDELINES.md`) + `capture`; behavioral jsdom tests (test-only dep;
runtime stays dependency-free); worker observability fix (logs a 0-segment cold-start,
[#271]). Promo/CUJ screencasts in `gs://khehsutra-rally-corpus/promo/`. ? Cold-start blocker
still open: native-WASB `worker infer` config that reproduces the engine session's 22-rally
result (see [[current-state]]). [[monetization-plan]] [[khehsutra-domain-launch]]
51

```

62. assistant (2026-06-21T09:23:54.511Z)

The `next-session-ux-localization.md` memory itself already **knows** `/api/health` exists (line 49-50: "engine M1 (`/api/health` + compile `download\_url`)") ? which confirms the LOCAL_APPLIANCE "does not exist" claim is the same drift-staleness class. Now I have the full surface. Let me cut the khehsutra branch from `origin/master` (safety protocol) and verify state.

63. user (2026-06-21T09:23:58.512Z)

```

=== origin/master ===
2421557 docs(dry-run): record Path-A local result (en+hi reel downloaded) + the engine fix it
surfaced (#54)
=== current ===
master
2421557 docs(dry-run): record Path-A local result (en+hi reel downloaded) + the engine fix it
surfaced (#54)
branch 'docs/correct-bucket-ingest-staleness' set up to track 'origin/master'.
Switched to a new branch 'docs/correct-bucket-ingest-staleness'
=== on branch ===
docs/correct-bucket-ingest-staleness
2421557 docs(dry-run): record Path-A local result (en+hi reel downloaded) + the engine fix it
surfaced (#54)

```

64. assistant (2026-06-21T09:24:31.131Z)

On branch `docs/correct-bucket-ingest-staleness` from `origin/master`. Now the edits. Starting with the primary stale doc ? `DRY\_RUN\_CUJ.md` §1 (the wrong "net-new" claim), with verified file:line evidence so the correction is self-proving.

65. user (2026-06-21T09:24:42.358Z)

The file C:\Users\avidu\Projects\khehsutra\docs\DRY_RUN_CUJ.md has been updated successfully. (file state is current in your context ? no need to Read it back)

66. assistant (2026-06-21T09:24:46.291Z)

Now §2 (Path B framing) and §4 (the scope decision):

67. user (2026-06-21T09:24:50.481Z)

The file C:\Users\avidu\Projects\khelsutra\docs\DRY_RUN_CUJ.md has been updated successfully.
(file state is current in your context ? no need to Read it back)

68. user (2026-06-21T09:24:55.559Z)

The file C:\Users\avidu\Projects\khelsutra\docs\DRY_RUN_CUJ.md has been updated successfully.
(file state is current in your context ? no need to Read it back)

69. assistant (2026-06-21T09:25:06.517Z)

Now the §6 DoD line for Path B, and then I'll append a §8 correction record:

70. user (2026-06-21T09:25:08.517Z)

The file C:\Users\avidu\Projects\khelsutra\docs\DRY_RUN_CUJ.md has been updated successfully.
(file state is current in your context ? no need to Read it back)

71. user (2026-06-21T09:25:13.457Z)

```
58      5. **Repeated with the UI in Hindi** (`?lang=hi`): the page + rally cards + "?? ?????"
+ "??? ???? ??" all in Hindi; reel compiled + **downloaded** (`highlights_local_hi.mp4`).
? multilingual workflow confirmed.
59
60      Screencast/proof: per-locale screenshots; both reels on the local machine. Screencast of
the i18n SPA is at `gs://khelsutra-rally-corpus/promo/khelsutra-i18n-screencast.mp4`.
61
62      ### ? Fix the dry run surfaced (engine) ? MERGED-PENDING
63      The async-job server **hung every job in `queued`** when run the documented way (`python
-m backend.main`): main.py runs as `__main__` (sets `__main__.db_instance`), but `job_worker`
reads `import backend.main` ? a **separate** module whose `db_instance` is `None` ? the drain
crashes silently. Latent because the unit tests monkeypatch the executor inline and the UI was
verified vs a stub engine. **Fix + a real-subprocess regression test ? engine
[#281](https://github.com/Khelsutra/badminton-highlight-indexer/pull/281)** (dispatch through
`backend.main`; FAILS-without / PASSES-with). Unblocked the run via the single-module
invocation; the fix applies to the droplet too.
64
```

72. assistant (2026-06-21T09:25:17.400Z)

Now append §8 ? the correction record with verified evidence and the anti-spiral convention:

73. user (2026-06-21T09:25:36.674Z)

The file C:\Users\avidu\Projects\khelsutra\docs\DRY_RUN_CUJ.md has been updated successfully.
(file state is current in your context ? no need to Read it back)

74. assistant (2026-06-21T09:25:43.497Z)

Now the sibling stale claims in `LOCAL\_APPLIANCE\_ARCHITECTURE.md`. First the `/api/health` "does not exist" claim (it shipped in M1 #268):

75. user (2026-06-21T09:25:46.599Z)

The file C:\Users\avidu\Projects\khelsutra\docs\LOCAL_APPLIANCE_ARCHITECTURE.md has been updated successfully. (file state is current in your context ? no need to Read it back)

76. assistant (2026-06-21T09:25:56.408Z)

The workflow confirms my finding independently: `alreadyworks: true`, `gaps: []`, with

exhaustive file:line evidence and only environmental runtime deps (google-cloud-storage + ADC).
Let me read the full result ? the adversarial verdicts, SPA map, and synthesis.

77. user (2026-06-21T09:25:57.561Z)

```
1      {
2      "summary": "Verify whether /api/process already accepts gcs:// ? API-DB video_id
(B-P1), map the test pattern + SPA front-half, adversarially refute the claim, then synthesize
the plan",
3      "agentCount": 7,
4      "logs": [],
5      "result": {
6      "trace": {
7      "alreadyWorks": true,
8      "evidence": [
9      {
10     "file": "backend/storage/ref.py",
11     "line": "60-61",
12     "what": "parse_uri normalizes scheme 'gs' -> 'gcs' (`if scheme == \"gs\":
scheme = \"gcs\"`), so gs://bucket/key becomes a gcs StorageRef(backend='gcs').",
13     },
14     {
15     "file": "backend/storage/ref.py",
16     "line": "87-88",
17     "what": "resolve_input_uri: explicit scheme wins first (`if
_SCHEME_RE.match(s): return s`) ? a gs:// / gcs:// input passes through unchanged regardless of
input_backend/input_root, so the DEFAULT local config still honors an explicit bucket URI."
18     },
19     {
20     "file": "backend/config/models.py",
21     "line": "315-316",
22     "what": "StorageConfig defaults: input_backend Literal default 'local',
input_root default ''. Under DEFAULT config a gs:// path is taken as an explicit-scheme bucket
ref (precedence rule 1), not forced local."
23     },
24     {
25     "file": "backend/config/models.py",
26     "line": "283",
27     "what": "SecurityConfig.allowed_video_root default None ? single-user mode;
validate_input_path only runs the null-byte check."
28     },
29     {
30     "file": "backend/utils/paths.py",
31     "line": "53-63",
32     "what": "validate_input_path(path, allowed_root=None): with allowed_root falsy
it ONLY rejects null bytes; it does NOT require a local/existing path, so a gs:// string passes
the submit-time validation."
33     },
34     {
35     "file": "backend/main.py",
36     "line": "269-276",
37     "what": "POST /api/process:
allowed_root=getattr(...security...allowed_video_root, None) (None by default) ->
validate_input_path passes; storage.resolve_input_ref(req.video_path, storage_cfg) classifies
the gs:// input."
38     },
39     {
40     "file": "backend/main.py",
41     "line": "283-284",
42     "what": "Submit-time existence fast-fail is GATED on local: `if
input_ref.backend == 'local' and not os.path.exists(input_ref.key)`. A gs:// input resolves to
backend 'gcs', so the os.path.exists 404 check is SKIPPED ? existence is deferred to the worker
fetch. No code gate blocks a bucket URI at submit."
43     },
44     {
```

```

45         "file": "backend/main.py",
46         "line": "292-296",
47         "what": "create_job(job_id, req.video_path, ...) persists the RAW
req.video_path (the gs:// string, unmodified) into the jobs table; then
_get_executor().submit(_drain_due_jobs) kicks the worker. Endpoint returns 202 + job_id."
48     },
49     {
50         "file": "backend/infrastructure/database.py",
51         "line": "486-499",
52         "what": "create_job INSERTs video_path verbatim (no normalization) into
jobs(video_path)."
53     },
54     {
55         "file": "backend/infrastructure/database.py",
56         "line": "588-608",
57         "what": "claim_due_job flips a queued job to running and returns get_job(id),
which does SELECT * ? so the stored gs:// video_path round-trips back to the worker intact."
58     },
59     {
60         "file": "backend/api/job_worker.py",
61         "line": "59-69",
62         "what": "_job_to_request rebuilds ProcessRequest(video_path=job['video_path'],
...) from the stored row ? the gs:// string is faithfully reconstructed."
63     },
64     {
65         "file": "backend/api/job_worker.py",
66         "line": "84-90",
67         "what": "_run_claimed_job calls _execute_processing(req); on success calls
set_job_video_id(job_id, result['video_id']) and mark_job_done ? the computed video_id is
recorded on the job for the poller."
68     },
69     {
70         "file": "backend/main.py",
71         "line": "159-161",
72         "what": "_execute_processing: local_video =
storage.acquire_local_input(req.video_path, global_config.storage) FETCHES the gs:// file to
scratch, then video_id = compute_video_id(local_video) (MD5 of the fetched bytes)."
73     },
74     {
75         "file": "backend/storage/__init__.py",
76         "line": "59-74",
77         "what": "acquire_local_input: for a non-local ref (gcs), mkdtemp scratch +
fetch(ref.uri, scratch/name) -> downloads and returns the LOCAL path; local refs are identity
passthrough."
78     },
79     {
80         "file": "backend/storage/__init__.py",
81         "line": "106-111",
82         "what": "fetch -> get_storage('gcs').fetch_to_local(ref, dst) for a gcs URI."
83     },
84     {
85         "file": "backend/storage/gcs.py",
86         "line": "30-39",
87         "what": "GCSStorage.fetch_to_local: makedirs +
self._blob(ref).download_to_filename(dst) ? the actual GCS download (lazy `from google.cloud
import storage`)."
88     },
89     {
90         "file": "backend/main.py",
91         "line": "190",
92         "what": "DB write is UNCONDITIONAL on the fetched local file: after validators
pass, db_instance.add_video(video_id, req.video_path, 'processed', ...) writes the videos row
(filepath = the ORIGINAL gs:// URI, status processed)."
93     },
94     {

```

```

95         "file": "backend/infrastructure/database.py",
96         "line": "287-304",
97         "what": "add_video is an UPSERT (INSERT ... ON CONFLICT(id) DO UPDATE) ?
unconditional write/update of the videos row keyed by video_id."
98     },
99     {
100         "file": "backend/main.py",
101         "line": "212-237",
102         "what": "segmenter.process_video(video_id, local_video, ...) runs on the
fetched local path; _finalize_reingest commits the produced play segments (destroy-after-
confirm).",
103     },
104     {
105         "file": "backend/pipeline/segmenters/motion.py",
106         "line": "201",
107         "what": "Segmenter persists each rally via self.db.add_play_segment(video_id,
start, end, ...) -> INSERT INTO play_segments (database.py:341-350). Same write path for
gemini.py:492, yolo_hybrid.py:346, trajectory_hybrid.py:367 ? the play_segments rows are keyed
by the MD5 video_id."
108     },
109     {
110         "file": "backend/main.py",
111         "line": "360-369",
112         "what": "POST /api/query -> db_instance.query_play_segments(video_id, filters)
returns the rows; combined with GET /api/jobs/{id} exposing video_id, the SPA can load the
result via ?video=<video_id>. Round-trip closed."
113     }
114 ],
115 "gaps": [],
116 "runtimeDeps": [
117     "google-cloud-storage SDK installed (the `storage-gcs` extra) ? imported lazily
in backend/storage/gcs.py:24 (`from google.cloud import storage`); absent => ImportError fails
the JOB (cloud), not the submit.",
118     "Application Default Credentials with read access to the bucket
(GOOGLE_APPLICATION_CREDENTIALS service-account JSON or workload identity) ? GCSStorage.__init__
does storage.Client(project=...) with no config-supplied secrets (gcs.py:18-25).",
119     "Network egress to storage.googleapis.com for download_to_filename
(gcs.py:38).",
120     "ffmpeg on PATH for the validators/segmenter (project-wide pipeline
prerequisite, unrelated to gs://).",
121 ],
122 "notes": "CLAIM HOLDS as code. Under DEFAULT single-user config
(security.allowed_video_root=None, storage.input_backend='local', input_root=''), POST
/api/process accepts a gs:// or gcs:// video_path end-to-end with NO code gate blocking
it:\n\n1. SUBMIT (main.py:269-300): validate_input_path with allowed_root=None only checks
null bytes (paths.py:60-61) ? a gs:// string passes. resolve_input_ref classifies it via
resolve_input_uri (explicit-scheme-wins, ref.py:87-88) and parse_uri (gs->gcs normalization,
ref.py:60-61) => backend 'gcs'. The os.path.exists 404 fast-fail is explicitly GATED behind
`input_ref.backend == 'local'` (main.py:283), so a bucket URI is NOT stat'd at submit.
create_job stores the RAW gs:// string (main.py:293, database.py:493).\n\n2. WORKER
(job_worker.py): claim_due_job -> get_job SELECT * round-trips the gs:// video_path;
_job_to_request rebuilds ProcessRequest(video_path=<gs://...>) verbatim (job_worker.py:64-69);
_run_claimed_job calls _execute_processing then set_job_video_id + mark_job_done
(job_worker.py:84-90).\n\n3. FETCH + PIPELINE (main.py:159-237): acquire_local_input downloads
the gs:// blob to scratch (storage/__init__.py:59-74 -> fetch -> gcs.py fetch_to_local:30-39),
computes video_id = MD5 of the fetched bytes, runs validators+segmenter on the LOCAL
path.\n\n4. DB WRITE (queryable): add_video UPSERTs the videos row unconditionally on
validation pass (main.py:190, database.py:287-304); the segmenter writes play_segments via
add_play_segment keyed by video_id (motion.py:201 etc., database.py:341-350). POST /api/query
reads them back (main.py:368). GET /api/jobs/{id} surfaces video_id (main.py:315), so the SPA
can deep-link ?video=<id>.\n\nThe gs:// download itself needs google-cloud-storage + ADC at
run time ? these are ENVIRONMENTAL runtime prerequisites, not code gaps: the import is lazy and
a missing dep / auth failure raises inside the worker and is recorded as a loud JOB failure
(job_worker.py:95-97 mark_job_failed), never silently dropped. The submit still returns 202. So
the integration is fully wired in code; only the live GCS fetch is gated on the deploy

```


environment.\\n\\nNote: the videos.filepath column stores the ORIGINAL gs:// URI (main.py:190 passes req.video_path), while the local scratch copy is ephemeral (OS temp). This is intentional per the M2 docstring (the source URI is kept for the DB record/report; only file-readers use the resolved local path). A later /api/compile on that video_id would re-need the source (video['filepath'] is the gs:// URI) ? but that is the compile path, outside this claim's scope (process -> queryable video_id), which is fully satisfied."

```
123     },
124     "testmap": {
125         "hasGcsProcessTest": true,
126         "patternFile": "C:/Users/avidu/Projects/badminton-highlight-
indexer/tests/test_api_endpoints.py",
127         "patternSummary": "The best end-to-end pattern for a gs://-into-API-DB guard is
the M2 block in tests/test_api_endpoints.py, specifically
test_process_nonlocal_uri_skips_exists_and_fetches (lines 375-398). It is in-process (FastAPI
TestClient, not subprocess) and drives a real job to completion deterministically via the shared
`env` fixture (tests/test_api_endpoints.py:27-35 and an identical copy in
tests/test_async_jobs.py:34-42). The `env` fixture: builds a real Database(tmp_path/'t.db');
monkeypatch.setattr(m, 'db_instance', db); monkeypatch.setattr(m, 'global_config', AppConfig()); and
crucially monkeypatch.setattr(m, '_get_executor', lambda: InlineExecutor()) where
InlineExecutor.submit(fn, *a, **k) just calls fn() synchronously ? so POST /api/process returns
202+job_id and the job is ALREADY 'done' by the time GET /api/jobs/{id} is read (no threads, no
polling, fully deterministic). The existing gs:// test monkeypatches the fetch seam
(m.storage.fetch -> a fake that writes bytes to dst), a capturing segmenter (_CaptureSeg via
m.segmenter_registry.get) to assert the segmenter received a LOCAL path (not the gs:// URI), and
m.registry.run_all_with_context -> (_pass(), {}) to pass validation. It then asserts:
r.status_code==202 (non-local skips the os.path.exists 404 gate), job['status']=='done', the
segmenter saw a fetched local x.mp4 (not gs://), AND the API-DB record
db.get_video(job['video_id'])['filepath'] keeps the SOURCE gs:// URI (provenance). The companion
that drives a REAL ThreadPoolExecutor with a deadline/poll loop (closer to #281's liveness
intent but still in-process) is test_real_executor_completes_job in
tests/test_async_jobs.py:332-352. The actual #281 subprocess pattern lives in
tests/test_server_entrypoint.py::test_python_m_backend_main_drains_submitted_jobs ? it Popen's
`python -m backend.main --server --port ...`, waits on /api/health, POSTs /api/process, and
polls /api/jobs/{id} asserting status != 'queued'; but it cannot inject fakes, so it only works
with a real local file, not a hermetic gs:// fetch.",
128         "mockStrategy": "Cleanest hermetic seam: monkeypatch.setattr(m.storage, 'fetch',
fake_fetch) where fake_fetch(uri, dst=None, *, config=None, overwrite=False) writes fixture
bytes to dst and returns dst. This is the EXACT seam the existing
test_process_nonlocal_uri_skips_exists_and_fetches uses (tests/test_api_endpoints.py:380-385)
and works because of the call chain: backend/main.py:159 calls
storage.acquire_local_input(req.video_path, global_config.storage); acquire_local_input
(backend/storage/_init__.py:59-74) resolves the ref, and for a non-local backend calls the BARE
name `fetch(ref.uri, dst, config=sc)` at line 74 ? which binds to the `fetch` attribute of the
backend.storage package module (defined at _init__.py:106). Because main.py imported `from
backend import storage` and references m.storage.fetch, patching m.storage.fetch (==
backend.storage.fetch) intercepts it before any GCS SDK or get_storage()/_backend_class('gcs')
is touched ? so NO google-cloud-storage import, no network. Avoid patching at lower seams for
this guard: GCSStorage(client=fake) (used by tests/test_storage.py:225
test_gcs_backend_roundtrip_with_fake_client) and a fake via get_storage/_backend_class are
correct for unit-testing the GCS backend but would force importing backend.storage.gcs (and the
real client construction path) and don't exercise the /api/process->DB flow. Patching
m.storage.fetch is the single hermetic seam that keeps the whole pipeline real
(resolve_input_ref, compute_video_id over the fetched bytes, validators, segmenter, db.add_video
keeping the gs:// URI) while faking only the network download. Note: this in-process seam does
NOT survive a subprocess, so a #281-style subprocess variant for gs:// is not hermetic without
an injected/fake backend or a no-SDK scheme like gdrive:// over a tmp mount.",
129         "testRunner": "python run_all_tests.py (repo root
C:/Users/avidu/Projects/badminton-highlight-indexer/run_all_tests.py). It defaults pytest to -q
and passes all extra args straight through (e.g. `python run_all_tests.py -k process`,
`--cov=backend --cov-report=term-missing`); it shells out to `[sys.executable, '-m', 'pytest',
*args]` and returns pytest's own exit code. The suite is fully mocked/offline (no
GPU/network/WSL). TestClient-based tests guard themselves with pytest.importorskip('httpx') at
module top (tests/test_api_endpoints.py:10, tests/test_async_jobs.py:18), so they skip cleanly
where httpx is absent ? install httpx so they actually run. obsreport is optional (reporting
tests importorskip it). There is NO conftest.py anywhere in the repo (the shared `env` fixture
```

```

is duplicated inline in each TestClient test file, not centralized).",
130     "gaps": [
131         "No conftest.py exists in the repo ? the `env` TestClient fixture (real Database
+ monkeypatched m.db_instance/m.global_config/m._get_executor=_InlineExecutor) and helpers
(_video, _pass, _CaptureSeg) are copy-pasted into both tests/test_api_endpoints.py and
tests/test_async_jobs.py; a new B-P1 guard would either duplicate the fixture again or live in
test_api_endpoints.py next to the existing M2 block to reuse it.",
132         "An assertion already exists
(test_process_nonlocal_uri_skips_exists_and_fetches): gs://-equivalent URI -> 202 (skips 404
gate) -> job done -> segmenter saw a LOCAL fetched path -> db.get_video(video_id)['filepath'] ==
the source gs:// URI. BUT it uses a gs:///style 'gs://bucket/videos/x.mp4' that is normalized
via parse_uri's gs->gcs mapping; the test string is literally 'gs://...'. To 'lock B-P1'
explicitly for BOTH gs:// AND gcs:// you'd want a parametrized assertion over
('gs://...', 'gcs://...') confirming both reach the API-DB with a queryable video_id
(db.query_play_segments(video_id, {})) non-empty) ? the current test asserts the source URI is
kept and status==done but does NOT additionally assert query_play_segments for the gs:// case
(that stronger DB-queryability assertion is only made for LOCAL inputs in
test_job_done_with_segments_and_video_id).",
133         "There is NO subprocess (#281-style) guard for the gs:///gcs:// path.
test_server_entrypoint.py only covers a real LOCAL dummy file. A true 'real regression guard
like #281' for gs:// cannot use the m.storage.fetch monkeypatch in a subprocess; the hermetic
options for a subprocess variant are (a) a gdrive:// URI over a tmp 'My Drive' mount (no SDK,
see tests/test_storage.py:252) routed through --video-path, or (b) wiring an injectable fake
backend via env/config ? neither exists yet.",
134         "The existing gs:// coverage asserts the segmenter received a local path and the
DB kept the URI, but does NOT assert the returned job['video_id'] is the MD5 of the FETCHED
bytes (compute_video_id(local_video)) ? worth pinning so a regression that hashes the URI string
instead of the fetched file is caught."
135     ]
136 },
137     "spamap": {
138         "currentFlow": "The SPA has NO router. `App.tsx:48` reads the video id ONCE at
mount: `const videoId = useMemo(() => new
URLSearchParams(window.location.search).get(`\"video\"`), [])`. That single value drives all
branching: it seeds `rallies/order` (DEMO_RALLIES if absent, [] if present) and `loadState`
(App.tsx:50-52). The back-half: an effect keyed on `[videoId]` (App.tsx:61-85) calls
`listRallies(videoId, cid)` = POST /api/query {video_id} ? returns play_segments ?
setRallies/setOrder + sel.apply(all ids) + loadState `\"loaded\"`. A second effect (App.tsx:90-98)
fetches GET /api/config for the min_shot_count floor. `onBuild` (App.tsx:109-142) does POST
/api/compile {video_id, segment_ids, output_filename} via compileReel(), then builds the
download/preview URL with highlightUrl(videoId, filename). The `\"back-half begins\"` the moment a
non-null videoId exists ? everything from /api/query onward already works. The gap is purely the
FRONT-half: nothing in the app ever SETS a video id (videoId is read-only from the URL and never
written), so there is no path to get from `\"I have a bucket URI\"` to `\"?video=<id>\"`.",
139         "apiClient": "All engine calls go through `web/src/api/client.ts`, a thin typed
wrapper. The private `request<T>(path, opts)` (client.ts:31-59) does same-origin
`fetch(`\"/api/\"+path)` with `content-type: application/json` + `x-correlation-id: cid` headers,
logs `api.request`/`api.response`/`api.http_error`/`api.network_error`, and throws a typed
`ApiError {status, body, cid}` on !res.ok or network failure. Public helpers wrap it:
`listRallies` (POST /query), `compileReel` (POST /compile), `getConfig` (GET /config), plus pure
`highlightUrl()`/`minShotCount()`. There is currently NO `processVideo` (POST /api/process) and
NO `getJob` (GET /api/jobs/{id}) helper ? those two must be added. Types live in
`web/src/api/types.ts` (RallySegment, QueryResponse, CompileResponse, EngineConfig); there is no
ProcessResponse / JobStatus type yet. Dev wiring: `vite.config.ts:11-14` proxies `/api` ?
http://localhost:8000; prod is one origin behind a reverse proxy.",
140         "filesToTouch": [
141             "web/src/api/types.ts ? add ProcessRequest body shape + ProcessResponse {job_id,
status, poll} + JobStatus {job_id, status: 'queued'|'running'|'done'|'failed', progress,
video_id, error, result}",
142             "web/src/api/client.ts ? add processVideo(videoPath, opts, cid) ? POST
/api/process (202) and getJob(jobId, cid) ? GET /api/jobs/{job_id}",
143             "web/src/hooks/useIngestJob.ts ? NEW hook: submit + poll loop with backoff,
cancellable, returns {phase, progress, error, submit, reset} (mirrors useRallySelection.ts
placement)",
144             "web/src/components/IngestPanel.tsx ? NEW: bucket-URI input + Process button +

```

```

live progress/status + error; on done calls onLoaded(video_id)",
145         "web/src/App.tsx ? make videoId STATE not a one-shot useMemo (seed from URL,
allow setVideoId); render IngestPanel when no videoId (above/instead of ModeBanner+QueryBar);
onLoaded sets videoId + history.replaceState('?video='+id) so reload is stable and the existing
[videoId] effect fires",
146         "web/src/i18n/locales/en/common.json ? add an `ingest.*` block",
147         "web/src/i18n/locales/hi/common.json ? same keys, Hindi",
148         "web/src/i18n/locales/kn/common.json ? same keys, Kannada",
149         "web/src/i18n/hygiene.test.ts ? already enforces key-parity across en/kn/hi, so
all three must stay in lockstep",
150         "web/src/components/IngestPanel.test.ts(x) + web/src/hooks/useIngestJob.test.ts
? NEW tests mirroring existing client.test.ts / *.test.ts conventions"
151     ],
152     "i18nKeysNeeded": [
153         "ingest.title",
154         "ingest.intro",
155         "ingest.uriLabel",
156         "ingest.uriPlaceholder (e.g. gs://bucket/match.mp4)",
157         "ingest.uriAriaLabel",
158         "ingest.submit",
159         "ingest.submitting",
160         "ingest.queued",
161         "ingest.running",
162         "ingest.progress (interpolates {progress} free-text phase from the job, e.g.
'starting'/'finished')",
163         "ingest.done",
164         "ingest.failed (interpolates {error})",
165         "ingest.emptyResult (job done but result.video_id absent / pipeline
result.status='failed')",
166         "ingest.invalidUri (client-side: empty/looks-not-a-uri)",
167         "ingest.networkError (fetch threw ? engine unreachable)",
168         "ingest.retry",
169         "ingest.cancel"
170     ],
171     "sketch": "Reuse the existing client+log+ApiError pattern exactly; do NOT hand-
roll fetch.\n\n1) api/types.ts ? add: `interface ProcessResponse { job_id: string; status:
string; poll: string }` and `interface JobStatus { job_id: string; status:
'queued'|'running'|'done'|'failed'; progress?: string|null; video_id?: string|null; error?:
string|null; result?: { status?: string; video_id?: string } | null }`. (Confirmed against
backend/main.py:297-322: POST returns 202 {job_id,status:'queued',poll}; GET returns
{job_id,status,progress,video_id,error,result,reports,...}. Engine `status` ?
queued|running|done|failed; on done the inner `result.status` ? success|failed and
`result.video_id` is the canonical id ? also surfaced top-level as job.video_id.)\n\n2)
api/client.ts ? add two helpers mirroring listRallies/compileReel:\n    `export async function
processVideo(videoPath: string, cid?: string): Promise<ProcessResponse> { return
request<ProcessResponse>('/process', { method:'POST', body: JSON.stringify({ video_path:
videoPath }), cid }); }`\n    `export async function getJob(jobId: string, cid?: string):
Promise<JobStatus> { return request<JobStatus>('/jobs/'+encodeURIComponent(jobId), {
method:'GET', cid }); }`\n    (ProcessRequest also supports optional
player_pool/max_frames/segmenter per main.py:91-95 ? leave them out of the v1 minimal UI; send
only video_path.)\n\n3) hooks/useIngestJob.ts ? a small state machine, placed next to
useRallySelection.ts:\n    phases: 'idle' | 'submitting' | 'polling' (carries job status+progress
text) | 'done' (video_id) | 'error' (message). `submit(uri)` : newCorrelationId() (reuse lib/log)
? log 'ingest.submit_start' ? processVideo(uri, cid) ? on 202 start a poll loop on job_id with
the SAME cid threaded through getJob so the whole journey is one correlatable id (matches the
App.tsx onBuild/listRallies cid discipline). Poll loop: setTimeout-based (NOT setInterval) so
each tick waits for the prior getJob to resolve; backoff e.g. 1s?1.5s??cap 5s; stop when
status==='done' or 'failed'. On 'done': if result.status==='failed' or no video_id ?
error('ingest.emptyResult'); else done(video_id). Errors: catch ApiError ? map via err.status
(404 unknown job, 400 bad/unsupported URI scheme ? ingest.invalidUri using parseDetail from
App.tsx, 5xx ? generic) and network throw ? 'ingest.networkError'. Return a cleanup that flips
an `alive` flag (same idiom as App.tsx:80-82) so unmount/cancel kills the loop.\n\n4)
components/IngestPanel.tsx ? structurally a sibling of QueryBar.tsx: useTranslation(); a <form
onSubmit> with a text <input> (value/onChange, aria-label={t('ingest.uriAriaLabel')},
placeholder={t('ingest.uriPlaceholder')}) + a submit button disabled while

```

phase==='submitting' || 'polling' (label flips t('ingest.submit')?t('ingest.submitting')); below it an aria-live='polite' region rendering the phase: queued/running show t('ingest.running') + the free-text job.progress via t('ingest.progress',{progress}); error shows the red banner style already used by ModeBanner/CompileResult (border-red-200 bg-red-50 text-red-800) with a Retry button. Props: `onLoaded(videoId: string)`. Client-side guard before submit: trim + reject empty (t('ingest.invalidateUri')) ? keep it light; the engine does the authoritative scheme/path validation (main.py:271-290 ? 400, surfaced via ApiError body).\n\n5) App.tsx handoff ? the ONE structural change: turn videoId from a one-shot `useMemo` into state seeded from the URL: `const [videoId, setVideoId] = useState(() => new URLSearchParams(window.location.search).get('video'))`. The existing `useEffect([videoId])` (currently the load-rallies effect) then fires automatically when setVideoId is called ? no other change to the back-half. Render: when `!videoId` show <IngestPanel onLoaded={ (id)=>{ setVideoId(id); window.history.replaceState(null, '', '?video='+encodeURIComponent(id)); } } /> in place of (or above) the demo ModeBanner; once videoId is set the existing query?grid?compile UI renders exactly as today and ?video= survives a reload. replaceState keeps the URL the single source of truth and reuses the proven [videoId] load path rather than inventing a parallel rally-loading code path.",

172 "risks": [

173 "Polling/backoff: use recursive setTimeout (await getJob each tick) not setInterval, else overlapping in-flight requests pile up if the engine is slow; cap backoff (~5s) and consider a max-attempts/elapsed ceiling so a wedged 'running' job doesn't poll forever. Thread one cid through submit + every poll (like onBuild) so the journey is traceable; log a loud 'ingest.poll_timeout' if the ceiling hits.",

174 "Cancellation/unmount: the poll loop MUST honor an `alive` flag (App.tsx:80-82 idiom) and a user Cancel; a leaked loop keeps calling getJob after the user navigated to ?video=. StrictMode double-mounts effects in dev (main.tsx:14-17) ? guard against a double submit.",

175 "Two status layers are easy to conflate: job.status (queued|running|done|failed = execution) vs result.status (success|failed = pipeline verdict). 'done' with result.status==='failed' or a null video_id is a SUCCESS-shaped HTTP 200 that is actually a failure ? must map to ingest.emptyResult, not silently hang or fake a load.",

176 "Error mapping: reuse ApiError + parseDetail (currently a private fn in App.tsx ? extract to lib so IngestPanel can reuse it). 400 from /api/process is the engine's path/scheme validation (unsupported bucket scheme, out-of-allowed-root, missing local file ? 404) ? surface the engine's `detail`, don't show a generic message. Network throw (engine down) is the common non-tech-savvy case ? friendly t('ingest.networkError'), not a stack string.",

177 "Offline-banner interplay: there is currently NO offline/navigator.onLine banner in the SPA (only self-contained assets + loud ApiError logging). The ingest screen is the FIRST screen a fresh user hits and it REQUIRES the engine ? so a friendly 'engine unreachable / is it running?' state matters more here than anywhere else. If an online/offline banner is added later, ensure it doesn't double-report with the IngestPanel's own networkError state.",

178 "Long jobs vs UX expectation: real video processing can take minutes (per memory: cv2 extraction bottleneck, 4K stitch). The progress text from the engine is free-form TEXT ('starting'/'finished'), not a percentage ? the UI can only show phase words, so set expectations (indeterminate spinner + the phase word), don't imply a percent bar.",

179 "i18n parity: hygiene.test.ts enforces identical key sets across en/kn/hi ? adding ingest.* to only en will fail the suite; all three catalogs must land together. kn/hi are AI-drafted/native-review-pending per i18n/index.ts, so flag the new strings for review.",

180 "URL handoff: replaceState (not a full reload) reuses the existing [videoId] effect, but the config-floor effect (App.tsx:90) runs once on mount ? confirm min_shot_count is still fetched for an ingest?load session (it is, since it's not keyed on videoId). Encode the id in the querystring (encodeURIComponent) to match how listRallies/highlightUrl already encode ids."

181]

182 },

183 "refutedCount": 0,

184 "totalSkeptics": 3,

185 "verdicts": [

186 {

187 "lens": "SUBMIT-TIME GATE (backend/main.py process_video submit path: validate_input_path, resolve_input_ref ValueError mapping, input_ref.backend=='local\' existence check, explicit-segmenter check)",

188 "refuted": false,

189 "reasoning": "I traced every branch of the submit path process_video() (backend/main.py:260-300) for req.video_path='gs://bucket/video.mp4'", req.segmenter=None,

under DEFAULT single-user config. Default config verified: config.json has NO `security` and NO `storage` sections, and no deployment.profile, so the loader's no-profile shallow merge (backend/config/loader.py:184) lets the Pydantic defaults supply SecurityConfig(allowed_video_root=None) and StorageConfig(input_backend="local", input_root=""); default_segmenter="gemini". Gate-by-gate: (1) validate_input_path("gs://...\\", allowed_root=None) at main.py:271 -> only the null-byte check runs (backend/utils/paths.py:60-63); a gs:// URI has no null byte -> PASS, no 400. (2) resolve_input_ref at main.py:274 -> resolve_input_uri sees an explicit scheme (ref.py:87, rule 1) and returns it as-is; parse_uri normalizes scheme "gs\\"->"gcs\\" (ref.py:60-61) and "gcs\\" is in _KNOWN (ref.py:22,65-67), returning StorageRef(backend="gcs\\", bucket="bucket\\", key="video.mp4\\"). NO ValueError is raised, so the except ValueError -> HTTPException(400) at main.py:275-276 never fires. (3) main.py:283 `if input\_ref.backend == "local\\" and not os.path.exists(...):` backend is "gcs\\", so the AND short-circuits and the 404 (main.py:284) is SKIPPED ? exactly as the code comment at main.py:277-282 intends (a bucket URI is not stat'd at submit). (4) main.py:285 `if req.segmenter and not segmenter\_registry.get(...):` req.segmenter is None (falsy), so the unknown-segmenter 400 is SKIPPED. After all four gates the endpoint calls create_job (which inserts unconditionally, inspecting nothing about the URI ? backend/infrastructure/database.py:486-499) and returns 202 + a queryable job_id. The downstream worker then resolves the same gs:// (acquire_local_input, main.py:159) and computes/persists video_id. I searched for any submit-path branch that 400s/404s a valid gs:// URI under default config and found NONE. The only failure modes are environmental (missing google-cloud-storage SDK or ADC at the worker fetch), explicitly excluded by the lens, and they occur on the worker, not the submit path. No concrete code-gate counterexample exists; the claim's submit-time portion holds.",

```

190         "counterexample": ""
191     },
192     {
193         "lens": "WORKER/EXECUTION + DB WRITE (post-202: job_worker drain,
_job_to_request, _execute_processing/acquire_local_input, add_video/get_video persistence)",
194         "refuted": false,
195         "reasoning": "I traced the entire post-202 worker/DB-write path and found NO
code gate that makes a valid gs://|gcs:// URI fail to yield a queryable API-DB video_id under
default single-user config (allowed_video_root=None, input_backend='local\\',
input_root='').\\n\\nSubmit acceptance: backend/main.py:269-296 ? validate_input_path
(utils/paths.py:53-63) only checks null bytes when allowed_root is None (default), so gs://
passes; storage.resolve_input_ref returns a gcs-backend ref (gs:// is normalized to gcs:// in
storage/ref.py:60-61); the local-existence os.path.exists fast-fail at main.py:283 is SKIPPED
because input_ref.backend != 'local\\"; create_job persists the row 'queued'; main.py:296
_get_executor().submit(_drain_due_jobs)` actually fires the worker by default.\\n\\nWorker drain:
job_worker.py:112 claim_due_job -> :116 _run_claimed_job -> :84 _job_to_request rebuilds
ProcessRequest from the stored video_path (the gs:// URI verbatim, identical handling to a local
path).\\n\\n_execute_processing (main.py:140-258): line 159 acquire_local_input(req.video_path,
global_config.storage) -> storage/_init_.py:59-74 resolves the gcs ref, mkdtemp's a scratch
dir, and fetch()->GCSStorage.fetch_to_local (gcs.py:30-39) downloads to scratch and returns a
LOCAL path. dst is a valid path: ref.name yields 'x.mp4\\" (ref.py:44-45). Line 160
compute_video_id(local_video) hashes the FETCHED bytes (hashing.py opens the local file), so the
id is well-formed exactly as in the local case. add_video(video_id, req.video_path, ...) is
invoked on every terminal branch (validation-fail main.py:179, processed main.py:190) ? it
UPSERTS a row keyed on id (database.py:287-304). Storing the gs:// URI in the text `filepath`
column is harmless to keying.\\n\\nget_video (database.py:418-425) selects WHERE id=? and returns
the row irrespective of the filepath column value, so db.get_video(video_id) returns a queryable
row. The worker also set_job_video_id + mark_job_done, surfaced via /api/jobs/{job_id}.video_id
(main.py:303-322).\\n\\nThe ONLY gs:// vs local divergence in this path is the fetch-to-scratch
network download, whose failure modes are a missing google-cloud-storage dep or missing ADC ?
explicitly excluded as environmental, not a code gate. The DB write is keyed on the content hash
and is byte-for-byte identical between the two cases. No concrete counterexample exists for my
lens.\\n\\nCaveat (out of my lens, does not refute): the shipped SPA in backend/api/static/app.js
has no ?video= query-param loader (no URLSearchParams/location.search), so the 'SPA can load
via ?video=' sub-claim is unsupported by the actual frontend ? but that is the frontend lens
and does not affect whether get_video(video_id) returns a queryable API-DB row, which it does.",
196         "counterexample": ""
197     },
198     {
199         "lens": "CONFIG & EDGE CASES (input_backend default, gs://?gcs:// normalization,
allowed_video_root default, gdrive://, and SPA ?video= back-half)",

```

```

200         "refuted": false,
201         "reasoning": "I traced the full path under DEFAULT single-user config
(storage.input_backend=\"local\", input_root=\"\", security.allowed_video_root=None ? confirmed
in backend/config/models.py:283,315-316) and found NO code gate that fails a valid gs://|gcs://
URI.\n\nROUTING / ACCEPTANCE (POST /api/process, backend/main.py:260-300):\n-
backend/utls/paths.py:53-63 validate_input_path: with allowed_root falsy (default), ONLY the
null-byte check runs, so gs:// passes. (A *configured* allowed_video_root would reject a non-
local URI via resolve_under_root ? but that is non-default; the lens asks about default config,
where it is None.)\n- backend/storage/ref.py:87-88 resolve_input_uri precedence rule 1: an
explicit scheme:// is returned AS-IS *before* input_backend/input_root are consulted ? so the
default input_backend=\"local\" is irrelevant for an explicit gs://. The URI routes to gcs
regardless.\n- backend/storage/ref.py:60-61 parse_uri: scheme is lowercased and gs?gcs
normalized; both gs:// and gcs:// produce an identical gcs StorageRef with non-empty bucket+key.
No edge here.\n- backend/main.py:283: the local-existence os.path.exists fast-fail is guarded by
`input_ref.backend == \"local\"`, so it is correctly SKIPPED for the gcs backend (no false 404).
Job row is created (create_job) and 202 returned.\n\nBACK-HALF (worker, backend/main.py:140-257
+ backend/storage/__init__.py:59-74):\n- acquire_local_input resolves the gcs ref, makes a
scratch dir, and calls fetch ? GCSStorage.fetch_to_local (backend/storage/gcs.py:30-39), which
downloads to dst. dst is always supplied (os.path.join(scratch, ref.name or \"in.mp4\"));
ref.name for gs://bucket/videos/x.mp4 = \"x.mp4\"). No code gate ? the only failure modes are
the lazy `from google.cloud import storage` import (gcs.py:24) or missing ADC, both
ENVIRONMENTAL and explicitly excluded.\n- compute_video_id hashes the fetched local file;
add_video (main.py:190) writes the row with the ORIGINAL gs:// string as the path column and the
MD5 as id; get_video (backend/infrastructure/database.py:418-425) returns that row by id.\n\nSPA
?video= back-half: the engine's bundled SPA (backend/api/static/app.js) has no ?video=
bootstrap, but the actual ?video=-consuming SPA is the khelsutra web app
(C:/Users/avidu/Projects/khelsutra/web/src/App.tsx:48 reads ?video=<id>; :66
listRallies(videoId) ? POST /api/query keyed purely by video_id). It loads strictly by the MD5
video_id and never touches the source-path scheme, so a gs://-sourced video is loadable byte-
identically to a local one. gdrive:// follows the same non-local path (resolve?fetch?hash?row)
with no extra gate.\n\nNo realistic DEFAULT config silently fails gs://?queryable video_id on a
code path; all failures are environmental (SDK/auth), which the task excludes.",
202         "counterexample": ""
203     }
204 ],
205     "synthesis": {
206         "bP1Verdict": "B-P1 ALREADY WORKS IN CODE ? the claim survives; 0 of 3 skeptics
refuted it, and I independently re-traced the path to confirm. Under DEFAULT single-user config
(security.allowed_video_root=None, storage.input_backend='local', input_root=''), POST
/api/process accepts a gs:// or gcs:// video_path end-to-end and produces a queryable API-DB
video_id with NO code gate blocking it. Strongest evidence chain (re-verified at file:line this
session): (1) SUBMIT backend/main.py:269-300 ? validate_input_path with allowed_root=None only
checks null bytes (utls/paths.py:53-63); resolve_input_ref classifies gs:// ? gcs backend
(ref.py:60-61 gs?gcs normalize, :87-88 explicit-scheme-wins); the os.path.exists 404 fast-fail
is explicitly GATED behind `input_ref.backend == \"local\"` (main.py:283), so a bucket URI is
NOT stat'd; create_job stores the raw URI and returns 202. (2) WORKER job_worker.py round-trips
the URI verbatim. (3) FETCH+PIPELINE main.py:159-160 ? acquire_local_input downloads the blob to
scratch, then `video_id = compute_video_id(local_video)` hashes the FETCHED bytes; add_video
UPSERts the videos row (main.py:179/190, keyed on MD5, filepath column keeps the source URI);
the segmenter writes play_segments keyed on video_id. POST /api/query reads them back. The
endpoint the doc calls 'net-new' EXISTS. The only gs://-vs-local divergence is the network
download, whose failure modes (missing google-cloud-storage SDK / missing ADC) are ENVIRONMENTAL
runtime prerequisites recorded as a LOUD job failure (job_worker.py mark_job_failed), never
silent ? not code gaps. IMPORTANT SCOPE PIN: this holds ONLY in default single-user mode; in
HOSTED mode (allowed_video_root set) gs:// is correctly rejected 400 ? already covered by
test_process_hosted_mode_rejects_nonlocal_uri (test_api_endpoints.py:432). I also ran the two
existing gs:// tests green (httpx 0.28.1 present). So B-P1 is a TEST + HONEST-DOC-CORRECTION PR,
not a code fix. The genuinely net-new work (the SPA ingest/progress front-half ? nothing in the
app ever SETS a video_id) is B-P2.",
207         "bP1PrPlan": {
208             "testFile": "C:/Users/avidu/Projects/badminton-highlight-
indexer/tests/test_api_endpoints.py (extend the existing M2 block in-place to reuse the `env`
fixture + _CaptureSeg + fake_fetch helpers; new test next to
test_process_nonlocal_uri_skips_exists_and_fetches at line 375)",
209             "testCases": [

```

```

210         "Parametrize over ('gs://bucket/videos/x.mp4', 'gcs://bucket/videos/x.mp4') so
BOTH the gs:// alias AND the canonical gcs:// scheme are locked through one assertion (the
existing test only exercises the literal 'gs://...').",
211         "Use the DEFAULT env fixture (AppConfig() => security.allowed_video_root=None,
storage.input_backend='local') so the test pins B-P1 in single-user mode; do NOT set
allowed_video_root.",
212         "Monkeypatch m.storage.fetch = fake_fetch that writes a FIXED byte payload to
dst (the same hermetic seam the existing test uses); patch m.registry.run_all_with_context ->
(_pass(),{}) and m.segmenter_registry.get -> _FakeSeg (the seg that actually persists a
play_segment, not _CaptureSeg) so query_play_segments is non-empty.",
213         "Assert POST returns 202 (non-local skips the os.path.exists 404 gate) and GET
/api/jobs/{id} reaches status=='done' (inline executor) with result['status']=='success'.",
214         "Assert job['video_id'] == compute_video_id(<the fetched bytes>) ? i.e. the id
is the MD5 of the FETCHED file, NOT a hash of the gs:// URI string (closes the gap the current
gs:// test leaves open; catches a regression that hashes the URI).",
215         "Assert db.get_video(job['video_id'])['filepath'] == the SOURCE gs://|gcs://
URI (provenance kept) AND status reads 'processed'.",
216         "Assert db.query_play_segments(job['video_id'], {}) is non-empty (the
QUERYABILITY guarantee ? the SPA's POST /api/query back-half returns rows) ? this stronger
assertion is currently only made for LOCAL inputs (test_job_done_with_segments_and_video_id),
never for the bucket case.",
217         "Keep test_process_hosted_mode_rejects_nonlocal_uri untouched as the explicit
counterweight (gs:// -> 400 when allowed_video_root is set); optionally add a one-line comment
cross-referencing the new test so the default-vs-hosted contract is legible."
218     ],
219     "docEdits": [
220         "DRY_RUN_CUJ.md §1 line 15: REMOVE the false claim 'Bridging \"gcs:// -> a
queryable video_id in the API DB\" is net-new'. Replace with [verified]: 'POST /api/process
ALREADY accepts a gs://|gcs:// video_path end-to-end under default single-user config
(allowed_video_root=None): submit skips the local-existence 404 (main.py:283), the worker
fetches the blob to scratch (acquire_local_input), hashes the FETCHED bytes into video_id, and
UPSERts queryable videos+play_segments rows (main.py:159-190). The endpoint is NOT net-new. (In
HOSTED mode, allowed_video_root rejects a bucket URI 400 by design.)'. Also fix the §1 line-14
bullet 'POST /api/process accepts only a local video_path' ? that is FALSE; correct to 'accepts
a local path OR an explicit bucket/Drive URI; the SPA simply never CALLS it (the real gap is the
missing front-half UI).'",
221         "DRY_RUN_CUJ.md §2 line 20 (Path B): the 'net-new HTTP bucket-ingest endpoint
(gcs:// -> process -> DB video_id)' is FALSE. Rewrite Path B as: 'The HTTP bucket-ingest
endpoint ALREADY exists (/api/process accepts gcs://, verified). Path B's net-new work is purely
the SPA front-half: an ingest URI input + a useIngestJob submit/poll hook + an App.tsx videoId
state handoff (appliance S1/S2). No engine endpoint is net-new.'",
222         "DRY_RUN_CUJ.md §6 DoD box 3 (line 48): once B-P2 lands, flip to checked; for
the B-P1 PR, add a parenthetical to box 3 noting 'engine accept-path verified+locked
(test_process_nonlocal_uri_*); remaining work is the SPA ingest screen.'",
223         "LOCAL_APPLIANCE_ARCHITECTURE.md line 87 (the Bucket-URI row): clarify the
half-true 'net-new (design)' tag ? the ENGINE endpoint is NOT net-new (/api/process accepts
gs:// today, verified main.py:269-300); only the SPA wiring that CALLS it is net-new. Change the
status note to 'engine: DONE (verified); SPA ingest call: design (B-P2)'.",
224     ],
225     "codeChanges": []
226 },
227     "bP2Sketch": "Follow-up PR (the genuinely net-new work): build the SPA FRONT-half
so a user points at a bucket URI in the UI and lands on ?video=<id>. The engine endpoints
already exist (B-P1) ? B-P2 is 100% khelsutra/web SPA + i18n, zero engine code. Files: (1)
web/src/api/types.ts ? add ProcessResponse {job_id; status; poll} and JobStatus {job_id;
status:'queued'|'running'|'done'|'failed'; progress?; video_id?; error?; result?:{status?;
video_id?}} (confirmed vs main.py:297-322). (2) web/src/api/client.ts ? add
processVideo(videoPath, cid) -> POST /api/process (send ONLY {video_path}; leave
player_pool/max_frames/segmenter out of v1) and getJob(jobId, cid) -> GET /api/jobs/{id}; reuse
the existing request<T>/ApiError/log pattern, do NOT hand-roll fetch. (3)
web/src/hooks/useIngestJob.ts ? NEW state machine (idle|submitting|polling|done|error) placed
next to useRallySelection.ts; submit(uri)->newCorrelationId()->processVideo->recursive
setTimeout poll loop (await getJob each tick, backoff 1s->5s cap, max-elapsed ceiling, thread
ONE cid through submit+every poll); on done map result.status==='failed'||no video_id ->
ingest.emptyResult else done(video_id); ApiError->status map (400 invalidUri via extracted

```

parseDetail, 404 unknown job, 5xx generic), network throw->networkError; honor an `alive` flag (App.tsx:80-82 idiom) for cancel/unmount + StrictMode double-mount guard. (4)

web/src/components/IngestPanel.tsx ? NEW sibling of QueryBar.tsx: <form> with URI <input> (aria-label, placeholder gs://bucket/match.mp4) + submit button (label flips submit->submitting, disabled while submitting/polling) + aria-live='polite' status region + red-banner error (border-red-200 bg-red-50 text-red-800, reuse ModeBanner style) + Retry/Cancel; prop onLoaded(videoId). (5) web/src/App.tsx ? the ONE structural change: turn videoId from one-shot useMemo (App.tsx:48) into useState(()=>new URLSearchParams(location.search).get('video')); render <IngestPanel onLoaded={id=>{setVideoId(id); history.replaceState(null, '', '?video='+encodeURIComponent(id))}}/> when !videoId; the existing useEffect([videoId]) load-rallies path then fires unchanged (config-floor effect is unkeyed so still runs). (6) i18n: add ingest.* block

(title,intro,uriLabel,uriPlaceholder,uriAriaLabel,submit,submitting,queued,running,progress {progress},done,failure {error},emptyResult,invalidUri,networkError,retry,cancel) to en/hi/kn TOGETHER (hygiene.test.ts enforces key-parity; flag hi/kn AI-drafted native-review-pending). (7) Tests: web/src/components/IngestPanel.test.tsx + web/src/hooks/useIngestJob.test.ts mirroring client.test.ts conventions. Key risks: recursive setTimeout not setInterval (no request pile-up); job.status (execution) vs result.status (pipeline verdict) ? done+result.status==='failed' is a success-shaped failure->emptyResult; friendly networkError on the FIRST screen a fresh user hits (engine-unreachable is the common non-tech case); progress is free-text phase words not a percent (indeterminate spinner). Long real jobs take minutes (cv2/4K). Update DRY_RUN_CUJ.md \$6 DoD box 3 to checked when B-P2 lands.",

228 "dropletNotes": "Option 3 (DoD \$6 box 4) = re-run the same CUJ on the CPU droplet 168.144.126.176 (Linux deploy target, ~2-core/3.8GB RAM) ? a DEPLOY/ops task, independent of B-P1/B-P2. Current state per \$6: droplet reachable but UN-provisioned (no engine checkout). What it entails: (1) From-scratch engine install: git clone the engine at B-P1-merged master, python venv, pip install requirements + httpx (TestClient suite) + the storage-gcs extra (google-cloud-storage) since the CUJ uses a gs:// source; apt-install ffmpeg + OpenCV system deps. (2) GCS auth on the box: GOOGLE_APPLICATION_CREDENTIALS service-account JSON with READ access to gs://khealsutra-rally-corpus (GCSStorage.__init__ does storage.Client(project=...)) with NO config-supplied secrets, gcs.py:18-25, so ADC must be in the env) + egress to storage.googleapis.com. (3) Detector on a 3.8GB CPU box: motion is key-free + CPU-only (safe); gemini needs GEMINI_API_KEY + uploads footage (paid+cloud, owner-gated); native WASB is GPU/WSL-only -> NOT available here. RAM is tight -> expect cv2 frame-extraction to be the bottleneck (gotcha-long-runs memory); pre-extract via ffmpeg / use the short staged clip (mahadevpura_smoke_180s.mp4, known-good). (4) Run the engine the FIXED way: `python -m backend.main --server` ? the #281 fix (drain async jobs under the backend.main module, MERGED-PENDING) is REQUIRED on the droplet else every job hangs in 'queued'; verify the #281 subprocess regression (test_server_entrypoint.py) is green on Linux. (5) Drive the CUJ: POST a gs:// (or local-fetched) video_path -> poll /api/jobs -> open the SPA at ?video=<id> -> select->compile->download the reel, repeat in Hindi. (6) Cost/hygiene: droplet is a SHARED owner-gated testbed (~\$24/mo) ? coordinate, clean frames/staged caches after (cache-hygiene policy). Net: B-P1 needs NO droplet to merge (hermetic in-process test); the droplet only ticks the Linux-deploy DoD box and is gated on provisioning + GCS creds + the #281 fix being present.",

229 "risks": [

230 "SCOPE TRAP: B-P1 holds ONLY under default single-user config. In HOSTED mode (security.allowed_video_root set) gs:// is correctly 400'd (validate_input_path->resolve_under_root rejects non-local). The new B-P1 test MUST use the default `env` fixture (AppConfig() => allowed_video_root=None) and must NOT claim the bridge works universally. Keep test_process_hosted_mode_rejects_nonlocal_uri as the counterweight; the doc correction must say 'default single-user' explicitly so a future hosted deploy isn't mis-read as broken.",

231 "ENVIRONMENTAL != CODE GAP, but a real run-time prerequisite: the live gs:// fetch needs google-cloud-storage (storage-gcs extra, lazy import gcs.py:24) + ADC + network egress. The hermetic test fakes m.storage.fetch so it proves NOTHING about the live download ? the doc/PR must be honest that the CODE path is wired but the LIVE fetch is gated on deploy env (verified on the droplet, option 3, not in CI).",

232 "Hermetic seam is in-process only: monkeypatch m.storage.fetch works because acquire_local_input calls the bare `fetch` name bound to the backend.storage package module ? it does NOT survive a subprocess. A true #281-style subprocess guard for gs:// is NOT possible with this seam (no fake injection across the process boundary). The B-P1 test is therefore an in-process TestClient lock (like the existing M2 block); the PR must NOT over-claim a '#281-style real-subprocess regression' for the gs:// path ? that would need a no-SDK scheme (gdrive:// over a tmp mount) or an injectable fake backend, neither of which exists yet.",


```

233         "Doc drift risk: LOCAL_APPLIANCE_ARCHITECTURE.md:87 ALSO calls the UI ingest-to-
process step 'net-new (design)'. That line is half-right ? the ENDPOINT is not net-new (accepts
gs:// today) but the SPA wiring to CALL it IS net-new. If only DRY_RUN_CUJ.md is corrected the
appliance doc stays subtly inconsistent; correct that line too (engine endpoint exists; only the
SPA ingest screen is net-new = B-P2).",
234         "Test must pin the RIGHT invariant: assert job['video_id'] ==
compute_video_id(fetched bytes) (NOT a hash of the URI string) AND
db.query_play_segments(video_id, {}) non-empty ? the current gs:// test asserts neither (only
segmenter-saw-local-path + filepath-keeps-URI). Without these two a regression that hashes the
URI or never persists segments would slip through. Use _FakeSeg (persists a segment), not
_CaptureSeg (only records the path), so query_play_segments is exercisable."
235     ]
236 }
237 }
238 }

```

78. assistant (2026-06-21T09:26:37.117Z)

The workflow confirms everything with rigor: **B-P1 already works, 0/3 skeptics refuted, gaps: []**. It adds three things I'll fold in:

1. **Scope pin**: the bridge holds only in *default single-user mode* (hosted mode correctly 400s gs://) ? the doc must say so.
2. **Honesty pin**: a true #281-*subprocess* guard isn't possible for gs:// (the ``m.storage.fetch`` seam is in-process) ? so the engine test is an in-process lock, not a subprocess guard.
3. The existing gs:// test never asserts ``video_id == MD5(fetched bytes)`` nor ``query_play_segments`` non-empty ? a genuine coverage gap.

Let me finish the LOCAL_APPLIANCE corrections (line 87 + T12):

79. user (2026-06-21T09:26:41.317Z)

The file C:\Users\avidu\Projects\khelsutra\docs\LOCAL_APPLIANCE_ARCHITECTURE.md has been updated successfully. (file state is current in your context ? no need to Read it back)

80. user (2026-06-21T09:26:45.844Z)

The file C:\Users\avidu\Projects\khelsutra\docs\LOCAL_APPLIANCE_ARCHITECTURE.md has been updated successfully. (file state is current in your context ? no need to Read it back)

81. assistant (2026-06-21T09:26:49.899Z)

Now the README index line (make it precise ? the *screen* is the gap, not the endpoint):

82. user (2026-06-21T09:26:51.860Z)

The file C:\Users\avidu\Projects\khelsutra\docs\README.md has been updated successfully. (file state is current in your context ? no need to Read it back)

83. assistant (2026-06-21T09:27:00.969Z)

Let me add the workflow's scope pin (default single-user vs hosted mode) to the §8 correction so a future hosted deploy isn't mis-read as broken:

84. user (2026-06-21T09:27:08.093Z)

The file C:\Users\avidu\Projects\khelsutra\docs\DRY_RUN_CUJ.md has been updated successfully. (file state is current in your context ? no need to Read it back)

85. assistant (2026-06-21T09:27:15.391Z)

Let me verify the khelsutra doc changes are coherent and scoped to the intended files:

86. user (2026-06-21T09:27:17.098Z)

```
=== changed files ===
M docs/DRY_RUN_CUJ.md
M docs/LOCAL_APPLIANCE_ARCHITECTURE.md
M docs/README.md
?? .claude/
?? site/access.db
=== diffstat ===
docs/DRY_RUN_CUJ.md          | 32 ++++++-----
docs/LOCAL_APPLIANCE_ARCHITECTURE.md | 6 +---
docs/README.md              | 3 +-
3 files changed, 31 insertions(+), 10 deletions(-)
```

87. user (2026-06-21T09:27:17.552Z)

```
10      ## 1. What already works vs the gap `[verified]`
11      **? Back half (shipped ? PER_RALLY_SELECTION M0):** the SPA loads `?video=<id>` ? `POST
/api/query` (rallies, default all-selected) ? deterministic query/select ? `POST /api/compile` ?
`<video>` preview + **`<a download>` saves the MP4 locally**. Reel download-to-local **works
today** (`GET /api/highlights/{video_id}/{filename}` streams it; `/api/compile` also returns
`download_url`). Fully **en/kn/hi** (the switcher / `?lang=`).
12
13      **? Front half ? the gap is the SPA, NOT the engine `[corrected 2026-06-21; see §8]`:**
14      - The SPA has **no ingest/process UI** ? it can't point at a video (bucket or local),
can't call `/api/process`, can't poll `/api/jobs`. It only gets a `video_id` from `?video=` in
the URL. **This SPA front-half is the *only* net-new work for Path B (= B-P2).**
15      - **HTTP bucket-ingest ALREADY EXISTS in the engine (shipped M2 / [#280]) `[verified
file:line]`:** `POST /api/process` accepts a `gs://`/`gcs://` (also `s3://`/`local://`)
`video_path` directly. Submit resolves the URI and **skips the local-existence check for non-
local backends** (`backend/main.py:274,283` ? `storage.resolve_input_ref`; `gs://`/`gcs`
normalized in `backend/storage/ref.py:60`); the job worker then fetches it to scratch
(`execute_processing` ? `storage.acquire_local_input`, `main.py:159`) and runs the **same**
hash ? validate ? segment ? DB pipeline as a local file ? writing a queryable `video_id` into
the API's SQLite `videos`/`play_segments`. Proven end-to-end by
`tests/test_api_endpoints.py::test_process_nonlocal_uri_skips_exists_and_fetches` (gs:// ? 202,
not 404 ? job `done` ? `db.get_video(video_id)` row with the **source URI preserved**), plus the
hosted-mode-rejection / unknown-scheme-400 / CLI-variant tests.
16      - *Where the earlier "net-new" claim went wrong:* it conflated the **API path**
(`/api/process`, which DID gain URI ingest in M2) with the separate **worker CLI** (`python -m
backend.worker infer --video-uri gcs://`), which writes `intervals.csv`/`report.json` to the
bucket ? **not** the API DB. The CLI distinction is real; the "the endpoint is local-only"
conclusion was not. ? **Engine-surface claims drift between sessions** (the engine ships PRs
independently) ? re-verify any `[gap]`/`net-new` engine claim against the cited `file:line`
before scoping it as work.
17
18      ## 2. Two paths to the goal
19      **Path A ? Cheapest dry run NOW (~zero new code).** Pre-process the bucket video out-of-
band to mint a `video_id` in the engine DB, then drive the existing SPA back-half. Proves the
**multilingual UI-driven workflow** end-to-end. The only non-UI step is "point at the bucket" (a
one-line CLI pre-process).
20
21      **Path B ? Full "point at a bucket video IN the UI" (SPA-only ? smaller than first
scoped).** The **HTTP bucket-ingest endpoint already exists** (§1) ? so Path B is **only the SPA
front-half** (appliance S1/S2): an **ingest** input (enter a `gs://` URI) ? `POST /api/process`
? an **ingest + progress** screen polling `GET /api/jobs/{id}` ? on `done`, load
`result.video_id` (which the existing back-half already renders). **No net-new engine endpoint**
? B-P1 is already done; the remaining work is **B-P2 (the SPA)**.
22
23      ## 3. Path A runbook (the cheapest dry run) `[verified-steps]`
24      1. **Engine host + deps:** run the engine FastAPI (`python -m backend.main --server`,
binds `127.0.0.1:8000`) on a box with `ffmpeg` + OpenCV. Gemini/motion paths are **CPU-only, no
WSL** ? local Windows works; the CPU droplet works too. (GPU/WSL only needed for native WASB.)
25      2. **Get the bucket video local + into the API DB:**
26      `gsutil cp gs://<bucket>/<match>.mp4 ./incoming/` (or the worker fetch), then `POST
```

```

/api/process {"video_path":"<abs local path>","segmenter":"gemini"}` ? poll `GET
/api/jobs/{job_id}` until `result.video_id`. *(`gemini` needs `GEMINI_API_KEY`; `motion` is key-
free but hides fabricated metadata.)*
27 3. **Open the SPA against the engine:** `npm run dev` (Vite proxies `/api` ? `:8000`,
same-origin) ? `http://localhost:5173/?video=<video_id>` ? rallies auto-load.
28 4. **Select ? Build ? download:** pick rallies ? Build ? `<video>` preview ? click the
download link ? MP4 saved local.
29 5. **Repeat in Hindi:** click **?????` (or `?lang=hi`) ? redo select?compile?download
? confirm the whole flow works with the UI in Hindi.
30
31 **Small unblocked SPA polish that makes Path A more "via the UI" (independent of A/B,
useful in both): a ***"Load a video by ID"*** input (paste the `video_id` ? load, instead of
editing the URL) + consume the engine's `download_url` (instead of hand-building it). One tight
PR.
32
33 ## 4. Open decisions ? OWNER (these gate the dry run)

```

88. assistant (2026-06-21T09:27:36.718Z)

\$1/\$2 read cleanly and the changes are scoped to the 3 intended docs. Let me re-verify branch/HEAD (safety protocol), stage explicit paths, and commit the khelsutra doc correction.

89. user (2026-06-21T09:27:39.074Z)

```

=== branch + HEAD (must be mine) ===
docs/correct-bucket-ingest-staleness
2421557 docs(dry-run): record Path-A local result (en+hi reel downloaded) + the engine fix it
surfaced (#54)
=== staging explicit paths ===
M docs/DRY_RUN_CUJ.md
M docs/LOCAL_APPLIANCE_ARCHITECTURE.md
M docs/README.md
?? .claude/
?? site/access.db

```

90. user (2026-06-21T09:27:49.308Z)

```

[docs/correct-bucket-ingest-staleness aa07c61] docs: correct stale "bucket ingest is net-new"
claims (B-P1 already shipped in M2/#280)
3 files changed, 31 insertions(+), 10 deletions(-)

```

91. user (2026-06-21T09:27:56.621Z)

```

=== commits on HEAD not in origin/master (must be ONLY mine) ===
aa07c61 docs: correct stale "bucket ingest is net-new" claims (B-P1 already shipped in M2/#280)
=== push ===
remote:
remote: Create a pull request for 'docs/correct-bucket-ingest-staleness' on GitHub by visiting:
remote: https://github.com/avidullu/khelsutra/pull/new/docs/correct-bucket-ingest-staleness
remote:
branch 'docs/correct-bucket-ingest-staleness' set up to track 'origin/docs/correct-bucket-
ingest-staleness'.
To https://github.com/avidullu/khelsutra.git
* [new branch] docs/correct-bucket-ingest-staleness -> docs/correct-bucket-ingest-
staleness

```

92. assistant (2026-06-21T09:28:02.443Z)

Clean ? only my commit. Let me open the PR:

93. user (2026-06-21T09:28:19.527Z)

<https://github.com/avidullu/khelsutra/pull/55>

94. assistant (2026-06-21T09:28:39.465Z)

Doc-correction PR is up: [khelsutra#55](https://github.com/avidullu/khelsutra/pull/55). Now the ****anti-spiral memory fixes**** ? the steering the next session reads first. Correcting the ``current-state.md`` NEXT SESSION FOCUS so B-P1 isn't re-scoped as a feature:

95. user (2026-06-21T09:28:49.384Z)

<tool_use_error>File has not been read yet. Read it first before writing to it.</tool_use_error>

96. user (2026-06-21T09:29:03.218Z)

14 ****?? NEXT SESSION FOCUS (2026-06-21, after the big localization+dry-run session):**** SPA is fully ****en+kn+hi (DONE)****; a real ****UI?engine?reel?download CUJ ran (en+hi)****; a ****critical engine bug fixed**** (#281). ****? NEXT BUILD (owner-picked, in order): (1) Path B ? point at a bucket video IN THE UI**** ? ****B-P1 = engine `gcs://` ingest endpoint**** (extend ``/api/process`` for a ``video_uri`` ? fetch via ``backend/storage`` ? process; today local-path-only). ? Scouted: ``_execute_processing`` (main.py:140) ALREADY calls ``storage.acquire_local_input(req.video_path,?)`` (:159) ? a ``gcs://`` ``video_path`` may already fetch ? ****verify that first****, then SPA ingest/progress screen (B-P2). **** (2) Droplet deploy (opt 3):**** deploy the FIXED engine to ``168.144.126.176`` (un-provisioned, 2-core/3.8GB) ? re-run CUJ on Linux. ****STILL OPEN (localization DoD):**** B3 (``site/`` marketing still English-only) + native-speaker review of the AI-drafted kn/hi. ****Ramp-up:**** ``khelsutra/docs/DRY_RUN_CUJ.md`` + ``FRIENDLY_AND_MULTILINGUAL.md`` §7 + the dry-run checkpoint below. ****Clean slate:**** both repos on master, ****0 open PRs****, no worktrees/VMs. khelsutra master ``2421557`` · engine master ``2afd54f``.

97. assistant (2026-06-21T09:29:06.394Z)

Now I can edit precisely:

98. user (2026-06-21T09:29:15.878Z)

The file `C:\Users\avidu\claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\memory\current-state.md` has been updated successfully. (file state is current in your context ? no need to Read it back)

99. assistant (2026-06-21T09:29:21.499Z)

Now the dry-run checkpoint (line 18) carries the same stale "local-path-only" claim ? let me correct its forward-looking note:

100. user (2026-06-21T09:29:22.292Z)

18 ****Checkpoint:**** 2026-06-21 (DRY-RUN CUJ + a CRITICAL ENGINE FIX) ? owner: "stitch the UI to the tool; dry-run a ****GCS-bucket video**** via the UI ? ****reel downloaded local****; repeat in Hindi. Run locally (opt 1) then droplet (opt 3); report both; identify fixes + merge." Chose ****A-now-then-B****, ****Gemini****. ****Tracked goal ? `khelsutra/docs/DRY\_RUN\_CUJ.md`**** ([#53]/[#54]). ****? PATH-A LOCAL RUN DONE (en + hi):**** isolated engine off origin/master (fresh DB/output worktree) + ``.env`` Gemini key ? ``gsutil cp gs://khelsutra-rally-corporus/videos/mahadevpura_smoke_180s.mp4`` ? ``POST /api/process {gemini}`` ? ****12 rallies**** (``video_id 05e0e577?``) ? SPA ``npm run dev`` (Vite proxies ``/api?:8000``) ``?video=05e0e577?`` ? loaded 12 real rallies ? ****Build ? /api/compile ? reel (1080p, 76.5s) DOWNLOADED to `~/Downloads/khelsutra-dryrun/highlights\_local\_{en,hi}.mp4`**** (50.9MB). ****Hindi repeat ?**** (page/cards/"??? ?????"/"???? ???? ????? ??" all Hindi; reel downloaded). ****?? CRITICAL BUG FOUND + FIXED ? engine [#281](https://github.com/Khelsutra/badminton-highlight-indexer/pull/281) MERGED:**** running the documented ``python -m backend.main --server`` runs main.py as ``__main__`` (sets ``__main__.db_instance``), but ``job_worker`/`uploads`` read ``import backend.main`` ? a SEPARATE module whose ``db_instance`` stays None ? the ``async-job drain`` crashes silently on ``None.claim_due_job()`` ? ****every /api/process job hangs in `queued` forever****. Latent: unit tests monkeypatch the executor inline; the UI was verified vs a STUB engine. Fix = dispatch ``__main__`?` ``backend.main``; + ``tests/test_server_entrypoint.py`` (real subprocess guard, FAILS-without/PASSES-with). Full engine suite green (2m23s). **\*\*Unblocked the run via the single-module invocation\*\*** (``python -c "...; from backend import main as m; m.main()"``). ? ****OPT-3 DROPLET (168.144.126.176) NOT YET RUN:**** reachable (Linux, py3.12, ffmpeg, 2-core/3.8GB) but ****un-**

provisioned** (no `~/rally` engine checkout) ? needs a from-scratch engine install OR a proper deploy of the FIXED engine (the fix applies there too). **? NEXT:** opt-3 droplet run (deploy fixed engine ? re-run CUJ); Path B (in-UI bucket ingest ? net-new HTTP gcs:// ingest + ingest/progress screens; `/api/process` is local-path-only today + the worker writes CSV to the bucket not the API DB). [[monetization-plan]]

101. user (2026-06-21T09:29:36.207Z)

The file C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\memory\current-state.md has been updated successfully. (file state is current in your context ? no need to Read it back)

102. assistant (2026-06-21T09:29:46.372Z)

Now the durable anti-spiral artifact ? a lesson memory so the *pattern* (not just this instance) is captured for future sessions:

103. assistant (2026-06-21T09:30:04.758Z)

name: lesson-verify-engine-surface-before-scoping

description: Anti-spiral ? before scoping a doc's "[gap]/net-new" engine claim as work, RE-VERIFY it against engine code at a cited file:line. The engine ships PRs between sessions, so its surface drifts and docs go stale. The B-P1 "gcs ingest is net-new" miss (2026-06-21).

metadata:

node_type: memory

type: feedback

The pattern that caused a near-spiral (2026-06-21): ``khelsutra/docs/DRY_RUN_CUJ.md` $1` claimed ? tagged ``[verified]`` ? that ``POST /api/process`` accepts only a **local** ``video_path`` and that bridging ``gcs://`` ? a queryable API-DB ``video_id`` was **net-new** (appliance T12/T14). The next-session steering then scoped **B-P1** = "build the engine gcs:// ingest endpoint."**It was already done and tested:** ``/api/process`` has accepted bucket/Drive URIs since engine **M2/#280** (``resolve_input_ref`` at ``main.py:274`` ? skips the local-exists-404 for non-local backends at ``main.py:283`` ? worker ``acquire_local_input`` at ``main.py:159`` fetches ? same pipeline ? queryable ``videos`/`play_segments``), with shipped tests (``test_process_nonlocal_uri_skips_exists_and_fetches`` + hosted-mode/unknown-scheme/CLI variants). A 3-skeptic verification workflow refuted **0/3**, ``gaps:[]``.

Why it went stale: the \$1 claim was inferred from the **worker CLI** (``backend.worker infer --video-uri``, which writes CSV to the bucket, NOT the API DB) and never checked against the **API endpoint's** routing code ? and M2/#280 had since wired URI ingest into ``/api/process``. A ``[verified]`` tag was applied to a hypothesis. **The engine is a sibling repo** (``Khelsutra/badminton-highlight-indexer``) that ships PRs independently between sessions ? its surface drifts, so any doc claim about it ages.

Why: scoping a "build it" task from a stale ``[gap]`` claim wastes a session re-discovering that the work is already shipped, and risks re-implementing something that exists. The owner explicitly asked to stop the spiral ("ensure the next sessions don't spiral in such territories again only to find the staleness").

How to apply:

- Before treating any ``[gap]`` / ``net-new`` / ``X-only`` / "does not exist" **engine-surface** claim in a doc as work-to-do, **re-verify it against engine code at a cited ``file:line``** (grep the endpoint + trace the call chain). Pairs with the *git-pull-before-reading* rule ? pull the engine first, since master moves between sessions.
- Treat ``[verified]`` without an accompanying ``file:line`` as a hypothesis, not a fact. When you do verify, leave the ``file:line`` so the next reader can re-check cheaply.
- When you find drift, fix it honestly *in place* (flip the claim + date it + add the file:line + keep the lineage) AND fix the memory steering, so it isn't re-discovered ? see [[doc-status-register]]'s honest-update discipline. Docs corrected here ? khelsutra [#55] (DRY_RUN_CUJ \$1/2/4/6/8 + LOCAL_APPLIANCE + README).
- A **test** is the most durable anti-staleness guard: an executable assertion of the contract

can't silently rot the way prose does. (Note: hermetic in-process locks ? subprocess guards ? be honest about which.)

Ties [[working-agreement-attribution]] (never assert unverified facts/provenance), [[gotcha-shared-checkout-branch-collisions]] (the engine checkout moves under you), and [[decision-rigor-norm]] (verify foundations, don't trust a label).

104. user (2026-06-21T09:30:04.790Z)

File created successfully at: C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\memory\lesson-verify-engine-surface-before-scoping.md (file state is current in your context ? no need to Read it back)

105. assistant (2026-06-21T09:30:13.052Z)

Now add the index pointer in `MEMORY.md`:

106. user (2026-06-21T09:30:14.211Z)

14 - [Working agreement: merging](working-agreement-merging.md) ? owner authorized Claude to MERGE its own good-engineering-fix PRs on verified-green CI (2026-06-12); product/strategy/licensing stay owner-gated.

15 - [Working agreement: tracked project docs](working-agreement-tracked-project-docs.md) ? substantial design/project work gets a **tracked project doc** (template + \$7 progress tracker + definition-of-done); freeze ? track across sessions ? mark done ? archive. Owner-loved framing (2026-06-16); codified in `CLAUDE.md` + `docs/PROJECT\_DOC\_TEMPLATE.md` (PR #186).

16 - [Doc-status register + archival due-diligence](doc-status-register.md) ? owner directive (2026-06-21): `docs/DOC\_STATUS.md` is the **living doc-lifecycle register**; **archive foundational docs only AFTER honestly updating them** (verify?flip status+completion note?update refs?keep lineage?move). Worked example = DECOUPLED #270. Codified in `CLAUDE.md` archive policy.

17 - [Launch Report convention](launch-report-convention.md) ? owner directive (2026-06-17): EVERY default-config/quality-bit flip ships with a documented **Launch Report** (total-quality-impact table on both rulers + over-seg guard + honest caveats), recorded for posterity BEFORE the flip merges. Born from the Gate-A #146 flip that regressed (#151). Pairs with [[eval-dual-set-regression]].

18 - [Feedback: launch quality bar](feedback-launch-quality-bar.md) ? approaching launch (2026-06-20): tightly **isolated one-item PRs + rising coverage**, and **serious observability** (full log analysis across every NEW code path; correlation/job IDs; loud failures, no silent drops). Make Observability a first-class doc section + DoD item.

19 - [Gotcha: shared-checkout branch collisions](gotcha-shared-checkout-branch-collisions.md) ? spawned tasks + sibling sessions SHARE this checkout; **branch from `origin/master` explicitly, re-verify HEAD before commit, stage explicit paths, serialize repo writes.** The #116/#117 collision (owner upset). Global form in ~/.claude/CLAUDE.md.

20 - [Gotcha: long GPU/WSL runs](gotcha-long-runs-gpu-wsl.md) ? running the WASB pipeline E2E on a big 4K video: Modern Standby kills it (keep-awake), cv2 extraction is the bottleneck (ffmpeg pre-extract), proxy-index/4K-stitch recipe + how to observe each phase.

21 - [GCP Ops Agent: always enable](gcp-ops-agent-enable.md) ? owner directive (missed TWICE): every rally GPU VM needs the Ops Agent for Observability/GPU graphs ? **3 pieces** (logging+monitoring APIs ON + SA metric/log-writer roles + agent install via startup-script). BAKE into `deploy/gcp` provisioning so it's automatic.

22 - [GCP VM: always DELETE, never just stop](gcp-vm-always-delete.md) ? owner cost rule (2026-06-20): a STOPPED VM still bills its disk (~\$34/mo for 200GB pd-ssd); **DELETE** ? \$0 (boot disk autoDelete=true). `teardown.sh` now defaults to `delete`; audit `gcloud compute disks list` for orphans. DO droplet `claude-do-rally-test` (~\$24/mo) is a shared testbed (owner-gated).

23 - [Compute = GCP only; DO dropped](compute-stack-gcp-only.md) ? **standing decision** (ADR-013, 2026-06-20): **GCP is the SOLE compute stack** (training + serving); **DigitalOcean dropped for compute** (GPU not enabled/credit-funded; Paperspace subscription-gated). Provision via `deploy/gcp/` only; don't relitigate DO/Paperspace without a new ADR; \$200 DO credits ? non-compute only.

107. user (2026-06-21T09:30:25.024Z)

The file C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\memory\MEMORY.md has been updated successfully. (file state is current in your context ? no need to Read it back)

108. assistant (2026-06-21T09:30:52.907Z)

Memory steering is corrected and the lesson is captured. Now the **strongest** anti-spiral guard: an executable test that locks the gs://?queryable-`video\_id` contract so it can't silently rot like the prose did. The workflow found the existing test never asserts `video\_id == MD5(fetched bytes)` nor `query\_play\_segments` non-empty. I'll add that as a tight engine PR ? in an **isolated worktree off `origin/master`** (the engine checkout is a concurrent session's branch; don't disturb it).

109. user (2026-06-21T09:30:56.860Z)

```
=== origin/master ===
cee4baa feat(serving): M4 ? DeviceContext + native WASB CPU-fallback (device="auto", cuda stays strict) (#283)
=== current checkout branch (leave it alone) ===
feat/serving-m4-device-context
=== create isolated worktree off origin/master ===
Preparing worktree (new branch 'test/gcs-process-queryable-coverage')
branch 'test/gcs-process-queryable-coverage' set up to track 'origin/master'.
HEAD is now at cee4baa feat(serving): M4 ? DeviceContext + native WASB CPU-fallback (device="auto", cuda stays strict) (#283)
```

110. assistant (2026-06-21T09:31:04.529Z)

Worktree ready off `origin/master`. Let me read the worktree's copy of the test region to anchor the new test precisely (and confirm helpers `\_FakeSeg`/`compute\_video\_id` are imported):

111. user (2026-06-21T09:31:05.356Z)

```
375     def test_process_nonlocal_uri_skips_exists_and_fetches(env):
376         """A gs:// input is NOT 404'd at submit (can't os.path.exists a bucket object); the
worker
377         fetches it to a local scratch file and the segmenter sees that LOCAL path, not the
URI."""
378         client, db, tmp_path, mp = env
379
380         def fake_fetch(uri, dst=None, *, config=None, overwrite=False):
381             with open(dst, "wb") as fh:
382                 fh.write(b"fetched-video-bytes")
383             return dst
384
385         mp.setattr(m.storage, "fetch", fake_fetch)
386         mp.setattr(m.registry, "run_all_with_context", lambda *a, **k: (_pass(), {}))
387         mp.setattr(m.segmenter_registry, "get", lambda name: _CaptureSeg)
388         _CaptureSeg.last_path = None
389         r = client.post("/api/process", json={"video_path": "gs://bucket/videos/x.mp4",
"segmenter": "x"})
390         assert r.status_code == 202                                     # NOT 404 ? non-local skips
the exists gate
391         job = client.get(f"/api/jobs/{r.json()['job_id']}").json()
392         assert job["status"] == "done"
393         assert _CaptureSeg.last_path is not None
394         assert not _CaptureSeg.last_path.startswith("gs://")          # a fetched LOCAL path, not
the URI
395         assert os.path.basename(_CaptureSeg.last_path) == "x.mp4"
396         # Provenance: the DB record keeps the SOURCE URI (not the ephemeral scratch path).
397         rec = db.get_video(job["video_id"])
398         assert rec is not None and rec["filepath"] == "gs://bucket/videos/x.mp4"
399
```

112. assistant (2026-06-21T09:31:19.725Z)

The worktree matches. Let me confirm `\_FakeSeg` (the segment-persisting fake) and `compute\_video\_id` are available in this test module before writing the new test:

113. user (2026-06-21T09:31:22.284Z)

```
=== _FakeSeg defined in test_api_endpoints.py? ===
52:class _FakeSeg:
=== compute_video_id accessible as m.compute_video_id? ===
30:from backend.utils.hashing import compute_video_id # canonical file-identity hashing
160:    video_id = compute_video_id(local_video)
=== _FakeSeg usage ===
52:class _FakeSeg:
73:    mp.setattr(m.segmenter_registry, "get", lambda name: _FakeSeg)
129:    mp.setattr(m.segmenter_registry, "get", lambda name: _FakeSeg)
322:    mp.setattr(m.segmenter_registry, "get", lambda name: _FakeSeg)
446:    mp.setattr(m.segmenter_registry, "get", lambda name: _FakeSeg)
Shell cwd was reset to C:\Users\avidu\Projects\badminton-highlight-indexer
```

114. assistant (2026-06-21T09:31:53.901Z)

`\_FakeSeg` (line 52, persists a segment) and `m.compute\_video\_id` are both available. Let me read the exact boundary after the existing test to anchor the insertion cleanly:

115. user (2026-06-21T09:31:54.773Z)

```
396         # Provenance: the DB record keeps the SOURCE URI (not the ephemeral scratch path).
397         rec = db.get_video(job["video_id"])
398         assert rec is not None and rec["filepath"] == "gs://bucket/videos/x.mp4"
399
400
401     def test_process_unknown_scheme_returns_400(env):
402         """An unsupported scheme (https/ftp/file) is a clean 400, never a 500 ? parse_uri's
ValueError
403         must not escape the request handler."""
404         client = env[0]
405         for bad in ("https://h/x.mp4", "ftp://h/x.mp4", "file:///etc/passwd"):
```

116. user (2026-06-21T09:32:11.301Z)

The file C:\Users\avidu\Projects\bhi-wt-gcs-test\tests\test_api_endpoints.py has been updated successfully. (file state is current in your context ? no need to Read it back)

117. assistant (2026-06-21T09:32:24.247Z)

Let me remove a sloppy no-op assertion I added and verify `compute\_video\_id` is purely content-based (my test hashes a different file path with the same bytes):

118. user (2026-06-21T09:32:27.145Z)

The file C:\Users\avidu\Projects\bhi-wt-gcs-test\tests\test_api_endpoints.py has been updated successfully. (file state is current in your context ? no need to Read it back)

119. user (2026-06-21T09:32:31.232Z)

```
=== compute_video_id impl (content-only?) ===
23:def compute_video_id(filepath: str) -> str:
24-    """Return the MD5 hex digest of the whole file ? the canonical video_id.
25-
26-    Streams the file in fixed-size chunks so multi-gigabyte recordings never load
27-    into memory at once.
28-    """
29-    hasher = hashlib.md5()
```



```
30-     with open(filepath, "rb") as f:
31-         for chunk in iter(lambda: f.read(_CHUNK_SIZE), b''):
32-             hasher.update(chunk)
33-     return hasher.hexdigest()
Shell cwd was reset to C:\Users\avidu\Projects\badminton-highlight-indexer
```

120. assistant (2026-06-21T09:32:37.165Z)

Confirmed content-only MD5. Now run the new test (plus the existing M2 block) in the worktree:

121. user (2026-06-21T09:32:47.342Z)

```
..... [100%]
5 passed, 20 deselected in 6.40s
Shell cwd was reset to C:\Users\avidu\Projects\badminton-highlight-indexer
```

122. assistant (2026-06-21T09:32:56.045Z)

Both parametrized variants pass alongside the existing gs:// tests. Let me lint the changed file and run the full worktree suite (the pre-PR gate):

123. user (2026-06-21T09:32:57.847Z)

```
=== ruff ===
All checks passed!
Shell cwd was reset to C:\Users\avidu\Projects\badminton-highlight-indexer
```

124. user (2026-06-21T09:32:58.717Z)

Command running in background with ID: bd5wysoe4. Output is being written to:
C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-indexer\9f6357d8-0e35-491a-a830-052e094c7f37\tasks\bd5wysoe4.output. You will be notified when it completes. To check interim output, use Read on that file path.

125. assistant (2026-06-21T09:33:05.218Z)

Ruff clean. While the suite runs, let me check whether the repo's CI runs mypy on `tests/` (to match the pre-PR gate):

126. user (2026-06-21T09:33:07.121Z)

```
=== mypy config (does it cover tests/?) ===
mypy.ini:7:# maps files to backend.* module names instead of duplicating top-level names.
pyproject.toml:5:# reason), so the wheel target packages are listed EXPLICITLY below ?
hatchling's
pyproject.toml:52:# tests skip via pytest.importorskip. For local dev against a sibling
checkout:
pyproject.toml:71:# MOUNTED Google Drive (Google Drive for Desktop) as plain files; see
backend/storage/gdrive.py.
pyproject.toml:88:packages = ["backend", "training"]
=== CI mypy invocation ===
.github/workflows/ci.yml:33: # Type gate (mypy.ini at repo root): lenient green baseline with
an explicit
.github/workflows/ci.yml:35: # up. Typed core deps are installed so mypy sees their real
signatures
.github/workflows/ci.yml:45: - name: Install mypy + typed core deps
.github/workflows/ci.yml:46:   run: python -m pip install mypy pydantic fastapi uvicorn
python-dotenv google-genai numpy
.github/workflows/ci.yml:48:   run: mypy backend training
.github/workflows/ci.yml:87:   # (pytest, pytest-cov, httpx for the fastapi TestClient,
ruff, mypy).
Shell cwd was reset to C:\Users\avidu\Projects\badminton-highlight-indexer
```

127. assistant (2026-06-21T09:33:14.412Z)

CI runs `mypy backend training` (not `tests/`), so my test-only change needs no mypy. Let me match CI's ruff scope (`ruff check .`) and note the coverage floor while the suite finishes:

128. user (2026-06-21T09:33:16.573Z)

```
=== CI ruff + cov invocation ===
16: # Lint gate (ruff.toml at repo root): pyflakes/pycodestyle + bugbear.
17: # Also the repo-wide syntax guard ? ruff reports invalid-syntax on every
28:     - name: Install ruff
29:       run: python -m pip install ruff
31:       run: ruff check . --output-format github
63:       run: python -m pip install pytest fastapi uvicorn pydantic python-dotenv google-genai
65:       run: python -m pytest tests/test_import_smoke.py -v
87:       # (pytest, pytest-cov, httpx for the fastapi TestClient, ruff, mypy).
90:       # pytest.importorskip ? run_all_tests.py prints which. The coverage floor
97:       run: python run_all_tests.py --cov=backend --cov=training --cov-report=term --cov-
fail-under=70
=== ruff check . (match CI) ===
All checks passed!
Shell cwd was reset to C:\Users\avidu\Projects\badminton-highlight-indexer
```

129. assistant (2026-06-21T09:33:28.739Z)

`ruff check .` matches CI and is clean; coverage floor is 70% and I'm only *adding* a test (can't regress). Let me wait for the full suite to confirm green, then commit. Checking its progress:

130. user (2026-06-21T09:33:30.292Z)

<system-reminder>Warning: the file exists but is shorter than the provided offset (1). The file has 1 lines.</system-reminder>

131. assistant (2026-06-21T09:33:59.012Z)

Suite still running. Let me confirm the worktree change is scoped to only my test file while it finishes:

132. user (2026-06-21T09:34:00.304Z)

```
M tests/test_api_endpoints.py
=== diffstat ===
 tests/test_api_endpoints.py | 43 ++++++++++++++++++++++++++++++++++++++
 1 file changed, 43 insertions(+)
Shell cwd was reset to C:\Users\avidu\Projects\badminton-highlight-indexer
```

133. assistant (2026-06-21T09:34:06.887Z)

Clean ? only the test file, +43 lines. Let me verify the engine remote (the repo moved to the Khelsutra org) while the suite finishes:

134. user (2026-06-21T09:34:07.730Z)

```
origin https://github.com/Khelsutra/badminton-highlight-indexer.git (fetch)
origin https://github.com/Khelsutra/badminton-highlight-indexer.git (push)
Shell cwd was reset to C:\Users\avidu\Projects\badminton-highlight-indexer
```

135. assistant (2026-06-21T09:34:13.164Z)

Let me check the suite result:

136. user (2026-06-21T09:34:15.244Z)

```

1      EXIT=0
2      ..... [ 95%]
3      ..... [100%]
4      1121 passed, 7 skipped in 72.48s (0:01:12)
5      =====
6      RESULT: PASS - all tests green
7      =====
8

```

137. assistant (2026-06-21T09:34:43.629Z)

Worktree suite is green: **1121 passed, 7 skipped** (clean `importorskip` skips; my 2 parametrized tests ran and passed). ruff clean, mypy N/A for tests, coverage can't regress. Let me commit the engine guard (safety protocol: verify HEAD, stage explicit path):

138. user (2026-06-21T09:34:50.004Z)

```

=== branch + HEAD ===
test/gcs-process-queryable-coverage
cee4baa feat(serving): M4 ? DeviceContext + native WASB CPU-fallback (device="auto", cuda stays
strict) (#283)
[test/gcs-process-queryable-coverage 2130416] test(api): lock gs://|gcs:// /api/process ->
queryable API-DB video_id (B-P1 guard)
  1 file changed, 43 insertions(+)
=== only my commit? ===
2130416 test(api): lock gs://|gcs:// /api/process -> queryable API-DB video_id (B-P1 guard)
Shell cwd was reset to C:\Users\avidu\Projects\badminton-highlight-indexer

```

139. assistant (2026-06-21T09:34:53.960Z)

Only my commit. Push and open the engine PR:

140. user (2026-06-21T09:35:11.872Z)

```

remote:
branch 'test/gcs-process-queryable-coverage' set up to track 'origin/test/gcs-process-queryable-
coverage'.
To https://github.com/Khelsutra/badminton-highlight-indexer.git
* [new branch]      test/gcs-process-queryable-coverage -> test/gcs-process-queryable-coverage
https://github.com/Khelsutra/badminton-highlight-indexer/pull/284
Shell cwd was reset to C:\Users\avidu\Projects\badminton-highlight-indexer

```

141. assistant (2026-06-21T09:35:29.339Z)

Engine guard is up: [badminton-highlight-indexer#284](https://github.com/Khelsutra/badminton-highlight-indexer/pull/284). Both PRs are good-engineering fixes (doc honesty + a test guard) ? within the merge authorization once CI is green, and merging completes the anti-spiral fix (master becomes truthful, not just a pending PR). Let me check CI on both:

142. user (2026-06-21T09:35:33.310Z)

```

=== engine PR #284 checks ===
Golden fixtures cross-OS (ubuntu-latest)          pending 0
https://github.com/Khelsutra/badminton-highlight-
indexer/actions/runs/27900239536/job/82559127889
Golden fixtures cross-OS (windows-latest)         pending 0
https://github.com/Khelsutra/badminton-highlight-
indexer/actions/runs/27900239536/job/82559127924
Ruff lint          pass    9s      https://github.com/Khelsutra/badminton-highlight-
indexer/actions/runs/27900239536/job/82559127914
Full test suite (offline, mocked)                 pending 0      https://github.com/Khelsutra/badminton-
highlight-indexer/actions/runs/27900239536/job/82559127888
Golden fixtures cross-OS (macos-latest)           pending 0      https://github.com/Khelsutra/badminton-
highlight-indexer/actions/runs/27900239536/job/82559127873

```

Mypy (lenient baseline) pending 0 <https://github.com/Khelsutra/badminton-highlight-indexer/actions/runs/27900239536/job/82559127875>
Syntax sweep + import smoke (no GPU stack) pending 0
<https://github.com/Khelsutra/badminton-highlight-indexer/actions/runs/27900239536/job/82559127884>

=== khelsutra PR #55 checks ===
Workers Builds: hidden-king-khelsutra pass 0
<https://dash.cloudflare.com/213c12cf60de9041dd8a3521aa206472/workers/services/view/hidden-king-khelsutra/production/builds/01852735-3811-4969-9f9a-055f214f210d>
site pass 12s
<https://github.com/avidullu/khelsutra/actions/runs/27900072361/job/82558687620>
web pass 19s
<https://github.com/avidullu/khelsutra/actions/runs/27900072361/job/82558687626>

143. assistant (2026-06-21T09:35:41.986Z)

khelsutra PR #55 is ****all green**** (site, web, Cloudflare build). Merging it completes the anti-spiral ? master becomes truthful, not just a pending PR (within the good-engineering-fix merge authorization). The engine PR's heavy checks are still pending. Let me merge #55:

144. user (2026-06-21T09:35:48.740Z)

=== khelsutra master now ===
d0461fd docs: correct stale "bucket ingest is net-new" claims (B-P1 already shipped in M2/#280) (#55)
2421557 docs(dry-run): record Path-A local result (en+hi reel downloaded) + the engine fix it surfaced (#54)

145. assistant (2026-06-21T09:35:55.314Z)

#55 merged to khelsutra master (`d0461fd`). Let me tidy the local khelsutra checkout (back to master) and update the memory to record the merge:

146. user (2026-06-21T09:35:59.409Z)

'refs/remotes/origin/docs/correct-bucket-ingest-staleness', but not yet merged to HEAD
Deleted branch docs/correct-bucket-ingest-staleness (was aa07c61).
=== khelsutra HEAD ===
d0461fd docs: correct stale "bucket ingest is net-new" claims (B-P1 already shipped in M2/#280) (#55)
=== local branches ===
* master

147. assistant (2026-06-21T09:36:05.317Z)

khelsutra is clean on master with the fix merged. Now let me check the engine PR's CI (the heavy checks were pending):

148. user (2026-06-21T09:36:07.075Z)

Golden fixtures cross-OS (ubuntu-latest) pass 21s
<https://github.com/Khelsutra/badminton-highlight-indexer/actions/runs/27900239536/job/82559127889>
Golden fixtures cross-OS (windows-latest) pending 0
<https://github.com/Khelsutra/badminton-highlight-indexer/actions/runs/27900239536/job/82559127924>
Mypy (lenient baseline) pass 28s <https://github.com/Khelsutra/badminton-highlight-indexer/actions/runs/27900239536/job/82559127875>
Ruff lint pass 9s <https://github.com/Khelsutra/badminton-highlight-indexer/actions/runs/27900239536/job/82559127914>
Syntax sweep + import smoke (no GPU stack) pass 20s
<https://github.com/Khelsutra/badminton-highlight-indexer/actions/runs/27900239536/job/82559127884>

Full test suite (offline, mocked) pending 0 <https://github.com/Khelsutra/badminton-highlight-indexer/actions/runs/27900239536/job/82559127888>
Golden fixtures cross-OS (macos-latest) pass 19s <https://github.com/Khelsutra/badminton-highlight-indexer/actions/runs/27900239536/job/82559127873>

149. assistant (2026-06-21T09:36:43.609Z)

Engine CI is progressing (ruff, mypy, syntax, golden-fixtures ubuntu/macos all pass; full suite + windows-golden still running). Let me record this session's checkpoint in memory while CI finishes:

150. user (2026-06-21T09:36:56.412Z)

The file C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\memory\current-state.md has been updated successfully. (file state is current in your context ? no need to Read it back)

151. assistant (2026-06-21T09:36:59.561Z)

Memory checkpoint recorded. Let me check the engine PR's remaining CI checks:

152. user (2026-06-21T09:37:01.561Z)

Golden fixtures cross-OS (ubuntu-latest) pass 21s
<https://github.com/Khelsutra/badminton-highlight-indexer/actions/runs/27900239536/job/82559127889>
Golden fixtures cross-OS (windows-latest) pass 50s
<https://github.com/Khelsutra/badminton-highlight-indexer/actions/runs/27900239536/job/82559127924>
Mypy (lenient baseline) pass 28s <https://github.com/Khelsutra/badminton-highlight-indexer/actions/runs/27900239536/job/82559127875>
Ruff lint pass 9s <https://github.com/Khelsutra/badminton-highlight-indexer/actions/runs/27900239536/job/82559127914>
Syntax sweep + import smoke (no GPU stack) pass 20s
<https://github.com/Khelsutra/badminton-highlight-indexer/actions/runs/27900239536/job/82559127884>
Full test suite (offline, mocked) pending 0 <https://github.com/Khelsutra/badminton-highlight-indexer/actions/runs/27900239536/job/82559127888>
Golden fixtures cross-OS (macos-latest) pass 19s <https://github.com/Khelsutra/badminton-highlight-indexer/actions/runs/27900239536/job/82559127873>

153. assistant (2026-06-21T09:37:12.578Z)

Only the full test suite remains (all else green). Let me watch it to completion, then merge:

154. user (2026-06-21T09:37:55.930Z)

Golden fixtures cross-OS (ubuntu-latest) pass 21s
<https://github.com/Khelsutra/badminton-highlight-indexer/actions/runs/27900239536/job/82559127889>
Golden fixtures cross-OS (windows-latest) pass 50s
<https://github.com/Khelsutra/badminton-highlight-indexer/actions/runs/27900239536/job/82559127924>
Mypy (lenient baseline) pass 28s <https://github.com/Khelsutra/badminton-highlight-indexer/actions/runs/27900239536/job/82559127875>
Ruff lint pass 9s <https://github.com/Khelsutra/badminton-highlight-indexer/actions/runs/27900239536/job/82559127914>
Syntax sweep + import smoke (no GPU stack) pass 20s
<https://github.com/Khelsutra/badminton-highlight-indexer/actions/runs/27900239536/job/82559127884>
Full test suite (offline, mocked) pass 2m32s <https://github.com/Khelsutra/badminton-highlight-indexer/actions/runs/27900239536/job/82559127888>
Golden fixtures cross-OS (macos-latest) pass 19s <https://github.com/Khelsutra/badminton-highlight-indexer/actions/runs/27900239536/job/82559127873>

```
Golden fixtures cross-OS (ubuntu-latest)      pass    21s
https://github.com/Khelsutra/badminton-highlight-
indexer/actions/runs/27900239536/job/82559127889
Golden fixtures cross-OS (windows-latest)     pass    50s
https://github.com/Khelsutra/badminton-highlight-
indexer/actions/runs/27900239536/job/82559127924
Mypy (lenient baseline) pass    28s    https://github.com/Khelsutra/badminton-highlight-
indexer/actions/runs/27900239536/job/82559127875
Ruff lint      pass    9s    https://github.com/Khelsutra/badminton-highlight-
indexer/actions/runs/27900239536/job/82559127914
Syntax sweep + import smoke (no GPU stack)     pass    20s
https://github.com/Khelsutra/badminton-highlight-
indexer/actions/runs/27900239536/job/82559127884
EXIT=0
```

155. assistant (2026-06-21T09:37:58.589Z)

All engine checks green (full suite 2m32s). Merging the guard, then cleaning up the worktree:

156. user (2026-06-21T09:38:05.544Z)

```
=== engine origin/master now ===
06ff4b7 test(api): lock gs://|gcs:// /api/process -> queryable API-DB video_id (B-P1 guard)
(#284)
cee4baa feat(serving): M4 ? DeviceContext + native WASB CPU-fallback (device="auto", cuda stays
strict) (#283)
```

157. assistant (2026-06-21T09:38:10.791Z)

Both PRs merged (engine master `06ff4b7`). Let me clean up the worktree and update the memory to reflect #284 merged:

158. user (2026-06-21T09:38:13.343Z)

```
Deleted branch test/gcs-process-queryable-coverage (was 2130416).
=== worktrees now ===
C:/Users/avidu/Projects/badminton-highlight-indexer
44c9f01 [feat/serving-m4-device-context]
C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/agent-a771ff1d85e18f2ed
7e0881a [perf/wasb-fast-frames]
C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/bhi-execpolicy
5cb7bbf [bhi-execpolicy]
C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/boxhill17
8736e2d [run/boxhill17-highlights]
C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/gap-doc
f04f3f3 [docs/rally-gap-closure]
C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/hardchop
614325d [feat/w1-hard-cap-fallback]
C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/pr-stitch
a6adcc5 [fix/runner-stitch-watermark]
C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/recdoc
9fd4215 [docs/recording-single-match]
C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/resweep
fe34013 [docs/lovo-resweep-n13]
C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/tier1-windowing
a3b209a [master]
C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/w2-safeguard
5ba6955 [feat/w2-never-empty-safeguard]
=== engine local checkout untouched? ===
feat/serving-m4-device-context
```

159. assistant (2026-06-21T09:38:18.140Z)

Worktree removed cleanly; the engine checkout and sibling worktrees are untouched. Let me flip

#284 to MERGED in the memory checkpoint:

160. user (2026-06-21T09:38:23.431Z)

The file C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\memory\current-state.md has been updated successfully. (file state is current in your context ? no need to Read it back)

161. assistant (2026-06-21T09:38:29.199Z)

Let me sweep the memory dir for any remaining stale "B-P1 = build the endpoint" / "local-path-only" framing I haven't caught (e.g. session-handoff):

162. user (2026-06-21T09:38:31.285Z)

C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\memory\current-state.md:14:[Omitted long matching line]
C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\memory\current-state.md:16:[Omitted long matching line]
C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\memory\current-state.md:20:[Omitted long matching line]
C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\memory\lesson-verify-engine-surface-before-scoping.md:10:[Omitted long matching line]

163. assistant (2026-06-21T09:38:42.852Z)

The matches at lines 14/20 are my *corrected* text and line 10 is the lesson quote. Let me verify line 16 (the localization checkpoint) doesn't hide an uncorrected stale claim:

164. user (2026-06-21T09:38:43.910Z)

14:[Omitted long matching line]
16:[Omitted long matching line]
20:[Omitted long matching line]

165. user (2026-06-21T09:38:52.206Z)

14:local-path-only
16:local-path-only
20:local-path-only

166. assistant (2026-06-21T09:39:10.013Z)

Line numbers shifted after my checkpoint insertion. Let me read the current structure to confirm line 16 is my corrected text, not a missed stale claim:

167. user (2026-06-21T09:39:11.005Z)

14 ***? NEXT SESSION FOCUS (2026-06-21, after the big localization+dry-run session):** SPA is fully **en+kn+hi (DONE)**; a real **UI?engine?reel?download CUJ ran (en+hi)**; a **critical engine bug fixed** (#281). ***? NEXT BUILD (owner-picked): (1) Path B ? point at a bucket video IN THE UI.** ? **B-P1 VERIFIED ALREADY DONE ? NO ENGINE CODE (do NOT re-scope it as a feature):** `POST /api/process` has accepted `gs://`/`gcs://` URIs since **M2/#280** (submit resolves the URI + skips the local-exists-404 for non-local backends `main.py:283`; the worker fetches to scratch `acquire\_local\_input main.py:159` ? same hash?validate?segment?DB pipeline ? queryable API-DB `video\_id`); **TESTED**
(`tests/test\_api\_endpoints.py::test\_process\_nonlocal\_uri\_skips\_exists\_and\_fetches` + hosted-mode-reject / unknown-scheme-400 / CLI variants). Confirmed by a **3-skeptic verification workflow (0 refuted, gaps:[])** . **Scope pin:** holds in DEFAULT single-user mode (hosted-mode `allowed\_video\_root` correctly **400s** a bucket URI ? by design). Stale "local-path-only / net-new" docs **CORRECTED ? khelsutra [#55]** (DRY_RUN-CUJ \$1/2/4/6/8 + LOCAL_APPLIANCE + README). ***? So Path B = ONLY B-P2 = the SPA ingest/progress screen** (enter a `gs://` URI ? `POST /api/process` ? poll `/api/jobs` ? load `result.video\_id`; back-half already works) ? **100%

khelsutra `web/` SPA + i18n (en/kn/hi), ZERO engine code.** Ready B-P2 sketch: `DRY\_RUN\_CUJ.md`
\$2 (new `web/src/{api/types,api/client,hooks/useIngestJob,components/IngestPanel}.tsx` +
`App.tsx` `videoId?state` handoff + `ingest.\*` keys). See [\[\[lesson-verify-engine-surface-before-scoping\]\]](#). ** (2) Droplet deploy (opt 3):** deploy the FIXED engine to `168.144.126.176` (un-
provisioned, 2-core/3.8GB) ? re-run CUJ on Linux. **STILL OPEN (localization DoD):** B3 (`site/`
marketing still English-only) + native-speaker review of the AI-drafted kn/hi. **Ramp-up:**
`khelsutra/docs/DRY\_RUN\_CUJ.md` + `FRIENDLY\_AND\_MULTILINGUAL.md` \$7 + the dry-run checkpoint
below. **Clean slate (at session start):** both repos on master, 0 open PRs, no worktrees/VMs.
khelsutra master `2421557` · engine master `2afd54f` (? origin/master `cee4baa`, #283 M4
squashed).

15

16 **Checkpoint:** 2026-06-21 (STALENESS-FIX / Path-B verify) ? owner: pick up Path B +
"fix all stale docs and stop next sessions spiralling into the staleness." **? Verified-FIRST
finding: B-P1 was ALREADY DONE** ? `POST /api/process` accepts `gs://`/`gcs://` since
M2/#280 (resolve?skip-exists-404?worker fetch?same pipeline?queryable API-DB `video\_id`) +
TESTED; confirmed by inline file:line trace AND a **3-skeptic verification workflow (0/3
refuted, gaps:[])** . So the doc-scoped "build the gcs ingest endpoint" was a phantom; Path B =
SPA-only (B-P2) . **DONE this session:** (1) **khelsutra [#55] MERGED** (master `d0461fd`) ?
corrected `DRY\_RUN\_CUJ` \$1/2/4/6/\$8 + `LOCAL\_APPLIANCE` (`/api/health` EXISTS now M1/#268;
bucket-URI ingest NOT a net-new endpoint; T12 engine-half ? done) + `README`; added the anti-
spiral convention "re-verify engine-surface `[gap]`/`net-new` claims at a cited file:line ? the
engine ships PRs between sessions." (2) **Memory steering corrected** (this NEXT-SESSION-FOCUS
block + the dry-run checkpoint were stale "local-path-only") + new lesson [\[\[lesson-verify-engine-surface-before-scoping\]\]](#) + MEMORY.md index. (3) **engine [#284] MERGED** (master
`06ff4b7`, all CI green incl. full suite 2m32s) ? regression GUARD
`test\_process\_bucket\_uri\_yields\_queryable\_video\_id` (parametrized gs://+gcs://) pinning
`video\_id==MD5(fetched bytes)` (not the URI) + `POST /api/query` queryability (the SPA contract;
previously only LOCAL had it); built+removed in an isolated worktree off `origin/master` (engine
checkout left on its concurrent-session branch). **? NEXT: B-P2** = the SPA ingest/progress
screen (enter gs:// URI ? /api/process ? poll /api/jobs ? load video_id; back-half already
works) ? **100% `web/` + i18n (en/kn/hi), ZERO engine code** ; ready sketch in `DRY\_RUN\_CUJ.md`
\$2 + the workflow synthesis
(`web/src/{api/types,api/client,hooks/useIngestJob,components/IngestPanel}.tsx` + `App.tsx`
`videoId?state` + `ingest.\*` keys). Then opt-3 droplet (deploy fixed engine + GCS creds + #281 fix
? re-run CUJ on Linux).

17

18 **Checkpoint:** 2026-06-21 (NEW SESSION ? UX+l10n focus STARTED) ? ramped up via a
research+verification **workflow** (web/+site/ string maps, Indic-l10n prior art, **5
adversarial verdicts:** both surfaces **ZERO i18n today** ; recording-edu **already shipped** on
the homepage gist + `/capture`; `I18N\_PLAN` targets the **dead** `backend/api/static` so its
P1/P3 wiring retargets to site/+web/). **? Isolated the work into ONE umbrella tracked project ?
`khelsutra/docs/FRIENDLY\_AND\_MULTILINGUAL.md`** (A: non-tech UX + B: kn/hi l10n; \$7 P-tracker =
source of truth; **archive when en+kn+hi render on BOTH surfaces**). Pointers wired: khelsutra
`docs/README`+`NEXT\_STEPS`; engine `I18N\_PLAN`+`UI\_PROPOSAL` banners + `DOC\_STATUS` \$3a/\$3b rows
(engine edits in an **isolated worktree off origin/master** ? engine checkout is a concurrent
session's branch; their findings-#1/#2 PR **#274 merged mid-session** ? master `03fa951`).
**Owner (AskUserQuestion) locked D6 (en+kn?hi) · D7 (OFL Noto self-host) · D8 (SPA-first) · D9
(catalogs in khelsutra); only D-BACKEND-DEPTH deferred. 3 PRs OPEN:** khelsutra docs
[#41](https://github.com/avidullu/khelsutra/pull/41); engine pointers
[#275](https://github.com/Khelsutra/badminton-highlight-indexer/pull/275); **B1 SPA i18n
scaffold + language switcher (en-only) ? [#42](https://github.com/avidullu/khelsutra/pull/42) ?
CI GREEN (web+site pass), verified LIVE in-browser** (switcher EN|???? persists; content falls
back to en with no kn catalog; active-highlight follows the *chosen* locale ? a bug caught by
browser verification + fixed). B1 = react-i18next + ICU + detector + `en` catalog + the SPA's
FIRST jsdom render tests (79 tests green); deps MIT/BSD-clean. Engine pointer PR built in an
isolated worktree (now removed); engine checkout kept flipping branches (concurrent session
active) ? never disturbed. **? NEXT slices (tracker \$7): B2** (full ~50-string extraction + key-
parity/no-bare-strings CI) ? **B3** (site `t()` shim) ? **B4/B5** (Noto fonts + kn catalog via
Gemini-draft?review). **Also (owner ask, 2026-06-21): the en/kn/hi academy flyers already
existed** (a sibling merged PR **#40** ? master `37053b9`; flyer.html / flyer-kn.html / flyer-
hi.html served at /flyer{,-kn,-hi}) but were UNLINKED ? **linked all three from the homepage
footer + added VM-host clean routes ? [#43](https://github.com/avidullu/khelsutra/pull/43)
(MERGED ? master `f121c7f`).** Then **owner asked for MORE flyer placement ?
[#44](https://github.com/avidullu/khelsutra/pull/44): "Flyer" in the desktop top-nav + the
3-language links on own-model's foot** (44 site tests green; verified live). (Flyers use Google-

Fonts Noto ? separate from the app's D7 self-host; not refactored.) **Owner #41 comment ? added a rigorous §6.5 "automatic language-negotiation" testing strategy** (matrix N1?N9: a kn/hi BROWSER auto-gets the respective version; DoD gate) ** concrete browser-language detection tests in B1 [#42]** (N1/N3?N7; 84 web tests green). #41/#42/#275/#44 all merged. **?? THEN owner: "take it to completion until you can run a screencast with a kannada-default browser ? right page renders ? switch hi ? en; merge PRs as you go." ? ? DONE for the `web/` SPA.** Merged in sequence (I merged each on green CI, owner-authorized): **B2 [#45]** (full ~50-string extraction + key-parity/no-bare-strings CI) . **B4 [#46]** (self-host OFL Noto Kannada/Devanagari, @fontsource-variable) . **B5 [#47]** (Kannada catalog) . **B6 [#48]** (Hindi catalog ? the extensibility proof: just a catalog + supported-list entry). **SPA now fully en+kn+hi; 99 web tests green** (incl. N1/N2 browser-detection, B5/B6 render, key-parity kn==hi==en, `` sync). **? SCREENCAST PRODUCED** (Playwright `locale: kn-IN` ? Kannada auto-renders ? switch hi ? en; `C:\Users\avidu\Projects\khelesutra-screencast\khelesutra-i18n-screencast.mp4` (9s) + per-locale frames). Tracker milestone ? [#49]. ? **kn/hi are AI-DRAFTED (me) ? native-speaker review pending** before wide public launch. **? REMAINING for full §9 DoD (NOT archive-ready):** **B3** (the `site/` marketing surface ? homepage/capture still English-only ? via the shared `t()` shim) . **A3** recording-education kn/hi . the **non-tech UX A-items** (A1/A2/A4/A6...) . native review. Playwright installed in isolated `C:\Users\avidu\Projects\khelesutra-screencast\` (kept out of web/ deps). **THEN owner polish batch (2026-06-21) ? all merged:** (1) **OFFLINE/self-contained verified + guarded ? [#50]:** the SPA is air-gap-renderable (Noto = local woff2 via @fontsource, `@font-face` url?local `/assets`, no Google-Fonts/CDN, `web/index.html` no external link); CI guard added so it can't regress. *(The marketing `site/` still uses Google-Fonts ? hosted/online surface, not the offline tool; self-host = optional follow-up.)* (2) **SCREENCAST uploaded ? `gs://khelesutra-rally-corpora/promo/khelesutra-i18n-screencast.mp4`** (+ local). (3) **Localized SAMPLE-QUERY PROMPT (en/kn/hi) ? [#51]:** `query.hint` = instructive "type `over 10s` for rallies >10s ? or tap a sample" + `query.try` labels pills "Sample queries"; verified live kn. (4) **§6.5 testing rule:** i18n testing is now a MANDATORY merge-gate for any future query-parsing/semantic-understanding/NL work (parser is English-grammar-only today, A9). **103 web tests green.** NO open PRs. [[next-session-ux-localization]] [[doc-status-register]] [[working-agreement-tracked-project-docs]]

19

20 ****Checkpoint:**** 2026-06-21 (DRY-RUN CUJ + a CRITICAL ENGINE FIX) ? owner: "stitch the UI to the tool; dry-run a **GCS-bucket video** via the UI ? **reel downloaded local**; repeat in Hindi. Run locally (opt 1) then droplet (opt 3); report both; identify fixes + merge." Chose ****A-now-then-B****, ****Gemini****. ****Tracked goal ? `khelesutra/docs/DRY\_RUN\_CUJ.md`**** ([#53]/[#54]). ****? PATH-A LOCAL RUN DONE (en + hi):**** isolated engine off origin/master (fresh DB/output worktree) + `.env` Gemini key ? `gsutil cp gs://khelesutra-rally-corpora/videos/mahadevpura\_smoke\_180s.mp4` ? `POST /api/process {gemini}` ? ****12 rallies**** (`video\_id 05e0e577?`) ? SPA `npm run dev` (Vite proxies /api?:8000) `?video=05e0e577?` ? loaded 12 real rallies ? ****Build ? /api/compile ? reel (1080p, 76.5s) DOWNLOADED to** `~/Downloads/khelesutra-dryrun/highlights\_local\_{en,hi}.mp4`** (50.9MB). ****Hindi repeat ?**** (page/cards/"??? ????"/"???? ??" all Hindi; reel downloaded). *****? CRITICAL BUG FOUND + FIXED ? engine [#281](https://github.com/Khelesutra/badminton-highlight-indexer/pull/281)** MERGED:****** running the documented `python -m backend.main --server` runs main.py as `\_\_main\_\_` (sets `\_\_main\_\_.db\_instance`), but `job\_worker`/`uploads` read `import backend.main` ? a SEPARATE module whose `db\_instance` stays None ? the async-job drain crashes silently on `None.claim\_due\_job()` ? ****every /api/process job hangs in `queued` forever****. Latent: unit tests monkeypatch the executor inline; the UI was verified vs a STUB engine. Fix = dispatch `\_\_main\_\_`?`backend.main`; + `tests/test\_server\_entrypoint.py` (real subprocess guard, FAILS-without/PASSES-with). Full engine suite green (2m23s). ****Unblocked the run via the single-module invocation**** (`python -c "...; from backend import main as m; m.main()"`). ? ****OPT-3 DROPLET (168.144.126.176) NOT YET RUN:**** reachable (Linux, py3.12, ffmpeg, 2-core/3.8GB) but ****un-provisioned**** (no `~/rally` engine checkout) ? needs a from-scratch engine install OR a proper deploy of the FIXED engine (the fix applies there too). ****? NEXT:**** opt-3 droplet run (deploy fixed engine ? re-run CUJ); Path B (in-UI bucket ingest). ? ****CORRECTION (next-session verify):** the "/api/process is local-path-only / net-new gcs ingest" note here was STALE** ? `api/process` already accepts `gs://` (M2/#280, tested); Path B is ****SPA-only (B-P2)****, no engine endpoint. (The "worker writes CSV to the bucket not the API DB" bit is the ***separate* `worker infer` CLI, not the API path ? that's what the stale claim conflated.**) Docs fixed ? khelesutra [#55]; see the NEXT SESSION FOCUS block above + [[lesson-verify-engine-surface-before-scoping]]. [[monetization-plan]]

All three `local-path-only` hits are within my *corrected* text (they quote the stale claim as fixed). The localization checkpoint is clean, and a memory-wide sweep found no remaining stale gcs/B-P1 framing. Everything is consistent.

Done ? staleness fixed at the root, with a guard so it can't recur

The finding (verified first, before touching anything): B-P1 was a phantom task. `POST /api/process` has accepted `gs://`/`gcs://` URIs since **engine M2/#280** ? it resolves the URI, skips the local-exists-404 for non-local backends ([main.py:283](backend/main.py:283)), the worker fetches to scratch ([main.py:159](backend/main.py:159)), runs the same pipeline, and writes a queryable API-DB `video\_id`. It's even already tested. Confirmed two ways: my inline file:line trace **and** a verification workflow with 3 adversarial skeptics (**0/3** refuted, `gaps: []`).

Stale docs corrected ? [khelsutra#55](https://github.com/avidullu/khelsutra/pull/55) (merged, `d0461fd`):

- `DRY\_RUN\_CUJ.md` ? reframed §1/§2/§4/§6 + a new **§8** correction record (file:line evidence, the *default-single-user* scope pin, the hosted-mode-400 counterweight, the hermetic-test caveat).

- `LOCAL\_APPLIANCE\_ARCHITECTURE.md` ? caught a **second** drift: `/api/health` "does not exist" (it shipped in M1/#268); bucket-URI ingest isn't a net-new endpoint; T12 engine-half marked done.

- `README.md` ? precise: no in-UI bucket-ingest *screen* yet (the endpoint already accepts `gs://`).

Anti-spiral guards (so the next session can't re-discover this cold):

1. **Memory steering corrected** ? the `current-state.md` NEXT-SESSION-FOCUS and the dry-run checkpoint both said "local-path-only"; now they say "B-P1 verified done, do NOT re-scope as a feature."
2. **New lesson memory** [lesson-verify-engine-surface-before-scoping](../../.claude/projects/C--Users-avidu-Projects-badminton-highlight-indexer/memory/lesson-verify-engine-surface-before-scoping.md) + MEMORY.md index ? the *pattern*: before scoping a doc's `[gap]`/`net-new` **engine** claim, re-verify at a cited `file:line`; the engine ships PRs between sessions, so its surface drifts. **[verified]** without a `file:line` is a hypothesis.
3. **Convention baked into the docs themselves** (the drift caveat in DRY_RUN §1/§8).
4. **An executable guard ?** [badminton-highlight-indexer#284](https://github.com/Khelsutra/badminton-highlight-indexer/pull/284) (merged, `06ff4b7`, all CI green) ? `test\_process\_bucket\_uri\_yields\_queryable\_video\_id` (parametrized gs://+gcs://) pins `video\_id == MD5(fetched bytes)` and `POST /api/query` queryability. A test can't silently rot the way prose did; a regression now fails CI loudly.

Both PRs verified locally first (khelsutra web+site green; engine full suite **1121?/7skip**, ruff clean, mypy excludes `tests/`). The engine guard was built in an isolated worktree off `origin/master` and removed ? the engine checkout stayed on its concurrent-session branch, untouched.

What's actually left for "bucket?reel fully UI-driven": only **B-P2** ? the SPA ingest/progress screen (enter a `gs://` URI ? `/api/process` ? poll `/api/jobs` ? load `video\_id`; the back-half already works). It's **100%** `web/` + i18n, zero engine code ? a ready, code-grounded build sketch is in `DRY\_RUN\_CUJ.md` §2 and the workflow synthesis. Want me to build B-P2 next?

169. user (2026-06-21T09:39:53.328Z)

can you make sure to sync to repo head

170. assistant (2026-06-21T09:40:33.225Z)

On it ? syncing both local checkouts to remote head. Let me check current state of both first (the engine checkout is on a feature branch, so I need to be careful not to disturb a possibly-concurrent branch).

171. user (2026-06-21T09:40:37.774Z)

=== KHELSUTRA ===

master...origin/master

?? .claude/

?? site/access.db

--- pull --ff-only ---

Already up to date.

d0461fd docs: correct stale "bucket ingest is net-new" claims (B-P1 already shipped in M2/#280) (#55)

172. user (2026-06-21T09:40:39.613Z)

=== ENGINE ===

--- current branch + tracked-tree cleanliness ---

master...origin/master

?? .claude/

?? site/access.db

--- origin/master head ---

d0461fd docs: correct stale "bucket ingest is net-new" claims (B-P1 already shipped in M2/#280) (#55)

--- does master track docs/KHELSUTRA_PRODUCT_OVERVIEW.md? (conflict check) ---

173. assistant (2026-06-21T09:40:52.612Z)

The second command inadvertently ran in the khelsutra dir (parallel cwd collision). khelsutra is confirmed at head (`d0461fd`). Let me check the engine repo correctly with an explicit path:

174. user (2026-06-21T09:40:57.331Z)

=== ENGINE pwd ===

/c/Users/avidu/Projects/badminton-highlight-indexer

--- branch + tree ---

feat/serving-m4-device-context...origin/feat/serving-m4-device-context [gone]

?? artifacts/

?? docs/KHELSUTRA_PRODUCT_OVERVIEW.md

--- origin/master head ---

06ff4b7 test(api): lock gs://|gcs:// /api/process -> queryable API-DB video_id (B-P1 guard) (#284)

--- HEAD ---

44c9f01 docs(serving): mark \$7 tracker M4 ? + changelog (PR #283)

--- does origin/master track docs/KHELSUTRA_PRODUCT_OVERVIEW.md? ---

(empty above = not tracked, no switch conflict)

175. assistant (2026-06-21T09:41:14.504Z)

Now the real engine state is clear: local is on `feat/serving-m4-device-context` (HEAD `44c9f01`, remote branch `[gone]` ? #283 merged+deleted), tracked tree clean, origin/master at head `06ff4b7`. The feature branch is fully merged and its remote is gone, and master doesn't track the untracked local files ? so switching to master is safe and is the correct "sync to head". Doing it:

176. user (2026-06-21T09:41:19.320Z)

--- switch to master ---

fatal: 'master' is already used by worktree at 'C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/tier1-windowing'

--- ff to head ---

Your configuration specifies to merge with the ref 'refs/heads/feat/serving-m4-device-context' from the remote, but no such ref was fetched.

=== ENGINE now ===

feat/serving-m4-device-context...origin/feat/serving-m4-device-context [gone]

?? artifacts/

44c9f01 docs(serving): mark \$7 tracker M4 ? + changelog (PR #283)

```
--- merged feat branch still local (remote gone): ---
* feat/serving-m4-device-context      44c9f01 [origin/feat/serving-m4-device-context:
gone] docs(serving): mark $7 tracker M4 ? + changelog (PR #283)
```

177. assistant (2026-06-21T09:42:51.580Z)

Now I see the real blocker: the engine main checkout is on `feat/serving-m4-device-context` (HEAD `44c9f01`), which **can't fast-forward** to master ? #283 squash-merged it, so the branch's pre-squash commits diverge ? and I **can't switch to `master`** because a sibling worktree (`tier1-windowing`) holds it.

Per the owner's hard rule I won't force blindly ? let me first prove it's safe: the tracked tree must be clean and the divergent commits must already be in master.

178. user (2026-06-21T09:42:55.365Z)

```
=== tracked-tree clean? (only untracked allowed) ===
?? artifacts/
?? docs/KHELSUTRA_PRODUCT_OVERVIEW.md
=== the 2 commits feat has that master lacks (by SHA) ===
44c9f01 docs(serving): mark $7 tracker M4 ? + changelog (PR #283)
6932d4a feat(serving): M4 ? DeviceContext + native WASB CPU-fallback (device="auto", cuda stays
strict)
=== are those commits' CONTENT already in master? (diff feat tree vs origin/master tree) ===
```

179. assistant (2026-06-21T09:43:04.971Z)

Proven safe: the tracked tree is clean (only untracked files), and excluding my just-merged test file, `feat@44c9f01`'s tree is **byte-identical** to `origin/master` ? the 2 divergent commits are the M4 work, already in master via the #283 squash. So a reset loses nothing (and is relog-reversible). Since `master` is worktree-locked, I'll fast-forward the current branch to head and repoint its upstream:

180. user (2026-06-21T09:43:08.090Z)

```
--- safety net: tag the old tip in case ---
--- reset working tree to origin/master head (untracked preserved) ---
HEAD is now at 06ff4b7 test(api): lock gs://gcs:// /api/process -> queryable API-DB video_id
(B-P1 guard) (#284)
--- repoint upstream so future git pull --ff-only works ---
branch 'feat/serving-m4-device-context' set up to track 'origin/master'.
=== ENGINE now ===
06ff4b7 test(api): lock gs://gcs:// /api/process -> queryable API-DB video_id (B-P1 guard)
(#284)
```

feat/serving-m4-device-context...origin/master

```
?? artifacts/
```

181. assistant (2026-06-21T09:43:30.943Z)

Both repos are now at remote head:

- **khelsutra**: `master` @ `d0461fd` (origin/master head) ?
- **engine**: working tree synced to `06ff4b7` (origin/master head) ? ? untracked files preserved, old tip tagged `\_pre\_sync\_m4\_44c9f01` + in relog for reversibility.

One honest caveat to record for the next session: the engine main checkout couldn't be put **on** `master` (a sibling worktree `tier1-windowing` holds that branch), so it's the `feat/serving-m4-device-context` branch fast-forwarded to master head and repointed to track `origin/master` (so `git pull --ff-only` works going forward). Let me note that in memory and do a final confirmation:

182. user (2026-06-21T09:43:36.262Z)

The file C:\Users\avidu\claude\projects\C--Users-avidu-Projects-badminton-highlight-

indexer\memory\current-state.md has been updated successfully. (file state is current in your context ? no need to Read it back)

183. assistant (2026-06-21T09:43:39.203Z)

Final confirmation that both working trees match remote head exactly:

184. user (2026-06-21T09:43:42.130Z)

=== ENGINE: working tree vs origin/master (empty = identical) ===

[engine HEAD] 06ff4b7 | [origin/master] 06ff4b7

=== KHELSUTRA: working tree vs origin/master ===

[khelsutra HEAD] d0461fd | [origin/master] d0461fd

185. assistant (2026-06-21T09:43:52.601Z)

Both repos are synced to remote head ? verified clean (zero diff against `origin/master` in each):

```
| Repo | Local HEAD | `origin/master` | Status |
|---|---|---|---|
| **engine** (`badminton-highlight-indexer`) | `06ff4b7` | `06ff4b7` | ? identical (includes #284) |
| **khelsutra** | `d0461fd` | `d0461fd` | ? identical (includes #55) |
```

What I did and why it was safe:

- **khelsutra** was already on `master` and fast-forwarded cleanly to head.
- **engine** was stuck on `feat/serving-m4-device-context` (HEAD `44c9f01`) ? a now-merged branch whose remote was deleted, which **couldn't fast-forward** (the #283 squash made it diverge) and **couldn't switch to `master`** (the sibling `tier1-windowing` worktree holds that branch). Before doing anything I proved it was lossless: the tracked tree was clean (only untracked `artifacts/` + `KHELSUTRA\_PRODUCT\_OVERVIEW.md`, both preserved), and excluding my just-merged test file the branch tree was **byte-identical** to `origin/master`. So I reset it to `origin/master` and repointed its upstream ? old tip tagged `\_pre\_sync\_m4\_44c9f01` and in reflow if you ever want it back.

One thing worth flagging: the engine main checkout is now at master head but still **named** `feat/serving-m4-device-context` (cosmetic ? `master` is worktree-locked). `git pull --ff-only` will keep it current going forward. If you'd prefer, I can free `master` by pruning the `tier1-windowing` worktree (I'd verify it's clean/abandoned first) so the main checkout sits on `master` properly ? but I left sibling worktrees untouched by default. I've recorded all of this in the memory checkpoint.

186. user (2026-06-21T09:51:52.636Z)

thanks for the cleanup! Yes can you start on B-P2 now and refresh my memory with the entire plan for once.

187. assistant (2026-06-21T09:52:33.266Z)

Let me lay out the full B-P2 plan, then start building it. First, refreshing the plan from the verification workflow's SPA map + my own read of the current code (verifying the map before I trust it ? per the lesson I just codified). Pulling khelsutra and reading the real surfaces:

188. user (2026-06-21T09:52:36.867Z)

HEAD=d0461fd

=== web/src structure ===

web/src/App.tsx

web/src/api/client.test.ts

web/src/api/client.ts

web/src/api/types.ts

web/src/components/CompileResult.tsx

```

web/src/components/LanguageSwitcher.tsx
web/src/components/QueryBar.examples.test.ts
web/src/components/QueryBar.tsx
web/src/components/RallyGrid.tsx
web/src/components/SelectionFooter.tsx
web/src/hooks/useRallySelection.ts
web/src/i18n/hygiene.test.ts
web/src/i18n/i18n.test.tsx
web/src/i18n/index.ts
web/src/i18n/locales/en/common.json
web/src/i18n/locales/hi/common.json
web/src/i18n/locales/kn/common.json
web/src/lib/floor.test.ts
web/src/lib/floor.ts
web/src/lib/format.test.ts
web/src/lib/format.ts
web/src/lib/log.ts
web/src/lib/query.test.ts
web/src/lib/query.ts
web/src/lib/selection.test.ts
web/src/lib/selection.ts
web/src/main.tsx
web/src/test/setup.ts

```

189. user (2026-06-21T09:52:42.956Z)

```

1      import { useEffect, useMemo, useState } from "react";
2      import { useTranslation, Trans } from "react-i18next";
3      import { useRallySelection } from "../hooks/useRallySelection";
4      import { RallyGrid } from "../components/RallyGrid";
5      import { QueryBar } from "../components/QueryBar";
6      import { SelectionFooter } from "../components/SelectionFooter";
7      import { LanguageSwitcher } from "../components/LanguageSwitcher";
8      import { CompileResult, type BuildState } from "../components/CompileResult";
9      import { runQuery, type ParsedQuery } from "../lib/query";
10     import { floorRisk } from "../lib/floor";
11     import { reelFilename } from "../lib/format";
12     import { log, newCorrelationId } from "../lib/log";
13     import { ApiError, compileReel, getConfig, highlightUrl, listRallies, minShotCount }
from "../api/client";
14     import type { EngineConfig, RallySegment } from "../api/types";
15
16     /** One compact, structured description of how a query was understood (for the trace
log). */
17     function parseTrace(q: ParsedQuery) {
18         return {
19             raw: q.raw,
20             recognized: q.recognized,
21             filters: q.filters.map((f) => `${f.field}${f.op}${f.value}`),
22             sort: q.sort ? `${q.sort.key}:${q.sort.dir}` : null,
23             limit: q.limit ?? null,
24             unavailable: q.unavailable.map((u) => `${u.kind}:${u.token}`),
25         };
26     }
27
28     // Demo data shown when no ?video=<id> is supplied. Shape matches the verified
play_segments
29     // contract; only honest fields are surfaced. A real session loads from POST /api/query.
30     const DEMO_RALLIES: RallySegment[] = [
31         { id: 1, start_time: 12, end_time: 20, duration: 8, server: null, receiver: null,
metadata: { shot_count: 9, shot_count_confidence: "High" } },
32         { id: 2, start_time: 41, end_time: 47, duration: 6, server: null, receiver: null,
metadata: { shot_count: 5 } },
33         { id: 3, start_time: 63, end_time: 84, duration: 21, server: null, receiver: null,
metadata: { shot_count: 19, shot_count_confidence: "Medium" } },

```

```

34      { id: 4, start_time: 95, end_time: 101, duration: 6, server: null, receiver: null,
metadata: {} },
35      { id: 5, start_time: 130, end_time: 142, duration: 12, server: null, receiver: null,
metadata: { shot_count: 11 } },
36      { id: 6, start_time: 158, end_time: 187, duration: 29, server: null, receiver: null,
metadata: { shot_count: 24, shot_count_confidence: "High" } },
37    ];
38
39    type LoadState =
40      | { status: "demo" }
41      | { status: "loading" }
42      | { status: "loaded"; count: number }
43      | { status: "error"; error: string };
44
45    export default function App() {
46      const { t } = useTranslation();
47      // Job-tethered (D10): a finished job's video_id arrives via ?video=<id>; absent ?
demo data.
48      const videoId = useMemo(() => new
URLSearchParams(window.location.search).get("video"), []);
49
50      const [rallies, setRallies] = useState<RallySegment[]>(videoId ? [] : DEMO_RALLIES);
51      const [order, setOrder] = useState<RallySegment[]>(videoId ? [] : DEMO_RALLIES);
52      const [loadState, setLoadState] = useState<LoadState>(videoId ? { status: "loading" }
: { status: "demo" });
53      const [reelName, setReelName] = useState("highlights");
54      const [build, setBuild] = useState<BuildState>({ status: "idle" });
55
56      // Display order is separate from selection: a query can re-sort the cards while the
shared
57      // selection set (sel) stays the single source of truth for what lands in the reel.
58      const sel = useRallySelection(rallies, "all");
59
60      // Load real rallies for a job's video (POST /api/query). Default ALL-selected (D8).
61      useEffect(() => {
62        if (!videoId) return;
63        const cid = newCorrelationId();
64        log("info", "rallies.load_start", { cid, video_id: videoId });
65        let alive = true;
66        listRallies(videoId, cid)
67          .then((rs) => {
68            if (!alive) return;
69            setRallies(rs);
70            setOrder(rs);
71            sel.apply(rs.map((r) => r.id));
72            setLoadState({ status: "loaded", count: rs.length });
73            log("info", "rallies.loaded", { cid, video_id: videoId, count: rs.length });
74          })
75          .catch((err) => {
76            if (!alive) return;
77            setLoadState({ status: "error", error: String(err) });
78            log("error", "rallies.load_failed", { cid, video_id: videoId, error: String(err)
});
79          });
80      return () => {
81        alive = false;
82      };
83      // videoId is stable for the page; sel.apply is a stable callback.
84      // eslint-disable-next-line react-hooks/exhaustive-deps
85    }, [videoId]);
86
87    // Read the engine's stitching.min_shot_count so the footer can WARN before a selected
rally is
88    // silently dropped at compile (P4 / D7). Offline (no engine) is fine: no config ? no
warning.

```

```

89     const [cfg, setCfg] = useState<EngineConfig | null>(null);
90     useEffect(() => {
91         let alive = true;
92         getConfig()
93             .then((c) => alive && setCfg(c))
94             .catch((err) => log("warn", "config.unavailable", { error: String(err) }));
95         return () => {
96             alive = false;
97         };
98     }, []);
99     const warning = floorRisk(rallies, sel.selected, cfg ? minShotCount(cfg) : undefined);
100
101     // Stable chronological ordinal (by start_time) so re-sorting never rennumbers a rally.
102     const ordinalOf = useMemo(() => {
103         const m = new Map<number, number>();
104         [...rallies].sort((a, b) => a.start_time - b.start_time).forEach((r, i) =>
105             m.set(r.id, i + 1));
106         return (id: number) => m.get(id) ?? 0;
107     }, [rallies]);
108     // Build ? POST /api/compile ? preview/download the stitched reel. One cid traces the
109     // whole call.
110     const onBuild = async () => {
111         if (build.status === "building") return;
112         const cid = newCorrelationId();
113         const filename = reelFilename(reelName);
114         const ids = [...sel.selected];
115         if (!videoId) {
116             log("info", "build.no_video", { cid, selected: ids.length });
117             setBuild({
118                 status: "error",
119                 error: t("build.noVideo"),
120             });
121             return;
122         }
123         setBuild({ status: "building" });
124         log("info", "build.start", {
125             cid,
126             video_id: videoId,
127             filename,
128             selected: ids.length,
129             total_sec: Math.round(sel.totalSec),
130             below_floor: warning?.atRisk.length ?? 0,
131         });
132         try {
133             const res = await compileReel(videoId, ids, filename, cid);
134             const url = highlightUrl(videoId, filename);
135             setBuild({ status: "done", url, filename });
136             log("info", "build.done", { cid, video_id: videoId, filename, filepath:
137                 res.filepath, url });
138         } catch (err) {
139             const status = err instanceof ApiError ? err.status : undefined;
140             const detail = err instanceof ApiError && err.body ? parseDetail(err.body) :
141                 String(err);
142             setBuild({ status: "error", error: detail });
143             log("error", "build.failed", { cid, video_id: videoId, filename, status, error:
144                 String(err) });
145         }
146     };
147
148     // Every query application is ONE correlatable journey: a `query.parse` line records
149     // exactly how
150     // the text was understood, then a single outcome line (apply / skip / empty) shares
151     // the same cid.
152     // Unsupported concepts and zero-result queries are logged loudly ? never silently

```



```

dropped.
147     const onApplyQuery = (q: ParsedQuery, source: "typed" | "example") => {
148         const cid = newCorrelationId();
149         log("info", "query.parse", { cid, source, ...parseTrace(q) });
150
151         // Demand signal + honest "we ignored this": surface every shot-type/player/score
the user asked
152         // for but we can't yet honour (PER_RALLY_SELECTION.md §3 roadmap fields).
153         if (q.unavailable.length > 0) {
154             log("info", "query.unavailable_request", { cid, source, raw: q.raw, unavailable:
q.unavailable });
155         }
156
157         if (!q.recognized) {
158             log("info", "query.skip", { cid, source, raw: q.raw, reason: "no actionable
clause" });
159             return; // empty / unparseable ? leave the user's curation untouched
160         }
161
162         try {
163             const { ordered, selectedIds } = runQuery(rallies, q);
164             setOrder(ordered);
165             sel.apply(selectedIds);
166             log("info", "query.apply", {
167                 cid,
168                 source,
169                 filters: q.filters.length,
170                 sort: q.sort ? `${q.sort.key}:${q.sort.dir}` : null,
171                 limit: q.limit ?? null,
172                 total: rallies.length,
173                 selected: selectedIds.length,
174                 selected_ids: selectedIds,
175                 dropped_unavailable: q.unavailable.map((u) => `${u.kind}:${u.token}`),
176             });
177             if (selectedIds.length === 0) {
178                 // Loud: a recognized query that matches nothing makes the grid look empty ? say
why.
179                 log("warn", "query.empty_result", { cid, source, ...parseTrace(q) });
180             }
181         } catch (err) {
182             // Defensive: the pure parser/selector shouldn't throw, but never fail silently if
it does.
183             log("error", "query.apply_failed", { cid, source, raw: q.raw, error: String(err)
});
184         }
185     };
186
187     // Manual curation edits are traced too, so a journey (query ? hand-tune) reads as one
story.
188     const onToggle = (id: number) => {
189         log("debug", "rally.toggle", { id, action: sel.isSelected(id) ? "remove" : "add" });
190         sel.toggle(id);
191     };
192     const onBulk = (op: "select_all" | "clear" | "invert") => {
193         const resultCount = op === "select_all" ? rallies.length : op === "clear" ? 0 :
rallies.length - sel.count;
194         log("info", "selection.bulk", { cid: newCorrelationId(), op, result_count:
resultCount });
195         if (op === "select_all") sel.selectAll();
196         else if (op === "clear") sel.clearAll();
197         else sel.invert();
198     };
199
200     return (
201         <main className="min-h-screen bg-paper text-ink pb-24">

```

```

202         <header className="bg-ink text-white">
203             <div className="mx-auto flex max-w-5xl items-center justify-between gap-3 px-6
py-4">
204                 <span className="text-xl font-extrabold tracking-tight">
205                     Khel<span className="text-green"?</span>Sutra <span className="font-
semibold text-lime">Studio</span>
206                 </span>
207                 <LanguageSwitcher />
208             </div>
209         </header>
210
211         <section className="mx-auto max-w-5xl px-6 py-10">
212             <span className="inline-flex rounded-full bg-green/15 px-3 py-1 text-xs font-
semibold text-green">
213                 {t("app.badge")}
214             </span>
215             <h1 className="mt-4 text-3xl font-extrabold tracking-
tight">{t("app.title")}</h1>
216             <p className="mt-2 max-w-2xl text-ink/70">
217                 <Trans i18nKey="app.intro" components={{ b: <strong />, i: <em /> }} />
218             </p>
219
220             <ModeBanner load={loadState} videoId={videoId} />
221
222             <div className="mt-6">
223                 <QueryBar onApply={onApplyQuery} />
224             </div>
225
226             <div className="mt-6 flex flex-wrap gap-2 text-sm">
227                 <button type="button" onClick={() => onBulk("select_all")} className="rounded-
full border border-black/15 px-3 py-1 hover:border-green">{t("bulk.selectAll")}</button>
228                 <button type="button" onClick={() => onBulk("clear")} className="rounded-full
border border-black/15 px-3 py-1 hover:border-green">{t("bulk.clear")}</button>
229                 <button type="button" onClick={() => onBulk("invert")} className="rounded-full
border border-black/15 px-3 py-1 hover:border-green">{t("bulk.invert")}</button>
230             </div>
231
232             <div className="mt-5">
233                 <RallyGrid rallies={order} isSelected={sel.isSelected} onToggle={onToggle}
ordinalOf={ordinalOf} />
234             </div>
235
236             <CompileResult build={build} onDismiss={() => setBuild({ status: "idle" })} />
237         </section>
238
239         <SelectionFooter
240             count={sel.count}
241             totalSec={sel.totalSec}
242             reelName={reelName}
243             onReelNameChange={setReelName}
244             onBuild={onBuild}
245             building={build.status === "building"}
246             warning={warning}
247         />
248     </main>
249 );
250 }
251
252 /** Pull FastAPI's {"detail": "..."} message out of an error body, else show it raw. */
253 function parseDetail(body: string): string {
254     try {
255         const j = JSON.parse(body);
256         if (j && typeof j.detail === "string") return j.detail;
257     } catch {
258         /* not JSON ? fall through */

```

```

259     }
260     return body;
261 }
262
263 function ModeBanner({ load, videoId }: { load: LoadState; videoId: string | null }) {
264     if (load.status === "demo") {
265         return (
266             <p className="mt-3 text-sm text-ink/50" data-testid="mode-banner">
267                 <Trans i18nKey="mode.demo" components={{ code: <code /> }} />
268             </p>
269         );
270     }
271     if (load.status === "loading") {
272         return (
273             <p className="mt-3 text-sm text-ink/60" data-testid="mode-banner">
274                 <Trans i18nKey="mode.loading" values={{ videoId }} components={{ code: <code />
275             </p>
276         );
277     }
278     if (load.status === "error") {
279         return (
280             <p className="mt-3 rounded-lg border border-red-200 bg-red-50 px-3 py-2 text-sm
281             text-red-800" data-testid="mode-banner">
282                 <Trans i18nKey="mode.error" values={{ videoId, error: load.error }}
283             components={{ code: <code /> }} />
284             </p>
285         );
286     }
287     return (
288         <p className="mt-3 text-sm text-ink/50" data-testid="mode-banner">
289             <Trans i18nKey="mode.loaded" values={{ count: load.count }} components={{ b:
290             <strong className="text-ink/70" /> }} />
291             </p>
292         );
293     }
294 }

```

190. user (2026-06-21T09:52:43.345Z)

```

1      // Typed client for the engine FastAPI. SAME-ORIGIN relative `/api/*` (the appliance
2      serves
3      // UI + engine behind one origin; in dev Vite proxies `/api` ? :8000). Every call
4      carries a
5      // correlation id (header + logs) so a journey is traceable end to end.
6      import { log, newCorrelationId } from "../lib/log";
7      import type { RallySegment, QueryResponse, CompileResponse, EngineConfig } from
8      "../types";
9      const BASE = "/api";
10
11      export class ApiError extends Error {
12          readonly cid?: string;
13          readonly status?: number;
14          readonly body?: string;
15          constructor(message: string, opts: { cid?: string; status?: number; body?: string;
16          cause?: unknown } = {}) {
17              super(message);
18              this.name = "ApiError";
19              this.cid = opts.cid;
20              this.status = opts.status;
21              this.body = opts.body;
22              if (opts.cause !== undefined) (this as { cause?: unknown }).cause = opts.cause;
23          }
24      }

```

```

22
23 interface RequestOptions {
24     method?: string;
25     body?: string;
26     headers?: Record<string, string>;
27     /** Correlation id to thread through this call; auto-generated if absent. */
28     cid?: string;
29 }
30
31 async function request<T>(path: string, opts: RequestOptions = {}): Promise<T> {
32     const cid = opts.cid ?? newCorrelationId();
33     const url = `${BASE}${path}`;
34     const method = opts.method ?? "GET";
35     log("info", "api.request", { cid, method, url });
36
37     let res: Response;
38     try {
39         res = await fetch(url, {
40             method,
41             body: opts.body,
42             headers: { "content-type": "application/json", "x-correlation-id": cid,
... (opts.headers ?? {}) },
43         });
44     } catch (err) {
45         // Loud failure, never silent (LOCAL_APPLIANCE_ARCHITECTURE.md §5).
46         log("error", "api.network_error", { cid, method, url, error: String(err) });
47         throw new ApiError(`Network error calling ${method} ${url}`, { cid, cause: err });
48     }
49
50     if (!res.ok) {
51         const body = await res.text().catch(() => "");
52         log("error", "api.http_error", { cid, method, url, status: res.status, body:
body.slice(0, 500) });
53         throw new ApiError(`${method} ${url} failed: ${res.status}`, { cid, status:
res.status, body });
54     }
55
56     const data = (await res.json()) as T;
57     log("info", "api.response", { cid, method, url, status: res.status });
58     return data;
59 }
60
61 /** List a video's detected rallies. POST /api/query {video_id} ? {play_segments}
[main.py:331]. */
62 export async function listRallies(videoId: string, cid?: string):
Promise<RallySegment[]> {
63     const data = await request<QueryResponse>("/query", {
64         method: "POST",
65         body: JSON.stringify({ video_id: videoId }),
66         cid,
67     });
68     return data.play_segments ?? [];
69 }
70
71 /** Stitch selected rallies into one reel. POST /api/compile {video_id, segment_ids,
output_filename} [main.py:342]. */
72 export async function compileReel(
73     videoId: string,
74     segmentIds: number[],
75     outputFilename: string,
76     cid?: string,
77 ): Promise<CompileResponse> {
78     return request<CompileResponse>("/compile", {
79         method: "POST",
80         body: JSON.stringify({ video_id: videoId, segment_ids: segmentIds, output_filename:

```

```

outputFilename }},
81     cid,
82     });
83 }
84
85 /** The stream/download URL for a compiled reel. GET
/api/highlights/{video_id}/{filename} [main.py:307]. */
86 export function highlightUrl(videoId: string, filename: string): string {
87     return
`${BASE}/highlights/${encodeURIComponent(videoId)}/${encodeURIComponent(filename)}`;
88 }
89
90 /** Read engine config (for the stitching floor). GET /api/config [main.py:121]. */
91 export async function getConfig(cid?: string): Promise<EngineConfig> {
92     return request<EngineConfig>("/config", { method: "GET", cid });
93 }
94
95 /** The shot-count floor /api/compile silently enforces ? the UI must warn, not hide
[main.py:362]. */
96 export function minShotCount(cfg: EngineConfig): number | undefined {
97     return cfg.stitching?.min_shot_count;
98 }
99

```

191. user (2026-06-21T09:52:43.790Z)

```

1 // Types mirror the engine's verified contract (badminton-highlight-indexer).
2 // Anchors: backend/infrastructure/database.py:124-135 (play_segments),
3 // backend/main.py:331-396 (query/compile), :307-328 (highlights), :121 (config).
4
5 export interface RallyMetadata {
6     /** Gemini's shuttle_exchange_count (fallback max(3, dur/0.8)); absent on some
detectors. */
7     shot_count?: number;
8     /** Gemini confidence STRING (often "Unknown") ? not a numeric meter. */
9     shot_count_confidence?: string;
10    detector?: string;
11    [key: string]: unknown;
12 }
13
14 export interface RallySegment {
15     /** The integer the UI selects and passes to /api/compile as segment_ids. */
16     id: number;
17     start_time: number;
18     end_time: number;
19     duration: number;
20     /** motion-segmenter FABRICATION (random) ? DO NOT surface (see
PER_RALLY_SELECTION.md). */
21     server: string | null;
22     /** motion-segmenter FABRICATION (random) ? DO NOT surface. */
23     receiver: string | null;
24     metadata: RallyMetadata;
25     /** motion-only fabricated sub-events ? DO NOT surface. */
26     events?: unknown[];
27 }
28
29 export interface QueryResponse {
30     play_segments: RallySegment[];
31 }
32
33 export interface CompileResponse {
34     status: string;
35     /** server-local path (NOT a URL) ? build the download URL with highlightUrl(). */
36     filepath: string;
37 }

```

```

38
39 export interface EngineConfig {
40   stitching?: {
41     /** The shot-count floor /api/compile silently enforces ? surface it, don't hide it.
42     */
43     min_shot_count?: number;
44     [key: string]: unknown;
45   };
46   [key: string]: unknown;
47 }

```

192. assistant (2026-06-21T09:53:01.978Z)

App.tsx, client.ts, types.ts confirm the workflow's map exactly. Now the i18n setup + sibling patterns (so IngestPanel/useIngestJob match house style and the catalogs stay key-parity-clean):

193. user (2026-06-21T09:53:02.865Z)

```

1  // i18n scaffold (tracker B1) ? the shared localization foundation for KhelSutra Studio.
2  //
3  // Per docs/FRIENDLY_AND_MULTILINGUAL.md + the engine's I18N_PLAN.md:
4  //   - keys, never prose (t("app.title"), enforced later by a no-bare-strings lint, B2);
5  //   - ICU MessageFormat for interpolation/plurals (Kannada/Hindi plural rules live in
the catalog,
6  //     not the code ? replaces the SPA's hand-rolled `count > 1 ? "s" : ""`);
7  //   - one negotiation order shared with the marketing site (B3);
8  //   - always-fallback to `en` so a missing translation degrades to English, never a raw
key.
9  import i18n from "i18next";
10 import { initReactI18next } from "react-i18next";
11 import LanguageDetector from "i18next-browser-languagedetector";
12 import ICU from "i18next-icu";
13 import enCommon from "../locales/en/common.json";
14 import knCommon from "../locales/kn/common.json";
15 import hiCommon from "../locales/hi/common.json";
16
17 // Supported locales (D6): `en` + `kn` + `hi`. Anything else falls back to `en`.
18 export const SUPPORTED_LOCALES = ["en", "kn", "hi"] as const;
19 export type Locale = (typeof SUPPORTED_LOCALES)[number];
20 export const DEFAULT_LOCALE: Locale = "en";
21
22 // ONE persisted-preference key, shared with the marketing site's t() shim (D9 / tracker
B3), so a
23 // visitor's language choice carries across both surfaces.
24 export const LANG_STORAGE_KEY = "kheelsutra.lang";
25
26 // Catalogs are co-located here (D9). `kn` + `hi` shipped (B5/B6, AI-drafted ? native
review
27 // pending). A missing key in any locale still falls back to `en` ? the by-design
invariant.
28 const resources = {
29   en: { common: enCommon },
30   kn: { common: knCommon },
31   hi: { common: hiCommon },
32 } as const;
33
34 void i18n
35   .use(ICU)
36   .use(LanguageDetector)
37   .use(initReactI18next)
38   .init({
39     resources,
40     fallbackLng: DEFAULT_LOCALE,
41     supportedLngs: [...SUPPORTED_LOCALES],

```

```

42     nonExplicitSupportedLngs: true, // en-IN -> en
43     defaultNS: "common",
44     ns: ["common"],
45     interpolation: { escapeValue: false }, // React already escapes
46     returnNull: false,
47     // Negotiation order (I18N_PLAN §2.3): persisted pref -> ?lang= -> browser -> en.
48     detection: {
49       order: ["localStorage", "querystring", "navigator"],
50       lookupLocalStorage: LANG_STORAGE_KEY,
51       lookupQuerystring: "lang",
52       caches: ["localStorage"],
53     },
54     react: { useSuspense: false },
55   });
56
57   // Keep <html lang> in sync with the active locale (screen readers + SEO use it).
Reflects the
58   // resolved content language, so an `en` fallback for an absent locale still reads as
`en`.
59   function syncDocumentLang(lng: string) {
60     if (typeof document !== "undefined") {
61       document.documentElement.lang = (i18n.resolvedLanguage || lng ||
DEFAULT_LOCALE).split("-")[0];
62     }
63   }
64   i18n.on("languageChanged", syncDocumentLang);
65   syncDocumentLang(i18n.language ?? DEFAULT_LOCALE);
66
67   export default i18n;
68

```

194. user (2026-06-21T09:53:03.346Z)

```

1     // CI guardrails that keep i18n "by design" (FRIENDLY_AND_MULTILINGUAL.md §4):
2     //   1. KEY-PARITY ? every locale catalog has EXACTLY the `en` key set (the
extensibility proof:
3     //     drop a stub locale and this lists every gap). Trivially green while only `en`
exists; it
4     //     starts enforcing the moment a kn/hi catalog lands (B5/B6).
5     //   2. NO-BARE-STRINGS (attribute level, v1) ? user-facing component attributes
(placeholder /
6     //     aria-label / title) must route through t(), never a hardcoded literal. A curated
start per
7     //     the doc; an AST-level sweep of JSX text can extend it later.
8     import { describe, it, expect } from "vitest";
9     import { readdirSync, readFileSync } from "node:fs";
10    import { join } from "node:path";
11
12    const LOCALES_DIR = join(import.meta.dirname, "locales");
13    const SRC_DIR = join(import.meta.dirname, "..");
14
15    function flatten(obj: Record<string, unknown>, prefix = ""): string[] {
16      const keys: string[] = [];
17      for (const [k, v] of Object.entries(obj)) {
18        const key = prefix ? `${prefix}.${k}` : k;
19        if (v && typeof v === "object" && !Array.isArray(v)) keys.push(...flatten(v as
Record<string, unknown>, key));
20        else keys.push(key);
21      }
22      return keys.sort();
23    }
24
25    const loadKeys = (locale: string): string[] =>
26      flatten(JSON.parse(readFileSync(join(LOCALES_DIR, locale, "common.json"), "utf8")) as
Record<string, unknown>);

```

```

27
28 describe("i18n key-parity ? every locale has EXACTLY the en keys", () => {
29     const enKeys = loadKeys("en");
30     const others = readdirSync(LOCALES_DIR, { withFileTypes: true })
31         .filter((d) => d.isDirectory() && d.name !== "en")
32         .map((d) => d.name);
33
34     it("en is the non-empty source catalog", () => {
35         expect(enKeys.length).toBeGreaterThan(20);
36     });
37
38     for (const loc of others) {
39         it(`${loc} matches en exactly (no missing, no extra keys)`, () => {
40             const keys = loadKeys(loc);
41             const missing = enKeys.filter((k) => !keys.includes(k));
42             const extra = keys.filter((k) => !enKeys.includes(k));
43             expect({ locale: loc, missing, extra }).toEqual({ locale: loc, missing: [], extra:
44 [] });
45         });
46     });
47
48 describe("no-bare-strings ? user-facing component attributes go through t()", () => {
49     const FILES = [
50         "App.tsx",
51         "components/QueryBar.tsx",
52         "components/RallyGrid.tsx",
53         "components/SelectionFooter.tsx",
54         "components/CompileResult.tsx",
55         "components/LanguageSwitcher.tsx",
56     ];
57     // A hardcoded placeholder / aria-label / title with letters is a bare-string
58     regression ? it
59     // must be an expression ({t(...)}), not a quoted literal.
60     const BARE_ATTR = /\b(?:placeholder|aria-label|title)\s*=\s*"[\^"]*[A-Za-z]{2}[\^"]*"\/g;
61
62     for (const f of FILES) {
63         it(`${f}: no hardcoded placeholder/aria-label/title`, () => {
64             const src = readFileSync(join(SRC_DIR, f), "utf8");
65             expect(src.match(BARE_ATTR) ?? []).toEqual([]);
66         });
67     }
68
69 describe("Indic fonts (B4) ? self-hosted OFL Noto, no runtime CDN", () => {
70     it("main.tsx bundles the OFL Noto Kannada + Devanagari fonts (self-hosted, not Google
71 Fonts)", () => {
72         const main = readFileSync(join(SRC_DIR, "main.tsx"), "utf8");
73         expect(main).toMatch(/@fontsource-variable\/noto-sans-kannada/);
74         expect(main).toMatch(/@fontsource-variable\/noto-sans-devanagari/);
75     });
76     it("the font stack falls back to Noto for Kannada + Devanagari glyphs", () => {
77         const css = readFileSync(join(SRC_DIR, "index.css"), "utf8");
78         expect(css).toMatch(/Noto Sans Kannada Variable/);
79         expect(css).toMatch(/Noto Sans Devanagari Variable/);
80     });
81
82     // The Studio SPA must render correctly on an AIR-GAPPED appliance (GPU/CPU/RAM, no
83     // internet) ?
84     // every asset (fonts, styles, scripts) is bundled local, nothing pulled from a
85     // CDN/Google Fonts.
86     describe("offline / self-contained ? no external runtime assets", () => {
87         const CDN = /googleapis|gstatic|fonts\.google|cdnjs|unpkg|jsdelivr|\.cdn\.[a-z]/i;
88         const FILES = [

```



```

87     "../index.html", // web/index.html
88     "main.tsx",
89     "index.css",
90 ];
91 for (const rel of FILES) {
92     it(`${rel}: no external font/CDN reference`, () => {
93         expect(readFileSync(join(SRC_DIR, rel), "utf8")).not.toMatch(CDN);
94     });
95 }
96 it("web/index.html loads no absolute-URL <link>/<script src> (everything is
bundled/local)", () => {
97     const html = readFileSync(join(SRC_DIR, "..", "index.html"), "utf8");
98     expect(html).not.toMatch(/(?:href|src)\s*=\s*["']https?:\/\//i);
99 });
100 });
101

```

195. user (2026-06-21T09:53:03.761Z)

```

1     // React state wrapper over the pure selection helpers in lib/selection.ts.
2     // Default = ALL selected (D8: curate-by-removal) ? the user removes warm-ups.
3     import { useCallback, useMemo, useState } from "react";
4     import type { RallySegment } from "../api/types";
5     import * as sel from "../lib/selection";
6
7     export function useRallySelection(rallies: RallySegment[], initial: "all" | "none" =
"all") {
8         const [selected, setSelected] = useState<Set<number>>(() =>
9             initial === "none" ? sel.clear() : sel.selectAll(rallies),
10         );
11
12         const toggle = useCallback((id: number) => setSelected((s) => sel.toggle(s, id)), []);
13         const selectAll = useCallback(() => setSelected(sel.selectAll(rallies)), [rallies]);
14         const clearAll = useCallback(() => setSelected(sel.clear()), []);
15         const invert = useCallback(() => setSelected((s) => sel.invert(s, rallies)),
[rallies]);
16         const isSelected = useCallback((id: number) => selected.has(id), [selected]);
17         /** Replace the whole selection (the query bar writes the shared set here ? D3). */
18         const apply = useCallback((ids: Iterable<number>) => setSelected(new Set(ids)), []);
19
20         const totalSec = useMemo(() => sel.selectedDurationSec(rallies, selected), [rallies,
selected]);
21
22         return { selected, count: selected.size, totalSec, isSelected, toggle, selectAll,
clearAll, invert, apply };
23     }
24

```

196. assistant (2026-06-21T09:53:18.348Z)

Now the en catalog structure + the QueryBar component idiom (the closest sibling to IngestPanel
? a localized form with an input):

197. user (2026-06-21T09:53:19.301Z)

```

1     {
2         "app": {
3             "badge": "per-rally selection · alpha",
4             "title": "Pick the rallies for your reel",
5             "intro": "Filter by <b>rally length</b> ? e.g. <i>?top 15 longest?</i> ? then fine-
tune by tapping cards. Everything starts <b>selected</b>; just drop the ones you don't want."
6         },
7         "lang": {
8             "label": "Language",

```

```

9      "en": "English",
10     "kn": "?????",
11     "hi": "??????"
12 },
13     "bulk": {
14         "selectAll": "Select all",
15         "clear": "Clear",
16         "invert": "Invert"
17     },
18     "mode": {
19         "demo": "Demo data ? open with <code>?video=?video_id?</code> against a running
engine to load a real session.",
20         "loading": "Loading rallies for <code>{videoId}</code>?",
21         "error": "Couldn't load rallies for <code>{videoId}</code>: {error}",
22         "loaded": "Loaded <b>{count}</b> rallies from this session."
23     },
24     "query": {
25         "placeholder": "e.g. over 10s · top 15 longest · shortest first",
26         "arialLabel": "Filter rallies by length, count or order",
27         "apply": "Apply",
28         "notAvailable": "not available yet",
29         "notAvailableTitle": "?{token}? needs {kind} detection ? not available yet",
30         "unreadable": "Couldn't read that ? try ?over 10s? or ?top 15 longest?.",
31         "hint": "Type over 10s for rallies longer than 10 seconds ? or tap a sample below.",
32         "try": "Sample queries:",
33         "support": "Filtering by <b>rally length</b> is fully supported. Shot-count is
<x>experimental</x> (under development).",
34     },
35     "grid": {
36         "empty": "No rallies to show yet.",
37         "rally": "Rally {n}",
38         "inReel": "Rally {n}, in reel",
39         "notInReel": "Rally {n}, not in reel",
40         "start": "start",
41         "long": "long",
42         "shots": "~{count} shots · experimental",
43         "shotsTitle": "Shot count is experimental (under development) and may be
inaccurate{conf}. Filter by rally length instead.",
44         "shotsConfidence": " ? confidence: {conf}"
45     },
46     "footer": {
47         "count": "{count, plural, one {# rally} other {# rallies}}",
48         "summaryRest": "selected · {total} total",
49         "reelName": "Reel name",
50         "reelNamePlaceholder": "my-highlights",
51         "build": "Build reel ({count})",
52         "building": "Building?",
53         "floorOne": "<b>Your only selected rally</b> has fewer than {min} shots ? the engine
would skip it, so the build would fail. Pick a longer rally.",
54         "floorAll": "<b>All {count} selected rallies</b> have fewer than {min} shots ? the
engine skips every one, so the build would fail. Pick longer rallies.",
55         "floorSomeLead": "{count, plural, one {# rally} other {# rallies}}",
56         "floorSomeRest": "with fewer than {min} shots will be skipped when building (the
engine's <code>min_shot_count</code> floor).",
57     },
58     "result": {
59         "ready": "Your reel is ready",
60         "failed": "Build failed",
61         "dismiss": "Dismiss",
62         "download": "Download {filename}"
63     },
64     "build": {
65         "noVideo": "Open with ?video=<video_id> against a running engine to build a reel ?
demo data can't be compiled."
66     },

```



```

50         className="min-w-[16rem] flex-1 rounded-xl border border-black/15 bg-white
px-4 py-2 text-sm text-ink outline-none focus:border-green"
51     />
52     <button
53         type="submit"
54         disabled={!parsed.recognized}
55         className="rounded-xl bg-ink px-4 py-2 text-sm font-semibold text-white
disabled:opacity-40"
56     >
57         {t("query.apply")}
58     </button>
59 </form>
60
61     {/* Live interpretation ? transparent + hand-correctable (no hidden NL guessing).
*/}
62     <div aria-live="polite" data-testid="query-pill" className="mt-2 flex flex-wrap
items-center gap-1.5">
63         {clauses.map((c) => (
64             <span key={c} className={CHIP + " bg-green/15 text-green"}>
65                 {c}
66             </span>
67         ))}
68         {parsed.unavailable.map((u) => (
69             <span
70                 key={u.token}
71                 title={t("query.notAvailableTitle", { token: u.token, kind: u.kind })}
72                 className={CHIP + " bg-black/5 text-ink/45"}
73             >
74                 <span className="line-through">{u.token}</span> · {t("query.notAvailable")}
75             </span>
76         ))}
77         {typed && !parsed.recognized && parsed.unavailable.length === 0 && (
78             <span className="text-xs text-ink/50">{t("query.unreadable")}</span>
79         )}
80         {!typed && <span className="text-xs text-ink/50">{t("query.hint")}</span>}
81     </div>
82
83     <div className="mt-3 flex flex-wrap gap-2 text-sm">
84         <span className="text-ink/40">{t("query.try")}</span>
85         {EXAMPLES.map((ex) => (
86             <button
87                 key={ex}
88                 type="button"
89                 onClick={() => applyText(ex)}
90                 className="rounded-full border border-black/15 px-3 py-1 text-xs
hover:border-green"
91             >
92                 {ex}
93             </button>
94         ))}
95     </div>
96
97     <p className="mt-2 text-xs text-ink/40">
98         <Trans i18nKey="query.support" components={{ b: <strong className="text-ink/60"
/>, x: <span className="text-amber-700" /> }} />
99     </p>
100 </div>
101 );
102 }
103

```

199. assistant (2026-06-21T09:54:57.676Z)

I've now read every load-bearing surface and the workflow's map checks out exactly. Here's the complete B-P2 plan, then I'll build it.

B-P2 ? "point at a bucket video in the UI" (the full plan)

****Goal:**** a user enters a ``gs://`` URI (or engine-local path) in the SPA ? it submits to ``/api/process`` ? shows live progress while polling ``/api/jobs/{id}`` ? on success loads that ``video_id`` into the existing rally-selection back-half. ****100% `web/` SPA + i18n, zero engine code**** (B-P1 already ships the endpoint).

****Engine contract (verified):**** ``POST /api/process {video_path}`` ? ``202 {job_id, status:"queued", poll}`` (fast-fails ``400`` bad/unsupported scheme, ``404`` missing local file); ``GET /api/jobs/{id}`` ? ``{status: queued|running|done|failed, progress, video_id, error, result:{status, video_id}}``. Two status layers: ``job.status`` (execution) vs ``result.status`` (pipeline verdict) ? ``done`` + ``result.status===`failed``/no ``video_id`` is a success-shaped failure.

****Files:****

```
| File | Change |
|---|---|
| `web/src/api/types.ts` | + `ProcessResponse`, `JobStatus` |
| `web/src/api/client.ts` | + `processVideo()` (POST /process), `getJob()` (GET /jobs/{id}) ?
reuse the `request<T>`/`ApiError`/log pattern |
| `web/src/lib/errors.ts` *(new)* | `parseDetail` (moved out of App.tsx, reused) +
`classifyError` ? error codes (no i18n in logic) |
| `web/src/hooks/useIngestJob.ts` *(new)* | state machine `idle?submitting?polling?done|error`;
recursive `setTimeout` poll w/ backoff + ceiling; one `cid` threaded; cancel/unmount-safe;
i18n-free (returns codes) |
| `web/src/components/IngestPanel.tsx` *(new)* | localized URI form + submit + `aria-live`
progress + red error banner + Retry/Cancel; prop `onLoaded(videoId)` |
| `web/src/App.tsx` | `videoId` `useMemo`?`useState`; render `` when `!videoId`;
`onLoaded` ? `setVideoId` + `history.replaceState('?video='+id)`; import `parseDetail` from lib
|
| `web/src/i18n/locales/{en,kn,hi}/common.json` | + `ingest.*` block (key-parity enforced; kn/hi
AI-drafted, native-review-pending) |
| `web/src/i18n/hygiene.test.ts` | add `IngestPanel.tsx` to the no-bare-strings file list |
| `web/src/hooks/useIngestJob.test.ts` + `components/IngestPanel.test.tsx` *(new)* | mirror
`client.test.ts` conventions |
```

****Decisions/risks handled:**** recursive ``setTimeout`` (no request pile-up) + backoff 1s?5s + a ~10-min ceiling; success-shaped-failure ? ``emptyResult``; friendly ``networkError`` (this is the first screen a fresh user hits); progress is free-text phase words (indeterminate spinner, not a fake %); ``gs://`` URI demo preserved below the panel.

****DoD:**** typecheck + all web tests green (incl. key-parity), live browser verification of submit?poll?load (mock-engine), then flip ``DRY_RUN_CUJ.md`` §6 DoD box 3.

Now building it ? branch off ``origin/master``:

200. user (2026-06-21T09:55:00.912Z)

branch 'feat/spa-bucket-ingest' set up to track 'origin/master'.
Switched to a new branch 'feat/spa-bucket-ingest'
feat/spa-bucket-ingest
d0461fd docs: correct stale "bucket ingest is net-new" claims (B-P1 already shipped in M2/#280) (#55)

201. assistant (2026-06-21T09:55:06.050Z)

On branch ``feat/spa-bucket-ingest``. Writing the API layer first ? types + client helpers:

202. user (2026-06-21T09:55:13.406Z)

The file `C:\Users\avidu\Projects\khelsutra\web\src\api\types.ts` has been updated successfully.
(file state is current in your context ? no need to Read it back)

203. user (2026-06-21T09:55:19.144Z)

The file C:\Users\avidu\Projects\khelsutra\web\src\api\client.ts has been updated successfully.
(file state is current in your context ? no need to Read it back)

204. user (2026-06-21T09:55:27.462Z)

The file C:\Users\avidu\Projects\khelsutra\web\src\api\client.ts has been updated successfully.
(file state is current in your context ? no need to Read it back)

205. assistant (2026-06-21T09:55:33.371Z)

Now `lib/errors.ts` ? `parseDetail` (moved from App.tsx for reuse) + a UI-agnostic error classifier:

206. assistant (2026-06-21T09:55:41.190Z)

```
// Error helpers shared by the ingest flow and the build flow. Kept i18n-FREE: callers map an
// IngestErrorCode to a localized string (so hooks stay testable without a translation context),
// while the engine's own `detail` text passes through verbatim (it's the most specific
message).
import { ApiError } from "../api/client";

/** Pull FastAPI's {"detail": "..."} message out of an error body, else return the raw body. */
export function parseDetail(body: string): string {
  try {
    const j = JSON.parse(body) as { detail?: unknown };
    if (j && typeof j.detail === "string") return j.detail;
  } catch {
    /* not JSON ? fall through to the raw body */
  }
  return body;
}

/** Stable, localizable failure categories surfaced by the ingest flow.
 * `engine` carries the engine's own message in `detail`; the rest are localized by code. */
export type IngestErrorCode = "invalidUri" | "networkError" | "emptyResult" | "timeout" |
"engine";

export interface IngestError {
  code: IngestErrorCode;
  /** Present for `engine` ? the verbatim FastAPI detail / status; rendered as-is, not
translated. */
  detail?: string;
}

/** Classify a thrown client error into a stable code. A network throw (ApiError with no
`status`)
 * is the common "engine unreachable" case ? its own friendly code; any HTTP error surfaces the
 * engine's `detail` (e.g. an unsupported-scheme 400, a not-found 404). */
export function classifyError(err: unknown): IngestError {
  if (err instanceof ApiError) {
    if (err.status === undefined) return { code: "networkError" };
    if (err.body) {
      const detail = parseDetail(err.body);
      if (detail) return { code: "engine", detail };
    }
    return { code: "engine", detail: `HTTP ${err.status}` };
  }
  return { code: "engine", detail: String(err) };
}
```

207. user (2026-06-21T09:55:41.207Z)

File created successfully at: C:\Users\avidu\Projects\khelsutra\web\src\lib\errors.ts (file
state is current in your context ? no need to Read it back)

208. assistant (2026-06-21T09:56:14.115Z)

Now the core ? `useIngestJob.ts` (i18n-free state machine, cancellable recursive-`setTimeout` poll with backoff + ceiling, one correlation id):

209. assistant (2026-06-21T09:56:33.275Z)

```
// Drives the front-half ingest: submit a video to POST /api/process, then poll GET
/api/jobs/{id}
// until it finishes, and hand the resulting video_id to the existing rally-loading back-half.
//
// Kept i18n-FREE (returns stable IngestErrorCodes, not prose) so it unit-tests without a
// translation context ? IngestPanel maps codes to localized strings. The poll loop is a
RECURSIVE
// setTimeout (never setInterval): each tick waits for the prior getJob to resolve, so a slow
engine
// can't pile up overlapping requests. It backs off, gives up after a ceiling, and is fully
// cancellable on unmount / user-cancel / a superseding submit.
import { useCallback, useEffect, useRef, useState } from "react";
import { getJob, processVideo } from "../api/client";
import { classifyError, type IngestError } from "../lib/errors";
import { log, newCorrelationId } from "../lib/log";
import type { JobStatus } from "../api/types";

const POLL_MIN_MS = 1000;
const POLL_MAX_MS = 5000;
const POLL_BACKOFF = 1.5;
const POLL_CEILING_MS = 10 * 60 * 1000; // a wedged 'running' job won't poll forever

export type IngestPhase =
  | { status: "idle" }
  | { status: "submitting" }
  | { status: "polling"; progress: string | null }
  | { status: "done"; videoId: string }
  | { status: "error"; error: IngestError };

interface Run {
  cancelled: boolean;
  timer?: ReturnType<typeof setTimeout>;
}

export function useIngestJob(onLoaded?: (videoId: string) => void) {
  const [phase, setPhase] = useState<IngestPhase>({ status: "idle" });
  const runRef = useRef<Run | null>(null);
  // Keep the latest onLoaded without making submit depend on it (a parent re-render won't
  restart polls).
  const onLoadedRef = useRef(onLoaded);
  onLoadedRef.current = onLoaded;

  const stop = useCallback(() => {
    const run = runRef.current;
    if (run) {
      run.cancelled = true;
      if (run.timer) clearTimeout(run.timer);
    }
    runRef.current = null;
  }, []);

  // Cancel any in-flight poll loop when the component unmounts (App.tsx:80-82 alive-flag
  idiom).
  useEffect(() => stop, [stop]);

  const submit = useCallback(
    (rawUri: string) => {
      const uri = rawUri.trim();
```

```

if (!uri) {
    setPhase({ status: "error", error: { code: "invalidUri" } });
    return;
}

stop(); // supersede any prior run
const run: Run = { cancelled: false };
runRef.current = run;

const cid = newCorrelationId();
const startedAt = Date.now();
setPhase({ status: "submitting" });
log("info", "ingest.submit_start", { cid, uri });

const fail = (error: IngestError, event: string) => {
    if (run.cancelled) return;
    setPhase({ status: "error", error });
    const level = error.code === "timeout" || error.code === "emptyResult" ? "warn" :
"error";
    log(level, event, { cid, uri, code: error.code, detail: error.detail });
};

const poll = (jobId: string, delay: number) => {
    run.timer = setTimeout(() => {
        void (async () => {
            if (run.cancelled) return;
            let job: JobStatus;
            try {
                job = await getJob(jobId, cid);
            } catch (err) {
                fail(classifyError(err), "ingest.poll_error");
                return;
            }
            if (run.cancelled) return;

            if (job.status === "done") {
                const videoId = job.video_id ?? job.result?.video_id ?? null;
                // Two status layers: a `done` job with a `failed` pipeline verdict (or no
video_id) is
                // a success-shaped FAILURE ? never load it, never hang.
                if (job.result?.status === "failed" || !videoId) {
                    fail({ code: "emptyResult" }, "ingest.empty_result");
                    return;
                }
                setPhase({ status: "done", videoId });
                log("info", "ingest.done", { cid, uri, video_id: videoId });
                onLoadedRef.current?.(videoId);
                return;
            }
            if (job.status === "failed") {
                fail({ code: "engine", detail: job.error ?? "job failed" }, "ingest.job_failed");
                return;
            }
            // queued | running ? reflect coarse progress (free-text phase) and keep polling.
            setPhase({ status: "polling", progress: job.progress ?? null });
            if (Date.now() - startedAt > POLL_CEILING_MS) {
                fail({ code: "timeout" }, "ingest.poll_timeout");
                return;
            }
            poll(jobId, Math.min(delay * POLL_BACKOFF, POLL_MAX_MS));
        })();
    }, delay);
};

void (async () => {

```



```

    let jobId: string;
    try {
      const resp = await processVideo(uri, cid);
      jobId = resp.job_id;
    } catch (err) {
      fail(classifyError(err), "ingest.submit_error");
      return;
    }
    if (run.cancelled) return;
    log("info", "ingest.queued", { cid, uri, job_id: jobId });
    setPhase({ status: "polling", progress: null });
    poll(jobId, POLL_MIN_MS);
  })();
},
[stop],
);

const cancel = useCallback(() => {
  stop();
  setPhase({ status: "idle" });
  log("info", "ingest.cancel", {});
}, [stop]);

const reset = useCallback(() => {
  stop();
  setPhase({ status: "idle" });
}, [stop]);

const busy = phase.status === "submitting" || phase.status === "polling";
return { phase, submit, cancel, reset, busy };
}

```

210. user (2026-06-21T09:56:33.295Z)

File created successfully at: C:\Users\avidu\Projects\khelsutra\web\src\hooks\useIngestJob.ts
(file state is current in your context ? no need to Read it back)

211. assistant (2026-06-21T09:57:11.231Z)

Now `IngestPanel.tsx` ? localized form mirroring the QueryBar idiom, with `aria-live` status + the house red error banner:

212. assistant (2026-06-21T09:57:27.173Z)

```

// The front-half ingest screen (appliance S1/S2, tracker B-P2). Point the tool at a video ? a
// `gs://` bucket URI or a path on the engine host ? and it submits to POST /api/process, shows
// live progress while polling GET /api/jobs/{id}, and hands the resulting video_id to the
existing
// rally-selection back-half (App lifts ?video=<id>). Honest by construction: the engine reports
// coarse phase WORDS (not a %), and a success-shaped failure (done but no rallies) is shown,
never
// silently treated as a load.
import { useState } from "react";
import type { FormEvent } from "react";
import { useTranslation } from "react-i18next";
import { useIngestJob } from "../hooks/useIngestJob";

interface Props {
  /** Called with the engine's video_id once a submitted job finishes successfully. */
  onLoaded: (videoId: string) => void;
}

export function IngestPanel({ onLoaded }: Props) {
  const { t } = useTranslation();

```

```

const [uri, setUri] = useState("");
const { phase, submit, cancel, reset, busy } = useIngestJob(onLoaded);

const onSubmit = (e: FormEvent) => {
  e.preventDefault();
  submit(uri);
};

// `engine` failures carry the engine's own message (a 400 scheme/path detail, a worker
error);
// every other code maps to a friendly localized line ? the network case especially, since
this
// is the first screen a fresh user hits and "is the engine running?" must read plainly.
const errorMsg =
  phase.status === "error"
    ? phase.error.code === "engine"
      ? (phase.error.detail ?? t("ingest.failed", { error: "" }))
      : t(`ingest.${phase.error.code}`)
    : null;

return (
  <section className="mt-6 rounded-2xl border border-black/10 bg-white p-5" data-
testid="ingest-panel">
    <h2 className="text-lg font-bold">{t("ingest.title")}</h2>
    <p className="mt-1 max-w-2xl text-sm text-ink/60">{t("ingest.intro")}</p>

    <form onSubmit={onSubmit} className="mt-3 flex flex-wrap items-center gap-2">
      <input
        type="text"
        value={uri}
        onChange={(e) => setUri(e.target.value)}
        placeholder={t("ingest.uriPlaceholder")}
        aria-label={t("ingest.uriAriaLabel")}
        disabled={busy}
        className="min-w-[18rem] flex-1 rounded-xl border border-black/15 bg-white px-4 py-2
text-sm text-ink outline-none focus:border-green disabled:opacity-50"
      />
      <button
        type="submit"
        disabled={busy}
        className="rounded-xl bg-ink px-4 py-2 text-sm font-semibold text-white
disabled:opacity-40"
      >
        {phase.status === "submitting" ? t("ingest.submitting") : t("ingest.submit")}
      </button>
      {busy && (
        <button
          type="button"
          onClick={cancel}
          className="rounded-xl border border-black/15 px-4 py-2 text-sm hover:border-green"
        >
          {t("ingest.cancel")}
        </button>
      )}
    </form>

    <p className="mt-2 text-xs text-ink/40">{t("ingest.hint")}</p>

    /* Live status (aria-live so screen readers hear progress). Indeterminate: the engine
reports
phase words ("hashing", "segmenting"), not a percentage ? so a spinner + the phase
word. */
    <div aria-live="polite" className="mt-3 text-sm" data-testid="ingest-status">
      {busy && (
        <p className="flex items-center gap-2 text-ink/70">

```

```

        <Spinner />
        <span>
          {t("ingest.working")}
          {phase.status === "polling" && phase.progress
            ? ` · ${t("ingest.progress", { progress: phase.progress })}`
            : ""}
        </span>
      </p>
    )}
    {phase.status === "error" && (
      <div className="rounded-lg border border-red-200 bg-red-50 px-3 py-2 text-red-800"
data-testid="ingest-error">
        <p>{errorMsg}</p>
        <button
          type="button"
          onClick={reset}
          className="mt-2 rounded-lg border border-red-300 px-3 py-1 text-xs hover:bg-
red-100"
        >
          {t("ingest.retry")}
        </button>
      </div>
    )}
  </div>
</section>
);
}

function Spinner() {
  return (
    <span
      className="inline-block h-3.5 w-3.5 animate-spin rounded-full border-2 border-ink/20
border-t-ink/70"
      aria-hidden="true"
    />
  );
}

```

213. user (2026-06-21T09:57:27.191Z)

File created successfully at:
C:\Users\avidu\Projects\khelsutra\web\src\components\IngestPanel.tsx (file state is current in
your context ? no need to Read it back)

214. assistant (2026-06-21T09:58:00.543Z)

Now wire it into `App.tsx`: lift `videoId` to state, render the panel when there's no video, and
reuse `parseDetail` from lib. First the imports and the `parseDetail` de-duplication:

215. user (2026-06-21T09:58:04.765Z)

The file C:\Users\avidu\Projects\khelsutra\web\src\App.tsx has been updated successfully. (file
state is current in your context ? no need to Read it back)

216. user (2026-06-21T09:58:16.118Z)

The file C:\Users\avidu\Projects\khelsutra\web\src\App.tsx has been updated successfully. (file
state is current in your context ? no need to Read it back)

217. assistant (2026-06-21T09:58:23.521Z)

Now make the rally-load effect set "loading" at the top (so an ingest?load handoff doesn't show
stale "demo"), and render the panel when there's no video:

218. user (2026-06-21T09:58:26.772Z)

The file C:\Users\avidu\Projects\khelsutra\web\src\App.tsx has been updated successfully. (file state is current in your context ? no need to Read it back)

219. user (2026-06-21T09:58:33.409Z)

The file C:\Users\avidu\Projects\khelsutra\web\src\App.tsx has been updated successfully. (file state is current in your context ? no need to Read it back)

220. assistant (2026-06-21T09:58:36.782Z)

Now remove the now-duplicated local `parseDetail` from App.tsx (it's imported from lib):

221. user (2026-06-21T09:58:39.265Z)

The file C:\Users\avidu\Projects\khelsutra\web\src\App.tsx has been updated successfully. (file state is current in your context ? no need to Read it back)

222. assistant (2026-06-21T09:59:51.609Z)

Now the i18n catalogs ? adding the `ingest.\*` block to all three (key-parity is CI-enforced). English first:

223. user (2026-06-21T09:59:56.173Z)

The file C:\Users\avidu\Projects\khelsutra\web\src\i18n\locales\en\common.json has been updated successfully. (file state is current in your context ? no need to Read it back)

224. assistant (2026-06-21T10:00:04.214Z)

Kannada (AI-draft, native-review-pending ? consistent with the existing kn/hi):

225. user (2026-06-21T10:00:34.110Z)

<tool_use_error>File has not been read yet. Read it first before writing to it.</tool_use_error>

226. assistant (2026-06-21T10:00:37.137Z)

Need to read the kn/hi catalogs first:

227. user (2026-06-21T10:00:38.094Z)

```
1      {
2        "app": {
3          "badge": "?????-????? ????? . ?????",
4          "title": "????? ?????????? ?????????????? ??????",
5          "intro": "<b>?????? ?????</b> ?????????? ?????? ? ????. <i>????? ?????? ???
15?</i> ? ???? ?????????????? ?????? ???? ?????????????? ??????????. ?????? <b>????????????</b>
????????????????????; ?????????????????? ?????? ??????????????"
6      },
7      "lang": {
8        "label": "?????",
9        "en": "English",
10       "kn": "?????",
11       "hi": "???????"
12     },
13     "bulk": {
14       "selectAll": "????????? ??????????",
15       "clear": "?????????????",
16       "invert": "?????????"
17     },
18     "mode": {
```

```

19      "demo": "???? ???? ? ?????? ????? ???? ????? ?????????????? ??????????????
<code>?video=?video_id?</code> ???? ??????????",
20      "loading": "<code>{videoId}</code> ???? ?????????????? ???? ?????????????????",
21      "error": "<code>{videoId}</code> ???? ?????????????? ???? ??????????????: {error}",
22      "loaded": "? ?????????? <b>{count}</b> ?????????????? ???? ??????????".
23  },
24  "query": {
25      "placeholder": "???. over 10s · top 15 longest · shortest first",
26      "ariaLabel": "????, ????? ???? ????? ?????????? ?????????????? ??????? ????",
27      "apply": "?????????",
28      "notAvailable": "????? ??????????",
29      "notAvailableTitle": "?{token}? ?? {kind} ????? ???? ? ????? ??????????",
30      "unreadable": "?????? ??????????? ? ?over 10s? ???? ?top 15 longest? ??????????",
31      "hint": "10 ?????????????? ?????? ?????????????? over 10s ???? ???? ???? ? ???? ??????
?????????? ?????? ????.",
32      "try": "????? ??????????:",
33      "support": "<b>????? ????</b>? ?????????? ??????? ?????????? ???????????
?????????????? ????-????? <x>??????????</x> (?????????????????)."
34  },
35  "grid": {
36      "empty": "???????? ?????? ?????? ??????????????",
37      "rally": "??????? {n}",
38      "inReel": "??????? {n}, ??????????",
39      "notInReel": "??????? {n}, ??????????????",
40      "start": "?????????",
41      "long": "?????",
42      "shots": "~{count} ?????????? · ??????????",
43      "shotsTitle": "???? ???? ?????????????????? (?????????????????) ????
????????????????{conf}. ?????? ?????? ?????? ?????????? ??????? ????.",
44      "shotsConfidence": " ? ??????: {conf}"
45  },
46  "footer": {
47      "count": "{count, plural, one {# ?????} other {# ?????????}}",
48      "summaryRest": "????????????? · ????? {total}",
49      "reelName": "???? ?????",
50      "reelNamePlaceholder": "my-highlights",
51      "build": "???? ????? ({count})",
52      "building": "?????????????????",
53      "floorOne": "<b>????? ???? ?????????? ??????</b>????? {min} ?????? ?????? ??????????
? ?????? ?????? ?????????????????, ?????????? ???? ??????????????. ?????? ?????????????? ??????",
54      "floorAll": "<b>????????? ???? {count} ??????????????</b> {min} ?????? ??????
????????????? ? ?????? ?????????????????? ?????????????????, ?????????? ???? ??????????????. ?????
????????????? ??????",
55      "floorSomeLead": "{count, plural, one {# ?????} other {# ?????????}}",
56      "floorSomeRest": "{min} ?????? ?????? ?????????????? ?????????? ?????????????????????
(????????? <code>min_shot_count</code> ????)."
57  },
58  "result": {
59      "ready": "????? ???? ??????????????",
60      "failed": "???? ??????????????",
61      "dismiss": "?????????????",
62      "download": "{filename} ?????????? ?????"
63  },
64  "build": {
65      "noVideo": "???? ?????? ?????????????? ?????????????? ?video=<video_id> ???? ?????????
? ???? ?????????? ?????? ??????????"
66  },
67  "selection": {
68      "count": "{count, plural, one {# ?????? ??????????????} other {# ??????????
?????????????}}?"
69  }
70  }
71

```

```
1      {
2      "app": {
3          "badge": "?????-???? ??? · ??????",
4          "title": "???? ??? ?? ??? ?????? ??????",
5          "intro": "<b>???? ?? ?????</b> ?? ????? ?? ?????? ???? ? ???? <i>????? ???? ???
15?</i> ? ??? ????? ??? ???? ?????? ?? ????????? ?????? ?? ??? <b>?????</b> ???? ???? ??; ?? ??
???? ?????? ?????? ??????"
6      },
7      "lang": {
8          "label": "????",
9          "en": "English",
10         "kn": "?????",
11         "hi": "?????"
12     },
13     "bulk": {
14         "selectAll": "??? ??????",
15         "clear": "???? ??????",
16         "invert": "?????"
17     },
18     "mode": {
19         "demo": "???? ???? ? ???? ???? ???? ???? ???? ???? ???? ???? ????
<code>?video=?video_id?</code> ?? ??????",
20         "loading": "<code>{videoId}</code> ?? ??? ?????? ???? ?? ??? ??????",
21         "error": "<code>{videoId}</code> ?? ??? ?????? ???? ???? ?? ?????: {error}",
22         "loaded": "?? ???? ?? <b>{count}</b> ?????? ???? ??????"
23     },
24     "query": {
25         "placeholder": "???. over 10s · top 15 longest · shortest first",
26         "ariaLabel": "?????, ?????? ?? ???? ?? ???? ?? ?????? ?????? ??????",
27         "apply": "???? ??????",
28         "notAvailable": "??? ?????? ??????",
29         "notAvailableTitle": "{token}? ?? ??? {kind} ?????? ?????? ? ??? ?????? ??????",
30         "unreadable": "?? ??? ???? ???? ? ?over 10s? ?? ?top 15 longest? ??????????",
31         "hint": "10 ?????? ?? ???? ?????????? ?? ??? over 10s ???? ???? ? ?? ???? ?? ??? ??????
??? ??????",
32         "try": "????? ??????:",
33         "support": "<b>???? ?? ?????</b> ?? ???? ?? ?????? ???? ???? ???? ?????? ????
???-????? <x>????????????</x> ?? (?????????)?"
34     },
35     "grid": {
36         "empty": "?????? ?? ??? ??? ??? ???? ???? ????",
37         "rally": "???? {n}",
38         "inReel": "???? {n}, ??? ???",
39         "notInReel": "???? {n}, ??? ??? ??????",
40         "start": "?????",
41         "long": "?????",
42         "shots": "~{count} ?????? · ??????????",
43         "shotsTitle": "??? ?????? ?????????????? ?? (?????????) ?? ??? ?? ???? ??{conf}? ???
???? ???? ?? ?????? ?? ?????? ??????",
44         "shotsConfidence": " ? ??????: {conf}"
45     },
46     "footer": {
47         "count": "{count, plural, one {#} other {#}}",
48         "summaryRest": "????? · ??? {total}",
49         "reelName": "??? ?? ???",
50         "reelNamePlaceholder": "my-highlights",
51         "build": "??? ?????? ({count})",
52         "building": "?? ??? ???",
53         "floorOne": "<b>???? ?????? ?????? ????</b> ??? {min} ?? ?? ?????? ??? ? ???? ????
???? ???? , ?????? ?????? ???? ???? ?? ???? ???? ??????",
54         "floorAll": "<b>???? {count} ?????? ??????</b> ??? {min} ?? ?? ?????? ??? ? ???? ?? ??
?? ???? ???? ?? , ?????? ?????? ???? ???? ???? ?????? ??????",
55         "floorSomeLead": "{count, plural, one {#} other {#}}",
56         "floorSomeRest": "????? {min} ?? ?? ?????? ??? , ?????? ???? ???? ???? ?? ?????? (????
?? <code>min_shot_count</code> ?????)"
57     }
```

```

57     },
58     "result": {
59         "ready": "???? ?? ????? ??",
60         "failed": "????? ????",
61         "dismiss": "????? ????",
62         "download": "{filename} ??????? ????"
63     },
64     "build": {
65         "noVideo": "??? ?????? ?? ??? ?????? ??? ? ?video=<video_id> ?? ????? ? ???? ????
?????? ???? ?? ??????"
66     },
67     "selection": {
68         "count": "{count, plural, one {# ???? ?????} other {# ?????? ?????}}}"
69     }
70 }
71

```

229. assistant (2026-06-21T10:00:42.099Z)

Now insert the `ingest` block into the Kannada and Hindi catalogs:

230. user (2026-06-21T10:01:14.195Z)

The file C:\Users\avidu\Projects\khelsutra\web\src\i18n\locales\kn\common.json has been updated successfully. (file state is current in your context ? no need to Read it back)

231. user (2026-06-21T10:01:28.122Z)

The file C:\Users\avidu\Projects\khelsutra\web\src\i18n\locales\hi\common.json has been updated successfully. (file state is current in your context ? no need to Read it back)

232. assistant (2026-06-21T10:01:33.077Z)

Now extend the no-bare-strings guard to cover the new component:

233. user (2026-06-21T10:01:35.556Z)

The file C:\Users\avidu\Projects\khelsutra\web\src\i18n\hygiene.test.ts has been updated successfully. (file state is current in your context ? no need to Read it back)

234. assistant (2026-06-21T10:01:41.404Z)

Code is wired. Before writing the new tests, let me see the test conventions (how `client.test.ts` mocks `fetch`) and the npm scripts, then typecheck:

235. user (2026-06-21T10:01:42.743Z)

```

1     import { describe, it, expect, vi, afterEach } from "vitest";
2     import { listRallies, compileReel, highlightUrl, getConfig, minShotCount, ApiError }
from "./client";
3
4     type FetchFn = typeof fetch;
5
6     function mockFetch(status: number, json: unknown, text?: string): FetchFn {
7         return vi.fn(async () => ({
8             ok: status >= 200 && status < 300,
9             status,
10            json: async () => json,
11            text: async () => text ?? JSON.stringify(json),
12        }))) as unknown as FetchFn;
13    }
14
15    function lastCall(f: FetchFn): [string, RequestInit] {
16        const calls = (f as unknown as { mock: { calls: [string, RequestInit][] } }

```

```

    }).mock.calls;
17     return calls[calls.length - 1];
18 }
19
20 afterEach(() => {
21     vi.restoreAllMocks();
22     vi.unstubAllGlobals();
23 });
24
25 describe("highlightUrl", () => {
26     it("builds an encoded same-origin path", () => {
27         expect(highlightUrl("abc123", "my
reel.mp4")).toBe("/api/highlights/abc123/my%20reel.mp4");
28     });
29 });
30
31 describe("listRallies", () => {
32     it("POSTs video_id to /api/query and returns play_segments", async () => {
33         const seg = { id: 1, start_time: 0, end_time: 5, duration: 5, server: null,
receiver: null, metadata: { shot_count: 7 } };
34         const f = mockFetch(200, { play_segments: [seg] });
35         vi.stubGlobal("fetch", f);
36
37         const rallies = await listRallies("vid");
38         expect(rallies).toHaveLength(1);
39         expect(rallies[0].id).toBe(1);
40
41         const [url, init] = lastCall(f);
42         expect(url).toBe("/api/query");
43         expect(init.method).toBe("POST");
44         expect(JSON.parse(init.body as string)).toEqual({ video_id: "vid" });
45         expect((init.headers as Record<string, string>)[ "x-correlation-id" ]).toBeTruthy();
46     });
47
48     it("returns [] when play_segments is missing", async () => {
49         vi.stubGlobal("fetch", mockFetch(200, {}));
50         expect(await listRallies("vid")).toEqual([]);
51     });
52 });
53
54 describe("compileReel", () => {
55     it("POSTs the selection to /api/compile", async () => {
56         const f = mockFetch(200, { status: "success", filepath: "/out/reel.mp4" });
57         vi.stubGlobal("fetch", f);
58
59         const res = await compileReel("vid", [1, 2, 3], "reel.mp4");
60         expect(res.status).toBe("success");
61
62         const [url, init] = lastCall(f);
63         expect(url).toBe("/api/compile");
64         expect(JSON.parse(init.body as string)).toEqual({
65             video_id: "vid",
66             segment_ids: [1, 2, 3],
67             output_filename: "reel.mp4",
68         });
69     });
70 });
71
72 describe("getConfig + minShotCount", () => {
73     it("reads the stitching floor", async () => {
74         vi.stubGlobal("fetch", mockFetch(200, { stitching: { min_shot_count: 4 } }));
75         const cfg = await getConfig();
76         expect(minShotCount(cfg)).toBe(4);
77     });
78

```



```

79     it("returns undefined when no floor is configured", async () => {
80         vi.stubGlobal("fetch", mockFetch(200, {}));
81         expect(minShotCount(await getConfig())).toBeUndefined();
82     });
83 });
84
85 describe("error handling (loud, never silent)", () => {
86     it("throws ApiError with status + correlation id on a non-ok response", async () => {
87         vi.stubGlobal("fetch", mockFetch(400, { detail: "bad" }, "bad request"));
88         await expect(listRallies("vid")).rejects.toMatchObject({ name: "ApiError", status:
400 });
89     });
90
91     it("throws ApiError on a network failure", async () => {
92         vi.stubGlobal("fetch", vi.fn(async () => { throw new Error("offline"); }) as unknown
as FetchFn);
93         await expect(getConfig()).rejects.toBeInstanceOf(ApiError);
94     });
95 });
96

```

236. user (2026-06-21T10:01:44.219Z)

```

=== scripts ===
{
  "dev": "vite",
  "build": "vite build",
  "preview": "vite preview",
  "typecheck": "tsc --noEmit",
  "test": "vitest run"
}
=== test setup ===
// Vitest setup (runs before each test file's imports). DOM-env only: jsdom under Node 26 does
not
// expose a Web Storage `localStorage`, which the i18n language detector and the persistence
test
// both need. Node-env test files (typeof window === "undefined") skip this entirely.
if (typeof window !== "undefined" && typeof globalThis.localStorage === "undefined") {
  let store: Record<string, string> = {};
  const mem: Storage = {
    get length() {
      return Object.keys(store).length;
    },
    clear() {
      store = {};
    },
    getItem(key: string) {
      return Object.prototype.hasOwnProperty.call(store, key) ? store[key] : null;
    },
    key(index: number) {
      return Object.keys(store)[index] ?? null;
    },
    removeItem(key: string) {
      delete store[key];
    },
    setItem(key: string, value: string) {
      store[key] = String(value);
    },
  };
  Object.defineProperty(window, "localStorage", { value: mem, configurable: true, writable: true
});
  Object.defineProperty(globalThis, "localStorage", { value: mem, configurable: true, writable:
true });
}

```

237. assistant (2026-06-21T10:01:57.380Z)

Now I understand the test conventions. Let me check what React testing utilities are available (for the hook/component tests):

238. user (2026-06-21T10:01:59.430Z)

```
devDeps: {
  "@tailwindcss/vite": "^4.0.0",
  "@testing-library/dom": "^10.4.1",
  "@testing-library/react": "^16.3.2",
  "@types/node": "^26.0.0",
  "@types/react": "^19.0.0",
  "@types/react-dom": "^19.0.0",
  "@vitejs/plugin-react": "^4.3.4",
  "jsdom": "^29.1.1",
  "tailwindcss": "^4.0.0",
  "typescript": "^5.7.0",
  "vite": "^6.0.0",
  "vitest": "^3.0.0"
}
=== how i18n.test.tsx renders ===
// @vitest-environment jsdom
//
// The SPA had NO render tests before this (verified) ? this file adds the jsdom safety net for
the
// i18n scaffold (tracker B1): real copy renders from the catalog, the switcher persists the
choice,
// missing-locale falls back to English, and ICU plurals format. It mounts the real <App />.
import { describe, it, expect, beforeEach, afterEach, vi } from "vitest";
import { render, screen, cleanup, fireEvent } from "@testing-library/react";
import i18n, { LANG_STORAGE_KEY } from "../index";
import App from "../App";

beforeEach(async () => {
  // No engine in the test env: stub fetch so App's getConfig()/listRallies effects reject
quietly
  // (App handles this gracefully ? cfg stays null) instead of hitting the network.
  vi.stubGlobal("fetch", vi.fn(() => Promise.reject(new Error("no engine in test"))));
  window.localStorage.clear();
  await i18n.changeLanguage("en"); // also guarantees init completed before the first render
});

afterEach(() => {
  cleanup();
})
```

239. assistant (2026-06-21T10:02:42.123Z)

`@testing-library/react` (v16) + jsdom are available. Let me typecheck the implementation before writing tests (catch type errors early):

240. user (2026-06-21T10:02:47.028Z)

EXIT=0

241. assistant (2026-06-21T10:02:50.684Z)

Typecheck clean. Let me run the existing suite to confirm the key-parity test passes with the new `ingest.\*` keys and nothing else broke:

242. user (2026-06-21T10:02:57.515Z)

```
{"ts":"2026-06-21T10:02:57.209Z","level":"error","event":"api.network_error","cid":"af2f54f6-79e1-491e-8d9b-
```

```

5fe76e98a37c","method":"GET","url":"/api/config","error":"Error: no engine in test"}

stderr | src/i18n/i18n.test.tsx > i18n scaffold (B1) > B6:
switching to Hindi renders the hi catalog (extensibility proof ? just a catalog)
{"ts":"2026-06-21T10:02:57.209Z","level":"warn","event":"config.unavailable","error":"ApiError: Network error calling GET /api/config"}

stdout | src/i18n/i18n.test.tsx > i18n scaffold (B1) > a
locale with no catalog falls back to English (never a raw key)
{"ts":"2026-06-21T10:02:57.222Z","level":"info","event":"api.request","cid":"322c5ff6-eae2-4742-b008-53cd1815f40d","method":"GET","url":"/api/config"}

stderr | src/i18n/i18n.test.tsx > i18n scaffold (B1) > a
locale with no catalog falls back to English (never a raw key)
{"ts":"2026-06-21T10:02:57.224Z","level":"error","event":"api.network_error","cid":"322c5ff6-eae2-4742-b008-53cd1815f40d","method":"GET","url":"/api/config","error":"Error: no engine in test"}

stderr | src/i18n/i18n.test.tsx > i18n scaffold (B1) > a
locale with no catalog falls back to English (never a raw key)
{"ts":"2026-06-21T10:02:57.224Z","level":"warn","event":"config.unavailable","error":"ApiError: Network error calling GET /api/config"}

stdout | src/i18n/i18n.test.tsx > browser-language negotiation (B1:
N1/N3?N7) > N6: an unsupported browser language (Tamil) serves English content ?
never a locale we can't render
{"ts":"2026-06-21T10:02:57.239Z","level":"info","event":"api.request","cid":"e33b3461-aa5d-4c92-85c3-3666cc84e473","method":"GET","url":"/api/config"}

stderr | src/i18n/i18n.test.tsx > browser-language negotiation (B1:
N1/N3?N7) > N6: an unsupported browser language (Tamil) serves English content ?
never a locale we can't render
{"ts":"2026-06-21T10:02:57.242Z","level":"error","event":"api.network_error","cid":"e33b3461-aa5d-4c92-85c3-3666cc84e473","method":"GET","url":"/api/config","error":"Error: no engine in test"}

stderr | src/i18n/i18n.test.tsx > browser-language negotiation (B1:
N1/N3?N7) > N6: an unsupported browser language (Tamil) serves English content ?
never a locale we can't render
{"ts":"2026-06-21T10:02:57.242Z","level":"warn","event":"config.unavailable","error":"ApiError: Network error calling GET /api/config"}

? src/i18n/i18n.test.tsx (13 tests)
450ms

Test Files  8 passed (8)
Tests      104 passed (104)
Start at   15:32:53
Duration   3.30s (transform 488ms, setup 110ms, collect 1.31s, tests 528ms,
environment 1.71s, prepare 1.89s)

```

243. assistant (2026-06-21T10:04:15.494Z)

All 104 pass ? the key-parity test now validates the new `ingest.\*` keys across en/kn/hi, and App still mounts cleanly with the panel. Now the new tests. First the pure error-helper unit tests:

244. assistant (2026-06-21T10:04:21.841Z)

```

import { describe, it, expect } from "vitest";
import { parseDetail, classifyError } from "../errors";
import { ApiError } from "../api/client";

```

```

describe("parseDetail", () => {
  it("extracts FastAPI's {detail}", () => {
    expect(parseDetail(JSON.stringify({ detail: "unsupported storage URI scheme" }))).toBe(
      "unsupported storage URI scheme",
    );
  });
  it("returns the raw body when it isn't JSON or has no string detail", () => {
    expect(parseDetail("plain text")).toBe("plain text");
    expect(parseDetail(JSON.stringify({ other: 1 }))).toBe(JSON.stringify({ other: 1 }));
    expect(parseDetail(JSON.stringify({ detail: { nested: true } }))).toBe(JSON.stringify({
detail: { nested: true } }));
  });
});

describe("classifyError", () => {
  it("a network throw (ApiError with no status) ? networkError", () => {
    expect(classifyError(new ApiError("net", { cid: "c" }))).toEqual({ code: "networkError" });
  });
  it("an HTTP error with a body ? engine + the FastAPI detail", () => {
    const err = new ApiError("x", { status: 400, body: JSON.stringify({ detail: "unsupported
storage URI scheme" }) });
    expect(classifyError(err)).toEqual({ code: "engine", detail: "unsupported storage URI
scheme" });
  });
  it("an HTTP error with no body ? engine + the HTTP status", () => {
    expect(classifyError(new ApiError("x", { status: 500 }))).toEqual({ code: "engine", detail:
"HTTP 500" });
  });
  it("a non-ApiError ? engine + the stringified error", () => {
    expect(classifyError(new Error("boom"))).toEqual({ code: "engine", detail: "Error: boom" });
  });
});

```

245. user (2026-06-21T10:04:21.873Z)

File created successfully at: C:\Users\avidu\Projects\khelsutra\web\src\lib\errors.test.ts (file state is current in your context ? no need to Read it back)

246. assistant (2026-06-21T10:04:27.434Z)

Now the hook state-machine tests (fake timers driving the poll loop):

247. assistant (2026-06-21T10:04:51.754Z)

```

// @vitest-environment jsdom
//
// The ingest state machine: submit ? poll (queued/running) ? done | error. Drives the recursive
// setTimeout poll loop with fake timers so each tick is deterministic, and pins the load-
bearing
// behaviours: the success-shaped-failure (done + no rallies ? emptyResult, never a load), the
// two error surfaces (engine detail vs friendly network), and cancellation stopping the loop.
import { describe, it, expect, vi, afterEach } from "vitest";
import { renderHook, act } from "@testing-library/react";
import { useIngestJob } from "../useIngestJob";
import type { JobStatus } from "../api/types";

type FetchFn = typeof fetch;

function httpOk(status: number, json: unknown) {
  return { ok: status >= 200 && status < 300, status, json: async () => json, text: async () =>
JSON.stringify(json) };
}

/** A fetch that answers POST /api/process with `process`, then walks `jobs` on each GET

```

```

/api/jobs/*. */
function fetchSeq(process: unknown, jobs: JobStatus[]): FetchFn {
  let i = 0;
  return vi.fn(async (url: string | URL, init?: RequestInit) => {
    const u = String(url);
    const method = init?.method ?? "GET";
    if (u === "/api/process" && method === "POST") return process;
    if (u.startsWith("/api/jobs/")) {
      const body = jobs[Math.min(i, jobs.length - 1)];
      i += 1;
      return httpOk(200, body);
    }
    throw new Error(`unexpected fetch ${method} ${u}`);
  }) as unknown as FetchFn;
}

/** Flush the submit microtasks, then walk the backoff ladder (1000 ? 1500 ? 2250 ? ?) for n
ticks. */
async function drain(ticks: number) {
  await act(async () => {
    await vi.advanceTimersByTimeAsync(0); // flush processVideo ? schedule the first poll
  });
  let delay = 1000;
  for (let k = 0; k < ticks; k += 1) {
    await act(async () => {
      await vi.advanceTimersByTimeAsync(delay);
    });
    delay = Math.min(delay * 1.5, 5000);
  }
}

afterEach(() => {
  vi.restoreAllMocks();
  vi.unstubAllGlobals();
  vi.useRealTimers();
});

describe("useIngestJob", () => {
  it("rejects an empty URI before any network call", () => {
    const f = vi.fn();
    vi.stubGlobal("fetch", f);
    const { result } = renderHook(() => useIngestJob());
    act(() => result.current.submit(""));
    expect(result.current.phase).toEqual({ status: "error", error: { code: "invalidUri" } });
    expect(f).not.toHaveBeenCalled();
  });

  it("submits, polls queued ? running ? done, and hands off the video_id", async () => {
    vi.useFakeTimers();
    vi.stubGlobal(
      "fetch",
      fetchSeq(httpOk(202, { job_id: "j1", status: "queued", poll: "/api/jobs/j1" }), [
        { job_id: "j1", status: "queued" },
        { job_id: "j1", status: "running", progress: "segmenting (gemini)" },
        { job_id: "j1", status: "done", video_id: "vid9", result: { status: "success", video_id:
"vid9" } },
      ]),
    );
    const onLoaded = vi.fn();
    const { result } = renderHook(() => useIngestJob(onLoaded));
    await act(async () => {
      result.current.submit("gs://b/x.mp4");
    });
    await drain(3);
    expect(onLoaded).toHaveBeenCalledWith("vid9");
  }
});

```

```

    expect(result.current.phase).toEqual({ status: "done", videoId: "vid9" });
  });

it("treats a done-but-failed pipeline verdict as emptyResult ? never a load", async () => {
  vi.useFakeTimers();
  vi.stubGlobal(
    "fetch",
    fetchSeq(httpOk(202, { job_id: "j2", status: "queued", poll: "/api/jobs/j2" })), [
      { job_id: "j2", status: "done", video_id: "vidX", result: { status: "failed" } },
    ],
  );
  const onLoaded = vi.fn();
  const { result } = renderHook(() => useIngestJob(onLoaded));
  await act(async () => {
    result.current.submit("gs://b/x.mp4");
  });
  await drain(1);
  expect(onLoaded).not.toHaveBeenCalled();
  expect(result.current.phase).toEqual({ status: "error", error: { code: "emptyResult" } });
});

it("surfaces an execution failure with the engine's error text", async () => {
  vi.useFakeTimers();
  vi.stubGlobal(
    "fetch",
    fetchSeq(httpOk(202, { job_id: "j3", status: "queued", poll: "/api/jobs/j3" })), [
      { job_id: "j3", status: "failed", error: "RuntimeError: boom" },
    ],
  );
  const { result } = renderHook(() => useIngestJob());
  await act(async () => {
    result.current.submit("gs://b/x.mp4");
  });
  await drain(1);
  expect(result.current.phase).toEqual({ status: "error", error: { code: "engine", detail:
"RuntimeError: boom" } });
});

it("maps a 400 from /api/process to the engine's detail", async () => {
  vi.useFakeTimers();
  vi.stubGlobal("fetch", fetchSeq(httpOk(400, { detail: "unsupported storage URI scheme" })),
[]);
  const { result } = renderHook(() => useIngestJob());
  await act(async () => {
    result.current.submit("ftp://nope");
  });
  await act(async () => {
    await vi.advanceTimersByTimeAsync(0);
  });
  expect(result.current.phase).toEqual({
    status: "error",
    error: { code: "engine", detail: "unsupported storage URI scheme" },
  });
});

it("maps an unreachable engine to a friendly networkError", async () => {
  vi.useFakeTimers();
  vi.stubGlobal("fetch", vi.fn(async () => {
    throw new Error("offline");
  }) as unknown as FetchFn);
  const { result } = renderHook(() => useIngestJob());
  await act(async () => {
    result.current.submit("gs://b/x.mp4");
  });
  await act(async () => {

```

```

    await vi.advanceTimersByTimeAsync(0);
  });
  expect(result.current.phase).toEqual({ status: "error", error: { code: "networkError" } });
});

it("cancel stops the poll loop and returns to idle", async () => {
  vi.useFakeTimers();
  vi.stubGlobal(
    "fetch",
    fetchSeq(httpOk(202, { job_id: "j4", status: "queued", poll: "/api/jobs/j4" })), [
      { job_id: "j4", status: "running", progress: "hashing" },
    ],
  );
  const onLoaded = vi.fn();
  const { result } = renderHook(() => useIngestJob(onLoaded));
  await act(async () => {
    result.current.submit("gs://b/x.mp4");
  });
  await drain(1);
  expect(result.current.phase.status).toBe("polling");
  act(() => result.current.cancel());
  expect(result.current.phase).toEqual({ status: "idle" });
  // Further time passes; nothing resumes, onLoaded never fires.
  await act(async () => {
    await vi.advanceTimersByTimeAsync(10000);
  });
  expect(result.current.phase).toEqual({ status: "idle" });
  expect(onLoaded).not.toHaveBeenCalled();
});
});
});

```

248. user (2026-06-21T10:04:51.773Z)

File created successfully at:

C:\Users\avidu\Projects\khelsutra\web\src\hooks\useIngestJob.test.ts (file state is current in your context ? no need to Read it back)

249. assistant (2026-06-21T10:04:57.329Z)

Let me run the hook test to validate the fake-timer approach before writing the panel test (which reuses it):

250. user (2026-06-21T10:05:03.400Z)

```

{"ts":"2026-06-21T10:05:03.121Z","level":"info","event":"ingest.submit_start","cid":"6694f97b-7f5f-4d61-b549-6a91c579b10f","uri":"gs://b/x.mp4"}
{"ts":"2026-06-21T10:05:03.121Z","level":"info","event":"api.request","cid":"6694f97b-7f5f-4d61-b549-6a91c579b10f","method":"POST","url":"/api/process"}

```

```

stdout | src/hooks/useIngestJob.test.ts > useIngestJob >
cancel stops the poll loop and returns to idle
{"ts":"2026-06-21T10:05:03.121Z","level":"info","event":"api.response","cid":"6694f97b-7f5f-4d61-b549-6a91c579b10f","method":"POST","url":"/api/process","status":202}

```

```

stdout | src/hooks/useIngestJob.test.ts > useIngestJob >
cancel stops the poll loop and returns to idle
{"ts":"2026-06-21T10:05:03.121Z","level":"info","event":"ingest.queued","cid":"6694f97b-7f5f-4d61-b549-6a91c579b10f","uri":"gs://b/x.mp4","job_id":"j4"}

```

```

stdout | src/hooks/useIngestJob.test.ts > useIngestJob >
cancel stops the poll loop and returns to idle

```

```
{"ts":"2026-06-21T10:05:04.121Z","level":"info","event":"api.request","cid":"6694f97b-7f5f-4d61-b549-6a91c579b10f","method":"GET","url":"/api/jobs/j4"}
```

```
stdout | src/hooks/useIngestJob.test.ts > useIngestJob >  
cancel stops the poll loop and returns to idle
```

```
{"ts":"2026-06-21T10:05:04.121Z","level":"info","event":"api.response","cid":"6694f97b-7f5f-4d61-b549-6a91c579b10f","method":"GET","url":"/api/jobs/j4","status":200}
```

```
stdout | src/hooks/useIngestJob.test.ts > useIngestJob >  
cancel stops the poll loop and returns to idle
```

```
{"ts":"2026-06-21T10:05:04.121Z","level":"info","event":"ingest.cancel"}
```

```
? src/hooks/useIngestJob.test.ts (7 tests)  
39ms
```

```
Test Files  2 passed (2)  
  Tests    13 passed (13)  
Start at   15:35:00  
Duration   2.43s (transform 103ms, setup 29ms, collect 342ms, tests 43ms,  
environment 1.54s, prepare 396ms)
```

251. assistant (2026-06-21T10:05:24.229Z)

13 pass ? the fake-timer drain is solid. Now the component integration test:

252. assistant (2026-06-21T10:05:40.399Z)

```
// @vitest-environment jsdom  
//  
// The ingest screen end to end: type a URI ? submit ? (mocked engine) ? onLoaded gets the  
video_id;  
// and the two failure surfaces a non-technical user actually hits ? the engine's own 400  
detail,  
// and a friendly "is the engine running?" when it's unreachable. Mounts the real i18n catalog.  
import { describe, it, expect, vi, beforeEach, afterEach } from "vitest";  
import { render, screen, fireEvent, cleanup, act } from "@testing-library/react";  
import i18n from "../i18n/index";  
import { IngestPanel } from "../IngestPanel";  
import type { JobStatus } from "../api/types";  
  
type FetchFn = typeof fetch;  
  
function httpOk(status: number, json: unknown) {  
  return { ok: status >= 200 && status < 300, status, json: async () => json, text: async () =>  
JSON.stringify(json) };  
}  
  
function fetchSeq(process: unknown, jobs: JobStatus[]): FetchFn {  
  let i = 0;  
  return vi.fn(async (url: string | URL, init?: RequestInit) => {  
    const u = String(url);  
    if (u === "/api/process" && (init?.method ?? "GET") === "POST") return process;  
    if (u.startsWith("/api/jobs/")) {  
      const body = jobs[Math.min(i, jobs.length - 1)];  
      i += 1;  
      return httpOk(200, body);  
    }  
    throw new Error(`unexpected fetch ${u}`);  
  }) as unknown as FetchFn;  
}  
  
beforeEach(async () => {  
  window.localStorage.clear();  
  await i18n.changeLanguage("en"); // guarantees init completed + a known locale for the
```



```

assertions
});

afterEach(() => {
  cleanup();
  vi.restoreAllMocks();
  vi.unstubAllGlobals();
  vi.useRealTimers();
});

const typeUri = (value: string) =>
  fireEvent.change(screen.getByLabelText(/video location/i), { target: { value } });
const clickProcess = () => fireEvent.click(screen.getByRole("button", { name: /process video/i }));

describe("IngestPanel", () => {
  it("submits a URI, polls, and hands the video_id to onLoaded", async () => {
    vi.useFakeTimers();
    vi.stubGlobal(
      "fetch",
      fetchSeq(httpOk(202, { job_id: "j", status: "queued", poll: "/api/jobs/j" }), [
        { job_id: "j", status: "running", progress: "segmenting" },
        { job_id: "j", status: "done", video_id: "vidZ", result: { status: "success", video_id:
"vidZ" } } ],
      ),
    );
    const onLoaded = vi.fn();
    render(<IngestPanel onLoaded={onLoaded} />);
    typeUri("gs://bucket/match.mp4");
    clickProcess();

    await act(async () => {
      await vi.advanceTimersByTimeAsync(0); // flush submit ? start polling
    });
    await act(async () => {
      await vi.advanceTimersByTimeAsync(1000); // running tick
    });
    await act(async () => {
      await vi.advanceTimersByTimeAsync(1500); // done tick
    });
    expect(onLoaded).toHaveBeenCalledWith("vidZ");
  });

  it("shows the engine's own 400 detail (e.g. an unsupported scheme)", async () => {
    vi.stubGlobal("fetch", fetchSeq(httpOk(400, { detail: "unsupported storage URI scheme:
ftp://" }), []));
    render(<IngestPanel onLoaded={vi.fn()} />);
    typeUri("ftp://nope");
    clickProcess();
    expect(await screen.findByText(/unsupported storage URI scheme/i)).toBeTruthy();
  });

  it("shows a friendly, localized message when the engine is unreachable", async () => {
    vi.stubGlobal("fetch", vi.fn(async () => {
      throw new Error("offline");
    }) as unknown as FetchFn);
    render(<IngestPanel onLoaded={vi.fn()} />);
    typeUri("gs://bucket/match.mp4");
    clickProcess();
    expect(await screen.findByText(/can't reach the engine/i)).toBeTruthy();
  });

  it("rejects an empty URI with a hint and never calls the engine", () => {
    const f = vi.fn();
    vi.stubGlobal("fetch", f);
  });

```

```

    render(<IngestPanel onLoaded={vi.fn()} />);
    clickProcess();
    expect(screen.getByTestId("ingest-error")).toBeTruthy();
    expect(f).not.toHaveBeenCalled();
  });
});

```

253. user (2026-06-21T10:05:40.419Z)

File created successfully at:
 C:\Users\avidu\Projects\khelsutra\web\src\components\IngestPanel.test.tsx (file state is current
 in your context ? no need to Read it back)

254. assistant (2026-06-21T10:05:49.645Z)

Run the panel test:

255. user (2026-06-21T10:05:55.335Z)

```

stderr | src/components/IngestPanel.test.tsx > IngestPanel >
shows the engine's own 400 detail (e.g. an unsupported scheme)
{"ts":"2026-06-21T10:05:55.023Z","level":"error","event":"ingest.submit_error","cid":"e78c77e3-884e-4cd1-a7d9-8a494d5fbdaa","uri":"ftp://nope","code":"engine","detail":"unsupported storage URI scheme: ftp://"}

stdout | src/components/IngestPanel.test.tsx > IngestPanel >
shows a friendly, localized message when the engine is unreachable
{"ts":"2026-06-21T10:05:55.047Z","level":"info","event":"ingest.submit_start","cid":"329ab314-38a9-4d09-934d-8c8719931089","uri":"gs://bucket/match.mp4"}
{"ts":"2026-06-21T10:05:55.048Z","level":"info","event":"api.request","cid":"329ab314-38a9-4d09-934d-8c8719931089","method":"POST","url":"/api/process"}

stderr | src/components/IngestPanel.test.tsx > IngestPanel >
shows a friendly, localized message when the engine is unreachable
{"ts":"2026-06-21T10:05:55.051Z","level":"error","event":"api.network_error","cid":"329ab314-38a9-4d09-934d-8c8719931089","method":"POST","url":"/api/process","error":"Error: offline"}

stderr | src/components/IngestPanel.test.tsx > IngestPanel >
shows a friendly, localized message when the engine is unreachable
{"ts":"2026-06-21T10:05:55.051Z","level":"error","event":"ingest.submit_error","cid":"329ab314-38a9-4d09-934d-8c8719931089","uri":"gs://bucket/match.mp4","code":"networkError"}

? src/components/IngestPanel.test.tsx (4 tests)
249ms

Test Files  1 passed (1)
Tests       4 passed (4)
Start at   15:35:52
Duration   2.64s (transform 95ms, setup 22ms, collect 435ms, tests 249ms,
environment 1.43s, prepare 197ms)

```

256. assistant (2026-06-21T10:05:59.763Z)

All new tests pass. Now the full gate ? typecheck + entire suite + build:

257. user (2026-06-21T10:06:13.653Z)

```

=== typecheck ===
TSC=0
=== full test suite ===

```

```
Test Files 11 passed (11)
Tests 121 passed (121)
Start at 15:36:06
Duration 3.31s (transform 558ms, setup 182ms, collect 2.12s, tests 913ms,
environment 5.09s, prepare 2.61s)
```

```
=== build ===
dist/assets/noto-sans-kannada-latin-ext-wght-normal-MdBBF41x.woff2
13.99 kB
dist/assets/noto-sans-devanagari-latin-wght-normal-Bxfr4UJH.woff2
25.38 kB
dist/assets/noto-sans-kannada-latin-wght-normal-DeLZCgFT.woff2
27.58 kB
dist/assets/noto-sans-kannada-kannada-wght-normal-DuwB-P8d.woff2
90.80 kB
dist/assets/noto-sans-devanagari-devanagari-wght-normal-DWI_L5Cu.woff2
121.19 kB
dist/assets/index-8AKPRqKx.css
20.64 kB ? gzip: 4.80 kB
dist/assets/index-DPkmKAZV.js
346.69 kB ? gzip: 109.43 kB
? built in 1.23s
```

258. assistant (2026-06-21T10:06:30.188Z)

Full gate is green: ****typecheck clean, 121 tests pass**** (+17 new), ****build succeeds****. Now the browser verification the workflow requires ? let me load the preview tools and start the dev server to see the panel live.

259. assistant (2026-06-21T10:06:48.517Z)

Let me check for an existing launch config (the dry-run session may have created one):

260. user (2026-06-21T10:06:50.206Z)

```
=== .claude contents ===
.claude/launch.json
=== launch.json? ===
{
  "version": "0.0.1",
  "configurations": [
    {
      "name": "khelsutra-web",
      "runtimeExecutable": "npm",
      "runtimeArgs": ["--prefix", "web", "run", "dev", "--", "--port", "5173", "--strictPort"],
      "port": 5173
    },
    {
      "name": "khelsutra-site",
      "runtimeExecutable": "node",
      "runtimeArgs": ["site/server.js"],
      "port": 8080
    }
  ]
}
```

261. assistant (2026-06-21T10:06:55.205Z)

There's already a `khelsutra-web` config. Starting it:

262. user (2026-06-21T10:06:58.254Z)

```
{
  "serverId": "c2c98e1a-af41-4922-b644-eb4e4aaf80b7",
```



```

[57] InlineTextBox: "????????? ? ? ? ?????? ?????? ? ? ???? ???? ? ? ????
????? ? ? ? ? ??????? ?????? ?????? ?????? ? ? ???? ? ? ???? "
[58] InlineTextBox: "?????"
[59] generic
[60] textbox: "?????? ? ? ????? ? ? ? gs:// ???? URL ? ? ???? ????? ? ? ? ? ??"
[63] button: "?????? ??????? ?????"
[64] StaticText: "?????? ??????? ??????"
[65] InlineTextBox: "?????? ??????? ??????"
[66] paragraph
[67] StaticText: "? ? gs:// ???? ???? , ? ? ???? ? ? ? ? ???? ? ? ???? ? ? ? ? ??"
?????"
[68] InlineTextBox: "? ? gs:// ???? ???? , ? ? ???? ? ? ? ? ???? ? ? ?????? ? ? ? ? ??"
?????"
[70] paragraph
[71] StaticText: "???? ???? ? ???? ???? ? ? ???? ? ? ? ? ???? ???? ? ? ? ? ?"
[72] InlineTextBox: "???? ???? ? ???? ???? ? ? ???? ? ? ? ? ???? ???? ? ? ? ? ?"
[73] code
[74] StaticText: "?video=?video_id?"
[75] InlineTextBox: "?video=?video_id?"
[76] StaticText: " ? ? ??????"
[77] InlineTextBox: " ? ? ??????"
[79] none
[80] generic
[81] textbox: "????, ?????? ? ? ???? ? ? ???? ? ? ??????? ??????? ??????"
[84] button: "???? ?????"
[85] StaticText: "???? ?????"
[86] InlineTextBox: "???? ?????"
[89] StaticText: "10 ????? ? ? ???? ??????? ? ? ? ? over 10s ???? ???? ? ? ???? ? ? ? ?
???? ? ? ? ? ??"
[90] InlineTextBox: "10 ????? ? ? ???? ??????? ? ? ? ? over 10s ???? ???? ? ? ???? ? ?
? ? ? ? ? ? ? ? ? ? ??"
[91] none
[93] StaticText: "????? ??????:"
[94] InlineTextBox: "????? ??????:"
[95] button: "over 10s"
[96] StaticText: "over 10s"
[97] InlineTextBox: "over 10s"
[98] button: "top 15 longest"
[99] StaticText: "top 15 longest"
[100] InlineTextBox: "top 15 longest"
[101] button: "shortest first"
[102] StaticText: "shortest first"
[103] InlineTextBox: "shortest first"
[104] button: "first 10"
[105] StaticText: "first 10"
[106] InlineTextBox: "first 10"
[107] paragraph
[108] strong
[109] StaticText: "???? ? ? ??????"
[110] InlineTextBox: "???? ? ? ??????"
[111] StaticText: " ? ? ???? ? ? ??????? ???? ???? ? ? ? ? ???? ? ? ? ?-????? "
[112] InlineTextBox: " ? ? ???? ? ? ??????? ???? ???? ? ? ? ? ???? ? ? ? ?-????? "
[113] StaticText: "?????????????"
[114] InlineTextBox: "?????????????"
[115] StaticText: " ? ? (?????????)?"
[116] InlineTextBox: " ? ? (?????????)?"
[117] none
[118] button: "??? ??????"
[119] StaticText: "??? ??????"
[120] InlineTextBox: "??? ??????"
[121] button: "???? ?????"
[122] StaticText: "???? ?????"
[123] InlineTextBox: "???? ?????"
[124] button: "?????"
[125] StaticText: "?????"

```

```
[126] InlineTextBox: "?????"
[128] list
[129] listitem
[130] button: "???? 1, ??? ???"
[133] StaticText: "???? 1"
[134] InlineTextBox: "???? 1"
[136] StaticText: "0:12"
[137] InlineTextBox: "0:12"
[138] StaticText: " "
[139] InlineTextBox: " "
[140] StaticText: "????"
[141] InlineTextBox: "????"
[142] StaticText: " . "
[143] InlineTextBox: " . "
[144] strong
[145] StaticText: "0:08"
[146] InlineTextBox: "0:08"
[147] StaticText: " "
[148] InlineTextBox: " "
[149] StaticText: "????"
[150] InlineTextBox: "????"
[152] StaticText: "~9 ????? . ??????????"
[153] InlineTextBox: "~9 ????? . ??????????"
[154] listitem
[155] button: "???? 2, ??? ???"
[158] StaticText: "???? 2"
[159] InlineTextBox: "???? 2"
[161] StaticText: "0:41"
[162] InlineTextBox: "0:41"
[163] StaticText: " "
[164] InlineTextBox: " "
[165] StaticText: "????"
[166] InlineTextBox: "????"
[167] StaticText: " . "
[168] InlineTextBox: " . "
[169] strong
[170] StaticText: "0:06"
[171] InlineTextBox: "0:06"
[172] StaticText: " "
[173] InlineTextBox: " "
[174] StaticText: "????"
[175] InlineTextBox: "????"
[177] StaticText: "~5 ????? . ??????????"
[178] InlineTextBox: "~5 ????? . ??????????"
[179] listitem
[180] button: "???? 3, ??? ???"
[183] StaticText: "???? 3"
[184] InlineTextBox: "???? 3"
[186] StaticText: "1:03"
[187] InlineTextBox: "1:03"
[188] StaticText: " "
[189] InlineTextBox: " "
[190] StaticText: "????"
[191] InlineTextBox: "????"
[192] StaticText: " . "
[193] InlineTextBox: " . "
[194] strong
[195] StaticText: "0:21"
[196] InlineTextBox: "0:21"
[197] StaticText: " "
[198] InlineTextBox: " "
[199] StaticText: "????"
[200] InlineTextBox: "????"
[202] StaticText: "~19 ????? . ??????????"
[203] InlineTextBox: "~19 ????? . ??????????"
```

[204] listitem
[205] button: "???? 4, ??? ???"
[208] StaticText: "???? 4"
[209] InlineTextBox: "???? 4"
[211] StaticText: "1:35"
[212] InlineTextBox: "1:35"
[213] StaticText: " "
[214] InlineTextBox: " "
[215] StaticText: "????"
[216] InlineTextBox: "????"
[217] StaticText: " . "
[218] InlineTextBox: " . "
[219] strong
[220] StaticText: "0:06"
[221] InlineTextBox: "0:06"
[222] StaticText: " "
[223] InlineTextBox: " "
[224] StaticText: "????"
[225] InlineTextBox: "????"
[226] listitem
[227] button: "???? 5, ??? ???"
[230] StaticText: "???? 5"
[231] InlineTextBox: "???? 5"
[233] StaticText: "2:10"
[234] InlineTextBox: "2:10"
[235] StaticText: " "
[236] InlineTextBox: " "
[237] StaticText: "????"
[238] InlineTextBox: "????"
[239] StaticText: " . "
[240] InlineTextBox: " . "
[241] strong
[242] StaticText: "0:12"
[243] InlineTextBox: "0:12"
[244] StaticText: " "
[245] InlineTextBox: " "
[246] StaticText: "????"
[247] InlineTextBox: "????"
[249] StaticText: "~11 ????? . ??????????"
[250] InlineTextBox: "~11 ????? . ??????????"
[251] listitem
[252] button: "???? 6, ??? ???"
[255] StaticText: "???? 6"
[256] InlineTextBox: "???? 6"
[258] StaticText: "2:38"
[259] InlineTextBox: "2:38"
[260] StaticText: " "
[261] InlineTextBox: " "
[262] StaticText: "????"
[263] InlineTextBox: "????"
[264] StaticText: " . "
[265] InlineTextBox: " . "
[266] strong
[267] StaticText: "0:29"
[268] InlineTextBox: "0:29"
[269] StaticText: " "
[270] InlineTextBox: " "
[271] StaticText: "????"
[272] InlineTextBox: "????"
[274] StaticText: "~24 ????? . ??????????"
[275] InlineTextBox: "~24 ????? . ??????????"
[276] sectionfooter
[277] none
[279] strong
[280] StaticText: "6 ??????"

```

[281] InlineTextBox: "6 ??????"
[282] StaticText: " "
[283] InlineTextBox: " "
[284] StaticText: "????? . ??? 1:22"
[285] InlineTextBox: "????? . ??? 1:22"
[286] LabelText
[288] StaticText: "??? ?? ???"
[289] InlineTextBox: "??? ?? ???"
[290] textbox: "??? ?? ???" (value: "highlights")
[293] StaticText: "highlights"
[294] InlineTextBox: "highlights"
[295] button: "??? ????? (6)"
[296] StaticText: "??? ????? (6)"
[297] InlineTextBox: "??? ????? (6)"

```

265. assistant (2026-06-21T10:07:36.760Z)

The ****IngestPanel renders live and fully localized**** (the page is in Hindi from a persisted pref): heading "?????? ?????? ?????", the URI textbox with Hindi aria-label, the "?????? ?????? ??????" button, and the hint ? all above the demo rallies. Let me exercise the live error path (no engine is running on :8000, so a submit should surface the friendly localized "is the engine running?" message):

266. user (2026-06-21T10:07:38.362Z)

Successfully filled: [data-testid="ingest-panel"] input[type="text"]

267. user (2026-06-21T10:07:41.384Z)

Successfully clicked: [data-testid="ingest-panel"] button[type="submit"]

268. assistant (2026-06-21T10:07:47.121Z)

Let me check the console (correlation-id observability) and read the rendered error text:

269. user (2026-06-21T10:07:48.087Z)

```

[error] {"ts":"2026-06-21T10:07:41.374Z","level":"error","event":"api.http_error","cid":"22435cbf-4df2-44e8-849b-7ed8a5a2daa9","method":"POST","url":"/api/process","status":500,"body":""}
[error] {"ts":"2026-06-21T10:07:41.374Z","level":"error","event":"api.http_error","cid":"22435cbf-4df2-44e8-849b-7ed8a5a2daa9","method":"POST","url":"/api/process","status":500,"body":""}
[error] {"ts":"2026-06-21T10:07:41.374Z","level":"error","event":"api.http_error","cid":"22435cbf-4df2-44e8-849b-7ed8a5a2daa9","method":"POST","url":"/api/process","status":500,"body":""}
[error] {"ts":"2026-06-21T10:07:41.374Z","level":"error","event":"api.http_error","cid":"22435cbf-4df2-44e8-849b-7ed8a5a2daa9","method":"POST","url":"/api/process","status":500,"body":""}
[error] {"ts":"2026-06-21T10:07:41.375Z","level":"error","event":"ingest.submit_error","cid":"22435cbf-4df2-44e8-849b-7ed8a5a2daa9","uri":"gs://khelsutra-rally-corpus/videos/mahadevpura_smoke_180s.mp4","code":"engine","detail":"HTTP 500"}
[error] {"ts":"2026-06-21T10:07:41.375Z","level":"error","event":"ingest.submit_error","cid":"22435cbf-4df2-44e8-849b-7ed8a5a2daa9","uri":"gs://khelsutra-rally-corpus/videos/mahadevpura_smoke_180s.mp4","code":"engine","detail":"HTTP 500"}
[error] {"ts":"2026-06-

```


(Showing last 10 of 72 entries. Use 'lines' parameter (max 200) to see more.)

285. assistant (2026-06-21T10:10:12.323Z)

Console is clean ? the only errors are the *intentional* structured `api.http\_error` logs (App's config effect failing gracefully with no engine; cfg stays null, no user-facing error). No React warnings or crashes.

****Verification complete:**** panel renders in en + hi, live submit produces the friendly localized "is the engine running?" message, observability logs carry the correlation id, and all 121 tests + typecheck + build are green.

Let me stop the preview and update the DRY_RUN tracker honestly (the SPA screen is built+tested, but the box only flips on a live e2e against a real engine):

286. user (2026-06-21T10:10:12.986Z)

Server c2c98e1a-af41-4922-b644-eb6e4aaf80b7 stopped

287. user (2026-06-21T10:10:21.631Z)

<tool_use_error>File has been modified since read, either by the user or by a linter. Read it again before attempting to write it.</tool_use_error>

288. assistant (2026-06-21T10:10:35.848Z)

The branch switch refreshed the file on disk; let me re-read that section then edit:

289. user (2026-06-21T10:10:36.514Z)

```
46      ## 6. Definition of done (the dry run)
47      - [x] A bucket video ? (engine processes) ? the **SPA loads its rallies** ? select ?
compile ? **reel downloaded to the local machine**, all driven from the UI (Path A allows a one-
line CLI pre-process). ? ? **Path A done (local, 2026-06-21)**, see §7.
48      - [x] The **same flow works with the UI in Hindi** ? multilingual workflow confirmed. ?
? (Kannada also works via the same switcher.)
49      - [ ] *(Path B = B-P2, SPA-only)* the bucket video is pointed-at **from the UI** (no CLI
pre-step) ? the engine endpoint already accepts the `gs://` URI (§1); remaining work is the SPA
ingest/progress screen.
50      - [ ] *(option 3)* the same CUJ confirmed on the **CPU droplet** (Linux deploy target) ?
droplet reachable but currently **un-provisioned** (no engine checkout; ~3.8GB RAM) ? needs a
from-scratch engine install or a proper deploy of the fixed engine.
51
```

290. user (2026-06-21T10:10:47.729Z)

The file C:\Users\avidu\Projects\khelsutra\docs\DRY_RUN_CUJ.md has been updated successfully.
(file state is current in your context ? no need to Read it back)

291. assistant (2026-06-21T10:10:58.619Z)

Final gate ? full typecheck + suite + build with all changes including the classifier refinement:

292. user (2026-06-21T10:11:12.276Z)

```
=== typecheck ===
TSC OK
=== full suite ===
Test Files  11 passed (11)
Tests      123 passed (123)
Start at   15:41:05
Duration   3.39s (transform 616ms, setup 152ms, collect 2.20s, tests 954ms,
```

environment 5.20s, prepare 2.77s)

```
=== build ===
dist/assets/index-8AKPRqKx.css
 20.64 kB ? gzip:   4.80 kB
dist/assets/index-BkCjBkEL.js
346.73 kB ? gzip: 109.44 kB
? built in 1.20s
```

293. assistant (2026-06-21T10:11:20.993Z)

All green: **123 tests pass** (+19 over baseline), typecheck + build clean. Let me verify branch/HEAD and stage explicit paths (safety protocol):

294. user (2026-06-21T10:11:22.504Z)

```
=== branch + HEAD ===
feat/spa-bucket-ingest
d0461fd docs: correct stale "bucket ingest is net-new" claims (B-P1 already shipped in M2/#280)
(#55)
=== changed files ===
 M docs/DRY_RUN_CUJ.md
  M web/src/App.tsx
  M web/src/api/client.ts
  M web/src/api/types.ts
  M web/src/i18n/hygiene.test.ts
  M web/src/i18n/locales/en/common.json
  M web/src/i18n/locales/hi/common.json
  M web/src/i18n/locales/kn/common.json
?? .claude/
?? site/access.db
?? web/src/components/IngestPanel.test.tsx
?? web/src/components/IngestPanel.tsx
?? web/src/hooks/useIngestJob.test.ts
?? web/src/hooks/useIngestJob.ts
?? web/src/lib/errors.test.ts
?? web/src/lib/errors.ts
```

295. assistant (2026-06-21T10:11:26.733Z)

All expected files (`.claude/` + `site/access.db` are pre-existing untracked, excluded). Staging explicit paths and committing:

296. user (2026-06-21T10:11:29.491Z)

```
=== staged ===
 M docs/DRY_RUN_CUJ.md
  M web/src/App.tsx
  M web/src/api/client.ts
  M web/src/api/types.ts
 A web/src/components/IngestPanel.test.tsx
 A web/src/components/IngestPanel.tsx
 A web/src/hooks/useIngestJob.test.ts
 A web/src/hooks/useIngestJob.ts
  M web/src/i18n/hygiene.test.ts
  M web/src/i18n/locales/en/common.json
  M web/src/i18n/locales/hi/common.json
  M web/src/i18n/locales/kn/common.json
 A web/src/lib/errors.test.ts
 A web/src/lib/errors.ts
```

297. user (2026-06-21T10:11:45.532Z)

[feat/spa-bucket-ingest d7f98e6] feat(web): B-P2 ? in-UI bucket ingest (point at a gs:// video ?

```
process ? load)
14 files changed, 735 insertions(+), 15 deletions(-)
create mode 100644 web/src/components/IngestPanel.test.tsx
create mode 100644 web/src/components/IngestPanel.tsx
create mode 100644 web/src/hooks/useIngestJob.test.ts
create mode 100644 web/src/hooks/useIngestJob.ts
create mode 100644 web/src/lib/errors.test.ts
create mode 100644 web/src/lib/errors.ts
=== only my commit vs origin/master? ===
d7f98e6 feat(web): B-P2 ? in-UI bucket ingest (point at a gs:// video ? process ? load)
```

298. assistant (2026-06-21T10:11:48.936Z)

Only my commit. Push and open the PR:

299. user (2026-06-21T10:12:10.307Z)

```
branch 'feat/spa-bucket-ingest' set up to track 'origin/feat/spa-bucket-ingest'.
To https://github.com/avidullu/khelsutra.git
 * [new branch]      feat/spa-bucket-ingest -> feat/spa-bucket-ingest
https://github.com/avidullu/khelsutra/pull/56
```

300. assistant (2026-06-21T10:12:30.615Z)

B-P2 PR is up: [khelsutra#56](https://github.com/avidullu/khelsutra/pull/56). Let me watch its CI:

301. user (2026-06-21T10:12:53.215Z)

```
site    pass    14s
https://github.com/avidullu/khelsutra/actions/runs/27901086426/job/82561400550
web      pending 0
https://github.com/avidullu/khelsutra/actions/runs/27901086426/job/82561400541
Workers Builds: hidden-king-khelsutra    pass    0
https://dash.cloudflare.com/213c12cf60de9041dd8a3521aa206472/workers/services/view/hidden-king-
khelsutra/production/builds/4d114c67-a582-45fd-97b5-30e658cb3112
site    pass    14s
https://github.com/avidullu/khelsutra/actions/runs/27901086426/job/82561400550
web      pass    25s
https://github.com/avidullu/khelsutra/actions/runs/27901086426/job/82561400541
Workers Builds: hidden-king-khelsutra    pass    0
https://dash.cloudflare.com/213c12cf60de9041dd8a3521aa206472/workers/services/view/hidden-king-
khelsutra/production/builds/4d114c67-a582-45fd-97b5-30e658cb3112
site    pass    14s
https://github.com/avidullu/khelsutra/actions/runs/27901086426/job/82561400550
web      pass    25s
https://github.com/avidullu/khelsutra/actions/runs/27901086426/job/82561400541
EXIT=0
```

302. assistant (2026-06-21T10:13:09.334Z)

CI all green on [khelsutra#56](https://github.com/avidullu/khelsutra/pull/56) (web 25s, site, Cloudflare build). Let me record the checkpoint in memory:

303. user (2026-06-21T10:13:20.932Z)

<tool_use_error>File has not been read yet. Read it first before writing to it.</tool_use_error>

304. user (2026-06-21T10:13:29.202Z)

```
16      **Checkpoint:** 2026-06-21 (ENGINE SERVING M5+ ? OWNER BATCH-AUTHORIZED the independent
path). Owner redirected: "first surface the owner-gated decisions so I can take them + unblock
max independent PRs." Ran a 6-agent grounded gate-map **workflow** (`wf_4849400f`) over README
$2/$7/$8 + code ? a decision menu splitting every M5?M12 gate into **(a) default-OFF code I can
```

```

merge now** vs **(b) thin owner residual**. **Owner answers (AskUserQuestion):** ** (1) BLANKET
YES** ? write+merge ALL default-OFF serving code (M5,M6,M8,M9-auth,M9-consent-
cols,M11-plumbing,M12), one isolated PR each; **+ SPEND OK up to $100/?10k** for due-diligence +
**production-grade testing/verification** (so the real M6/M7 cloud runs are ALSO unblocked
within the cap ? must respect it, DELETE resources after, confirm cmd+cost before each spend per
[[gcp-vm-always-delete]]). **(2) MERGE each on green CI** (OAuth real-testing deferred to a
later rigorous pass). **(3) ALL recommended design picks** "with generous budget in mind": **co-
located stitch** (D4) . **libx264 ENCODE + NVDEC DECODE-only** (protect parity) . **hard-reject
`<2GB` v1** (config knob) . confirm `COMPUTE_DECOUPLED_SERVING/` name . **M6 = Cloud Run JOBS +
bucket-poll completion**. **? PLANNED PR ORDER:** A(doc: $8 resolutions + P3 reconcile + tracker
housekeeping) ? B(M5 preset+startup-checks+L1?L4 parity test) ? C(M6 `backend/compute/` registry
+ Dockerfile + mocked-dispatch) ? D(M8 v6 migration + pricebook[$0 placeholder] + USD/INR
aggregation + /cost API) ? E(M9-auth Cf-Access JWT verify + tenant scoping; M9-consent inert
NULL cols EXTEND D's v6 migration) ? F(M11 capture?store?shadow-eval?goldenize plumbing,
`flywheel.*`=OFF) ? G(M12 `GDriveApiStorage`+`gdrive-api://`, fake-Drive tests) + Ops-Agent
codify into `deploy/gcp/provision.sh`. Coordinate: D/E/M9-consent share ONE v6 migration (author
in D, extend in E). **Genuinely owner-only residual:** real GPU run (M5 runlog) [or I run on
cloud GPU within cap], counsel ToS/DPA (M9-consent text), Google OAuth app (M12 real), real GCP
prices+FX (M8 calibrate), CF team-domain/AUD+ingress-lock, repo rename, DO droplet/token retire.
? shared checkout ? sibling merged **#284** (`06ff4b7`, bucket-URI regression guard) mid/just-
before this session; RE-FETCH origin/master before each branch. [[working-agreement-merging]]
[[feedback-launch-quality-bar]] [[compute-stack-gcp-only]]

```

305. assistant (2026-06-21T10:13:47.233Z)

The shared memory file was concurrently edited ? a sibling session added an "ENGINE SERVING M5+" checkpoint (a different track) at line 16. I must not clobber it. Let me re-read the current structure to find a safe insertion point and confirm my earlier edits survived:

306. user (2026-06-21T10:13:48.645Z)

```

10      # Current state ? updated continuously (see [[working-agreement-checkpoints]])
11
12      > **Scope of this file = LIVE only:** the current handoff + *today's* checkpoints + open
PRs/decisions. Keep it read-first-short (?1 page). Older checkpoints are archived in [[current-
state-archive]] ? when today's checkpoints age out (next clean-slate day), move them there,
don't let them pile up here.
13
14      ***? NEXT SESSION FOCUS (2026-06-21, after the big localization+dry-run session):** SPA
is fully **en+kn+hi (DONE)**; a real **UI?engine?reel?download CUJ ran (en+hi)**; a **critical
engine bug fixed** (#281). *** NEXT BUILD (owner-picked): (1) Path B ? point at a bucket video
IN THE UI.** ? **B-P1 VERIFIED ALREADY DONE ? NO ENGINE CODE (do NOT re-scope it as a
feature):** `POST /api/process` has accepted `gs://`/`gcs://` URIs since **M2/#280** (submit
resolves the URI + skips the local-exists-404 for non-local backends `main.py:283`; the worker
fetches to scratch `acquire_local_input main.py:159` ? same hash?validate?segment?DB pipeline ?
queryable API-DB `video_id`); **TESTED**
(`tests/test_api_endpoints.py::test_process_nonlocal_uri_skips_exists_and_fetches` + hosted-
mode-reject / unknown-scheme-400 / CLI variants). Confirmed by a **3-skeptic verification
workflow (0 refuted, gaps:[])**. **Scope pin:** holds in DEFAULT single-user mode (hosted-mode
`allowed_video_root` correctly **400s** a bucket URI ? by design). Stale "local-path-only / net-
new" docs **CORRECTED ? khelsutra [#55]** (DRY_RUN_CUJ $1/2/4/6/8 + LOCAL_APPLIANCE + README).
***? So Path B = ONLY B-P2 = the SPA ingest/progress screen** (enter a `gs://` URI ? `POST
/api/process` ? poll `/api/jobs` ? load `result.video_id`; back-half already works) ? **100%
khelsutra `web/` SPA + i18n (en/kn/hi), ZERO engine code.** Ready B-P2 sketch: `DRY_RUN_CUJ.md`
$2 (new `web/src/{api/types,api/client,hooks/useIngestJob,components/IngestPanel}.tsx` +
`App.tsx` `videoId?state handoff + `ingest.*` keys). See [[lesson-verify-engine-surface-before-
scoping]]. **(2) Droplet deploy (opt 3):** deploy the FIXED engine to `168.144.126.176` (un-
provisioned, 2-core/3.8GB) ? re-run CUJ on Linux. **STILL OPEN (localization DoD):** B3 (`site/`
marketing still English-only) + native-speaker review of the AI-drafted kn/hi. **Ramp-up:**
`khelsutra/docs/DRY_RUN_CUJ.md` + `FRIENDLY_AND_MULTILINGUAL.md` $7 + the dry-run checkpoint
below. **Clean slate (at session start):** both repos on master, 0 open PRs, no worktrees/VMs.
khelsutra master `2421557` . engine master `2afd54f` (? origin/master `cee4baa`, #283 M4
squashed).
15
16      **Checkpoint:** 2026-06-21 (ENGINE SERVING M5+ ? OWNER BATCH-AUTHORIZED the independent

```

path). Owner redirected: "first surface the owner-gated decisions so I can take them + unblock max independent PRs." Ran a 6-agent grounded gate-map ****workflow**** (`wf\_4849400f`) over README §2/§7/§8 + code ? a decision menu splitting every M5?M12 gate into ****a) default-OFF code I can merge now**** vs ****b) thin owner residual****. ****Owner answers (AskUserQuestion):**** ****1) BLANKET YES**** ? write+merge ALL default-OFF serving code (M5,M6,M8,M9-auth,M9-consent-cols,M11-plumbing,M12), one isolated PR each; ****SPEND OK up to \$100/?10k**** for due-diligence + ****production-grade testing/verification**** (so the real M6/M7 cloud runs are ALSO unblocked within the cap ? must respect it, DELETE resources after, confirm cmd+cost before each spend per `[[gcp-vm-always-delete]]`). ****2) MERGE each on green CI**** (OAuth real-testing deferred to a later rigorous pass). ****3) ALL recommended design picks**** "with generous budget in mind": ****co-located stitch**** (D4) . ****libx264 ENCODE + NVDEC DECODE-only**** (protect parity) . ****hard-reject <2GB v1**** (config knob) . confirm `COMPUTE_DECOUPLED_SERVING/`` name . ****M6 = Cloud Run JOBS + bucket-poll completion****. ****? PLANNED PR ORDER:**** A(doc: §8 resolutions + P3 reconcile + tracker housekeeping) ? B(M5 preset+startup-checks+L1?L4 parity test) ? C(M6 `backend/compute/` registry + Dockerfile + mocked-dispatch) ? D(M8 v6 migration + pricebook[\$0 placeholder] + USD/INR aggregation + /cost API) ? E(M9-auth Cf-Access JWT verify + tenant scoping; M9-consent inert NULL cols EXTEND D's v6 migration) ? F(M11 capture?store?shadow-eval?goldenize plumbing, `flywheel.\*`=OFF) ? G(M12 `GDriveApiStorage`+`gdrive-api://`, fake-Drive tests) + Ops-Agent codify into `deploy/gcp/provision.sh`. Coordinate: D/E/M9-consent share ONE v6 migration (author in D, extend in E). ****Genuinely owner-only residual:**** real GPU run (M5 runlog) [or I run on cloud GPU within cap], counsel ToS/DPA (M9-consent text), Google OAuth app (M12 real), real GCP prices+FX (M8 calibrate), CF team-domain/AUD+ingress-lock, repo rename, D0 droplet/token retire. ? shared checkout ? sibling merged ****#284**** (`06ff4b7`, bucket-URI regression guard) mid/just-before this session; RE-FETCH origin/master before each branch. `[[working-agreement-merging]]` `[[feedback-launch-quality-bar]]` `[[compute-stack-gcp-only]]`

17
18 ****Checkpoint:**** 2026-06-21 (STALENESS-FIX / Path-B verify) ? owner: pick up Path B + "fix all stale docs and stop next sessions spiralling into the staleness." ****? Verified-FIRST finding: B-P1 was ALREADY DONE**** ? `POST /api/process` accepts `gs://`/`gcs://` since ****M2/#280**** (resolve?skip-exists-404?worker fetch?same pipeline?queryable API-DB `video\_id`) + TESTED; confirmed by inline file:line trace AND a ****3-skeptic verification workflow** (0/3 refuted, gaps:[]). So the doc-scoped "build the gcs ingest endpoint" was a phantom; Path B = ****SPA-only (B-P2)****. ****DONE this session:**** (1) ****khelsutra [#55] MERGED**** (master `d0461fd`) ? corrected `DRY\_RUN\_CUJ` §1/2/4/6/§8 + `LOCAL\_APPLIANCE` (`/api/health` EXISTS now M1/#268; bucket-URI ingest NOT a net-new endpoint; T12 engine-half ? done) + `README`; added the anti-spiral convention ****"re-verify engine-surface [gap]`/`net-new` claims at a cited file:line ? the engine ships PRs between sessions." (2) \*\*Memory steering corrected\*\*** (this NEXT-SESSION-FOCUS block + the dry-run checkpoint were stale "local-path-only") + new lesson `[[lesson-verify-engine-surface-before-scoping]]` + MEMORY.md index. (3) **\*\*engine [#284] MERGED\*\*** (master `06ff4b7`, all CI green incl. full suite 2m32s) ? regression GUARD ``test_process_bucket_uri_yields_queryable_video_id`` (parametrized `gs://+gcs://`) pinning ``video_id`=MD5(fetched bytes)`` (not the URI) + `POST /api/query` queryability (the SPA contract; previously only LOCAL had it); built+removed in an isolated worktree off `origin/master` (engine checkout left on its concurrent-session branch). ****? NEXT: B-P2**** = the SPA ingest/progress screen (enter `gs:// URI ? /api/process ? poll /api/jobs ? load video_id`; back-half already works) ? ****100% `web/` + i18n (en/kn/hi), ZERO engine code****; ready sketch in `DRY\_RUN\_CUJ.md` §2 + the workflow synthesis `(`web/src/{api/types,api/client,hooks/useIngestJob,components/IngestPanel}.tsx` + `App.tsx` videoId?state + `ingest.` keys)`. Then opt-3 droplet (deploy fixed engine + GCS creds + #281 fix ? re-run CUJ on Linux). ****? Repos SYNCED TO HEAD at session end:**** khelsutra master `d0461fd`; ****engine**** working tree `06ff4b7` ? NOTE the engine main checkout couldn't switch to `master` (sibling worktree `tier1-windowing` holds it), so it's the merged-but-remote-gone `feat/serving-m4-device-context` branch ****reset --hard to origin/master + upstream repointed to origin/master**** (safe: tracked tree clean + the 2 divergent commits were the M4 #283 squash, byte-identical to master; old tip tagged `pre\_sync\_m4\_44c9f01` + reflog). So `git pull --ff-only` works there now; the branch NAME is cosmetic (points at master head).

307. assistant (2026-06-21T10:14:02.385Z)

My content survived intact ? the sibling inserted their M5+ checkpoint ****between**** my NEXT-SESSION-FOCUS (line 14) and my staleness-fix checkpoint (line 18). I'll append the B-P2 status to the end of ****my**** checkpoint (a unique anchor that won't touch the sibling's block):

308. user (2026-06-21T10:14:10.712Z)

The file C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\memory\current-state.md has been updated successfully. (file state is current in your context ? no need to Read it back)

309. assistant (2026-06-21T10:14:33.372Z)

Done. **B-P2 is built, verified, and up as [khelsutra#56](https://github.com/avidullu/khelsutra/pull/56) with green CI.** The full plan is at the top of this turn; here's what landed against it.

What shipped

A user can now point at a bucket video **entirely in the UI** ? no CLI pre-step:

```
`IngestPanel` (enter `gs://?`) ? `POST /api/process` ? poll `GET /api/jobs/{id}` (recursive `setTimeout`, backoff + 10-min ceiling, one correlation id) ? on `done`, hand `result.video_id` to `App` ? `?video=` is set ? the existing rally-selection back-half loads. **100% `web/` + i18n, zero engine code** (it rides B-P1's already-shipped endpoint).
```

```
New: [api/client.ts](web/src/api/client.ts) (`processVideo`/`getJob`),  
[lib/errors.ts](web/src/lib/errors.ts) (`parseDetail` de-duped + `classifyError`),  
[hooks/useIngestJob.ts](web/src/hooks/useIngestJob.ts) (the state machine),  
[components/IngestPanel.tsx](web/src/components/IngestPanel.tsx), plus `App.tsx` wiring and `ingest.*` in en/kn/hi.
```

Verification

- **123 web tests** (+19: error classifier, hook state-machine with a fake-timer poll loop, panel integration), `tsc` clean, `vite build` green, key-parity green.
- **Browser-verified** live: panel renders in **en + hi**, and a submit with no engine shows the friendly localized message (?????: "???? ?" / EN: "Can't reach the engine. Is it running?"), with structured correlation-id logs.
- **One UX nuance fixed from the live run:** a bare 5xx / dev-proxy-down was showing a cryptic "HTTP 500" ? now a 5xx with no body maps to the friendly `networkError`, while a real engine detail (e.g. a 400 unsupported-scheme) still passes through verbatim.

Honest status

- I left **#56 for you to merge** ? it's a user-facing product change, and the working agreement keeps product gated to you (there's precedent for me merging SPA PRs on your "merge as you go," so just say the word). CI is green.
- `DRY\_RUN\_CUJ.md` §6 box 3 stays **?* ? the screen is built + tested, but it only flips on a **live e2e-from-UI run against a real engine** (local, or the droplet ? option 3).
- kn/hi `ingest.\*` strings are **AI-drafts pending native review** (consistent with the existing catalogs).

Heads-up

A **concurrent session is actively working the engine-serving M5+ track** and writing to the shared memory/checkout (it's aware of my #284). No conflict ? I appended to my own memory block and stayed off its branches.

Want me to **merge #56** and then move to the **droplet e2e** (deploy the fixed engine + GCS creds ? run the bucket?reel CUJ from the UI on Linux, which flips the last DoD box)?

310. user (2026-06-21T10:14:36.738Z)

thanks for the snippet, make sure to merge green PRs and consistently improve coverage and also if possible leave a screencast of the entire CUJ it builds! Add them to the gs:// bucket in promo directory where other screencasts also live. I want this to be a "proof and validation driven" development!

311. assistant (2026-06-21T10:15:50.412Z)

Love it ? proof-and-validation-driven it is. The screencast of the in-UI CUJ *is* the live e2e validation that flips the last DoD box. Let me (1) merge #56, then (2) run the real engine and

record the entire bucket?reel CUJ driven from the UI, and (3) upload it to the promo bucket.

First, merge the green B-P2 PR:

312. user (2026-06-21T10:15:58.728Z)

```
Workers Builds: hidden-king-khelsutra pass 0
https://dash.cloudflare.com/213c12cf60de9041dd8a3521aa206472/workers/services/view/hidden-king-
khelsutra/production/builds/4d114c67-a582-45fd-97b5-30e658cb3112
site pass 14s
https://github.com/avidullu/khelsutra/actions/runs/27901086426/job/82561400550
web pass 25s
https://github.com/avidullu/khelsutra/actions/runs/27901086426/job/82561400541
--- merging ---
khelsutra master now: 4525c5d feat(web): B-P2 ? in-UI bucket ingest (point at a gs:// video ?
process ? load) (#56)
```

313. assistant (2026-06-21T10:16:10.990Z)

#56 merged (khelsutra master `4525c5d`). Now let me survey the engine environment for a real CUJ run ? Python/deps, ffmpeg, an available video, GCS fetch capability, and the screencast tooling:

314. user (2026-06-21T10:16:31.410Z)

```
Deleted branch feat/spa-bucket-ingest (was d7f98e6).
=== engine python + key deps ===
py 3.13.13 | fastapi 0.136.1 | cv2 4.13.0
Traceback (most recent call last):
=== ffmpeg ===
ffmpeg version 8.1.1-full_build-www.gyan.dev Copyright (c) 2000-2026 the FFmpeg developers
=== gcloud/gsutil auth ===
avi.dullu@gmail.com
=== candidate local videos ===
-rw-r--r-- 1 avidu 197609 22383681 Jun 20 23:46 artifacts/promo/curated_top10_longest.mp4
-rw-r--r-- 1 avidu 197609 776632 Jun 20 23:46 artifacts/promo/full_cuj_real.mp4
-rw-r--r-- 1 avidu 197609 709229 Jun 20 23:02 artifacts/promo/promo_cuj.mp4
-rw-r--r-- 1 avidu 197609 22383681 Jun 20 23:52 artifacts/review/reel_CURATED_top10.mp4
-rw-r--r-- 1 avidu 197609 31749944 Jun 20 23:52 artifacts/review/reel_FULL_22rallies.mp4
-rw-r--r-- 1 avidu 197609 90194321 Jun 20 23:52
artifacts/review/source_mahadevpura_smoke_180s.mp4
total 99596
drwxr-xr-x 1 avidu 197609 0 Jun 21 13:42 .
drwxr-xr-x 1 avidu 197609 0 Jun 21 13:40 ..
-rw-r--r-- 1 avidu 197609 50985465 Jun 21 13:41 highlights_local_en.mp4
-rw-r--r-- 1 avidu 197609 50985465 Jun 21 13:42 highlights_local_hi.mp4
=== bucket promo dir (existing screencasts) ===
gs://khelsutra-rally-corpus/promo/full_cuj_real.mp4
gs://khelsutra-rally-corpus/promo/khelsutra-i18n-screencast.mp4
gs://khelsutra-rally-corpus/promo/promo_cuj.gif
gs://khelsutra-rally-corpus/promo/promo_cuj.mp4
```

315. assistant (2026-06-21T10:17:17.641Z)

Engine env is ready (Python 3.13, opencv, ffmpeg all present), gcloud is authed, and the known-good 180s clip is local. The one gap for a true **bucket**?reel run: `google-cloud-storage` isn't installed in the engine env (so it can't fetch `gs://` yet). Let me check ADC availability and install it:

316. user (2026-06-21T10:17:29.671Z)

```
=== SA key present? ===
-rw-r--r-- 1 avidu 197609 2396 Jun 20 14:47 /c/Users/avidu/.gcp/rally-worker-sa.json
=== ADC present? ===
-rw-r--r-- 1 avidu 197609 358 Jun 6 11:57
```

```
C:\Users\avidu\AppData\Roaming\gcloud\application_default_credentials.json
=== ADC token works? ===
ya29.a0AT3oNZ8iXyRZsP4S9m...
=== verify the bucket video exists ===
  90194321  2026-06-20T14:20:32Z  gs://khelsutra-rally-corpus/videos/mahadevpura_smoke_180s.mp4
TOTAL: 1 objects, 90194321 bytes (86.02 MiB)
```

317. assistant (2026-06-21T10:17:49.438Z)

All set for a true bucket fetch: SA key present, ADC works, and the 86 MiB clip is in the bucket. Let me install `google-cloud-storage` into the engine env and confirm config + the key-free `motion` detector (no paid/cloud action for the screencast):

318. user (2026-06-21T10:18:15.637Z)

```
=== install google-cloud-storage ===
```

```
[notice] A new release of pip is available: 26.0.1 -> 26.1.2
[notice] To update, run: python.exe -m pip install --upgrade pip
google-cloud-storage OK 3.12.0
=== config.json present? default segmenter? ===
sport badminton | default_segmenter gemini
=== motion segmenter registered? ===
available: ['motion', 'gemini', 'yolo_hybrid', 'tracknet_hybrid', 'wasb_hybrid',
'fusion_hybrid']
```

319. assistant (2026-06-21T10:19:10.992Z)

All set ? `google-cloud-storage 3.12.0` installed, `motion` detector available (key-free, no paid/cloud action). Let me run the engine in an **isolated worktree** (fresh DB/output, the dry-run discipline) and validate the full gs:///reel CUJ via the API *before* recording ? proof first:

320. user (2026-06-21T10:19:14.538Z)

```
Preparing worktree (detached HEAD fefa412)
HEAD is now at fefa412 docs(serving): resolve S8 design picks (D16/D17) + reconcile per-video P3
vs M3 (#285)
worktree ready: fefa412 docs(serving): resolve S8 design picks (D16/D17) + reconcile per-video
P3 vs M3 (#285)
```

321. assistant (2026-06-21T10:19:28.342Z)

origin/master moved to `fefa412` (sibling merged serving PR #285) ? my worktree is off the latest, which includes my #284. Now start the engine from the worktree with the SA credential, in the background:

322. user (2026-06-21T10:19:39.021Z)

```
engine starting (bg pid 1945)
=== health ===
{"status":"ok","sport":"badminton","default_segmenter":"gemini"}
=== log tail ===
INFO:      Started server process [35256]
INFO:      Waiting for application startup.
INFO:      Application startup complete.
INFO:      Uvicorn running on http://127.0.0.1:8000 (Press CTRL+C to quit)
[SERVER] Starting local FastAPI server on http://localhost:8000...
INFO:      127.0.0.1:52924 - "GET /api/health HTTP/1.1" 200 OK
```

323. assistant (2026-06-21T10:19:46.571Z)

Engine is healthy. Now validate the full gs:///reel CUJ via the API before recording (proof first). Submit the bucket video with the key-free `motion` detector and poll:

324. user (2026-06-21T10:20:29.325Z)

```
=== submit gs:// + motion ===
{"job_id":"27e865fa6adb416fb5e118ee6f06d0e5","status":"queued","poll":"/api/jobs/27e865fa6adb416fb5e118ee6f06d0e5"}
job_id=27e865fa6adb416fb5e118ee6f06d0e5
=== poll (motion processing) ===
[5s] running validating None
[10s] running validating None
[15s] running validating None
[20s] running validating None
[25s] running segmenting (motion) None
[30s] running segmenting (motion) None
[35s] done finished 05e0e577029daafc425ca70c0c4f0a3
FINAL: {"job_id":"27e865fa6adb416fb5e118ee6f06d0e5","status":"done","progress":"finished","video_id":"05e0e577029daafc425ca70c0c4f0a3","error":null,"result":{"video_id":"05e0e577029daafc425ca70c0c4f0a3","status":"success","message":"Successfully indexed 0 play segments using 'motion' segmenter.", "results":[{"validator_name":"format_check","passed":true,"message":"Video format is valid MP4 container.", "details":{"format_name":"mov,mp4,m4a,3gp,3g2,mj2","duration":180.046167,"size_bytes":90194321,"codec_name":"h264"}},{ "validator_name":"resolution_fps_check","passed":true,"message":"Video resolution and frame rate are valid."}}]}
Shell cwd was reset to C:\Users\avidu\Projects\badminton-highlight-indexer
```

325. assistant (2026-06-21T10:21:34.276Z)

The gs:// fetch + processing worked end-to-end (video_id `05e0e577?`), **but** `motion` found 0 rallies on this clip ? so it can't build a reel. The dry-run got real rallies only with **Gemini** (12) or GPU WASB (22). Let me check whether the Gemini key is available before deciding how to proceed:

326. user (2026-06-21T10:21:39.827Z)

```
=== .env GEMINI key present? ===
.env
(no GEMINI_API_KEY in .env files)
=== env var? ===
GEMINI_API_KEY set in env: False
=== does the MAIN checkout DB already have rallies for 05e0e577? (reuse option) ===
no local DB has rallies for 05e0e577
```

327. assistant (2026-06-21T10:22:09.767Z)

The `.env` mentions `GEMINI\_API\_KEY` (the engine loads it via dotenv), but I need to confirm it's a real key before offering it. Let me verify it's non-empty without printing the secret:

328. user (2026-06-21T10:22:12.008Z)

GEMINI_API_KEY present: True | length: 53 | prefix: AQ.A?

329. assistant (2026-06-21T10:22:33.127Z)

The Gemini key is present and real (the engine loaded it via dotenv). But Gemini uploads the clip to Google and is a paid action ? your standing guardrail requires explicit OK each time, so I'll confirm rather than assume. The gs://?reel plumbing is already proven working (the engine fetched the bucket clip and produced a `video\_id`); the only question is the detector for real rallies in the screencast:

330. user (2026-06-21T13:00:59.241Z)

Your questions have been answered: "motion (key-free) found 0 rallies on the smoke clip, so it can't build a reel. For a real end-to-end bucket?reel screencast, which detector should the engine use?"="Gemini (recommended)". You can now continue with these answers in mind.

331. assistant (2026-06-21T13:01:32.492Z)

Gemini it is (now explicitly authorized for this screencast). Note: my B-P2 IngestPanel submits `{video\_path}` only, so the UI ingest uses the engine's **default segmenter (gemini)** automatically ? exactly what we want. Let me do one API validation run with Gemini first (confirm real rallies + measure timing so I set the recording waits right), then record the UI flow:

332. user (2026-06-21T13:02:01.003Z)

```
=== submit gs:// + gemini (default) ===
{"job_id":"20d4dddc607d4607991bbca88ea02f48","status":"queued","poll":"/api/jobs/20d4dddc607d4607991bbca88ea02f48"}
=== poll ===
[5s] running validating
[10s] running validating
[15s] running validating
[20s] done finished
video_id=05e0e577029daafc425ca70c0c4f0a3
=== rallies via /api/query ===
rally count: 0
Shell cwd was reset to C:\Users\avidu\Projects\badminton-highlight-indexer
```

333. assistant (2026-06-21T13:02:22.612Z)

20s with 0 rallies ? too fast for real Gemini. The likely cause: the engine started from the **worktree** (`bhi-wt-screencast`), whose dir has **no `.env`** (it's gitignored, so not in a fresh checkout) ? the `GEMINI\_API\_KEY` wasn't loaded. Let me confirm from the engine log:

334. user (2026-06-21T13:02:24.674Z)

```
=== worktree has .env? ===
(NO .env in worktree ? no GEMINI_API_KEY loaded)
=== engine log: gemini/key errors ===
18:32:00 [INFO] backend.main: [API] Using segmenter: gemini
18:32:00 [ERROR] backend.pipeline.segmenters.gemini: GEMINI_API_KEY environment variable is not set! To run the gemini segmenter, set your API key. In PowerShell:
$env:GEMINI_API_KEY="your_api_key_here"
```

335. assistant (2026-06-21T13:02:41.028Z)

Confirmed ? the worktree has no `.env`, so Gemini ran keyless and returned 0. Let me stop the engine, give the worktree the `.env` (so dotenv loads the key), and restart:

336. user (2026-06-21T13:02:47.789Z)

```
=== find + kill engine on :8000 ===
listener pid: 35256
SUCCESS: The process with PID 35256 has been terminated.
=== copy .env into worktree (gitignored, local only) ===
worktree .env GEMINI key len: 53
```

337. assistant (2026-06-21T13:02:54.122Z)

Engine stopped, worktree now has the `.env` with the key. Restart with both credentials (Gemini key via dotenv + GCS SA for the bucket fetch):

338. user (2026-06-21T13:06:10.689Z)

```
engine restarting (bg 597)
=== health ===
{"status":"ok","sport":"badminton","default_segmenter":"gemini"}
=== re-run gemini CUJ ===
job=07d383c72a814993a0bc70e7f3661c50
[5s] running validating
[10s] running validating
```

```
[15s] running validating
[20s] running validating
[25s] running segmenting (gemini)
[30s] running segmenting (gemini)
[35s] running segmenting (gemini)
[40s] running segmenting (gemini)
[45s] running segmenting (gemini)
[50s] running segmenting (gemini)
[55s] running segmenting (gemini)
[60s] running segmenting (gemini)
[65s] running segmenting (gemini)
[70s] running segmenting (gemini)
[75s] running segmenting (gemini)
[80s] running segmenting (gemini)
[85s] running segmenting (gemini)
[90s] running segmenting (gemini)
[95s] running segmenting (gemini)
[100s] running segmenting (gemini)
[105s] running segmenting (gemini)
[110s] running segmenting (gemini)
[115s] running segmenting (gemini)
[120s] running segmenting (gemini)
[125s] running segmenting (gemini)
[130s] running segmenting (gemini)
[135s] running segmenting (gemini)
[140s] running segmenting (gemini)
[145s] running segmenting (gemini)
[150s] running segmenting (gemini)
[155s] running segmenting (gemini)
[160s] running segmenting (gemini)
[165s] running segmenting (gemini)
[170s] done finished
video_id=05e0e577029daafc425ca70c0c4f0a3
rally count: 12
```

339. assistant (2026-06-21T13:07:00.399Z)

****Real proof achieved:**** gs:// fetch ? Gemini ? ****12 rallies**** (matching the dry-run),
`/api/query` returns them. Gemini took ~145s. Now let me build the screencast of the full UI
CUJ. Let me inspect the existing Playwright harness (used for the i18n screencast) to reuse it:

340. user (2026-06-21T13:07:02.917Z)

```
=== dir contents ===
total 629
drwxr-xr-x 1 avidu 197609      0 Jun 21 12:45 .
drwxr-xr-x 1 avidu 197609      0 Jun 21 15:49 ..
-rw-r--r-- 1 avidu 197609 342162 Jun 21 12:45 khelsutra-i18n-screencast.mp4
drwxr-xr-x 1 avidu 197609      0 Jun 21 12:43 node_modules
-rw-r--r-- 1 avidu 197609  1714 Jun 21 12:43 package-lock.json
-rw-r--r-- 1 avidu 197609   309 Jun 21 12:43 package.json
-rw-r--r-- 1 avidu 197609  2090 Jun 21 12:44 screencast.mjs
-rw-r--r-- 1 avidu 197609 123371 Jun 21 12:44 shot-1-kn.png
-rw-r--r-- 1 avidu 197609  83368 Jun 21 12:44 shot-2-hi.png
-rw-r--r-- 1 avidu 197609  62302 Jun 21 12:44 shot-3-en.png
drwxr-xr-x 1 avidu 197609      0 Jun 21 12:44 video
=== existing scripts ===
screencast.mjs
DIR MISSING
Shell cwd was reset to C:\Users\avidu\Projects\badminton-highlight-indexer
```

341. assistant (2026-06-21T13:07:09.566Z)

Let me read the existing harness to adapt it for the CUJ:

342. user (2026-06-21T13:07:10.048Z)

```
1      // KhelSutra Studio i18n screencast: a Kannada-DEFAULT browser auto-renders Kannada,
2      // then the switcher walks kn -> hi -> en. Proves the owner's success criterion.
3      import { chromium } from "playwright";
4
5      const URL = process.env.URL || "http://localhost:5180";
6      const KN_TITLE = "????? ?????????? ?????????????? ??????";
7      const HI_TITLE = "????? ??? ?? ??? ?????????? ??????";
8      const EN_TITLE = "Pick the rallies for your reel";
9      const HOLD = 2600;
10
11     const browser = await chromium.launch();
12     // A Kannada-configured browser: locale sets navigator.language=kn-IN. Fresh context =>
no stored
13     // preference => the SPA negotiates from the BROWSER language (the whole point).
14     const ctx = await browser.newContext({
15       locale: "kn-IN",
16       viewport: { width: 1280, height: 900 },
17       recordVideo: { dir: "video", size: { width: 1280, height: 900 } },
18     });
19     const page = await ctx.newPage();
20     await page.goto(URL, { waitUntil: "domcontentloaded" });
21
22     // 1) Kannada renders automatically (browser language ? kn).
23     await page.getByText(KN_TITLE, { exact: true }).waitFor({ timeout: 15000 });
24     console.log("kn rendered from browser language:", await page.evaluate(() =>
document.documentElement.lang));
25     await page.waitForTimeout(HOLD);
26     await page.screenshot({ path: "shot-1-kn.png" });
27
28     // 2) Switch to Hindi.
29     await page.locator('button[lang="hi"]').click();
30     await page.getByText(HI_TITLE, { exact: true }).waitFor({ timeout: 8000 });
31     console.log("hi rendered:", await page.evaluate(() => document.documentElement.lang));
32     await page.waitForTimeout(HOLD);
33     await page.screenshot({ path: "shot-2-hi.png" });
34
35     // 3) Switch to English.
36     await page.locator('button[lang="en"]').click();
37     await page.getByText(EN_TITLE, { exact: true }).waitFor({ timeout: 8000 });
38     console.log("en rendered:", await page.evaluate(() => document.documentElement.lang));
39     await page.waitForTimeout(HOLD);
40     await page.screenshot({ path: "shot-3-en.png" });
41
42     await ctx.close(); // flushes the video
43     await browser.close();
44     console.log("DONE");
45
```

343. user (2026-06-21T13:07:10.405Z)

```
1      {
2        "name": "kheelsutra-screencast",
3        "version": "1.0.0",
4        "description": "",
5        "main": "index.js",
6        "scripts": {
7          "test": "echo \"Error: no test specified\" && exit 1"
8        },
9        "keywords": [],
10       "author": "",
11       "license": "ISC",
12       "type": "commonjs",
13       "dependencies": {
```

```

14     "playwright": "^1.61.0"
15   }
16 }
17

```

344. assistant (2026-06-21T13:07:47.673Z)

Good ? the harness uses `playwright` chromium, records to `video/`, against the SPA dev server. Let me read `CompileResult.tsx` (to target the download link) and start the SPA dev server pointing at the live engine:

345. user (2026-06-21T13:07:48.154Z)

```

1    // The compiled-reel result (PER_RALLY_SELECTION.md P5): once Build ? POST /api/compile
succeeds,
2    // preview the reel inline (<video> over GET /api/highlights/...) and offer a download.
The engine
3    // returns a server-local filepath, NOT a URL (main.py:394), so the SPA builds the URL
itself. On a
4    // failure (e.g. the min_shot_count 400) we show the engine's reason ? loud, never
swallowed.
5
6    import { useTranslation } from "react-i18next";
7
8    export type BuildState =
9      | { status: "idle" }
10     | { status: "building" }
11     | { status: "done"; url: string; filename: string }
12     | { status: "error"; error: string };
13
14    interface Props {
15      build: BuildState;
16      onDismiss: () => void;
17    }
18
19    export function CompileResult({ build, onDismiss }: Props) {
20      const { t } = useTranslation();
21      if (build.status === "idle" || build.status === "building") return null;
22
23      return (
24        <section
25          data-testid="compile-result"
26          className={
27            "mt-6 rounded-xl border p-4 " +
28            (build.status === "error" ? "border-red-200 bg-red-50" : "border-green/30 bg-
green/5")
29        >
30          <div className="flex items-start justify-between gap-3">
31            <h2 className="font-semibold text-ink">
32              {build.status === "done" ? t("result.ready") : t("result.failed")}
33            </h2>
34            <button
35              type="button"
36              onClick={onDismiss}
37              aria-label={t("result.dismiss")}
38              className="rounded-full px-2 text-ink/50 hover:text-ink"
39            >
40              ?
41            </button>
42          </div>
43
44          {build.status === "done" ? (
45            <div className="mt-3">
46              /* eslint-disable-next-line jsx-a11y/media-has-caption ? a stitched rally

```

```

reel has no captions */}
48         <video
49             data-testid="reel-video"
50             src={build.url}
51             controls
52             playsInline
53             className="w-full max-w-xl rounded-lg bg-black"
54         />
55         <a
56             href={build.url}
57             download={build.filename}
58             className="mt-3 inline-block rounded-full bg-green px-4 py-2 text-sm font-
semibold text-[#04140D]"
59         >
60             {t("result.download", { filename: build.filename })}
61         </a>
62     </div>
63 ) : (
64     <p className="mt-2 text-sm text-red-800" data-testid="compile-error">
65         {build.error}
66     </p>
67 )}
68 </section>
69 );
70 }
71

```

346. user (2026-06-21T13:07:57.618Z)

```
=== vite log ===
```

```

VITE v6.4.3 ready in 551 ms

? Local: http://localhost:5180/
? Network: use --host to expose
=== SPA reachable? ===
200 (http code)
=== /api proxied to engine? ===
{"status":"ok","sport":"badminton","default_segmenter":"gemini"}

```

347. assistant (2026-06-21T13:08:22.744Z)

SPA dev server is up on :5180 and proxies `/api` to the live engine. The download is an `` link, reel preview is `[data-testid="reel-video"]`. Let me write the CUJ recording script:

348. assistant (2026-06-21T13:08:38.034Z)

```

// KhelSutra Studio ? FULL bucket?reel CUJ, driven entirely from the UI (B-P2).
// Point at a gs:// bucket video ? engine processes (real Gemini) ? review rallies ? build ?
download.
// Records the whole journey. Run against the SPA dev server (proxies /api ? the live engine on
:8000).
import { chromium } from "playwright";

const URL = process.env.URL || "http://localhost:5180";
const GS_URI = "gs://khelesutra-rally-corpus/videos/mahadevpura_smoke_180s.mp4";
const PROCESS_TIMEOUT = 300000; // up to 5 min for the real Gemini run + gs:// fetch
const HOLD = 1800;

const browser = await chromium.launch();
// Fresh context, English browser ? the SPA negotiates `en` (no persisted pref).
const ctx = await browser.newContext({

```



```

    locale: "en-US",
    viewport: { width: 1280, height: 900 },
    recordVideo: { dir: "video-cuj", size: { width: 1280, height: 900 } },
  });
  const page = await ctx.newPage();
  page.on("console", (m) => { if (m.type() === "error") console.log(" [page error]",
  m.text().slice(0, 160)); });

  await page.goto(URL, { waitUntil: "domcontentloaded" });

  // 1) Land on the ingest screen (no ?video= ? the "Process a video" panel).
  await page.getByText("Process a video", { exact: true }).waitFor({ timeout: 15000 });
  console.log("ingest panel rendered; lang =", await page.evaluate(() =>
  document.documentElement.lang));
  await page.waitForTimeout(HOLD);
  await page.screenshot({ path: "cuj-1-ingest.png" });

  // 2) Paste the bucket URI (typed for the camera) and submit.
  const input = page.getByLabel(/video location/i);
  await input.click();
  await input.pressSequentially(GS_URI, { delay: 28 });
  await page.waitForTimeout(700);
  await page.screenshot({ path: "cuj-2-uri.png" });
  await page.getByRole("button", { name: /process video/i }).click();

  // 3) Live progress while the engine fetches the bucket video + runs Gemini.
  await page.getByText(/Processing your video/i).waitFor({ timeout: 15000 }).catch(() => {});
  await page.waitForTimeout(HOLD);
  await page.screenshot({ path: "cuj-3-processing.png" });

  // 4) Rallies load into the existing back-half (the ingest panel hands off video_id ? ?video=).
  await page.getByText(/rallies from this session/i).waitFor({ timeout: PROCESS_TIMEOUT });
  const loaded = await page.getByText(/rallies from this session/i).textContent();
  console.log("rallies loaded:", loaded?.trim());
  await page.waitForTimeout(HOLD);
  await page.screenshot({ path: "cuj-4-rallies.png" });

  // 5) Build the reel from the (default all-selected) rallies.
  const buildBtn = page.getByRole("button", { name: /build reel/i });
  await buildBtn.scrollIntoViewIfNeeded();
  await buildBtn.click();

  // 6) Reel ready ? preview + download.
  await page.getByText("Your reel is ready", { exact: true }).waitFor({ timeout: 120000 });
  await page.getByTestId("compile-result").scrollIntoViewIfNeeded();
  await page.waitForTimeout(HOLD);
  await page.screenshot({ path: "cuj-5-reel-ready.png" });

  // 7) Download to the local machine (the real end of the CUJ).
  const [download] = await Promise.all([
    page.waitForEvent("download"),
    page.getByRole("link", { name: /download/i }).click(),
  ]);
  await download.saveAs("cuj-downloaded-reel.mp4");
  console.log("downloaded reel:", download.suggestedFilename());
  await page.waitForTimeout(1200);

  await ctx.close(); // flushes the video
  await browser.close();
  console.log("DONE");

```

349. user (2026-06-21T13:08:38.062Z)

File created successfully at: C:\Users\avidu\Projects\kheIsutra-screencast\cuj-screencast.mjs
 (file state is current in your context ? no need to Read it back)

350. assistant (2026-06-21T13:08:45.005Z)

Now run the full CUJ recording (real engine, real Gemini ~3 min). This is the live e2e proof:

351. user (2026-06-21T13:13:54.555Z)

```
engine alive? 200
ingest panel rendered; lang = en
rallies loaded: Loaded 12 rallies from this session.
[page error] Failed to load resource: the server responded with a status of 500 (Internal
Server Error)
[page error] {"ts":"2026-06-
21T13:11:53.911Z","level":"error","event":"api.http_error","cid":"446c80c5-9f97-4e24-aab6-
d76616c45a54","method":"POST","url":"/api/compile","sta
[page error] {"ts":"2026-06-
21T13:11:53.912Z","level":"error","event":"build.failed","cid":"446c80c5-9f97-4e24-aab6-
d76616c45a54","video_id":"05e0e577029daafc425ca70c0c4f0a
node:internal/modules/run_main:107
  triggerUncaughtException(
    ^
locator.waitFor: Timeout 120000ms exceeded.
Call log:
  - waiting for getByText('Your reel is ready', { exact: true }) to be visible

    at C:\Users\avidu\Projects\khelsutra-screencast\cuj-screencast.mjs:55:61 {
  log: [
    " - waiting for getByText('Your reel is ready', { exact: true }) to be visible"
  ],
  name: 'TimeoutError'
}

Node.js v26.1.0
Shell cwd was reset to C:\Users\avidu\Projects\badminton-highlight-indexer
```

352. assistant (2026-06-21T13:14:14.735Z)

****Proof-driven development just caught a real gap!**** The CUJ got all the way through ? ingest ? process ? ****12 rallies loaded**** ? but ``/api/compile`` returned ****500****. Let me confirm the root cause in the engine log:

353. user (2026-06-21T13:14:15.844Z)

```
18:33:14 [INFO] backend.main: Running validations for gs://khelsutra-rally-
corpus/videos/mahadevpura_smoke_180s.mp4...
18:38:57 [INFO] backend.main: Running validations for gs://khelsutra-rally-
corpus/videos/mahadevpura_smoke_180s.mp4...
INFO:      127.0.0.1:61123 - "POST /api/compile HTTP/1.1" 500 Internal Server Error
```

354. assistant (2026-06-21T13:14:24.031Z)

The 500 has no traceback in the log. Let me read the compile endpoint + compiler to confirm it's reading the ``gs://`` filepath as a local file:

355. user (2026-06-21T13:14:25.353Z)

```
399         sc = meta.get("shot_count") if isinstance(meta, dict) else None
400         try:
401             if sc is not None and int(sc) < stitching.min_shot_count:
402                 continue
403         except (TypeError, ValueError):
404             pass # unparseable count -> treat as unknown, keep
405         kept.append(s)
406     if len(kept) < len(matched_segments):
```

```

407         logger.info(f"[compile] stitching.min_shot_count={stitching.min_shot_count}:
dropped "
408             f"{len(matched_segments) - len(kept)}/{len(matched_segments)}
segments below the floor.")
409         if not kept:
410             raise HTTPException(status_code=400,
411                 detail=f"All {len(matched_segments)} matching segments fall
below "
412 f"stitching.min_shot_count={stitching.min_shot_count}.")
413
414         # Extract start/end time pairs
415         time_pairs = [(s["start_time"], s["end_time"]) for s in kept]
416
417         # Run compiler with the configured stitching paddings + optional branding watermark
418         output_dir = getattr(getattr(global_config, "output", None), "output_dir", "output")
or "output"
419         compiler = VideoCompiler(output_dir, begin_padding=stitching.begin_padding,
420             end_padding=stitching.end_padding,
watermark=stitching.watermark)
421         try:
422             out_path = compiler.compile_highlights(video["filepath"], time_pairs, out_name)
423             # `filepath` is server-local (not reachable by the browser); also return the
ready-to-use
424             # download URL so the SPA needn't hand-build it (P6 / PER_RALLY_SELECTION.md
$4.3).
425             download_url = f"/api/highlights/{quote(req.video_id)}/{quote(out_name)}"
426             return {"status": "success", "filepath": out_path, "download_url": download_url}
427         except Exception as e:
428             raise HTTPException(status_code=500, detail=f"Highlight stitching failed:
{str(e)}") from e
429
430     # --- CLI Debug Utility & Orchestration ---
431     def run_cli_diagnose_clip(video_path: str, timestamp: float):
432         """CLI debugging utility to examine segment properties around a timestamp."""
433         print("-" * 60)
434         print(f"DEBUGGING VIDEO CLIP: {video_path} at {timestamp}s")
435         print("-" * 60)
436
437         cfg = load_config() # legacy wrapper still works, returns dict for now
438
439         # 1. Validation check diagnostics
440         print("[DIAGNOSTIC] Running validation framework...")
441         results = registry.run_all(video_path, cfg)
442         for r in results:
443             status_symbol = "[PASS]" if r.passed else "[FAIL]"
444             print(f"    {status_symbol} {r.validator_name}: {r.message}")
445             if r.details:
446                 print(f"        Details: {r.details}")
447
448         # 2. Local segment analysis around the timestamp (sampling +/- 5 seconds)
449         print("-" * 60)
450         print("[DIAGNOSTIC] Analyzing local motion around the timestamp...")
451         import cv2
452         cap = cv2.VideoCapture(video_path)
453         if not cap.isOpened():

```

356. assistant (2026-06-21T13:14:55.411Z)

Confirmed the bug: [main.py:422](backend/main.py:422) passes `video["filepath"]` (the stored `gs://` URI for a bucket-ingested video) straight to the compiler, which needs a **local** file ? ffmpeg can't open `gs://` ? 500. The process path resolves URIs via `acquire\_local\_input`; the compile path doesn't. **This is the real gap that completes the bucket?reel CUJ** ? exactly what proof-driven validation is for.

Let me fix it in a clean branch (engine PR), with a regression test. First the worktree off the latest master:

357. user (2026-06-21T13:14:59.449Z)

```
origin/master: 43eccc1 feat(serving): M8 ? per-job cost ledger + versioned pricebook + /cost API
(default-OFF) (#288)
branch 'fix/compile-resolve-bucket-source' set up to track 'origin/master'.
HEAD is now at 43eccc1 feat(serving): M8 ? per-job cost ledger + versioned pricebook + /cost API
(default-OFF) (#288)
branch ready
```

358. assistant (2026-06-21T13:15:05.492Z)

origin/master moved to `43eccc1` (#288) ? sibling is merging fast; my fix branch is off the latest. Let me locate the exact compile call in this worktree (line numbers may have shifted) and apply the fix:

359. user (2026-06-21T13:15:07.546Z)

```
448- # Extract start/end time pairs
449- time_pairs = [(s["start_time"], s["end_time"]) for s in kept]
450-
451- # Run compiler with the configured stitching paddings + optional branding watermark
452- output_dir = getattr(getattr(global_config, "output", None), "output_dir", "output") or
"output"
453- compiler = VideoCompiler(output_dir, begin_padding=stitching.begin_padding,
454-                           end_padding=stitching.end_padding,
watermark=stitching.watermark)
455- try:
456-     out_path = compiler.compile_highlights(video["filepath"], time_pairs, out_name)
457-     # `filepath` is server-local (not reachable by the browser); also return the ready-
to-use
458-     # download URL so the SPA needn't hand-build it (P6 / PER_RALLY_SELECTION.md §4.3).
459-     download_url = f"/api/highlights/{quote(req.video_id)}/{quote(out_name)}"
460-     return {"status": "success", "filepath": out_path, "download_url": download_url}
461- except Exception as e:
462-     raise HTTPException(status_code=500, detail=f"Highlight stitching failed: {str(e)}")
from e
463-
464-# --- CLI Debug Utility & Orchestration ---
```

360. assistant (2026-06-21T13:15:22.100Z)

Applying the fix ? resolve the source to a local path before stitching (identity for local, fetch for bucket URIs):

361. user (2026-06-21T13:15:26.641Z)

<tool_use_error>File has not been read yet. Read it first before writing to it.</tool_use_error>

362. user (2026-06-21T13:15:32.915Z)

```
451 # Run compiler with the configured stitching paddings + optional branding watermark
452 output_dir = getattr(getattr(global_config, "output", None), "output_dir", "output")
or "output"
453 compiler = VideoCompiler(output_dir, begin_padding=stitching.begin_padding,
454                           end_padding=stitching.end_padding,
watermark=stitching.watermark)
455 try:
456     out_path = compiler.compile_highlights(video["filepath"], time_pairs, out_name)
457     # `filepath` is server-local (not reachable by the browser); also return the
ready-to-use
458     # download URL so the SPA needn't hand-build it (P6 / PER_RALLY_SELECTION.md
```

```

$4.3).
459         download_url = f"/api/highlights/{quote(req.video_id)}/{quote(out_name)}"
460         return {"status": "success", "filepath": out_path, "download_url": download_url}
461     except Exception as e:
462         raise HTTPException(status_code=500, detail=f"Highlight stitching failed:
{str(e)}") from e

```

363. user (2026-06-21T13:15:40.432Z)

The file C:\Users\avidu\Projects\bhi-wt-compile-fix\backend\main.py has been updated successfully. (file state is current in your context ? no need to Read it back)

364. assistant (2026-06-21T13:15:45.297Z)

Now a regression test. Let me see how `/api/compile` is currently tested (mocking the compiler):

365. user (2026-06-21T13:15:47.018Z)

```

146: def test_compiler_and_reingest_safety_honor_output_dir_override(env, tmp_path, monkeypatch):
193: def test_compile_rejects_traversal_output_filename(env):
197:     r = client.post("/api/compile", json={
202: def test_compile_success_with_mocked_compiler(env):
211:     def compile_highlights(self, raw, pairs, name):
214:         mp.setattr(m, "VideoCompiler", _FakeCompiler)
215:         r = client.post("/api/compile", json={
234: def test_compile_enforces_config_min_shot_count(env):
252:     def compile_highlights(self, raw, pairs, name):
256:         mp.setattr(m, "VideoCompiler", _FakeCompiler)
257:         r = client.post("/api/compile", json={
265: def test_compile_400_when_all_below_min_shot_count(env):
270:     r = client.post("/api/compile", json={
276: def test_compile_uses_configured_stitching_paddings(env):
278:     Regression for the 2026-06-10 audit finding (d): /api/compile constructed
279:     VideoCompiler without the paddings, so reels always used the constructor
302:         r = client.post("/api/compile", json={

```

366. user (2026-06-21T13:15:52.268Z)

```

202     def test_compile_success_with_mocked_compiler(env):
203         client, db, tmp_path, mp = env
204         db.add_video("vid", str(_video(tmp_path)), "processed", [])
205         sid = db.add_play_segment("vid", 1.0, 4.0)
206
207         class _FakeCompiler:
208             def __init__(self, out_dir, watermark=None, **kwargs):
209                 pass
210
211             def compile_highlights(self, raw, pairs, name):
212                 return f"output/{name}"
213
214         mp.setattr(m, "VideoCompiler", _FakeCompiler)
215         r = client.post("/api/compile", json={
216             "video_id": "vid", "segment_ids": [sid], "output_filename": "out.mp4"})
217         assert r.status_code == 200
218         body = r.json()
219         assert body["filepath"] == "output/out.mp4"
220         # the browser-reachable URL is returned so the SPA needn't hand-build it (P6)
221         assert body["download_url"] == "/api/highlights/vid/out.mp4"
222
223
224     def test_health_ok(env):
225         client = env[0]
226         r = client.get("/api/health")
227         assert r.status_code == 200

```

```
228         body = r.json()
229         assert body["status"] == "ok"
230         assert body["sport"] == "badminton"
231         assert "default_segmenter" in body # readiness fields present for the SPA banner
232
233
234     def test_compile_enforces_config_min_shot_count(env):
```

367. assistant (2026-06-21T13:16:11.690Z)

Now add the regression test (mirrors the existing compile test + the bucket-fetch seam):

368. user (2026-06-21T13:16:19.418Z)

The file C:\Users\avidu\Projects\bhi-wt-compile-fix\tests\test_api_endpoints.py has been updated successfully. (file state is current in your context ? no need to Read it back)

369. assistant (2026-06-21T13:16:24.076Z)

Run the new + existing compile tests and lint in the fix worktree:

370. user (2026-06-21T13:16:37.838Z)

```
..... [100%]
7 passed, 23 deselected in 9.55s
=== ruff ===
All checks passed!
Shell cwd was reset to C:\Users\avidu\Projects\badminton-highlight-indexer
```

371. assistant (2026-06-21T13:16:57.547Z)

7 compile tests pass (incl. the new regression) + ruff clean. Let me run the full engine suite in the background (PR gate) while I swap the running engine to the fixed one and re-record the now-complete CUJ:

372. user (2026-06-21T13:16:59.052Z)

Command running in background with ID: bk56fo34o. Output is being written to:
C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-indexer\9f6357d8-0e35-491a-a830-052e094c7f37\tasks\bk56fo34o.output. You will be notified when it completes. To check interim output, use Read on that file path.

373. assistant (2026-06-21T13:17:20.020Z)

The launcher returned (the suite runs detached, writing to `tmp/compilefix\_suite.log` ? I'll check it before the PR). Now swap to the fixed engine and re-record the full CUJ:

374. user (2026-06-21T13:17:34.263Z)

```
killing old engine pid 29644
SUCCESS: The process with PID 29644 has been terminated.
fixed engine starting (1408)
health: {"status":"ok","sport":"badminton","default_segmenter":"gemini"}
```

375. assistant (2026-06-21T13:17:43.319Z)

Fixed engine is live on :8000 (SPA proxies to it). Re-recording the full CUJ ? this time compile should resolve the bucket source and produce the reel:

376. user (2026-06-21T13:21:48.917Z)

```
ingest panel rendered; lang = en
rallies loaded: Loaded 12 rallies from this session.
```

downloaded reel: highlights.mp4
DONE
Shell cwd was reset to C:\Users\avidu\Projects\badminton-highlight-indexer

377. assistant (2026-06-21T13:21:58.177Z)

****Full CUJ succeeded end-to-end**** ? ingest ? 12 rallies ? build ? ****reel downloaded****
(`highlights.mp4`). The compile fix completes the bucket?reel loop. Let me verify the recording artifacts and the downloaded reel:

378. user (2026-06-21T13:22:01.006Z)

```
=== recorded video ===
total 17308
drwxr-xr-x 1 avidu 197609      0 Jun 21 18:47 .
drwxr-xr-x 1 avidu 197609      0 Jun 21 18:51 ..
-rw-r--r-- 1 avidu 197609 17718627 Jun 21 18:51 page@648c2a87421a5177387fac2c2997ce11.webm
=== downloaded reel (proof of the full CUJ) ===
-rw-r--r-- 1 avidu 197609 50604038 Jun 21 18:51 cuj-downloaded-reel.mp4
codec_name=h264
width=1920
height=1080
codec_name=aac
codec_name=unknown
duration=76.029167
size=50604038
=== screenshots ===
-rw-r--r-- 1 avidu 197609 75790 Jun 21 18:47 cuj-1-ingest.png
-rw-r--r-- 1 avidu 197609 77670 Jun 21 18:47 cuj-2-uri.png
-rw-r--r-- 1 avidu 197609 79594 Jun 21 18:47 cuj-3-processing.png
-rw-r--r-- 1 avidu 197609 76136 Jun 21 18:51 cuj-4-rallies.png
-rw-r--r-- 1 avidu 197609 266133 Jun 21 18:51 cuj-5-reel-ready.png
Shell cwd was reset to C:\Users\avidu\Projects\badminton-highlight-indexer
```

379. assistant (2026-06-21T13:22:17.825Z)

Everything is real: the ****downloaded reel is a genuine 1920×1080 / 76s h264 reel**** stitched from the 12 Gemini rallies ? proof the full UI CUJ works. The raw recording includes the ~170s Gemini wait, so let me check its duration and produce a watchable promo:

380. user (2026-06-21T13:22:19.412Z)

```
raw webm: video-cuj/page@648c2a87421a5177387fac2c2997ce11.webm
raw duration: 242.240000s
Shell cwd was reset to C:\Users\avidu\Projects\badminton-highlight-indexer
```

381. assistant (2026-06-21T13:23:29.757Z)

Raw is 242s (mostly the ~170s Gemini wait). Let me produce a watchable promo ? keep the ingest/typing and the rallies/build/reel/download at readable speed, fast-forward the processing wait:

382. user (2026-06-21T13:23:37.007Z)

```
[libx264 @ 0000028437aaa400] ref B L0: 62.8% 35.7% 1.6%
[libx264 @ 0000028437aaa400] ref B L1: 97.8% 2.2%
[libx264 @ 0000028437aaa400] kb/s:306.66
=== result ===
duration=54.200000
size=2095294
-rw-r--r-- 1 avidu 197609 2095294 Jun 21 18:53 khelsutra-bucket-cuj-screencast.mp4
Shell cwd was reset to C:\Users\avidu\Projects\badminton-highlight-indexer
```

383. assistant (2026-06-21T13:23:45.812Z)

54s / 2MB ? good promo length. Let me verify the content by extracting key frames and viewing them (proof the screencast shows the right stages):

384. user (2026-06-21T13:23:48.682Z)

```
-rw-r--r-- 1 avidu 197609 495236 Jun 21 18:53 frame-ingest.png
-rw-r--r-- 1 avidu 197609 438734 Jun 21 18:53 frame-rallies.png
-rw-r--r-- 1 avidu 197609 401624 Jun 21 18:53 frame-reel.png
Shell cwd was reset to C:\Users\avidu\Projects\badminton-highlight-indexer
```

385. assistant (2026-06-21T13:24:51.274Z)

The frames confirm the flow (ingest with gs:// URI typed ? 12 rallies loaded ? building), but the ****reel-ready climax is rushed at the very end**** and the rallies-reveal got caught in the fast-forward. Let me pinpoint the real stage boundaries from the raw to re-encode precisely:

386. user (2026-06-21T13:24:54.104Z)

```
-rw-r--r-- 1 avidu 197609 433109 Jun 21 18:54 raw-175.png
-rw-r--r-- 1 avidu 197609 435318 Jun 21 18:54 raw-188.png
-rw-r--r-- 1 avidu 197609 515252 Jun 21 18:54 raw-241.png
=== cuj-5 reel-ready screenshot (the climax) size ===
-rw-r--r-- 1 avidu 197609 266133 Jun 21 18:51 cuj-5-reel-ready.png
Shell cwd was reset to C:\Users\avidu\Projects\badminton-highlight-indexer
```

387. assistant (2026-06-21T13:25:47.577Z)

raw-188 is still "segmenting (gemini)" ? the Gemini run was longer (~190-215s), so the rallies-reveal/build/reel are all in the final ~25-50s. And `cuj-5-reel-ready.png` is a perfect climax frame (reel preview + download visible). Let me re-encode with the correct boundaries (fast-forward only the spinner) and freeze the reel-ready climax at the end:

388. user (2026-06-21T13:26:00.119Z)

```
[libx264 @ 0000017a6cde1b00] ref B L1: 98.1% 1.9%
[libx264 @ 0000017a6cde1b00] kb/s:254.63
=== result ===
duration=63.566667
size=2047580
frames extracted
Shell cwd was reset to C:\Users\avidu\Projects\badminton-highlight-indexer
```

389. assistant (2026-06-21T13:26:08.668Z)

63.6s promo. Let me verify the key moments ? the rallies reveal, the build, and the frozen climax:

390. user (2026-06-21T13:26:09.949Z)

```
ok
Shell cwd was reset to C:\Users\avidu\Projects\badminton-highlight-indexer
```

391. assistant (2026-06-21T13:26:40.542Z)

v2-reveal (22s) still shows processing, and the ****climax freeze is perfect**** (reel preview + download + 12 rallies). But there's too much spinner mid-video because Gemini ran past 185s. Let me pinpoint exactly when the 12 rallies replace the demo (the "6 rallies" ? "12 rallies" transition) to fast-forward the spinner cleanly:

392. user (2026-06-21T13:26:42.807Z)

```
done
Shell cwd was reset to C:\Users\avidu\Projects\badminton-highlight-indexer
```


393. assistant (2026-06-21T13:27:18.847Z)

Pinpointed: rallies load ~210-215s (raw-205 = processing, raw-218 = 12 rallies + building). Now I can encode cleanly ? keep the **rallies reveal** at full speed, fast-forward only the spinner and the build wait, and freeze the climax:

394. user (2026-06-21T13:27:29.815Z)

```
[libx264 @ 0000022d68dca940] ref B L1: 98.0% 2.0%
[libx264 @ 0000022d68dca940] kb/s:308.67
=== final ===
duration=42.866667
size=1670821
verify frames extracted
Shell cwd was reset to C:\Users\avidu\Projects\badminton-highlight-indexer
```

395. assistant (2026-06-21T13:27:35.261Z)

42.9s / 1.67MB ? ideal promo length. Let me verify the reveal and ending frames:

396. assistant (2026-06-21T13:27:58.491Z)

The promo is correct end-to-end: ingest ? gs:// URI ? real Gemini processing ? **12 rallies reveal** ? build ? **reel ready + preview + download** (frozen climax). 42.9s, honest and watchable. Let me check the engine-fix test suite and upload the screencast to the promo bucket:

397. user (2026-06-21T13:28:10.257Z)

```
=== full engine suite (compile fix) result ===
1182 passed, 7 skipped in 101.55s (0:01:41)
=====
RESULT: PASS - all tests green
=====
=== upload screencast to promo bucket ===
Copying file://khelsutra-bucket-cuj-screencast.mp4 [Content-Type=video/mp4]...
/ [0 files][ 0.0 B/ 1.6 MiB]
/ [1 files][ 1.6 MiB/ 1.6 MiB]
-

Operation completed over 1 objects/1.6 MiB.
=== promo dir now ===
  776632 2026-06-20T18:16:41Z gs://khelsutra-rally-corpus/promo/full_cuj_real.mp4
 1670821 2026-06-21T13:28:04Z gs://khelsutra-rally-corpus/promo/khelsutra-bucket-cuj-
screencast.mp4
 4082564 2026-06-20T17:32:49Z gs://khelsutra-rally-corpus/promo/promo_cuj.gif
 709229 2026-06-20T17:32:49Z gs://khelsutra-rally-corpus/promo/promo_cuj.mp4
Shell cwd was reset to C:\Users\avidu\Projects\badminton-highlight-indexer
```

398. assistant (2026-06-21T13:28:20.976Z)

Engine suite green: **1182 passed, 7 skipped**. Screencast is in the promo bucket. Now finalize the engine-fix PR ? mypy (matching CI), then commit:

399. user (2026-06-21T13:28:59.384Z)

```
=== mypy (CI scope) ===
backend\pipeline\validators\base.py:36: note: By default the bodies of untyped functions are not
checked, consider using --check-untyped-defs [annotation-unchecked]
backend\providers\__init__.py:7: note: By default the bodies of untyped functions are not
checked, consider using --check-untyped-defs [annotation-unchecked]
Success: no issues found in 130 source files
=== branch + HEAD ===
fix/compile-resolve-bucket-source
43eccc1 feat(serving): M8 ? per-job cost ledger + versioned pricebook + /cost API (default-OFF)
```

```
(#288)
=== changed files ===
  M backend/main.py
  M tests/test_api_endpoints.py
Shell cwd was reset to C:\Users\avidu\Projects\badminton-highlight-indexer
```

400. assistant (2026-06-21T13:29:11.617Z)

mypy clean (130 files), ruff clean, suite green. Commit the fix (stage explicit paths):

401. user (2026-06-21T13:29:16.931Z)

```
[fix/compile-resolve-bucket-source 7bbdd19] fix(api): resolve a bucket-ingested video's source
to a local file before stitching
 2 files changed, 43 insertions(+), 1 deletion(-)
=== only my commit? ===
7bbdd19 fix(api): resolve a bucket-ingested video's source to a local file before stitching
Shell cwd was reset to C:\Users\avidu\Projects\badminton-highlight-indexer
```

402. assistant (2026-06-21T13:29:22.250Z)

Only my commit. Push and open the engine PR:

403. user (2026-06-21T13:29:36.683Z)

```
branch 'fix/compile-resolve-bucket-source' set up to track 'origin/fix/compile-resolve-bucket-
source'.
To https://github.com/Khelsutra/badminton-highlight-indexer.git
 * [new branch]      fix/compile-resolve-bucket-source -> fix/compile-resolve-bucket-source
https://github.com/Khelsutra/badminton-highlight-indexer/pull/289
Shell cwd was reset to C:\Users\avidu\Projects\badminton-highlight-indexer
```

404. assistant (2026-06-21T13:29:54.809Z)

Engine fix PR up: [badminton-highlight-indexer#289](https://github.com/Khelsutra/badminton-highlight-indexer/pull/289). Let me watch its CI and merge on green (per your standing instruction):

405. user (2026-06-21T13:32:25.574Z)

```
Syntax sweep + import smoke (no GPU stack)      pass    22s
https://github.com/Khelsutra/badminton-highlight-
indexer/actions/runs/27905882954/job/82574320209
Full test suite (offline, mocked)                pass    2m23s  https://github.com/Khelsutra/badminton-
highlight-indexer/actions/runs/27905882954/job/82574320181
Golden fixtures cross-OS (macos-latest)          pass    17s    https://github.com/Khelsutra/badminton-
highlight-indexer/actions/runs/27905882954/job/82574320152
Golden fixtures cross-OS (ubuntu-latest)         pass    23s    https://github.com/Khelsutra/badminton-highlight-
indexer/actions/runs/27905882954/job/82574320270
Golden fixtures cross-OS (windows-latest)        pass    55s    https://github.com/Khelsutra/badminton-highlight-
indexer/actions/runs/27905882954/job/82574320216
Mypy (lenient baseline) pass    32s    https://github.com/Khelsutra/badminton-highlight-
indexer/actions/runs/27905882954/job/82574320148
Ruff lint pass    9s    https://github.com/Khelsutra/badminton-highlight-
indexer/actions/runs/27905882954/job/82574320150
Syntax sweep + import smoke (no GPU stack)      pass    22s
https://github.com/Khelsutra/badminton-highlight-
indexer/actions/runs/27905882954/job/82574320209
EXIT=0
```

406. assistant (2026-06-21T13:32:30.606Z)

All CI green on #289. Merging:

407. user (2026-06-21T13:32:38.216Z)

--- engine origin/master now ---
e68b583 fix(api): resolve a bucket-ingested video's source to a local file before stitching
(#289)

408. assistant (2026-06-21T13:32:56.879Z)

#289 merged (engine master `e68b583`) ? the bucket?reel CUJ now works permanently. Let me flip the DRY_RUN DoD box (live e2e proven) and record the Path-B run result. Branch off khelsutra master:

409. user (2026-06-21T13:33:00.602Z)

Switched to a new branch 'docs/dry-run-pathb-done'
on docs/dry-run-pathb-done @ 4525c5d feat(web): B-P2 ? in-UI bucket ingest (point at a gs:// video ? process ? load) (#56)

410. user (2026-06-21T13:33:04.808Z)

```
46      ## 6. Definition of done (the dry run)
47      - [x] A bucket video ? (engine processes) ? the **SPA loads its rallies** ? select ?
compile ? **reel downloaded to the local machine**, all driven from the UI (Path A allows a one-
line CLI pre-process). ? ? **Path A done (local, 2026-06-21)**, see §7.
48      - [x] The **same flow works with the UI in Hindi** ? multilingual workflow confirmed. ?
? (Kannada also works via the same switcher.)
49      - [ ] *(Path B = B-P2, SPA-only)* the bucket video is pointed-at **from the UI** (no CLI
pre-step). ? **SPA ingest/progress screen BUILT + unit/jsdom-tested** (`web/`: enter a `gs://`
URI ? `POST /api/process` ? poll `/api/jobs` ? load `result.video_id`; localized en/kn/hi;
friendly engine-unreachable copy) ? browser-verified rendering + live error path. **Box flips on
a live e2e-from-UI run against a real engine** (local or the droplet, option 3).
50      - [ ] *(option 3)* the same CUJ confirmed on the **CPU droplet** (Linux deploy target) ?
droplet reachable but currently **un-provisioned** (no engine checkout; ~3.8GB RAM) ? needs a
from-scratch engine install or a proper deploy of the fixed engine.
51
```

411. user (2026-06-21T13:33:15.930Z)

The file C:\Users\avidu\Projects\khelsutra\docs\DRY_RUN_CUJ.md has been updated successfully.
(file state is current in your context ? no need to Read it back)

412. assistant (2026-06-21T13:33:20.696Z)

Now append a §9 run-result section. Let me read the doc tail to find the insertion point:

413. user (2026-06-21T13:33:22.145Z)

****Why it went stale (anti-spiral lesson):**** the §1 claim was tagged `[verified]` but was inferred from the ****worker CLI**** (`backend.worker infer`, which writes CSV to the bucket) rather than checked against the ****API endpoint's**** routing code ? and M2/#280 had since wired URI ingest into `/api/process`. ****The engine ships PRs between sessions, so its surface drifts.**** Convention going forward (also baked into the memory steering): ****before scoping any** `[gap]`/`net-new` engine claim as work, re-verify it against engine code at a cited `file:line` ? a `[verified]` tag without a `file:line` is a hypothesis, not a fact.

Changelog

- `2026-06-21` ? created; Path-A run recorded (§7).
- `2026-06-21` ? ****§8 correction:**** verified B-P1 (`gs://`/`api/process`?API-DB `video\_id`) is already shipped + tested (M2/#280); §1/§2/§4/§6 reframed so Path B = SPA-only (B-P2). Sibling stale claims fixed in `LOCAL\_APPLIANCE\_ARCHITECTURE.md` (`api/health` exists; bucket-URI ingest not a net-new endpoint; T12 engine-half done) + `README.md`.

---total lines: 82

414. assistant (2026-06-21T13:33:43.507Z)

Append §9 (Path-B run result) before the changelog, and add a changelog entry:

415. user (2026-06-21T13:33:54.398Z)

The file C:\Users\avidu\Projects\khelsutra\docs\DRY_RUN_CUJ.md has been updated successfully.
(file state is current in your context ? no need to Read it back)

416. assistant (2026-06-21T13:34:00.341Z)

Commit + push + open the doc PR:

417. user (2026-06-21T13:34:15.578Z)

```
[docs/dry-run-pathb-done ee52d5d] docs(dry-run): Path-B live e2e DONE ? in-UI bucket?reel CUJ +
screencast (DoD §6 box 3 ?)
  1 file changed, 12 insertions(+), 1 deletion(-)
ee52d5d docs(dry-run): Path-B live e2e DONE ? in-UI bucket?reel CUJ + screencast (DoD §6 box 3
?)
To https://github.com/avidullu/khelsutra.git
 * [new branch]      docs/dry-run-pathb-done -> docs/dry-run-pathb-done
https://github.com/avidullu/khelsutra/pull/57
```

418. assistant (2026-06-21T13:34:22.188Z)

Doc PR up: [khelsutra#57](https://github.com/avidullu/khelsutra/pull/57). Let me watch CI and merge:

419. user (2026-06-21T13:34:32.233Z)

```
Workers Builds: hidden-king-khelsutra   pending 0
https://dash.cloudflare.com/213c12cf60de9041dd8a3521aa206472/workers/services/view/hidden-king-
khelsutra/production/builds/cfbd45d4-0d01-4ce0-87de-1ba16c0afb80
site      pending 0
https://github.com/avidullu/khelsutra/actions/runs/27906008924/job/82574657037
web       pending 0
https://github.com/avidullu/khelsutra/actions/runs/27906008924/job/82574657040
```

420. user (2026-06-21T13:35:01.768Z)

```
web      pass    26s
https://github.com/avidullu/khelsutra/actions/runs/27906008924/job/82574657040
Workers Builds: hidden-king-khelsutra   pass    0
https://dash.cloudflare.com/213c12cf60de9041dd8a3521aa206472/workers/services/view/hidden-king-
khelsutra/production/builds/cfbd45d4-0d01-4ce0-87de-1ba16c0afb80
site     pass    11s
https://github.com/avidullu/khelsutra/actions/runs/27906008924/job/82574657037
web      pass    26s
https://github.com/avidullu/khelsutra/actions/runs/27906008924/job/82574657040
---
```

Deleted branch docs/dry-run-pathb-done (was ee52d5d).
khelsutra master: 7750afc docs(dry-run): Path-B live e2e DONE ? in-UI bucket?reel CUJ +
screencast (DoD §6 box 3 ?) (#57)

421. assistant (2026-06-21T13:35:22.057Z)

All three PRs merged on green this session (B-P2 #56, engine fix #289, DoD doc #57). Now clean up the runtime + worktrees:

422. user (2026-06-21T13:35:30.168Z)

```
=== kill engine :8000 + SPA :5180 ===
kill 8000 -> pid 37848
SUCCESS: The process with PID 37848 has been terminated.
kill 5180 -> pid 6612
SUCCESS: The process with PID 6612 has been terminated.
=== remove engine worktrees ===
removed bhi-wt-screencast
removed bhi-wt-compile-fix
=== remaining worktrees (mine gone; siblings untouched) ===
(my worktrees cleaned)
```

423. assistant (2026-06-21T13:35:38.859Z)

Cleanup complete ? engine + SPA stopped, both my worktrees removed, siblings untouched. Now update the memory checkpoint (the shared file may have shifted again ? let me locate my B-P2 note):

424. user (2026-06-21T13:35:40.096Z)

19:B-P2 BUILT

425. user (2026-06-21T13:35:45.673Z)

```
19      - **? B-P2 BUILT (in-UI bucket ingest) ? khelsutra
[#56](https://github.com/avidullu/khelsutra/pull/56) OPEN, CI GREEN (web+site+CF build),
awaiting owner merge** (feature PR ? left for owner; precedent exists for me merging SPA PRs on
owner's "merge as you go"). 100% `web/` + i18n, ZERO engine code (rides B-P1). New `web/src/{api
/types,api/client(processVideo/getJob),lib/errors(parseDetail+classifyError),hooks/useIngestJob(
idle?submitting?polling?done|error; recursive-setTimeout poll backoff+10min-ceiling+1 cid;
success-shaped-failure?emptyResult),components/IngestPanel}` + `App.tsx` videoId
useMemo?useState + IngestPanel-when-!videoId + history.replaceState; `ingest.*` en/kn/hi (AI-
draft, native-review-pending). **123 web tests (+19), tsc/build green, browser-verified** (panel
renders en+hi; live submit?friendly localized "is the engine running?"; cid logs). Live-fixed
UX: bare 5xx/proxy-down ? friendly `networkError`, not "HTTP 500". `DRY_RUN_CUJ $6` box3 = ?
(flips on a live **e2e-from-UI** vs a real engine). **? NEXT:** merge #56 ? live e2e-from-UI
(local or droplet opt-3) ? flip box3; then localization DoD tail (B3 site `t()` shim; native
kn/hi review).
```

426. user (2026-06-21T13:36:01.241Z)

The file C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\memory\current-state.md has been updated successfully. (file state is current in your context ? no need to Read it back)

427. assistant (2026-06-21T13:36:31.699Z)

Everything's done and recorded in memory. Here's the proof-and-validation-driven arc this turn:

? All green PRs merged

```
| PR | What | Result |
|---|---|---|
| [khelsutra#56](https://github.com/avidullu/khelsutra/pull/56) | B-P2 in-UI bucket ingest |
merged `4525c5d` |
| [engine#289](https://github.com/Khelsutra/badminton-highlight-indexer/pull/289) | compile-
from-bucket fix | merged `e68b583` |
| [khelsutra#57](https://github.com/avidullu/khelsutra/pull/57) | DoD $6 box 3 ? + $9 run result
| merged `7750afc` |
```

? Screencast of the entire CUJ ? in the promo bucket

`gs://khelsutra-rally-corpus/promo/khelsutra-bucket-cuj-screencast.mp4` (42s, next to the other promo clips). It shows the real flow, driven entirely from the UI: paste a `gs://` bucket URI ? live Gemini processing ? **12 rallies load** ? Build ? **reel preview + download** (frozen climax). I verified each stage by reading extracted frames.

? Proof-driven development paid off immediately

Recording the live CUJ **surfaced a real bug** that unit tests and the prior Path-A run had missed: ``/api/compile` **500'd` on a bucket-ingested video ? ``videos.filepath`` keeps the ``gs://`` source URI (M2 provenance), and the stitcher was handed that directly, so ffmpeg couldn't open it. I fixed it (compile now resolves the source via ``acquire_local_input`` first ? identity for local, fetch-to-scratch for a bucket URI), added a regression test, **re-ran the CUJ to confirm the reel builds + downloads**, and only then recorded the final screencast. That's the loop working as intended: **the attempt to prove it is what found the gap.**

? Coverage up

+19 web tests (B-P2) and +1 engine regression test
(``test_compile_resolves_bucket_source_to_local_before_stitching``). Engine suite **1182 passed / 7 skipped**; web **123 tests**; ruff + mypy clean across both.

Validation summary

- **Real local engine** (Gemini + ``google-cloud-storage``+ADC for the bucket fetch) ? the `gs://` fetch, segmentation, query, compile, and highlight-download all exercised live.
- **DoD \$6 box 3 is now** (live e2e-from-UI). The remaining DoD box (option 3) is the **droplet** run ? deploy the fixed engine to ``168.144.126.176`` + GCS creds and re-run the same CUJ on Linux.

Cleanup done (engine + SPA stopped, my worktrees removed, siblings untouched). One note: ``google-cloud-storage`` is now pip-installed in the local engine env (it enables local ``gs://`` fetch ? a useful addition). Want me to take on the **droplet e2e** next to close the final DoD box?

428. user (2026-06-21T13:52:22.620Z)

lets park the droplet e2e since getting a gpu only machine there requires some subscription, Can you scan the repo and see where does the work around making the tool Cloud Run friendly lies and if thats already being worked upon

429. assistant (2026-06-21T13:52:47.274Z)

Good call parking the droplet. Let me scan the engine repo for the Cloud Run work. First, sync to head (the serving track has been merging fast) and scout the key surfaces ? docs, code, and active PRs:

430. user (2026-06-21T13:52:54.371Z)

=== sync engine to head ===

HEAD: e68b583 fix(api): resolve a bucket-ingested video's source to a local file before stitching (#289)

=== serving / cloud-run / compute docs ===
docs/COMPUTE_DECOUPLED_SERVING

=== backend/compute or compute backend code? ===
__init__.py
__pycache__
base.py
cloud_run.py

=== Dockerfile(s)? ===
./deploy/cloudrun/Dockerfile

=== deploy/gcp Cloud Run bits ===
README.md
create_gpu_vm.sh
ops-agent-startup.sh
provision.sh
teardown.sh

431. assistant (2026-06-21T13:53:18.595Z)

Found it ? the Cloud Run work is real and present: ``docs/COMPUTE_DECOUPLED_SERVING/``, `**`backend/compute/cloud_run.py`**` (a compute backend!), and `**`deploy/cloudrun/Dockerfile`**`. This is the concurrent serving session's track. Let me fan out a workflow to map exactly what's built vs planned and confirm it's actively in progress:

432. user (2026-06-21T13:54:04.211Z)

Workflow launched in background. Task ID: w0zlnwswf

Summary: Map where the Cloud-Run-friendly serving work lives (docs + code + deploy), its milestone status (done/in-progress/planned), and whether it is actively being worked on
Transcript dir: C:\Users\avidu\claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\9f6357d8-0e35-491a-a830-052e094c7f37\subagents\workflows\wf_635348d6-51c
Script file: C:\Users\avidu\claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\9f6357d8-0e35-491a-a830-052e094c7f37\workflows\scripts\cloudrun-work-scan-wf_635348d6-51c.js

(Edit this file with Write/Edit and re-invoke Workflow with {scriptPath:

"C:\Users\avidu\claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\9f6357d8-0e35-491a-a830-052e094c7f37\workflows\scripts\cloudrun-work-scan-wf_635348d6-51c.js"} to iterate without resending the script.)

Run ID: wf_635348d6-51c

To resume after editing the script: Workflow({scriptPath: "C:\Users\avidu\claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\9f6357d8-0e35-491a-a830-052e094c7f37\workflows\scripts\cloudrun-work-scan-wf_635348d6-51c.js", resumeFromRunId: "wf_635348d6-51c"}) ? completed agents return cached results.

You will be notified when it completes. Use /workflows to watch live progress.

433. assistant (2026-06-21T13:54:17.725Z)

The workflow is mapping the details. The quick scout already answers the headline: `**yes, the Cloud-Run-friendly work exists and is actively being built**` ? there's a dedicated compute-backend module (``backend/compute/cloud_run.py`` + ``base.py``), a container (``deploy/cloudrun/Dockerfile``), and a design track (``docs/COMPUTE_DECOUPLED_SERVING/``). This is the concurrent serving session's M5?M12 line (it merged M8 / #288 and the doc #285 earlier today, and ADR-012 ratified Cloud Run + GPU as the serving model).

I'm letting the workflow finish the precise mapping ? what's real vs scaffolded in ``cloud_run.py``, whether the Dockerfile builds, the per-milestone status, and the open/merged PR trail ? so I give you a grounded "here's exactly where it lives and how far along it is" rather than just the headline. Standing by for it.

434. user (2026-06-21T13:54:44.261Z)

can you scout all the docs and find out the next steps based on them or the track that you have been following.

435. assistant (2026-06-21T13:55:34.729Z)

I'll scout the full doc corpus across both repos and extract the next steps per track. The Cloud Run scan is already running, so I'll point this one at the `*other*` tracks (appliance/CUJ, localization, rally-quality, owned-model, cross-cutting) and fold the serving result in at synthesis.

436. user (2026-06-21T13:56:13.240Z)

Workflow launched in background. Task ID: w7ydrkzjj

Summary: Scout all live docs across both repos, extract per-track status + next steps, and synthesize a prioritized roadmap (what is actionable vs owner-gated)

Transcript dir: C:\Users\avidu\claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\9f6357d8-0e35-491a-a830-052e094c7f37\subagents\workflows\wf_b6ef6275-de5

Script file: C:\Users\avidu\claude\projects\C--Users-avidu-Projects-badminton-highlight-

```
indexer\9f6357d8-0e35-491a-a830-052e094c7f37\workflows\scripts\docs-next-steps-scan-  
wf_b6ef6275-de5.js
```

```
(Edit this file with Write/Edit and re-invoke Workflow with {scriptPath:  
"C:\Users\avidu\claude\projects\C--Users-avidu-Projects-badminton-highlight-  
indexer\9f6357d8-0e35-491a-a830-052e094c7f37\workflows\scripts\docs-next-steps-scan-  
wf_b6ef6275-de5.js"} to iterate without resending the script.)
```

```
Run ID: wf_b6ef6275-de5
```

```
To resume after editing the script: Workflow({scriptPath: "C:\Users\avidu\claude\projects\C--  
Users-avidu-Projects-badminton-highlight-  
indexer\9f6357d8-0e35-491a-a830-052e094c7f37\workflows\scripts\docs-next-steps-scan-  
wf_b6ef6275-de5.js", resumeFromRunId: "wf_b6ef6275-de5"}) ? completed agents return cached  
results.
```

You will be notified when it completes. Use /workflows to watch live progress.

437. assistant (2026-06-21T13:56:24.387Z)

Both scans are now running in parallel:

- **Cloud Run / serving track** (`wf\_635348d6`) ? where the Cloud-Run-friendly work lives + how far along.

- **All-docs next-steps** (`wf\_b6ef6275`) ? appliance/CUJ, localization, rally-quality, owned-model, and the cross-cutting registers (DOC_STATUS, NEXT_STEPS).

When both land I'll merge them into one prioritized roadmap ? what's next on the track we've been driving (in-UI CUJ + localization), what's actionable now vs owner-gated, and a clean pointer to the serving track so we don't collide with the concurrent session. Standing by for the results.

438. user (2026-06-21T13:59:12.350Z)

```
<task-notification>
```

```
<task-id>w0zlnwswf</task-id>
```

```
<tool-use-id>toolu_01CDLYX1ckumxwSK8GWMGGt3</tool-use-id>
```

```
<output-file>C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-  
highlight-indexer\9f6357d8-0e35-491a-a830-052e094c7f37\tasks\w0zlnwswf.output</output-file>
```

```
<status>completed</status>
```

```
<summary>Dynamic workflow "Map where the Cloud-Run-friendly serving work lives (docs + code +  
deploy), its milestone status (done/in-progress/planned), and whether it is actively being  
worked on" completed</summary>
```

```
<result>{"docs":{"adr":{"Cloud Run + GPU as the serving model was ratified across a 4-ADR chain  
in C:/Users/avidu/Projects/badminton-highlight-  
indexer/docs/archives/decisions/DECISIONS.md:\n\n ADR-011 (2026-06-20, RATIFIED) ? overrode the  
prior \"rent nothing / not a service\" posture and DEFERRED M3 (serving endpoint + per-video  
billing) + M4 (scale-to-zero + cost guard). Carries the owner-gated $06 gates: DPA/consent for  
non-owner uploads, collector md5 keying, WASB-C3 (weights provenance).\n\n ADR-012 (2026-06-20,  
RATIFIED, DECISIONS.md:598) ? the substantive serving ratification: \"Cloud Run + GPU (L4),  
scale-to-zero, is the supported SERVING deployment model\" for pay-per-video economics (per-  
request GPU-seconds = the per-video bill, zero idle cost, co-located with gs://khelesutra-rally-  
corpus). Stay GPU-native (L4/T4); TPU deferred; NVDEC named as the decode lever. First work item  
= containerize python -m backend.worker (calls out the WASB torch-1.11 conda env as the known  
hard part). Edge auth named a hard prerequisite before public serving.\n\n ADR-013 (2026-06-20,  
RATIFIED, DECISIONS.md:661) ? GCP is the SOLE compute stack (training + serving); DigitalOcean  
dropped for compute; provision via deploy/gcp/ only.\n\n ADR-014 (2026-06-21, RATIFIED ? M0  
design sign-off, build M1+, DECISIONS.md:686) ? \"Compute-decoupled rally serving\": one pure  
3-stage core (DETECT/WINDOW/STITCH) never branched on profile; ComputeBackend (where) +  
StorageBackend (what-bytes) selected by one startup config; ships truly-local + cloud-serving +  
cpu-mock; stitch runs on Cloud Run so its cost is metered. REFINES (does not supersede)  
ADR-011/012 by realizing their deferred M3/M4; builds on ADR-013.\n\nOwner gates (per the $7  
tracker): M5 owner real-GPU; M6/M7 owner GCP spend (within the D17 $100/?10k cap, delete-after);  
M8 owner real prices/billing; M9 owner privacy/counsel (DPA/consent + edge auth); M10/M11/M12  
owner-gated (BYO approval, consent/privacy, OAuth/consent). The 5 originally-gating open  
questions are locked as D7?D12; the smaller picks as D16; the verification budget as  
D17.\"},\"cloudRunDesign\":\"SOURCE OF TRUTH: docs/COMPUTE_DECOUPLED_SERVING/README.md $4 + $2  
decisions, plus deploy/cloudrun/{Dockerfile,README.md}.\n\nJOBS vs SERVICE: Cloud Run JOBS, NOT  
Services-with-GPU (D16). One Job execution runs the detect+stitch unit. Rationale: it fits the
```


async upload?poll?reel flow + scale-to-zero, and reuses the existing async job control plane verbatim (JobWaiting / claim_due_job / ExecutionPolicy WAIT_AND_RESUME). Implemented as backend/compute/CloudRunCompute (backend/compute/cloud_run.py): get_compute_backend(config) returns a backend ONLY when compute_target=cloud_run (else None ? in-process path, default-OFF).\n\nGPU vs CPU: GPU-native, NVIDIA L4 (cuda), detector_impl=native (the Linux-native WASB runner). DETECT runs on GPU. WINDOW is a CPU-pure function (trajectory_to_action_windows). STITCH is real CPU/ffmpeg work and is CO-LOCATED in the SAME Cloud Run GPU job (D4/D16) so gpu_active_seconds (detect) + wall_seconds (detect+stitch) + bytes_out (reel egress) all land on ONE metered invocation ? the per-job cost ledger. Stitch-on-laptop would hide that cost. Codec (D16): libx264 ENCODE (protects the D5 byte-determinism / parity guarantee) + NVDEC for DECODE only (speed lever, never touches output bytes); NVENC encode is DEFERRED behind a config knob. cpu-mock profile (detector_impl=stub) runs the whole thing GPU-free for \$0 CI.\n\nSCALE-TO-ZERO: D7 ? accept the ~1?3 min cold-start (amortized over a multi-minute async job); NO warm pool (a ~\$500/mo idle pool would break per-video billing). Revisit a warm lane only on a UX complaint.\n\nDISPATCH + COMPLETION DETECTION: The control plane (DEPLOYMENT_PROFILE=cloud-serving ? compute_target=cloud_run) dispatches a Cloud Run Job execution (injectable client), then BUCKET-POLLS a done-marker for completion ? NOT the Cloud Run API status. The marker is <out_uri>/_dispatch/<token>.done, polled via execution_policy.poll_until + the storage layer, so it reuses JobWaiting/claim_due_job and survives a control-plane restart (D16). The dispatched container runs INLINE ? CloudRunCompute forces compute_target=local_gpu + indexing.detector_impl=native in the per-execution args so the Job never re-dispatches itself; the worker writes the --done-uri marker on success. Known gap (follow-up): a FAILED job currently writes NO marker, so the dispatcher's bounded poll times out (ExecutionPolicy.max_wait_sec ? clean rc 4); a failure-marker carrying the real non-zero rc (to distinguish failed from hung/cold-start-slow) is deferred. Reels returned via signed URL. Input cap: hard-reject <2GB for v1 (a config cap, raise to 4GB later in one line, matching the M2 chunked-upload threshold).\n\nCONTAINERIZATION + THE WASB conda-env CONCERN: deploy/cloudrun/Dockerfile builds the L4 image FROM nvidia/cuda:12.4.1-runtime-ubuntu22.04, adds ffmpeg (stitch/decode) + DejaVu fonts (watermark filter). It deliberately carries TWO separate python stacks to avoid a framework conflict (same split as the native GPU box, deploy/digitalocean/gpu_setup.sh): (1) the MAIN app env (this repo + ffmpeg) drives the pipeline + stitching; (2) a conda 'wasb' env (python 3.8 + torch 1.11.0+cu113 / torchvision 0.12.0+cu113, from nttcom/WASB-SBDT) runs the WASB shuttle detector as a SUBPROCESS ? the native runner shells out to \$WASB_PYTHON so the app's torch never co-installs with WASB's torch-1.11. ENV wires WASB_PYTHON=/opt/conda/envs/wasb/bin/python, WASB_REPO_DIR, WASB_WEIGHTS, RALLY_DETECTOR_IMPL=native; ENTRYPOINT is python3 -m backend.worker. This dual-runtime packaging is exactly the \"known hard part\" ADR-012 flagged. WASB-C3 concern: the pretrained badminton weights are academic-data-trained, provenance NOT cleared for commercial sharing ? they are NOT baked into the image; they are STAGED AT RUNTIME (bucket fetch or mount) with \$WASB_WEIGHTS pointed at them. WASB-C3 must be resolved before any public, non-owner serving. CI does NOT build this image (a CUDA+conda image is too heavy); the dispatch shim is proven with a fake client (tests/test_compute_backend.py). The real build/push/L4 run is the M7 owner step under the D17 budget.\",\"trackerFile\":\"C:/Users/avidu/Projects/badminton-highlight-indexer/docs/COMPUTE_DECOUPLED_SERVING/README.md ? \$7 \"Deliverables & progress tracker\" is the explicit source of truth (legend: ? Todo . ? In progress . ? Done . ? Gated). The \$11 Changelog gives the per-milestone landing notes, and \$2 (Decisions D1?D17) + \$4 (Design) back it. Supporting: deploy/cloudrun/{Dockerfile,README.md}, docs/COMPUTE_DECOUPLED_SERVING/runlogs/RUNLOG_truly-local_TEMPLATE.md, and ADR-011/012/013/014 in docs/archives/decisions/DECISIONS.md.\",\"milestones\": [{\"id\":\"M0\",\"title\":\"Design doc + project dir + runlogs/ + ADR-014\",\"cloudRunRelevant\":false,\"status\":\"? Done (? APPROVED 2026-06-21, owner sign-off)\"},{\"id\":\"M1\",\"title\":\"deployment.profile + compute_target config + preset bundles + precedence/auto-detect (no behaviour change, profile='default)\",\"cloudRunRelevant\":true,\"status\":\"? Done (PR #279)\"},{\"id\":\"M2\",\"title\":\"input_backend/input_root; wire main CLI/API input through StorageRef.parse_uri (local=identity)\",\"cloudRunRelevant\":true,\"status\":\"? Done (PR #280)\"},{\"id\":\"M3\",\"title\":\"Configurable output/scratch base ? kill hardcoded output/ (default stays output/, byte-identical)\",\"cloudRunRelevant\":true,\"status\":\"? Done (PR #282) ? was ? default-OFF + Launch Report on flip; landed default-OFF so no Launch Report needed\"},{\"id\":\"M4\",\"title\":\"DeviceContext + WASB CPU-fallback (device='auto'); unified healthcheck/device wiring\",\"cloudRunRelevant\":true,\"status\":\"? Done (PR #283) ? safe, cuda default unchanged\"},{\"id\":\"M5\",\"title\":\"truly-local profile end-to-end (preset + startup checks) + first committed runlog\",\"cloudRunRelevant\":true,\"status\":\"? In progress (PR #286: preset+startup-checks+L4 parity test ?; real-GPU truly-local runlog ? owne ... (truncated 36446 chars, full result in C:\\Users\\avidu\\AppData\\Local\\Temp\\claude\\C--Users-avidu-Projects-badminton-highlight-

```
indexer\9f6357d8-0e35-491a-a830-052e094c7f37\tasks\w0zlnwswf.output)</result>
<usage><agent_count>5</agent_count><subagent_tokens>324733</subagent_tokens><tool_uses>65</tool_
uses><duration_ms>308121</duration_ms></usage>
</task-notification>
```

439. assistant (2026-06-21T13:59:20.521Z)

The Cloud Run scan is done and detailed. Let me read the full result to get the milestone status, code/deploy findings, activity, and synthesis:

440. user (2026-06-21T13:59:21.730Z)

```
1      {
2      "summary": "Map where the Cloud-Run-friendly serving work lives (docs + code +
deploy), its milestone status (done/in-progress/planned), and whether it is actively being
worked on",
3      "agentCount": 5,
4      "logs": [],
5      "result": {
6      "docs": {
7      "adr": "Cloud Run + GPU as the serving model was ratified across a 4-ADR chain in
C:/Users/avidu/Projects/badminton-highlight-indexer/docs/archives/decisions/DECISIONS.md:\n\n-
ADR-011 (2026-06-20, RATIFIED) ? overrode the prior \"rent nothing / not a service\" posture and
DEFERRED M3 (serving endpoint + per-video billing) + M4 (scale-to-zero + cost guard). Carries
the owner-gated $06 gates: DPA/consent for non-owner uploads, collector md5 keying, WASB-C3
(weights provenance).\n- ADR-012 (2026-06-20, RATIFIED, DECISIONS.md:598) ? the substantive
serving ratification: \"Cloud Run + GPU (L4), scale-to-zero, is the supported SERVING deployment
model\" for pay-per-video economics (per-request GPU-seconds = the per-video bill, zero idle
cost, co-located with gs://khelsutra-rally-corpus). Stay GPU-native (L4/T4); TPU deferred; NVDEC
named as the decode lever. First work item = containerize python -m backend.worker (calls out
the WASB torch-1.11 conda env as the known hard part). Edge auth named a hard prerequisite
before public serving.\n- ADR-013 (2026-06-20, RATIFIED, DECISIONS.md:661) ? GCP is the SOLE
compute stack (training + serving); DigitalOcean dropped for compute; provision via deploy/gcp/
only.\n- ADR-014 (2026-06-21, RATIFIED ? M0 design sign-off, build M1+, DECISIONS.md:686) ?
\"Compute-decoupled rally serving\": one pure 3-stage core (DETECT/WINDOW/STITCH) never branched
on profile; ComputeBackend (where) + StorageBackend (what-bytes) selected by one startup config;
ships truly-local + cloud-serving + cpu-mock; stitch runs on Cloud Run so its cost is metered.
REFINES (does not supersede) ADR-011/012 by realizing their deferred M3/M4; builds on
ADR-013.\n\nOwner gates (per the $7 tracker): M5 owner real-GPU; M6/M7 owner GCP spend (within
the D17 $100/?10k cap, delete-after); M8 owner real prices/billing; M9 owner privacy/counsel
(DPA/consent + edge auth); M10/M11/M12 owner-gated (BYO approval, consent/privacy,
OAuth/consent). The 5 originally-gating open questions are locked as D7?D12; the smaller picks
as D16; the verification budget as D17.",
8      "cloudRunDesign": "SOURCE OF TRUTH: docs/COMPUTE_DECOUPLED_SERVING/README.md $4 +
$2 decisions, plus deploy/cloudrun/{Dockerfile,README.md}.\n\nJOBS vs SERVICE: Cloud Run JOBS,
NOT Services-with-GPU (D16). One Job execution runs the detect+stitch unit. Rationale: it fits
the async upload?poll?reel flow + scale-to-zero, and reuses the existing async job control plane
verbatim (JobWaiting / claim_due_job / ExecutionPolicy WAIT_AND_RESUME). Implemented as
backend/compute/CloudRunCompute (backend/compute/cloud_run.py): get_compute_backend(config)
returns a backend ONLY when compute_target=cloud_run (else None ? in-process path, default-
OFF).\n\nGPU vs CPU: GPU-native, NVIDIA L4 (cuda), detector_impl=native (the Linux-native WASB
runner). DETECT runs on GPU. WINDOW is a CPU-pure function (trajectory_to_action_windows).
STITCH is real CPU/ffmpeg work and is CO-LOCATED in the SAME Cloud Run GPU job (D4/D16) so
gpu_active_seconds (detect) + wall_seconds (detect+stitch) + bytes_out (reel egress) all land on
ONE metered invocation ? the per-job cost ledger. Stitch-on-laptop would hide that cost. Codec
(D16): libx264 ENCODE (protects the D5 byte-determinism / parity guarantee) + NVDEC for DECODE
only (speed lever, never touches output bytes); NVENC encode is DEFERRED behind a config knob.
cpu-mock profile (detector_impl=stub) runs the whole thing GPU-free for $0 CI.\n\nSCALE-TO-ZERO:
D7 ? accept the ~1?3 min cold-start (amortized over a multi-minute async job); NO warm pool (a
~$500/mo idle pool would break per-video billing). Revisit a warm lane only on a UX
complaint.\n\nDISPATCH + COMPLETION DETECTION: The control plane (DEPLOYMENT_PROFILE=cloud-
serving ? compute_target=cloud_run) dispatches a Cloud Run Job execution (injectable client),
then BUCKET-POLLS a done-marker for completion ? NOT the Cloud Run API status. The marker is
<out_uri>/_dispatch/<token>.done, polled via execution_policy.poll_until + the storage layer, so
it reuses JobWaiting/claim_due_job and survives a control-plane restart (D16). The dispatched
```

container runs `INLINE ? CloudRunCompute` forces `compute_target=local_gpu + indexing.detector_impl=native` in the per-execution args so the Job never re-dispatches itself; the worker writes the `--done-uri` marker on success. Known gap (follow-up): a FAILED job currently writes NO marker, so the dispatcher's bounded poll times out (`ExecutionPolicy.max_wait_sec ? clean rc 4`); a failure-marker carrying the real non-zero rc (to distinguish failed from hung/cold-start-slow) is deferred. Reels returned via signed URL. Input cap: hard-reject <2GB for v1 (a config cap, raise to 4GB later in one line, matching the M2 chunked-upload threshold).

CONTAINERIZATION + THE WASB conda-env CONCERN:

`deploy/cloudrun/Dockerfile` builds the L4 image FROM `nvidia/cuda:12.4.1-runtime-ubuntu22.04`, adds `ffmpeg` (stitch/decode) + `DejaVu` fonts (watermark filter). It deliberately carries TWO separate python stacks to avoid a framework conflict (same split as the native GPU box, `deploy/digitalocean/gpu_setup.sh`): (1) the MAIN app env (this repo + `ffmpeg`) drives the pipeline + stitching; (2) a conda 'wasb' env (python 3.8 + torch 1.11.0+cu113 / torchvision 0.12.0+cu113, from `nttcom/WASB-SBDT`) runs the WASB shuttle detector as a SUBPROCESS ? the native runner shells out to `$WASB_PYTHON` so the app's torch never co-installs with WASB's torch-1.11. ENV wires `WASB_PYTHON=/opt/conda/envs/wasb/bin/python`, `WASB_REPO_DIR`, `WASB_WEIGHTS`, `RALLY_DETECTOR_IMPL=native`; `ENTRYPOINT` is `python3 -m backend.worker`. This dual-runtime packaging is exactly the \"known hard part\" ADR-012 flagged. WASB-C3 concern: the pretrained badminton weights are academic-data-trained, provenance NOT cleared for commercial sharing ? they are NOT baked into the image; they are STAGED AT RUNTIME (bucket fetch or mount) with `$WASB_WEIGHTS` pointed at them. WASB-C3 must be resolved before any public, non-owner serving. CI does NOT build this image (a CUDA+conda image is too heavy); the dispatch shim is proven with a fake client (`tests/test_compute_backend.py`). The real build/push/L4 run is the M7 owner step under the D17 budget.",

9 "trackerFile": "C:/Users/avidu/Projects/badminton-highlight-indexer/docs/COMPUTE_DECOUPLED_SERVING/README.md ? §7 \"Deliverables & progress tracker\" is the explicit source of truth (legend: ? Todo · ? In progress · ? Done · ? Gated). The §11 Changelog gives the per-milestone landing notes, and §2 (Decisions D1?D17) + §4 (Design) back it. Supporting: `deploy/cloudrun/{Dockerfile,README.md}`, `docs/COMPUTE_DECOUPLED_SERVING/runlogs/RUNLOG-truly-local_TEMPLATE.md`, and ADR-011/012/013/014 in `docs/archives/decisions/DECISIONS.md`.",

```
10           "milestones": [
11           {
12            "id": "M0",
13            "title": "Design doc + project dir + runlogs/ + ADR-014",
14            "cloudRunRelevant": false,
15            "status": "? Done (? APPROVED 2026-06-21, owner sign-off)"
16           },
17           {
18            "id": "M1",
19            "title": "deployment.profile + compute_target config + preset bundles +
precedence/auto-detect (no behaviour change, profile='' default)",
20            "cloudRunRelevant": true,
21            "status": "? Done (PR #279)"
22           },
23           {
24            "id": "M2",
25            "title": "input_backend/input_root; wire main CLI/API input through
StorageRef.parse_uri (local=identity)",
26            "cloudRunRelevant": true,
27            "status": "? Done (PR #280)"
28           },
29           {
30            "id": "M3",
31            "title": "Configurable output/scratch base ? kill hardcoded output/ (default
stays output/, byte-identical)",
32            "cloudRunRelevant": true,
33            "status": "? Done (PR #282) ? was ? default-OFF + Launch Report on flip;
landed default-OFF so no Launch Report needed"
34           },
35           {
36            "id": "M4",
37            "title": "DeviceContext + WASB CPU-fallback (device='auto'); unified
healthcheck/device wiring",
38            "cloudRunRelevant": true,
```

```

39         "status": "? Done (PR #283) ? safe, cuda default unchanged"
40     },
41     {
42         "id": "M5",
43         "title": "truly-local profile end-to-end (preset + startup checks) + first
committed runlog",
44         "cloudRunRelevant": true,
45         "status": "? In progress (PR #286: preset+startup-checks+L4 parity test ?;
real-GPU truly-local runlog ? owner GPU)"
46     },
47     {
48         "id": "M6",
49         "title": "Cloud Run GPU dispatch shim (control plane ? job via storage ? re-
claim) + containerized worker image (ffmpeg+CUDA+WASB+fonts); mocked-dispatch tests",
50         "cloudRunRelevant": true,
51         "status": "? In progress (PR #287: shim+image+mocked tests ?; real build/run =
M7 ? owner GCP spend)"
52     },
53     {
54         "id": "M7",
55         "title": "cloud-serving profile e2e on L4: upload<2GB + gdrive/bucket ?
detect+stitch on Cloud Run ? reel via signed URL; DEPLOY_REPORT",
56         "cloudRunRelevant": true,
57         "status": "? Planned / ? Gated (owner spend, D17 $100/?10k budget)"
58     },
59     {
60         "id": "M8",
61         "title": "Per-job cost ledger (v6 migration) + pricebook + aggregation +
/api/.../cost",
62         "cloudRunRelevant": true,
63         "status": "? In progress (PR #288: ledger+pricebook+/cost ? default-OFF $0;
real GCP prices + worker usage-emission ? owner/follow-up)"
64     },
65     {
66         "id": "M9",
67         "title": "Edge auth (Cf-Access extraction) + per-tenant scoping on user_id;
DPA/consent gate wiring",
68         "cloudRunRelevant": true,
69         "status": "? Planned / ? Gated (owner privacy/counsel)"
70     },
71     {
72         "id": "M10",
73         "title": "byo_worker / zero-infra-BYO ? design-only, DEFERRED",
74         "cloudRunRelevant": true,
75         "status": "? Planned / ? Gated (explicit owner approval; design-only,
deferred)"
76     },
77     {
78         "id": "M11",
79         "title": "Data-flywheel harness (D12): capture upload + Gemini reel/rally
output ? reliably store ? shadow-eval owned-vs-Gemini ? promote to golden TRAINING set",
80         "cloudRunRelevant": false,
81         "status": "? Planned / ? Gated (owner consent/privacy)"
82     },
83     {
84         "id": "M12",
85         "title": "gdrive-api (OAuth) StorageBackend (D14 Northstar): read source from
+ write reel back to user's Google Drive next to source",
86         "cloudRunRelevant": true,
87         "status": "? Planned / ? Gated (owner OAuth/consent)"
88     }
89 ],
90     "notes": "SCOPE OF \"Cloud-Run-friendly\": Milestones flagged cloudRunRelevant
cover the chain that makes the tool deployable on Cloud Run with per-video billing ?
config/profile layer (M1), URI-aware input (M2), de-hardcoded output base needed for bucket

```

workspaces (M3), device portability incl. CPU fallback (M4), the truly-local twin profile + parity test that proves the core is profile-invariant (M5), the actual Cloud Run dispatch shim + container image (M6), the live L4 e2e (M7), per-job cost ledger / per-video billing (M8), edge auth/tenancy (M9), BYO compute (M10), and the OAuth Drive round-trip storage backend (M12). M11 (data-flywheel) is a quality/training-loop harness riding on cloud jobs, not Cloud-Run plumbing itself, so it is flagged false.\n\nSTATUS SUMMARY as the tracker marks it: M0?M4 ? Done (the config/input/output/device foundation is COMPLETE, all merged #279/#280/#282/#283). M5/M6/M8 ? In progress ? the default-OFF CODE has landed (#286/#287/#288) but each retains an owner gate for the real run/spend/calibration. M7/M9/M10/M11/M12 ? Planned, all ? owner-gated. Per the 2026-06-21 changelog, the owner gave a blanket go to WRITE+MERGE the default-OFF code for every remaining gated milestone (one isolated PR each), reserving the gate only for the real run/spend/counsel/calibration ? which is why M5/M6/M8 read as code-done-but-?.\n\nCURRENT BRANCH note: the working branch is feat/serving-m4-device-context; HEAD (44c9f01) is the M4 (#283) tracker/changelog commit. The README on disk already reflects M5?M8 PRs (#286?#288) landing, so the doc is ahead of this branch's git history ? read the README §7 tracker + §11 changelog as authoritative, not the local branch's commit log. NOTE: `git pull --ff-only` FAILED at session start ("Cannot fast-forward to multiple branches") ? I did NOT force anything (per global convention) and read local state only; the local checkout may lag remote.\n\nKEY HONEST CAVEATS (README §6): green CI proves plumbing + window/stitch determinism + profile-parity for \$0 but NOT field quality or that real-GPU detection is good; cross-GPU detection is NOT byte-identical (parity is a window-level claim); cloud-serving ships with Gemini handoff ON (D11) until the owned model clears the #172 floor; WASB-C3 (weight sharing) + DPA/consent remain owner/counsel-gated before public non-owner serving."

```

91         },
92         "code": {
93             "computeRegistry": "backend/compute/base.py defines the seam as an ABC +
dataclasses, NOT a Registry-singleton (unlike segmenters/storage). ComputeBackend(ABC) has a
single abstractmethod run_infer(self, spec: ComputeJobSpec) -> ComputeResult (base.py:46-53);
ComputeJobSpec frozen dataclass (base.py:20-30) carries video_uri, out_uri, optional segmenter,
config_overrides (a tuple so the spec stays frozen/hashable); ComputeResult frozen dataclass
(base.py:33-43) carries returncode (0 ok / 2 validation-config / 3 segmenter-fail / 4 = remote
dispatch timeout or empty/unreadable marker), outputs_uri, video_id, report. The selection logic
lives in backend/compute/__init__.py as a lazy FACTORY get_compute_backend(config, *,
run_client=None, policy=DEFAULT_POLICY, token=None) -> Optional[ComputeBackend]
(__init__.py:32-61): it returns None for any compute_target other than "cloud_run"
(__init__.py:47-49) and only lazily imports + builds CloudRunCompute for "cloud_run"
(__init__.py:51-61). So there is NO ComputeBackend Registry singleton with @register decorators
? it is a single-branch factory function modeled on backend/storage's lazy factory idiom
(base.py:3-10, __init__.py:1-7).",
94             "cloudRunBackend": "backend/compute/cloud_run.py implements
CloudRunCompute(ComputeBackend) (cloud_run.py:82-156). DISPATCH/SUBMIT: run_infer
(cloud_run.py:103-132) builds worker argv ["infer", "--video-uri", ..., "--out-uri", ..., "--
done-uri", f"{out_root}/dispatch/{token}.done"] (token = injected self._token or fresh uuid4
per call, cloud_run.py:105-108), appends --segmenter and every --set k=v from
spec.config_overrides PLUS _DEFAULT_CONTAINER_OVERRIDES =
("deployment.compute_target=local_gpu", "indexing.detector_impl=native") (cloud_run.py:41-44,
109-112) so the dispatched container runs INLINE and never re-dispatches, then calls
self._client.run_job(job_name, argv, region, project) (cloud_run.py:114-115). POLL/AWAIT: marker
= poll_until(lambda: self._read_marker(done_uri), self._policy, label="cloud-run-infer")
(cloud_run.py:119-120) using execution_policy.poll_until (execution_policy.py:125-151; default
poll_every_n_sec=10s, max_wait_sec=1800s, raises PolicyTimeout on exhaustion). COMPLETION
DETECTION IS A BUCKET-POLL of a storage marker, NOT the Cloud Run API status: _read_marker
(cloud_run.py:135-140) parse_uri(done_uri) then get_storage(ref.backend).exists(ref), returning
None (keep polling) until the worker writes the marker; once present it reads the JSON
(_read_json, cloud_run.py:142-155, downloads to a temp local dst first since gcs/s3 require an
explicit dst). Then it extracts video_id + returncode (a MISSING returncode defaults to FAILURE
1, never 0; an empty/unreadable marker is treated as failure, cloud_run.py:121-128), composes
outputs_uri = f"{out_root}/outputs/{video_id}/", reads report.json, returns ComputeResult
(cloud_run.py:129-132). The marker is written by the worker via _write_done_marker
(backend/worker/__main__.py:140-153, {video_id, returncode, segment_count, status} pushed with
storage.put), which CloudRunCompute consumes ? so the completion signal is storage-backed (D16),
reusing the existing park/resume primitives.",
95             "wiring": "The compute seam is wired into the STANDALONE CLI worker ONLY, NOT into
/api/process. backend/worker/__main__.py::cmd_infer (worker/__main__.py:156-168) calls compute =
get_compute_backend(config) and, if non-None, returns _dispatch_remote(compute, args)

```

(worker/___main___py:116-137), which builds a ComputeJobSpec from argv and calls compute.run_infer(spec), mapping PolicyTimeout -> rc 4 (worker/___main___py:124-137); if None it falls through to the unchanged inline PULL->hash->validate->segment path (worker/___main___py:170+). This worker is `python -m backend.worker infer ...` ? the same binary the Cloud Run container runs. CRITICAL GAP: the HTTP request/job flow does NOT use this seam. POST /api/process (backend/main.py:268-308) only fast-fails + creates a 'queued' jobs-table row and calls _get_executor().submit(_drain_due_jobs) (main.py:304). _drain_due_jobs (backend/api/job_worker.py:132-153) -> _run_claimed_job (job_worker.py:72-98) -> backend.main._execute_processing (main.py:148+) runs the pipeline IN-PROCESS on the ThreadPoolExecutor (job_worker.py:30-38, max_workers=1). _execute_processing NEVER calls get_compute_backend / compute.run_infer ? it always hashes/validates/segments locally in the API process. So today every /api/process job runs in-process; the Cloud Run dispatch path is reachable only by invoking the backend.worker CLI directly with a cloud_run profile. No dispatcher bridges the API job worker to the ComputeBackend yet.",

96 "flagsDefault": "Selection is config.deployment.compute_target (Literal[\"\", \"local_gpu\", \"cloud_run\", \"cpu_mock\"], DEFAULT \"\" ? backend/config/models.py:351). get_compute_backend returns non-None ONLY for compute_target == \"cloud_run\" (___init___py:47-49); \"\", \"local_gpu\", \"cpu_mock\" all return None => in-process/inline. compute_target is normally set via deployment.profile (Literal[\"\", \"truly-local\", \"cloud-serving\", \"cpu-mock\"], DEFAULT \"\" ? models.py:348), whose presets (backend/config/presets.py:43-63; COMPUTE_TARGET_FOR_PROFILE presets.py:32-36) map cloud-serving->cloud_run, truly-local->local_gpu, cpu-mock->cpu_mock; an explicitly-blanked compute_target under an active profile is back-filled to the profile's canonical target by the loader (backend/config/loader.py:167-181). CONFIRMED DEFAULT-OFF: with no profile selected both profile and compute_target are \"\" (models.py:348,351; presets.py:11-16), so get_compute_backend returns None and the worker runs inline ? Cloud Run is default-OFF and in-process is the default. Cloud Run coordinates come from ENV not config: CLOUD_RUN_JOB (default \"rally-worker\"), CLOUD_RUN_REGION (default \"asia-south1\"), GOOGLE_CLOUD_PROJECT (___init___py:54-57). Related M8 BillingConfig.enabled is also default-OFF (models.py:367).",

97 "mockedVsReal": "REAL: CloudRunCompute orchestration (argv build, container-override injection, bucket-poll loop, marker parsing with fail-safe returncode, report fetch) is real and unit-testable; storage layer (parse_uri/get_storage/fetch/put/exists) is real; the worker's _write_done_marker (worker/___main___py:140-153) is real; config/profile/loader plumbing is real. The real client GoogleCloudRunJobClient.run_job (cloud_run.py:60-79) DOES contain real Cloud Run API calls when reached ? it lazy-imports google.cloud.run_v2, builds JobsClient().job_path(project,region,job_name), wraps argv in RunJobRequest.Overrides.ContainerOverride(args=argv), and calls client.run_job(RunJobRequest(...)) ? BUT it is marked '# pragma: no cover - exercised only on the real M7 run' and google-cloud-run is intentionally NOT a CI/test dependency (cloud_run.py:60-72), so this path has never actually run; owner-verification is deferred to M7 (___init___py:45, cloud_run.py:18-19). MOCKED IN PRACTICE: the trigger is an injected abstraction CloudRunJobClient(ABC, cloud_run.py:47-57); tests pass a FAKE client (no GCP) and that is the ONLY exercised path. A real cloud dispatch still needs: (1) install google-cloud-run + valid GOOGLE_CLOUD_PROJECT/region creds on the control plane; (2) a deployed Cloud Run GPU Job named per CLOUD_RUN_JOB whose container runs `python -m backend.worker infer`; (3) a real gcs storage backend reachable by both dispatcher and container for the gs:// video/out/done URIs; (4) owner-verification that ContainerOverride(args=argv) truly overrides job args and the container writes the done-marker; (5) crucially, an API-side dispatcher to route /api/process jobs through the ComputeBackend (bypassed today); (6) M8 usage-emission hooks (cost stays 0 without them, job_worker.py:101-129).",

98 "gaps": [

99 "WIRING GAP: get_compute_backend is consulted ONLY in backend/worker/___main___py::cmd_infer (worker/___main___py:166); the HTTP path POST /api/process -> _drain_due_jobs -> _run_claimed_job -> backend.main._execute_processing (main.py:148, job_worker.py:86) NEVER calls it, so all API jobs run in-process on the ThreadPoolExecutor (job_worker.py:30-38). Cloud Run dispatch is reachable only by invoking `python -m backend.worker infer` directly with a cloud_run config; no dispatcher bridges the async job worker to the ComputeBackend.",

100 "REAL CLOUD CALLS UNVERIFIED: GoogleCloudRunJobClient.run_job (cloud_run.py:60-79) is the only real-API code but is pragma-no-cover and google-cloud-run is intentionally excluded from test deps; the real dispatch has never been exercised (owner-gated to M7). CI only runs the injected fake client.",

101 "RETURN-PATH ASYMMETRY: ComputeResult carries returncode + outputs_uri + report, but _dispatch_remote (worker/___main___py:116-137) only propagates result.returncode as the process exit code; outputs_uri/report are logged, not surfaced; there is no API consumer of

```

ComputeResult yet.",
102         "COMPLETION SIGNAL IS A STORAGE MARKER, NOT CLOUD RUN STATUS (by design, D16): a
container that dies BEFORE writing the marker is detected only by poll_until exhausting
max_wait_sec=1800s -> PolicyTimeout -> rc 4 (cloud_run.py:119-120, worker/__main__.py:124-134);
the Cloud Run execution's own failure status is never consulted, so a fast-failing job still
costs the full ~30min poll budget before being reported failed.",
103         "COST METERING ABSENT: even with billing.enabled + a real pricebook, recorded
cost is 0 because the pipeline does not emit result['usage'] yet (job_worker.py:101-129,
models.py:363-365).",
104         "compute_target='cpu_mock' selects NO dedicated backend: get_compute_backend
treats cpu_mock the same as '/local_gpu (returns None, __init__.py:49); the cpu-mock profile
relies on indexing.detector_impl=stub, not a ComputeBackend implementation."
105     ]
106 },
107     "deploy": {
108         "dockerfile": "deploy/cloudrun/Dockerfile (C:/Users/avidu/Projects/badminton-
highlight-indexer/deploy/cloudrun/Dockerfile, 62 lines) builds a single image for the rally GPU
**Job** worker (COMPUTE_DECOUPLED_SERVING M6/M7, ADR-012/D16).\n\nBASE IMAGE:
`nvidia/cuda:12.4.1-runtime-ubuntu22.04` (CUDA 12.4 runtime, Ubuntu 22.04) ? a GPU/CUDA base,
not a slim/CPU base.\n\nOS DEPS: apt-installs `ffmpeg` (decode/stitch), `fonts-dejavu-core`
(watermark filter), `git`/`curl`/`ca-certificates`/`build-essential`, and the distro
`python3`/`python3-pip`/`python3-venv`.\n\nTWO DELIBERATELY-SEPARATE PYTHON STACKS (same split
as the native GPU box deploy/digitalocean/gpu_setup.sh, to avoid a framework/torch conflict):\n
1) MAIN APP ENV ? system python3; `pip install .` of this repo from pyproject.toml (COPY
pyproject.toml, README.md, backend/, training/). Drives the pipeline + ffmpeg stitching. Notably
the main env installs NO torch (detection is delegated out to the WASB subprocess).\n
2) NATIVE
WASB CONDA ENV ? installs Miniconda to /opt/conda, accepts conda TOS for main+r channels,
creates conda env `wasb` with python=3.8, clones https://github.com/nttcom/WASB-SBDT.git into
/opt/models/WASB-SBDT, and pip-installs torch==1.11.0+cu113 / torchvision==0.12.0+cu113 (CUDA
11.3 wheels, via the pytorch cu113 extra-index) plus hydra-core, omegaconf, pandas, opencv-
python-headless, pillow, scipy, tqdm, pyyaml, matplotlib, numpy<1.24, and the WASB repo's own
requirements.txt if present. The native runner shells out to $WASB_PYTHON as a subprocess, so
the app's torch never co-installs with the detector's torch-1.11 stack.\n\nWASB WEIGHTS: NOT
baked in (WASB-C3 ? pretrained badminton weights are academic-data-trained, provenance not
cleared for commercial sharing). The image only sets $WASB_WEIGHTS to a default path
(/opt/models/WASB-SBDT/pretrained_weights/wasb_badminton_best.pth.tar); weights are expected to
be staged at runtime (bucket fetch or mount) and pointed at via $WASB_WEIGHTS.\n\nENV WIRING:
WASB_PYTHON=/opt/conda/envs/wasb/bin/python, WASB_REPO_DIR=/opt/models/WASB-SBDT,
WASB_WEIGHTS=<default path>, RALLY_DETECTOR_IMPL=native (so the container uses the Linux-native
WASB runner, not the WSL/cuda path). The native runner finds the env+repo purely by these env
vars (no `conda activate`).\n\nENTRYPOINT: `[\"python3\", \"-m\", \"backend.worker\"]` ? i.e.
the WORKER, not the FastAPI server. The dispatcher supplies per-execution args at run time:
`infer --video-uri gs:/// --out-uri gs:/// --done-uri gs:/// --segmenter wasb_hybrid --set ?`.
backend/worker/__main__.py cmd_infer runs detect+window+stitch INLINE (co-located so
CPU/wall/egress meter into one job, D4), then writes the report and (when --done-uri is passed)
a tiny JSON completion marker {video_id, returncode, ...} to the bucket.",
109         "buildable": "SCAFFOLD / owner-built-once, NOT built by CI and not part of the
automated test loop. Findings: (1) NO CI builds it ? the only GitHub Actions workflow is
.github/workflows/ci.yml, which runs the mocked offline test suite on a CPU runner (it even
installs CPU-only torch and explicitly notes torch is not a pyproject dep); a repo-wide grep for
`gcloud builds submit`/`gcloud run`/`buildx`/`docker build` finds zero hits in CI. The
Dockerfile header states plainly: `\"CI does NOT build this (a CUDA image is too heavy)\"`. (2) NO
build/push/deploy SCRIPT exists ? there is no shell script that builds or pushes this image. The
deploy/gcp/*.sh scripts (provision.sh, create_gpu_vm.sh, ops-agent-startup.sh, teardown.sh) only
provision a GPU VM + GCS bucket + IAM + budget; none reference Cloud Run, the Dockerfile, or
`gcloud run`/`gcloud builds`. The only build/push/deploy instructions are COPY-PASTE commands
documented in deploy/cloudrun/README.md (gcloud builds submit; gcloud run jobs create), to be
run by the owner. (3) The control-plane dispatch shim IS real + landed + mock-tested
(backend/compute/cloud_run.py CloudRunCompute, tests/test_compute_backend.py with a fake
CloudRunJobClient ? the real GoogleCloudRunJobClient lazy-imports google-cloud-run, which is
intentionally NOT a test dependency, so that path is never exercised in CI). Per the
README/Dockerfile, the real build+push+run on an L4 is the **M7 owner step** within a $100/?10k
budget; status in deploy/cloudrun/README.md is `\"dispatch shim + image are landed + mock-tested
(M6); the real build + push + run on an L4 is the M7 owner step.\"` So: the image is intended to
be buildable and the worker/dispatch logic is exercised by tests, but the image itself is

```

verified only by a manual owner run, not by any automated build/push/deploy/CI.",

110 "gpuStory": "The image is GPU-FIRST and Cloud-Run-Job-only. Base is the CUDA 12.4 runtime and the WASB conda env pins torch 1.11.0+cu113 / torchvision 0.12.0+cu113 (CUDA wheels), so the detector subprocess expects an NVIDIA GPU. The Cloud Run resource it targets is a GPU **Job** (not a Service): deploy/cloudrun/README.md \$2 creates it with `gcloud run jobs create rally-worker --gpu 1 --gpu-type nvidia-l4 --cpu 4 --memory 16Gi --max-retries 0 --task-timeout 3600`, scale-to-zero (D7, no warm pool); the control plane bucket-polls a done-marker rather than the Cloud Run API. CPU-vs-GPU at the IMAGE/deploy level is NOT a configurable axis here ? there is no CPU variant of this image, no `--gpu`-optional path, and ffmpeg is NVDEC-decode + libx264-encode (the image keeps ffmpeg's default libx264 for parity, D16). A CPU FALLBACK does exist but only at the CODE level, not in the container build: the current branch's M4 work (feat/serving-m4-device-context, commit 6932d4a) added a DeviceContext (backend/pipeline/detectors/device.py) and native WASB CPU-fallback where device="auto" can degrade to CPU while device="cuda" stays strict. That fallback is a property of the native runner / config, not something this Dockerfile's cu113 torch build or the L4 Job spec is set up to exploit ? the cu113 wheels would still run on CPU only via torch's CPU code paths if no GPU were present, but the documented serving path always requests an L4. In short: native WASB packaging into a CUDA container is feasible and is exactly what this Dockerfile does (conda py3.8 + torch-1.11+cu113 mirroring the verified D0/native GPU box); the open caveat is weights provenance (WASB-C3), not GPU packaging.",

111 "deployScripts": "NONE beyond the Dockerfile + manual README commands. deploy/cloudrun/ contains only Dockerfile + README.md ? no build.sh/deploy.sh/cloudbuild.yaml. The actual Cloud Run lifecycle lives as documented copy-paste commands in deploy/cloudrun/README.md: \$1 `gcloud artifacts repositories create ? && gcloud builds submit --tag ? :latest -f deploy/cloudrun/Dockerfile .`; \$2 `gcloud run jobs create rally-worker --gpu 1 --gpu-type nvidia-l4 --cpu 4 --memory 16Gi --max-retries 0 --task-timeout 3600 --set-env-vars WASB\_WEIGHTS=?`; \$3 wiring the dispatcher via env (DEPLOYMENT_PROFILE=cloud-serving ? compute_target=cloud_run, CLOUD_RUN_JOB / CLOUD_RUN_REGION / GOOGLE_CLOUD_PROJECT). The deploy/gcp/ scripts are unrelated to Cloud Run ? provision.sh (GCS bucket + rally-worker SA + Storage Object Admin + Ops-Agent writer roles + \$100 budget + GPU-quota check), create_gpu_vm.sh (spins up a g2-standard-4 L4 / n1+t4 VM with the Ops Agent ? a plain GPU VM, NOT Cloud Run), ops-agent-startup.sh, teardown.sh (defaults to delete). There is no `gcloud run deploy` or `gcloud run jobs` anywhere in deploy/gcp/ (grep-confirmed). M7 acceptance is a manual owner runlog (DEPLOY_REPORT_cloud-serving_<date>.md under docs/COMPUTE_DECOUPLED_SERVING/runlogs/), not an automated deploy.",

112 "gaps": [

113 "No CI builds, lints, or smoke-tests the image ? the CUDA+conda image is deemed too heavy for CI (ci.yml is CPU-only mocked tests), so image rot can go undetected until the owner's manual M7 build.",

114 "No build/push/deploy script ? only copy-paste gcloud commands in deploy/cloudrun/README.md; nothing automates `gcloud builds submit` or `gcloud run jobs create`, and deploy/gcp/ has zero Cloud Run wiring.",

115 "WASB weights are not baked in (WASB-C3 provenance not cleared for commercial sharing) and there is NO staging step in the image or scripts ? the README assumes a runtime bucket-fetch/mount that is not yet automated; serving cannot actually run until weights are staged and WASB-C3 is resolved.",

116 "The real Cloud Run client (GoogleCloudRunJobClient) and the actual L4 build/run are owner-verified only (M7), never exercised by tests ? google-cloud-run is intentionally excluded from test deps, so that whole real path is untested in CI.",

117 "Known correctness gap documented in the README: a FAILED Job writes no done-marker, so the dispatcher's bounded bucket-poll just times out to a generic rc 4 and cannot distinguish a real failure from a slow cold-start; a failure-marker carrying the real rc is a noted follow-up.",

118 "No CPU variant / GPU-optional image: the image is hard-pinned to CUDA (cu113 torch) and the Job spec always requests an L4; the M4 DeviceContext CPU-fallback exists only in code (backend/pipeline/detectors/device.py + native_wasb_runner.py), not as a buildable CPU deploy target, so there is no documented way to serve on Cloud Run without a GPU.",

119 "WORKDIR/COPY caching: `pip install .` runs after COPYing backend/ + training/, so any source change re-runs the heavy main-env install; the expensive conda/WASB layer is earlier so it caches, but there is no requirements-only pre-copy step for the app env."

120]

121 },

122 "activity": {

123 "mergedPRs": [

124 "#288 feat(serving): M8 ? per-job cost ledger + versioned pricebook + /cost API


```

(default-OFF)",
125     "#287 feat(serving): M6 ? Cloud Run GPU dispatch shim + worker image (default-
OFF, mocked)",
126     "#286 feat(serving): M5 ? truly-local profile per-profile startup checks + L4
profile-parity test (default-OFF)",
127     "#285 docs(serving): resolve $8 design picks (D16/D17) + reconcile per-video P3
vs M3",
128     "#284 test(api): lock gs://|gcs:// /api/process -> queryable API-DB video_id
(B-P1 guard)",
129     "#283 feat(serving): M4 ? DeviceContext + native WASB CPU-fallback (device=auto,
cuda strict)",
130     "#282 feat(serving): M3 ? configurable output/scratch base (de-hardcode output/,
default-OFF)",
131     "#281 fix(server): drain async jobs under `python -m backend.main` (double-
module hang)",
132     "#280 feat(serving): M2 ? URI-aware input (input_backend/input_root,
local=identity)",
133     "#279 feat(serving): M1 ? deployment.profile + compute_target config (pure,
additive, default-OFF)",
134     "#278 docs(serving): M0 APPROVED + lock D13-D15 (switchable profiles, Drive
Northstar, consent-not-surprise)",
135     "#277 docs(serving): lock the 5 M0 gating decisions (D7-D12) + data-flywheel
harness (M11)",
136     "#289 fix(api): resolve a bucket-ingested video's source to a local file before
stitching (compile path)",
137     "#268 feat(api): GET /api/health + return download_url from /api/compile
(M1/P6)",
138     "#263 docs(adr): ADR-012 ? GPU-native compute; TPU deferred; Cloud Run+GPU is
the serving model"
139 ],
140 "openPRs": [
141     "#290 feat(serving): M9-auth ? Cloudflare Access edge-auth + per-tenant owner_id
(default-OFF) ? opened 2026-06-21T13:49Z, not draft, +406/-13"
142 ],
143 "activelyWorked": true,
144 "whoAndCadence": "Extremely active, concentrated in a single burst on 2026-06-21
(author: Avi Dullu / current session). The serving track went from M0 approval to M8 code merged
in roughly half a day: #278 (M0 APPROVED) at 06:59Z, #279 (M1) 07:34Z, #280 (M2) 08:15Z, #281
(fix) 08:22Z, #282 (M3) 08:39Z, #283 (M4) 09:12Z, #284 (test) 09:35Z, #285 (design picks)
09:58Z, #286 (M5) 10:27Z, #287 (M6) 10:53Z, #288 (M8) 11:16Z, #289 (compile fix) 13:29Z ? then
#290 (M9-auth) opened 13:49Z and still OPEN. That is ~13 serving PRs in ~7 hours, on top of the
earlier decoupled-compute/deploy run (#250-#268, GCP L4 deploy report) over 2026-06-20. Recent
origin/master HEAD = e68b583 (#289). The work is one isolated default-OFF PR per milestone, off
origin/master, no stacking.",
145 "nextPlanned": "Per the live tracker docs/COMPUTE_DECOUPLED_SERVING/README.md $7:
code for M1-M8 + M5/M6 plumbing is merged default-OFF, and M9-auth (#290) is the open PR. The
next planned Cloud-Run steps are: (1) M7 ? the real cloud-serving e2e on a GCP L4 (upload<2GB /
gdrive / bucket -> detect AND stitch co-located on a Cloud Run JOB -> reel via signed URL) with
a committed DEPLOY_REPORT, the key owner-gated real-spend/real-build milestone (D17 $100/?10k
budget; M6 already shipped the Cloud Run dispatch shim + deploy/cloudrun Dockerfile/worker
image, mocked, awaiting the real build/push/run). (2) finish M8 (owner inserts real GCP prices
as a new pricebook version + wire worker usage-emission). (3) M9 consent/DPA gate wiring
(counsel-gated) alongside the edge-auth in #290. (4) M11 data-flywheel harness
(capture->store->shadow-eval->goldenize consented adult jobs). (5) M12 gdrive-api OAuth
StorageBackend (the Northstar Drive round-trip). M10 byo_worker stays design-only/deferred."
146 ],
147 "synthesis": {
148     "headline": "The Cloud-Run-friendly work is a single, formally-tracked subproject
? \"Compute-Decoupled Rally Serving\" ? whose source of truth is
docs/COMPUTE_DECOUPLED_SERVING/README.md ($7 progress tracker + $11 changelog), backed by
ADR-011/012/013/014 in docs/archives/decisions/DECISIONS.md. It realizes Cloud Run + GPU (L4)
scale-to-zero JOBS for pay-per-video serving via one pure DETECT/WINDOW/STITCH core selected by
a ComputeBackend (where) + StorageBackend (what-bytes) at startup config. The code seam lives in
backend/compute/ (the CloudRunCompute dispatch shim) and the container in deploy/cloudrun/. And
yes ? it is very actively being worked: this is the current concurrent serving session's primary

```

focus, which shipped ~13 default-OFF PRs (M0?M8, #278?#288) in roughly 7 hours on 2026-06-21 and has PR #290 (M9 edge-auth) open and non-draft right now. The config/input/output/device foundation (M1?M4) is merged and Done; the Cloud Run shim, container image, truly-local profile, and cost ledger (M5/M6/M8) are code-landed default-OFF but retain owner gates for the real GPU run/spend; M7 (the real L4 e2e) and M9?M12 are planned and owner-gated.",

```
149     "whereItLives": [  
150         "docs/COMPUTE_DECOUPLED_SERVING/README.md ? THE source of truth: $7 progress  
tracker (M0?M12, legend ??/?/?/?), $2 decisions D1?D17, $4 design, $11 changelog. Read $7+$11 as  
authoritative (the on-disk README is ahead of older branch git logs).",  
151         "docs/archives/decisions/DECISIONS.md ? the 4-ADR ratification chain: ADR-011  
(deferred M3/M4), ADR-012 (Cloud Run+GPU L4 is the serving model), ADR-013 (GCP is sole  
compute), ADR-014 (compute-decoupled serving design sign-off).",  
152         "backend/compute/__init__.py ? the default-OFF factory  
get_compute_backend(config); returns a backend ONLY for compute_target=='cloud_run', else None  
(in-process). Single-branch factory, not a Registry singleton.",  
153         "backend/compute/cloud_run.py ? CloudRunCompute(ComputeBackend): builds worker  
argv, injects container-overrides so the dispatched job runs inline, dispatches via injectable  
CloudRunJobClient, and bucket-polls a done-marker (NOT the Cloud Run API) for completion.  
GoogleCloudRunJobClient.run_job is the only real-API code (pragma no-cover, never exercised in  
CI).",  
154         "backend/compute/base.py ? the seam contract: ComputeBackend ABC + frozen  
ComputeJobSpec / ComputeResult dataclasses (returncode 0/2/3/4).",  
155         "backend/worker/__main__.py ? the ONLY consumer of the seam today: cmd_infer  
calls get_compute_backend and _dispatch_remote; also writes the done-marker (_write_done_marker)  
on success. This is the `python -m backend.worker` binary the container runs.",  
156         "deploy/cloudrun/Dockerfile ? the L4 Job image: FROM nvidia/cuda:12.4.1-runtime  
+ ffmpeg + DejaVu fonts; TWO python stacks (main app env, no torch + a conda 'wasb' py3.8  
torch-1.11+cu113 subprocess); ENTRYPOINT python3 -m backend.worker. The dual-runtime packaging  
ADR-012 flagged as the hard part.",  
157         "deploy/cloudrun/README.md ? the manual M7 lifecycle: copy-paste gcloud builds  
submit + gcloud run jobs create --gpu 1 --gpu-type nvidia-l4 + dispatcher env wiring. No  
build/deploy script exists.",  
158         "backend/config/{models.py,presets.py,loader.py} ? deployment.profile (truly-  
local/cloud-serving/cpu-mock) + compute_target selection, both default '=OFF.",  
159         "docs/COMPUTE_DECOUPLED_SERVING/runlogs/ ? where the M5 truly-local RUNLOG and  
M7 cloud-serving DEPLOY_REPORT (owner real-GPU runs) are to be committed.",  
160         "deploy/gcp/*.sh ? provision.sh/create_gpu_vm.sh/teardown.sh: GCS bucket +  
SA/IAM + budget + a plain L4 GPU VM. NOTE: these have ZERO Cloud Run wiring (grep-confirmed) ?  
they are not the Cloud Run path."  
161     ],  
162     "statusByLayer": "DESIGN: Done ? ADR-011/012/013/014 ratified; M0 design doc  
owner-APPROVED 2026-06-21; decisions D1?D17 locked. CODE (compute backend): Foundation Done ? M1  
config/profile (#279), M2 URI-aware input (#280), M3 de-hardcoded output base (#282), M4  
DeviceContext+WASB CPU-fallback (#283) all merged default-OFF. The CloudRunCompute dispatch shim  
+ cost ledger are code-landed default-OFF (M6 #287, M8 #288) but ? In-progress (owner gate on  
real run/spend/calibration). CRITICAL: the seam is wired ONLY into the standalone  
`backend.worker` CLI ? the HTTP path (POST /api/process ? _drain_due_jobs ? _execute_processing)  
NEVER calls get_compute_backend and always runs in-process (verified: zero references in  
backend/main.py and backend/api/job_worker.py). CONTAINER: Scaffold built once / not CI-built ?  
deploy/cloudrun/Dockerfile exists and is the real image, but CI (ci.yml) is CPU-only mocked  
tests and never builds it; no CPU-variant/GPU-optional image exists; WASB weights are NOT baked  
in (WASB-C3 provenance) and have no automated runtime-staging step. DEPLOY: Planned/Gated ? no  
build/push/deploy script and no `gcloud run` in deploy/gcp/; the real L4 build+push+run is the  
M7 owner step (D17 $100/?10k budget). M5/M6/M8 ?; M7/M9/M10/M11/M12 ? all ? owner-gated (real-  
GPU, spend, privacy/counsel, BYO approval, OAuth/consent).",  
163     "alreadyWorkedVerdict": "YES ? decisively, and right now by the concurrent serving  
session (author Avi Dullu / current owner session). Evidence: ~13 isolated default-OFF PRs took  
the track from M0 approval to M8 code-merged in roughly 7 hours on 2026-06-21 (#278 06:59Z ?  
#288 11:16Z ? #289 compile fix 13:29Z), and PR #290 (M9-auth: Cloudflare Access edge-auth + per-  
tenant owner_id, default-OFF) is OPEN and non-draft, updated 2026-06-21T13:49Z, on branch  
feat/serving-m9-auth (gh-confirmed live). origin/master HEAD == local master == e68b583 (#289);  
the on-disk README $7 already shows M5?M8 landed and $11 changelog records the burst. So the  
Cloud-Run-friendly work is not just planned ? it is the active head-of-line work item this  
session, advancing one milestone-PR at a time off origin/master with no stacking.",  
164     "whatsLeft": [  

```

```

165         "WIRE THE API TO THE SEAM (biggest functional gap): POST /api/process ?
_drain_due_jobs ? _run_claimed_job ? backend.main._execute_processing runs the pipeline in-
process on a max_workers=1 ThreadPoolExecutor and never calls get_compute_backend. Cloud Run
dispatch is reachable today ONLY by invoking `python -m backend.worker infer` directly. A
dispatcher bridging the async API job worker to ComputeBackend.run_infer does not exist yet, and
there is no API consumer of ComputeResult (outputs_uri/report are only logged).",
166         "M7 ? the real cloud-serving e2e on a GCP L4 (upload<2GB / gdrive / bucket ?
detect+stitch on a Cloud Run GPU Job ? reel via signed URL) with a committed DEPLOY_REPORT.
Owner-gated on real GCP spend (D17 $100/??10k). The real GoogleCloudRunJobClient.run_job path is
pragma-no-cover and has never executed; google-cloud-run is intentionally excluded from test
deps.",
167         "Container productionization: no CI build/lint/smoke of the CUDA+conda image
(rot undetected until M7); no CPU-variant/GPU-optional image; WASB weights not baked in (WASB-C3
not cleared for commercial sharing) and no automated runtime staging step ? serving can't
actually run until weights are staged AND WASB-C3 is resolved.",
168         "Deploy automation: no build.sh/deploy.sh/cloudbuild.yaml; only copy-paste
gcloud commands in deploy/cloudrun/README.md; deploy/gcp/ has zero Cloud Run wiring.",
169         "M8 residual: real GCP prices as a new pricebook version + worker usage-emission
hooks (recorded cost is 0 until the pipeline emits result['usage']).",
170         "Failure-marker gap (documented): a FAILED Job writes no done-marker, so the
dispatcher's bounded bucket-poll just times out to a generic rc 4 (~30 min) and can't
distinguish a real failure from a slow cold-start.",
171         "M5 residual: the real-GPU truly-local runlog (owner GPU).",
172         "M9 ? edge auth + DPA/consent gate wiring (code in open PR #290; the
consent/counsel gate remains owner/privacy-gated).",
173         "M10 byo_worker (design-only/deferred), M11 data-flywheel harness, M12 gdrive-
api OAuth StorageBackend (Northstar Drive round-trip) ? all planned + owner-gated."
174     ],
175     "dontDuplicate": "DO NOT re-open or re-implement anything M0?M8: the design
(ADR-011/012/013/014), the config/profile/input/output/device foundation (M1?M4, merged), and
the Cloud Run dispatch shim + container image + cost ledger (M5/M6/M8, code-landed default-OFF)
are all done by the active session ? re-doing them would collide. DO NOT start M9 edge-auth: it
is in OPEN PR #290 (feat/serving-m9-auth) right now. DO NOT hand-roll a Cloud Run client, a new
ComputeBackend Registry, or a second container image ? the seam is CloudRunCompute in
backend/compute/cloud_run.py (a single-branch factory in backend/compute/__init__.py, by design
NOT a Registry singleton) and the image is deploy/cloudrun/Dockerfile. COORDINATION: the $7
tracker in docs/COMPUTE_DECOUPLED_SERVING/README.md is the single source of truth ? check a
row's ??? status and its PR link before touching it. Follow the session's established
discipline: ONE isolated default-OFF PR per milestone, branched explicitly from origin/master
(git fetch origin && git checkout -b <name> origin/master), re-verify HEAD before commit, stage
explicit paths (never git add -A), and serialize repo writes (don't push while the active
session may be mid-PR ? this checkout is shared, the #116/#117 collision happened here). The
genuinely un-owned, non-colliding next pieces a separate worker could pick up are: (1) the
API?ComputeBackend dispatcher (the wiring gap), and (2) the failure-marker carrying the real rc
? but coordinate with the owner first since both touch the serving track the session is driving.
Anything requiring real GCP spend, a real GPU, weights/WASB-C3, or privacy/counsel
(M7/M9-gate/M10/M11/M12) is owner-gated ? do not attempt it."
176     }
177 }
178 }

```

441. assistant (2026-06-21T13:59:58.910Z)

The Cloud Run scan is back with a definitive answer. The broader all-docs next-steps scan is still finishing ? here's the serving-track piece now, then I'll fold it into the full roadmap.

Where the Cloud-Run work lives ? and yes, it's actively being built

****Source of truth:****

[`docs/COMPUTE\_DECOUPLED\_SERVING/README.md`](docs/COMPUTE_DECOUPLED_SERVING/README.md) \$7
tracker + \$11 changelog, ratified by ****ADR-011/012/013/014****
([DECISIONS.md](docs/archives/decisions/DECISIONS.md)).

| Layer | Where | Status |

```
|---|---|---|
| **Design** | ADR-011/012/014 + COMPUTE_DECOUPLED_SERVING/README | ? ratified (Cloud Run
**Jobs** + L4 GPU, scale-to-zero, per-video billing) |
| **Code seam** | `backend/compute/{base.py, __init__.py, cloud_run.py}` | `CloudRunCompute`
dispatch shim ? **landed default-OFF** (M6 #287); dispatches a Job + bucket-polls a `.done`
marker |
| **Container** | `deploy/cloudrun/{Dockerfile, README.md}` | image exists (CUDA 12.4 + ffmpeg +
a conda `wasb` torch-1.11 subprocess) ? **not CI-built**; real build = M7 |
| **Config** | `backend/config/{models, presets, loader}.py` | `deployment.profile` /
`compute_target` ? default `""` = **in-process (OFF)** |
```

****Actively worked? Decisively yes ? it's the concurrent session's head-of-line work.**** ~13
isolated default-OFF PRs took it from ****M0 ? M8** in ~7 hours today** (#278?#288, +my #289 fix),
and ****M9 edge-auth is OPEN** right now as [#290](https://github.com/Khelsutra/badminton-highlight-indexer/pull/290)**.

****Milestone status:**** M1?M4 ? (config/input/output/device foundation, merged) . M5/M6/M8 ? (code
landed default-OFF, gated on the real GPU run/spend) . ****M7**** (real L4 e2e) + M9?M12 ? owner-
gated.

****The one big functional gap**** (and the most interesting un-owned piece): the ****HTTP path isn't**
wired to the seam yet**. `POST /api/process ? \_drain\_due\_jobs ? \_execute\_processing` always runs
****in-process****; `get\_compute\_backend()` is consulted ****only**** by the standalone `python -m
backend.worker infer` CLI. So Cloud Run dispatch works from the worker CLI but not from the API
job worker ? bridging that (an API?`ComputeBackend` dispatcher) is net-new and currently un-
owned.

****? Don't duplicate / coordinate:**** M0?M8 are done by the active session and M9 is in open PR
#290 ? re-doing any of it (or hand-rolling a second Cloud Run client / container) would collide
on this shared checkout. The serving track is owner-driven, one default-OFF PR per milestone off
`origin/master`.

The full cross-track next-steps roadmap (appliance/CUJ, localization, rally-quality, owned-
model) is landing from the second scan ? I'll consolidate everything into one prioritized list
when it completes.

442. user (2026-06-21T14:00:47.902Z)

```
<task-notification>
<task-id>w7ydrkzjj</task-id>
<tool-use-id>toolu_01FeZGgtUYhQHL2zPD9hBmHy</tool-use-id>
<output-file>C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-
highlight-indexer\9f6357d8-0e35-491a-a830-052e094c7f37\tasks\w7ydrkzjj.output</output-file>
<status>completed</status>
<summary>Dynamic workflow "Scout all live docs across both repos, extract per-track status +
next steps, and synthesize a prioritized roadmap (what is actionable vs owner-gated)"
completed</summary>
<result>{"tracks":[{"track":"appliance/CUJ","docsRead":["C:/Users/avidu/Projects/khelsutra/docs/
LOCAL_APPLIANCE_ARCHITECTURE.md","C:/Users/avidu/Projects/khelsutra/docs/DRY_RUN_CUJ.md","C:/Use
rs/avidu/Projects/khelsutra/docs/PER_RALLY_SELECTION.md","C:/Users/avidu/Projects/khelsutra/docs
/REQUEST_ACCESS_FORM.md","C:/Users/avidu/Projects/khelsutra/docs/PORTABILITY.md","C:/Users/avidu
/Projects/khelsutra/NEXT_STEPS.md"],"statusSummary":"The appliance/CUJ track spans 5 docs plus
NEXT_STEPS. SHIPPED: (1) The dry-run CUJ goal (DRY_RUN_CUJ.md) is largely DONE ? $6 DoD boxes
1-3 are ? (Path A local en+hi reel-download done 2026-06-21; Path B/B-P2 in-UI
gs://bucket-&gt;process-&gt;12 rallies-&gt;reel-download live e2e done 2026-06-21 with
screencast). Only box 4 (option 3 ? same CUJ on the CPU droplet) remains, and the OWNER JUST
PARKED it (droplet un-provisioned, ~3.8GB RAM, no engine checkout; a real GPU machine needs a
subscription). The two engine bugs the runs surfaced are both fixed (engine #281 queued-job
hang; engine #289 compile-from-bucket 500). (2) PER_RALLY_SELECTION (the SPA back-half the CUJ
rides on) is M0/MVP COMPLETE ? P0-P5 ? (pick-&gt;query-&gt;select-&gt;compile-&gt;download on
existing endpoints, verified e2e + CUJ replay). (3) REQUEST_ACCESS_FORM: P0-P1 ?, P2 ?
(Worker+D1 built+tested, owner-deploy pending ? but NEXT_STEPS says the form is already LIVE on
Cloudflare with D1+email, so P2 status in REQUEST_ACCESS_FORM.md is stale). (4) PORTABILITY:
design + proven (Cloudflare&lt;-&gt;Node/SQLite VM adapter swap, provision/teardown scripts
tested live) ? no open tracker rows. IN PROGRESS / NOT STARTED: the broader
```

LOCAL_APPLIANCE_ARCHITECTURE (KhelSutra Studio) is still DRAFT/owner-gated ? \$9 tracker has only T0 ? and T12 ? (engine half done); T1-T11 are ? (none started); T13 (GPU host) and T14 (bucket-worker bridge) are ? owner-gated. Note the broader appliance is NOT the current active focus ? NEXT_STEPS names 'Friendly & Multilingual' (FRIENDLY_AND_MULTILINGUAL.md) as the active owner directive, a separate track.", "nextSteps": [{"step": "Confirm the bucket-&reel CUJ on the CPU droplet (Linux deploy target) ? from-scratch engine install or a proper deploy of the fixed engine (#281/#289), then replay the in-UI gs://&process-&rallies-&compile-&download flow through the gate.", "sourceDoc": "DRY_RUN_CUJ.md \$6 DoD box 4 (option 3) + \$10/\$4.3 droplet reality", "gated": "Blocked (PARKED by owner ? droplet un-provisioned, ~3.8GB RAM, no GPU; a real GPU machine needs a subscription)", "effort": "medium"}, {"step": "T1 ? Build the L1 mock engine + capability fixtures + browser-replay CUJ scripts (the hermetic &is 4-CUJ harness).", "sourceDoc": "LOCAL_APPLIANCE_ARCHITECTURE.md \$9 tracker T1 (depends T0; not gated)", "gated": "No", "effort": "medium"}, {"step": "T2 ? Appliance shell: Caddyfile + systemd units + firewall + edge auth + rotate the chat-shared GEMINI_API_KEY (P1).", "sourceDoc": "LOCAL_APPLIANCE_ARCHITECTURE.md \$9 tracker T2", "gated": "Owner (edge-auth choice \$10 Q1: Caddy basicauth vs cloudflared+Cloudflare Access)", "effort": "medium"}, {"step": "T3 ? L3 live posture test: assert raw :8000 / :9000/:9001/:9023 are refused from outside.", "sourceDoc": "LOCAL_APPLIANCE_ARCHITECTURE.md \$9 tracker T3 (depends T2; not gated)", "gated": "Blocked (on T2 + a deployed box)", "effort": "small"}, {"step": "T4 ? S1 ingest screen: local-upload wired (existing /api/upload/init|part|complete) + segmenter picker from /api/capabilities.", "sourceDoc": "LOCAL_APPLIANCE_ARCHITECTURE.md \$9 tracker T4 (depends T1; not gated)", "gated": "No", "effort": "medium"}, {"step": "T5 ? S2 progress screen: 3s poll of free-text stage + 'one job at a time' notice + restart-failure terminal state.", "sourceDoc": "LOCAL_APPLIANCE_ARCHITECTURE.md \$9 tracker T5 (depends T4)", "gated": "No", "effort": "small"}, {"step": "T6 ? S3: embed RallyPicker verbatim + compile + download (per-rally is one console screen).", "sourceDoc": "LOCAL_APPLIANCE_ARCHITECTURE.md \$9 tracker T6 (depends T4)", "gated": "No", "effort": "small"}, {"step": "T7 ? (engine) GET /api/capabilities + GET /api/health + UI capability banner. Note GET /api/health already EXISTS (engine M1/#268) per the 2026-06-21 correction, so the remaining net-new is /api/capabilities + the banner. (= PER_RALLY_SELECTION P6 family.)", "sourceDoc": "LOCAL_APPLIANCE_ARCHITECTURE.md \$9 T7 + PER_RALLY_SELECTION.md \$7 P6", "gated": "Owner (coord with active engine sessions)", "effort": "medium"}, {"step": "T8 ? (engine) GET /api/videos + GET /api/reels/{video_id} for S4 reload-survival (only single-row get_video exists today). (= PER_RALLY_SELECTION P8 family; needs identity scoping.)", "sourceDoc": "LOCAL_APPLIANCE_ARCHITECTURE.md \$9 T8 + PER_RALLY_SELECTION.md \$7 P8", "gated": "Owner (coord + identity-scoping gate)", "effort": "medium"}, {"step": "T9 ? S5 Settings screen: render config + capabilities; show the compute/storage target selector DISABLED ('single-box: local').", "sourceDoc": "LOCAL_APPLIANCE_ARCHITECTURE.md \$9 tracker T9 (depends T7,T8)", "gated": "Blocked (on T7,T8)", "effort": "small"}, {"step": "T10 ? Observability pass: correlation id (job_id/upload_id/request_id) threaded end-to-end + structured logs at every new path + a thin metrics surface; logs inspected during CUJ replay (DoD item).", "sourceDoc": "LOCAL_APPLIANCE_ARCHITECTURE.md \$9 T10 + \$5", "gated": "No", "effort": "medium"}, {"step": "T11 ? Retention sweep for output/ + uploads/ originals (currently never deleted) before any non-owner exposure.", "sourceDoc": "LOCAL_APPLIANCE_ARCHITECTURE.md \$9 tracker T11", "gated": "Owner (retention defaults \$10 Q2: auto-delete window / keep proxies+intervals only)", "effort": "small"}, {"step": "T12 ? Ingest from bucket URI (s3://, gs://) ? engine half DONE (/api/process accepts URIs, M2/#280, verified); remaining is the SPA ingest call (= B-P2). NOTE: B-P2 SPA ingest was SHIPPED 2026-06-21 (engine #56 + DRY_RUN_CUJ \$9), so T12 is effectively complete though the \$9 tracker still shows ? ? flag for status update.", "sourceDoc": "LOCAL_APPLIANCE_ARCHITECTURE.md \$9 tracker T12 (?) vs DRY_RUN_CUJ.md \$9 (B-P2 done)", "gated": "No (engine+SPA done; tracker row stale)", "effort": "small"}, {"step": "T13 ? Stand up a provider-agnostic GPU host + wire detector_impl=native (NativeWasbRunner); owner byte-parity gate (WSL CSV == native CSV) on a real GPU box.", "sourceDoc": "LOCAL_APPLIANCE_ARCHITECTURE.md \$9 tracker T13 (? owner; elevated to run in parallel) + \$10 Q4", "gated": "Blocked (? owner ? owner provisions host+token; native WASB on engine branch feat/native-wasb-runner, trajectory_hybrid still raises; PARKED-adjacent: needs a paid GPU machine/subscription)", "effort": "large"}, {"step": "T14 ? (reserved) Bucket-worker bridge: 'remote GPU as a setting' ? a net-new orchestration endpoint reflecting worker.infer outputs into the jobs/poll contract. (= PER_RALLY_SELECTION P9/P10-adjacent; reserved scale path.)", "sourceDoc": "LOCAL_APPLIANCE_ARCHITECTURE.md \$9 tracker T14 (? owner; depends T8,T12) + \$10 Q5", "gated": "Blocked (? owner ? P5 bridge ownership undecided)", "effort": "large"}, {"step": "PER_RALLY_SELECTION P6 ? (engine) GET /api/health + offline banner; /api/compile returns download_url (additive, low-risk; today returns a filepath only).", "sourceDoc": "PER_RALLY_SELECTION.md \$7 P6 (not

```

gated)","gated":"No","effort":"small"},{"step":"PER_RALLY_SELECTION P7 ? (engine) GET
/api/videos/{id}/rallies RESTful/cacheable alias returning the same {play_segments}
shape.", "sourceDoc":"PER_RALLY_SELECTION.md §7 P7 (not
gated)","gated":"No","effort":"small"},{"step":"PER_RALLY_SELECTION P9 ? (engine) source/clip
stream (Range or /clip?start=&end=) + timeline + per-rally preview/thumbnail (no streamable
source exists today, only the compiled reel).","sourceDoc":"PER_RALLY_SELECTION.md §7 P9
(depends P8)","gated":"Blocked (? owner; on P8)","effort":"large"}, {"step":"PER_RALLY_SELECTION
P10 ? (engine) PATCH /api/segments/{id} corpus correction loop (writes the dead
candidate_segments.human_label/notes columns).","sourceDoc
... (truncated 65367 chars, full result in C:\Users\avidu\AppData\Local\Temp\claude\C--Users-
avidu-Projects-badminton-highlight-
indexer\9f6357d8-0e35-491a-a830-052e094c7f37\tasks\w7ydrkzjj.output)</result>
<usage><agent_count>6</agent_count><subagent_tokens>529065</subagent_tokens><tool_uses>70</tool_
uses><duration_ms>274134</duration_ms></usage>
</task-notification>

```

443. assistant (2026-06-21T14:00:56.673Z)

The all-docs scan is back. Let me read the full result ? the synthesis and the tracks I haven't seen yet:

444. user (2026-06-21T14:00:57.882Z)

```

1      {
2      "summary": "Scout all live docs across both repos, extract per-track status + next
steps, and synthesize a prioritized roadmap (what is actionable vs owner-gated)",
3      "agentCount": 6,
4      "logs": [],
5      "result": {
6      "tracks": [
7      {
8      "track": "appliance/CUJ",
9      "docsRead": [
10     "C:/Users/avidu/Projects/khelsutra/docs/LOCAL_APPLIANCE_ARCHITECTURE.md",
11     "C:/Users/avidu/Projects/khelsutra/docs/DRY_RUN_CUJ.md",
12     "C:/Users/avidu/Projects/khelsutra/docs/PER_RALLY_SELECTION.md",
13     "C:/Users/avidu/Projects/khelsutra/docs/REQUEST_ACCESS_FORM.md",
14     "C:/Users/avidu/Projects/khelsutra/docs/PORTABILITY.md",
15     "C:/Users/avidu/Projects/khelsutra/NEXT_STEPS.md"
16     ],
17     "statusSummary": "The appliance/CUJ track spans 5 docs plus NEXT_STEPS. SHIPPED:
(1) The dry-run CUJ goal (DRY_RUN_CUJ.md) is largely DONE ? §6 DoD boxes 1-3 are ? (Path A local
en+hi reel-download done 2026-06-21; Path B/B-P2 in-UI gs://bucket->process->12 rallies->reel-
download live e2e done 2026-06-21 with screencast). Only box 4 (option 3 ? same CUJ on the CPU
droplet) remains, and the OWNER JUST PARKED it (droplet un-provisioned, ~3.8GB RAM, no engine
checkout; a real GPU machine needs a subscription). The two engine bugs the runs surfaced are
both fixed (engine #281 queued-job hang; engine #289 compile-from-bucket 500). (2)
PER_RALLY_SELECTION (the SPA back-half the CUJ rides on) is M0/MVP COMPLETE ? P0-P5 ?
(pick->query->select->compile->download on existing endpoints, verified e2e + CUJ replay). (3)
REQUEST_ACCESS_FORM: P0-P1 ?, P2 ? (Worker+D1 built+tested, owner-deploy pending ? but
NEXT_STEPS says the form is already LIVE on Cloudflare with D1+email, so P2 status in
REQUEST_ACCESS_FORM.md is stale). (4) PORTABILITY: design + proven (Cloudflare<->Node/SQLite VM
adapter swap, provision/teardown scripts tested live) ? no open tracker rows. IN PROGRESS / NOT
STARTED: the broader LOCAL_APPLIANCE_ARCHITECTURE (KhelSutra Studio) is still DRAFT/owner-gated
? §9 tracker has only T0 ? and T12 ? (engine half done); T1-T11 are ? (none started); T13 (GPU
host) and T14 (bucket-worker bridge) are ? owner-gated. Note the broader appliance is NOT the
current active focus ? NEXT_STEPS names 'Friendly & Multilingual' (FRIENDLY_AND_MULTILINGUAL.md)
as the active owner directive, a separate track.",
18     "nextSteps": [
19     {
20     "step": "Confirm the bucket->reel CUJ on the CPU droplet (Linux deploy
target) ? from-scratch engine install or a proper deploy of the fixed engine (#281/#289), then
replay the in-UI gs://->process->rallies->compile->download flow through the gate.",
21     "sourceDoc": "DRY_RUN_CUJ.md §6 DoD box 4 (option 3) + §10/§4.3 droplet
reality",

```

```

22         "gated": "Blocked (PARKED by owner ? droplet un-provisioned, ~3.8GB RAM, no
GPU; a real GPU machine needs a subscription)",
23         "effort": "medium"
24     },
25     {
26         "step": "T1 ? Build the L1 mock engine + capability fixtures + browser-
replay CUJ scripts (the hermetic <1s 4-CUJ harness).",
27         "sourceDoc": "LOCAL_APPLIANCE_ARCHITECTURE.md §9 tracker T1 (depends T0; not
gated)",
28         "gated": "No",
29         "effort": "medium"
30     },
31     {
32         "step": "T2 ? Appliance shell: Caddyfile + systemd units + firewall + edge
auth + rotate the chat-shared GEMINI_API_KEY (P1).",
33         "sourceDoc": "LOCAL_APPLIANCE_ARCHITECTURE.md §9 tracker T2",
34         "gated": "Owner (edge-auth choice §10 Q1: Caddy basicauth vs
cloudflared+Cloudflare Access)",
35         "effort": "medium"
36     },
37     {
38         "step": "T3 ? L3 live posture test: assert raw :8000 / :9000/:9001/:9023 are
refused from outside.",
39         "sourceDoc": "LOCAL_APPLIANCE_ARCHITECTURE.md §9 tracker T3 (depends T2; not
gated)",
40         "gated": "Blocked (on T2 + a deployed box)",
41         "effort": "small"
42     },
43     {
44         "step": "T4 ? S1 ingest screen: local-upload wired (existing
/api/upload/init|part|complete) + segmenter picker from /api/capabilities.",
45         "sourceDoc": "LOCAL_APPLIANCE_ARCHITECTURE.md §9 tracker T4 (depends T1; not
gated)",
46         "gated": "No",
47         "effort": "medium"
48     },
49     {
50         "step": "T5 ? S2 progress screen: 3s poll of free-text stage + 'one job at a
time' notice + restart-failure terminal state.",
51         "sourceDoc": "LOCAL_APPLIANCE_ARCHITECTURE.md §9 tracker T5 (depends T4)",
52         "gated": "No",
53         "effort": "small"
54     },
55     {
56         "step": "T6 ? S3: embed RallyPicker verbatim + compile + download (per-rally
is one console screen).",
57         "sourceDoc": "LOCAL_APPLIANCE_ARCHITECTURE.md §9 tracker T6 (depends T4)",
58         "gated": "No",
59         "effort": "small"
60     },
61     {
62         "step": "T7 ? (engine) GET /api/capabilities + GET /api/health + UI
capability banner. Note GET /api/health already EXISTS (engine M1/#268) per the 2026-06-21
correction, so the remaining net-new is /api/capabilities + the banner. (= PER_RALLY_SELECTION
P6 family.)",
63         "sourceDoc": "LOCAL_APPLIANCE_ARCHITECTURE.md §9 T7 + PER_RALLY_SELECTION.md
§7 P6",
64         "gated": "Owner (coord with active engine sessions)",
65         "effort": "medium"
66     },
67     {
68         "step": "T8 ? (engine) GET /api/videos + GET /api/reels/{video_id} for S4
reload-survival (only single-row get_video exists today). (= PER_RALLY_SELECTION P8 family;
needs identity scoping.)",
69         "sourceDoc": "LOCAL_APPLIANCE_ARCHITECTURE.md §9 T8 + PER_RALLY_SELECTION.md

```

```

§7 P8",
70         "gated": "Owner (coord + identity-scoping gate)",
71         "effort": "medium"
72     },
73     {
74         "step": "T9 ? S5 Settings screen: render config + capabilities; show the
compute/storage target selector DISABLED ('single-box: local').",
75         "sourceDoc": "LOCAL_APPLIANCE_ARCHITECTURE.md §9 tracker T9 (depends
T7,T8)",
76         "gated": "Blocked (on T7,T8)",
77         "effort": "small"
78     },
79     {
80         "step": "T10 ? Observability pass: correlation id
(job_id/upload_id/request_id) threaded end-to-end + structured logs at every new path + a thin
metrics surface; logs inspected during CUJ replay (DoD item).",
81         "sourceDoc": "LOCAL_APPLIANCE_ARCHITECTURE.md §9 T10 + §5",
82         "gated": "No",
83         "effort": "medium"
84     },
85     {
86         "step": "T11 ? Retention sweep for output/ + uploads/ originals (currently
never deleted) before any non-owner exposure.",
87         "sourceDoc": "LOCAL_APPLIANCE_ARCHITECTURE.md §9 tracker T11",
88         "gated": "Owner (retention defaults §10 Q2: auto-delete window / keep
proxies+intervals only)",
89         "effort": "small"
90     },

```

445. assistant (2026-06-21T14:01:04.492Z)

Let me jump to the synthesis (the prioritized roadmap) and the other-track summaries:

446. user (2026-06-21T14:01:09.683Z)

```

8:         "track": "appliance/CUJ",
177:        "track": "localization+UX",
277:        "track": "rally-quality",
378:        "track": "owned-model+strategy",
497:        "track": "cross-cutting",
570:        "synthesis": {
571:[Omitted long matching line]
572:[Omitted long matching line]
573:        "prioritizedNext": [
577:            "track": "localization+UX",
584:            "track": "localization+UX",
591:            "track": "localization+UX",
598:            "track": "cross-cutting",
605:            "track": "localization+UX",
612:            "track": "localization+UX",
619:            "track": "localization+UX",
626:            "track": "cross-cutting/serving",
633:            "track": "localization+UX",
640:            "track": "cross-cutting",
647:            "track": "owned-model",
654:            "track": "rally-quality",
661:            "track": "rally-quality",
668:            "track": "localization+UX",
675:            "track": "localization+UX",
682:            "track": "rally-quality + owned-model",
689:            "track": "rally-quality",
696:            "track": "owned-model + rally-quality",
703:            "track": "rally-quality + owned-model",
710:            "track": "appliance/CUJ",
717:            "track": "owned-model/strategy",

```



```
722:      "ownerDecisionsNeeded": [  
733:[Omitted long matching line]
```

447. user (2026-06-21T14:01:15.328Z)

```
570      "synthesis": {  
571        "headline": "Across five tracks the picture is: lots of slices have shipped but  
the umbrella/register docs lag reality, and most of the heaviest remaining work is  
owner/GPU/corpus-gated. On the track I've been driving (appliance/CUJ + localization), the dry-  
run CUJ is essentially DONE end-to-end ? Path A (local en/hi reel download) and Path B/B-P2 (in-  
UI gs:// bucket -> process -> 12 rallies -> reel download) are both live with a screencast  
(khelsutra/docs/DRY_RUN_CUJ.md §6), the two engine bugs are fixed (#281, #289), and the web SPA  
is fully trilingual en+kn+hi (FRIENDLY_AND_MULTILINGUAL.md §7). The only CUJ box left (option 3,  
CPU droplet) is PARKED by the owner. The single most sensible next move is therefore to pivot  
off the parked droplet and onto the active owner directive ? the \"Friendly & Multilingual\"  
track ? starting with the two ungated, high-leverage rows that the SPA milestone unblocked: B3  
(the shared t() shim that carries the trilingual win to the site/ surface) and A1 (the plain-  
language en-catalog copy pass). Everything bigger (broader KhelSutra Studio appliance T1-T11,  
rally-quality, owned-model) stays owner-gated or corpus/GPU-blocked.",  
572      "theTrackIveBeenOn": "DONE now on this track: (1) Dry-run CUJ ? Path A (local  
en+hi reel download) and Path B/B-P2 (in-UI gs:// bucket -> process -> 12 rallies -> compile ->  
reel download) are both live e2e with screencast; engine #281 (queued-job hang) and #289  
(compile-from-bucket 500) fixed (khelsutra/docs/DRY_RUN_CUJ.md §6 boxes 1-3 ?, §9). (2)  
PER_RALLY_SELECTION M0/MVP complete, P0-P5 ? (the SPA back-half the CUJ rides on). (3) SPA fully  
trilingual en+kn+hi ? Kannada-default browser auto-renders Kannada, switcher walks kn->hi->en,  
proven by Playwright screencast; B1/B2/B4/B5/B6 ? (FRIENDLY_AND_MULTILINGUAL.md §7). (4)  
PORTABILITY design proven (Cloudflare->Node/SQLite adapter swap). The ONE remaining CUJ box  
(option 3 ? same CUJ on the CPU droplet) is PARKED by the owner (droplet un-provisioned, ~3.8GB  
RAM, no GPU; real GPU needs a subscription) ? do not pursue it. Concrete next steps in order: 1)  
B3 ? build the tiny shared t(key,vars)/data-i18n shim for site/ (en-only first), proving the  
shared catalogs work across both surfaces (FRIENDLY_AND_MULTILINGUAL.md §7 B3); ungated,  
unblocks A3 and A6 and is the only thing between the SPA milestone and a trilingual site/. NOTE:  
catalogs currently live at web/src/i18n/locales/{en,kn,hi}/common.json, NOT a shared top-level  
dir ? B3 must hoist them or have the site shim import from there (cross-surface sharing premise  
not yet realized). 2) A1 ? plain-language copy pass on the en catalog (de-jargon banners/footer,  
icon+text, non-color selection indicator); ungated, high-impact/small-effort (§7 A1, §4 UX-  
levers). 3) A2 ? localized WhatsApp share for the downloaded reel, SPA+site (§7 A2; no wa.me  
wiring exists today). 4) A4 ? mobile-first touch targets + make tooltip text visible on phones  
(§7 A4). 5) A5 ? friendly engine-error mapping over raw FastAPI detail (§7 A5). 6) A8 ?  
background-processing reassurance + icon-chip status (§7 A8). 7) A9 tap-only filter controls  
(large; §7 A9 + §6.5 query-parsing merge-gate). 8) A7 (camera-placement diagram) then A3/A6 once  
B3 lands. THEN owner-gated quality gates before any public launch: native-speaker review of the  
AI-drafted kn/hi catalogs, and owner-machine font eyeball (cloud CI can't render Noto fonts).  
The broader appliance (T1-T11 in LOCAL_APPLIANCE_ARCHITECTURE.md §9) is NOT the active focus and  
stays DRAFT/owner-gated.",  
573      "prioritizedNext": [  
574        {  
575          "rank": 1,  
576          "step": "B3 ? shared t(key,vars)/data-i18n shim for site/ (en-only first),  
hoisting or importing the existing web/src/i18n/locales catalogs so both surfaces share them.",  
577          "track": "localization+UX",  
578          "gated": "No",  
579          "why": "Highest-priority remaining UX row; ungated; unblocks A3/A6 and carries  
the shipped trilingual SPA win to the English-only site/ funnel (index/capture/own-model.html).  
Source: FRIENDLY_AND_MULTILINGUAL.md §7 B3.",  
580        },  
581        {  
582          "rank": 2,  
583          "step": "A1 ? plain-language copy pass on the en catalog (de-jargon,  
icon+text, non-color selection indicator).",  
584          "track": "localization+UX",  
585          "gated": "No",  
586          "why": "High impact / small effort, depends only on B2 (done). Directly serves  
the non-tech-savvy directive. Source: FRIENDLY_AND_MULTILINGUAL.md §7 A1, §4 UX-levers."  
587        },
```

```

588         {
589             "rank": 3,
590             "step": "A2 ? localized WhatsApp share for the downloaded reel (SPA + site).",
591             "track": "localization+UX",
592             "gated": "No",
593             "why": "High/small; WhatsApp is THE distribution channel for the persona; no
wa.me wiring exists yet. Source: FRIENDLY_AND_MULTILINGUAL.md §7 A2."
594         },
595         {
596             "rank": 4,
597             "step": "Refresh DOC_STATUS §3a/§4 + NEXT_STEPS to reflect
COMPUTE_DECOUPLED_SERVING M5/M6/M8 + bucket-ingest B-P1/B-P2 merged on origin/master; mark
khelsutra DRY_RUN_CUJ delivered in khelsutra README.",
598             "track": "cross-cutting",
599             "gated": "No",
600             "why": "Cheap doc hygiene; the register's 'M1-M4 SHIPPED / M5+ authorized'
line and khelsutra README '? active goal' are now factually wrong, misleading every cold start.
Source: engine docs/DOC_STATUS.md §3a row 33, khelsutra docs/README.md."
601         },
602         {
603             "rank": 5,
604             "step": "A4 ? mobile-first touch targets + make tooltip text visible on phones
(SPA).",
605             "track": "localization+UX",
606             "gated": "No",
607             "why": "Ungated, core to non-tech mobile users; no min-h/touch-target sizing
in web/src yet. Source: FRIENDLY_AND_MULTILINGUAL.md §7 A4."
608         },
609         {
610             "rank": 6,
611             "step": "A5 ? friendly engine-error mapping over raw FastAPI detail (SPA
client-side map).",
612             "track": "localization+UX",
613             "gated": "No",
614             "why": "Ungated; stops CompileResult from surfacing untranslated raw engine
text to non-tech users. Source: FRIENDLY_AND_MULTILINGUAL.md §7 A5."
615         },
616         {
617             "rank": 7,
618             "step": "A8 ? background-processing reassurance + verdict/status as icon chips
(SPA).",
619             "track": "localization+UX",
620             "gated": "No",
621             "why": "Ungated; sets honest expectations on slow Indian uplinks. Source:
FRIENDLY_AND_MULTILINGUAL.md §7 A8."
622         },
623         {
624             "rank": 8,
625             "step": "Codify the GCP Ops Agent into deploy/gcp (finish + PR the current
local branch feat/gcp-ops-agent-and-dryrun-guide).",
626             "track": "cross-cutting/serving",
627             "gated": "No",
628             "why": "Ungated, already in-flight locally (1 ahead of origin/master); owner
directive missed twice ? bake observability into every VM provision. Source: engine NEXT_STEPS
P1 #6."
629         },
630         {
631             "rank": 9,
632             "step": "A7 ? add a camera-placement diagram to fix English-source recording
gaps before localizing.",
633             "track": "localization+UX",
634             "gated": "No",
635             "why": "Ungated, no dep; diagram is reusable across all 3 languages so do it
before A3. Source: FRIENDLY_AND_MULTILINGUAL.md §7 A7."
636         },

```

```

637         {
638             "rank": 10,
639             "step": "Per-video workspace P1-P6 ? author the v6 DB migration ONCE jointly
with M8's v6 cost-ledger migration; route detector/+tmp/ per video.",
640             "track": "cross-cutting",
641             "gated": "No",
642             "why": "Ungated engine work; must be co-authored with M8 to avoid a migration-
ladder collision (both want user_version=6). Source: engine NEXT_STEPS P1 #5, DOC_STATUS $3a
PER_VIDEO_WORKSPACE."
643         },
644         {
645             "rank": 11,
646             "step": "M0.3 compliance cleanup ? purge the Gemini-silver training path in
backend/eval/distill_local.py + calibrate_local.py threshold distillation; retarget to human +
open-teacher labels.",
647             "track": "owned-model",
648             "gated": "No (Claude-doable code; sequence before any training run)",
649             "why": "Hard guardrail blocker (paid-LLM-as-training-source) that is pure
local code work, unlike the rest of M0.x; verified distill_local.py still present. Source:
OWNED_MODEL_IMPLEMENTATION_PLAN.md M0.3(i).\"
650         },
651         {
652             "rank": 12,
653             "step": "W3 ? record the standing decision to keep the scorer parameter-light
(numpy LogReg; no learned temporal net until corpus >> now); doc-only.",
654             "track": "rally-quality",
655             "gated": "No",
656             "why": "Actionable now, zero risk, captures TemporalMaxer evidence so it isn't
relitigated. Source: RALLY_QUALITY_RESEARCH.md §7 W3.\"
657         },
658         {
659             "rank": 13,
660             "step": "Golden Regression Fixtures P0 + M4 ? leak remediation (scrub real
player/venue names + personal paths in eval_baselines + add CI leak-scan) and the quality-floor
gate / first regression_baseline.json.",
661             "track": "rally-quality",
662             "gated": "No (owned-data hygiene)",
663             "why": "Actionable now, owned-data hygiene the docs flag as already-
actionable; P3 real Tier-0 commit stays owner-gated. Source: GOLDEN_REGRESSION_FIXTURES.md §7
P0/M4.\"
664         },
665         {
666             "rank": 14,
667             "step": "Native-speaker review of the AI-drafted kn + hi catalogs
(web/src/i18n/locales/{kn,hi}/common.json) before any wide public launch.",
668             "track": "localization+UX",
669             "gated": "Owner (native-speaker-paced)",
670             "why": "Green CI proves key-parity+render, NOT translation quality (Gemini
drafts are inference-only per guardrail); blocks public launch but not internal progress.
Source: FRIENDLY_AND_MULTILINGUAL.md §6/$9.\"
671         },
672         {
673             "rank": 15,
674             "step": "Owner-machine font eyeball ? set OS/browser to Kannada then Hindi,
confirm both surfaces render the right language AND Noto fonts display.",
675             "track": "localization+UX",
676             "gated": "Owner (owner-machine)",
677             "why": "Final font gate; cloud CI containers can't render fonts. Source:
FRIENDLY_AND_MULTILINGUAL.md §6.5/$9, I18N_PLAN.md §6.\"
678         },
679         {
680             "rank": 16,
681             "step": "Grow the golden corpus toward n>=16 (multi-court badminton + >=3
matches/sport, table-tennis first; adults-only).\",
682             "track": "rally-quality + owned-model\",

```

```

683         "gated": "Owner (owner labels; GPU/WSL owner-side)",
684         "why": "Single highest-leverage lever ? gates the multi-feature scorer, M2
calibration, the R2-as-gate swap, full-corpus training, and all Tier-3 bets. Source: NEXT_STEPS
P0 #2, RALLY_QUALITY_RESEARCH §7 row C."
685     },
686     {
687         "rank": 17,
688         "step": "Wire rally_gate Gate-A (net-crossings >=2) into the served path as a
config knob (fusion.min_crossings exists default-OFF); promote only on measured golden lift.",
689         "track": "rally-quality",
690         "gated": "Owner (needs golden re-measure to promote)",
691         "why": "Named the cheapest, highest-confidence precision (G2) win for the
held-shuttle/service-feed structural bug. Source: RALLY_DETECTION_GAP_CLOSURE.md §7 P2,
MULTI_SIGNAL_FUSION_PLAN.md §10 P2."
692     },
693     {
694         "rank": 18,
695         "step": "Set the #172 ship-gate target (e.g. LOVO F1@tIoU0.5 >= 0.70 on multi-
court folds, paired-sign p<0.05 vs heuristic).",
696         "track": "owned-model + rally-quality",
697         "gated": "Owner (needs corpus growth to inform)",
698         "why": "Currently undefined; both the owned-model DoD and the rally-quality
gap-closure % depend on a ratified bar. Source: OWNED_MODEL_IMPLEMENTATION_PLAN.md M0.4,
NEXT_STEPS #2/#7."
699     },
700     {
701         "rank": 19,
702         "step": "Extract GPU golden person/audio features + .rallies.csv labels, then
run the W2/P3 multi-feature LOVO ablation (per-cue ?F1 + CI + sign-test) on served windowing.",
703         "track": "rally-quality + owned-model",
704         "gated": "Blocked (GPU-extracted golden features not in checkout; owner-
side)",
705         "why": "The next measured detection step but physically blocked on owner
GPU/WSL artifacts; copy JSONs to output/ to satisfy the skipif litmus. Source:
RALLY_DETECTION_GAP_CLOSURE.md §7 P3, NEXT_STEPS GPU-box pickup."
706     },
707     {
708         "rank": 20,
709         "step": "Greenlight + execute the broader appliance T1-T11 (mock-engine CUJ
harness, appliance shell/edge-auth, ingest/progress/picker screens, observability, retention).",
710         "track": "appliance/CUJ",
711         "gated": "Owner (LOCAL_APPLIANCE_ARCHITECTURE DRAFT; several rows need §10
Q1/Q2 answers)",
712         "why": "The KhelSutra Studio umbrella is owner-gated and explicitly NOT the
active focus; defer behind the localization/UX directive. Source:
LOCAL_APPLIANCE_ARCHITECTURE.md §9 T1-T11."
713     },
714     {
715         "rank": 21,
716         "step": "Rename the repo off 'badminton' (multisport); resolve WASB C3 weight-
provenance before any model sharing.",
717         "track": "owned-model/strategy",
718         "gated": "Owner (+counsel for C3)",
719         "why": "Owner-emphatic ASAP before broad external sharing; C3 is a commercial-
ship blocker that multiplies per sport. Source: OWNED_MODEL_IMPLEMENTATION_PLAN.md Decisions #5,
COMMERCIALIZATION C3."
720     }
721 ],
722 "ownerDecisionsNeeded": [
723     "Confirm the localization/UX directive remains the sole active focus now that
the dry-run CUJ is delivered and the droplet (option 3) is PARKED ? i.e. proceed B3 ->
A1/A2/A4/A5/A8 -> A7 -> A3/A6, and treat the broader KhelSutra Studio appliance (T1-T11) as
deferred.",
724     "Native-speaker review of the AI-drafted kn + hi catalogs
(web/src/i18n/locales/{kn,hi}/common.json) ? owner/native-speaker-paced; the only thing between

```

the trilingual milestone and a public launch on the translation-quality axis.",

725 "Run the owner-machine font eyeball (Kannada then Hindi, both surfaces) ? cloud CI can't render Noto fonts; final §9 DoD gate.",

726 "Set the #172 ship-gate F1@tIoU target (proposed LOV0 F1@tIoU0.5 >= 0.70 on multi-court folds, paired-sign p<0.05) ? gates both rally-quality DoD and owned-model M0.4; needs corpus growth to inform.",

727 "Grow the golden corpus (owner labels multi-court badminton + >=3 matches/sport, TT first, adults-only) ? unblocks the multi-feature scorer, M2 calibration, the R2-as-gate swap, full-corpus training, and all Tier-3 bets simultaneously.",

728 "Greenlight the next Phase-0 owned-model milestone(s) (plan recommends M0.1 event-index DB migration then M0.4 scorer swap, prefer TemporalMaxer) ? nothing beyond greenlit M0.x starts without explicit go.",

729 "D-BACKEND-DEPTH (B7): how deep engine-side customer-facing backend localization goes for v1 ? recommendation is customer_* report text + validation-rejection reasons first, deferred phase-two engine-repo PR.",

730 "Approve starting RALLY_DETECTION_GAP_CLOSURE P1-P6 (Gate-A wiring + person-feature extraction + Lever A/B/C), one PR per row, default-OFF, promote only on measured golden lift.",

731 "Rename the repo off 'badminton' (owner action, before broad external sharing) and authorize the WASB C3 weight-provenance resolution path (own-trained vs clean MIT/Apache/BSD detector swap, counsel sign-off)."

732],

733 "servingNote": "The Cloud-Run/serving track (COMPUTE_DECOUPLED_SERVING ? M1-M6+M8 merged on origin/master, M5/M6/M8 = ? residual, M7/M9/M10/M11/M12 = ?) is being scanned in a separate session/scan; defer to that for the serving milestone detail. The only serving items surfaced here are the two cheap ungated touch-points that overlap this register: (a) refreshing DOC_STATUS §3a/§4 + NEXT_STEPS so the stale 'M1-M4 SHIPPED / M5+ authorized' line matches the merged reality, and (b) finishing/PR-ing the in-flight GCP Ops Agent codification on feat/gcp-ops-agent-and-dryrun-guide. Do not duplicate the M5-M12 driving here. Source: engine COMPUTE_DECOUPLED_SERVING/README.md §7, DOC_STATUS.md §3a row 33.",

734 "staleFlags": [

735 "LOCAL_APPLIANCE_ARCHITECTURE.md §9 tracker T12 still ? ('engine done; SPA pending') but the SPA half (B-P2 in-UI bucket ingest) shipped 2026-06-21 (web PR #56, recorded in DRY_RUN_CUJ.md §9 + git 4525c5d) ? T12 should be ?. The whole doc header still says 'Status: DRAFT' / T0 ? while shipped slices (T12, RallyPicker M0) understate real progress.",

736 "REQUEST_ACCESS_FORM.md §6 P2 marked ? ('owner runs wrangler d1 create + deploy') but NEXT_STEPS 'Where we are' says the form is already LIVE on Cloudflare (Worker hidden-king-khelsutra + D1 khelsutra-access) persisting + emailing per registration ? P2 (and the email part of P3) read as already-deployed; tracker stale.",

737 "LOCAL_APPLIANCE_ARCHITECTURE.md §4.6 reuse map / §11 DoD still list GET /api/health under 'net-new' though the §3 inline 2026-06-21 correction confirms it now EXISTS (engine M1/#268) ? residual inconsistency. Both PER_RALLY_SELECTION.md and LOCAL_APPLIANCE_ARCHITECTURE.md carry stale 'Last updated: 2026-06-20' headers despite 2026-06-21 bodies.",

738 "FRIENDLY_AND_MULTILINGUAL.md §7 tracker stops at PR #48-#49; newer adjacent work merged unrecorded ? #51 (localized sample-query prompt + i18n query-parse test rule, partially advances A9) and #52/#44 (localized academy flyer-kn/flyer-hi.html + nav). A9 should be noted partially-started; decide whether the standalone flyers count toward B3/A3.",

739 "Catalog-location drift: I18N_PLAN.md §2.2 and FRIENDLY_AND_MULTILINGUAL.md §4 diagram locales/ at repo root (D-CATALOG-HOME), but shipped catalogs live under web/src/i18n/locales/{en,kn,hi}/common.json and are NOT yet a shared top-level dir ? B3's cross-surface-sharing premise is unrealized (flag, B3 must hoist or import).",

740 "I18N_PLAN.md §3 phase table + §5 key-files still name backend/api/static as the P1 target though that surface is retired (UI_PROPOSAL B5 / D3); only the top banner reconciles it.",

741 "engine DOC_STATUS.md §3a row 33 + §4 #6 say COMPUTE_DECOUPLED_SERVING is 'M1-M4 SHIPPED; M5+ authorized', but origin/master merged M5 (#286), M6 (#287), M8 (#288), B-P1 guard (#284), bucket-ingest fix (#289); last full review (2026-06-21) predates them. NEXT_STEPS P0 #1 / P1 #4 still describe M5-M12 as upcoming. The project's own §7 tracker IS current.",

742 "khelsutra docs/README.md still bills DRY_RUN_CUJ.md as the open '? active goal' though its §6 DoD is fully ? (Path A + Path B live, #57); also does not index docs/ANNOTATE_AND_TRAIN.md (DRAFT, owner-gated).",

743 "RALLY_QUALITY_RESEARCH.md §7 says golden 'n=6' with W1.5/W2 ? and 'multi-feature scorer next', but QUALITY_ITERATIONS / R1_MILESTONE_LAUNCH_REPORT / GOLDEN_REGRESSION_FIXTURES all state n=15, R1 milestone (W1 cap6 + R4) shipped DEFAULT-ON

(#204), R2 metric implemented, R5/P7 shipped ? §7 should add R2/R3/R4/R5/P7 rows or point to RALLY_DETECTION_GAP_CLOSURE.md §7 as the live tracker. Bare corpus 'n' is cited inconsistently (6 / 13 / 15) across docs ? needs one canonical statement distinguishing manifest-size vs GPU-feature-extracted vs golden-scored.",
744 "NEXT_STEPS.md rally-quality detail body is dated 2026-06-13/06-15 (pre-R1/R2/R5/P7) though the top 'Where we are' was refreshed 2026-06-21 ? reads as if W1/W2 are the frontier when the live frontier is RALLY_DETECTION_GAP_CLOSURE.md (2026-06-18). RALLY_QUALITY_RESEARCH §7.1 still carries a 'TODO: compute exact % gap-to-Gemini closed' and Gemini golden intervals live only in the owner-side scorer DB, so the DoD's headline gap-closure % is uncomputed.",

448. assistant (2026-06-21T14:02:19.921Z)

Both scans are in. Here's the consolidated picture and the next steps ? distilled from the full doc corpus across both repos.

Where things stand on the track we've been driving

The ****dry-run CUJ is done end-to-end**** (Path A + Path B, screencast, both engine bugs fixed), the ****SPA is fully trilingual****, and PER_RALLY_SELECTION M0 is complete. The ****only remaining CUJ box (droplet) you just parked.**** So the track naturally pivots to its sibling ? the ****"Friendly & Multilingual" directive**** (`FRIENDLY\_AND\_MULTILINGUAL.md`), which ****NEXT_STEPS**** names as the active owner focus and which the trilingual-SPA milestone just unblocked.

Next steps ? actionable now (ungated), in priority order

#	Step	Track	Source
1	**B3** ? shared `t()`/`data-i18n` shim so the **`site/` marketing pages go trilingual** (homepage/capture/own-model are still English-only). *Note: catalogs live at `web/src/i18n/locales/` ? B3 must hoist/import them; the "shared top-level `locales/`" premise isn't realized yet.* l10n/UX FRIENDLY §7 B3		
2	**A1** ? plain-language copy pass on the `en` catalog (de-jargon, icon+text) l10n/UX FRIENDLY §7 A1		
3	**A2** ? localized WhatsApp share for the reel (the persona's channel; no `wa.me` wiring exists) l10n/UX FRIENDLY §7 A2		
4	**Doc-hygiene PR** ? fix stale trackers (see below) cross-cutting DOC_STATUS/READMEs		
5	A4/A5/A8 ? mobile touch targets · friendly engine-error mapping · status chips l10n/UX FRIENDLY §7		
?	*(engine, but coordinate ? busy serving session)* M0.3 compliance cleanup (purge the Gemini-silver training path); finish the in-flight Ops-Agent codification PR owned-model / serving OWNED_MODEL M0.3 · NEXT_STEPS		

Owner-gated (your call unblocks the most)

- ****Native-speaker review**** of the AI-drafted kn/hi + an ****owner-machine font eyeball**** ? the two launch gates on the localization DoD.
- ****Grow the golden corpus**** (multi-court, ?3 matches/sport, adults-only) ? the single highest-leverage lever; unblocks the multi-feature scorer, M2 calibration, full-corpus training.
- Set the ****#172 ship-gate F1 target****; greenlight the next ****Phase-0 owned-model**** milestone; the broader ****KheIsutra Studio appliance (T1?T11)****; the ****repo rename + WASB-C3**** weights provenance.

Serving / Cloud Run ? don't touch

It's the ****concurrent session's active line**** (M1?M8 merged, M9 open as #290). The only overlap for us is the doc-hygiene refresh below ? no duplicating M5?M12.

? Stale docs the scan flagged (aligns with your earlier "stop the staleness" directive)

- `LOCAL\_APPLIANCE\_ARCHITECTURE.md` T12 still ? ? but ****B-P2 shipped (#56)**** ? should be ?; header still "DRAFT".
- `kheIsutra/docs/README.md` still bills `DRY\_RUN\_CUJ.md` as the open "? active goal" ? its DoD is now fully ?.
- engine `DOC\_STATUS.md`/`NEXT\_STEPS` still say serving is "M1?M4 shipped, M5+ authorized" ? but ****M5/M6/M8 merged****.
- `REQUEST\_ACCESS\_FORM.md` P2 ? ? but the form is ****live on Cloudflare****.

- `RALLY\_QUALITY\_RESEARCH.md` §7 says golden **n=6** while other docs say **n=15**; bare "n" cited as 6/13/15 inconsistently.

My recommendation: start with **B3** (highest-leverage ungated UX win ? carries the trilingual milestone to the public site), and bundle the **doc-hygiene cleanup** as a quick parallel PR since it directly serves your anti-staleness rule. Which would you like me to pick up?

449. user (2026-06-21T14:06:40.338Z)

Just so I understand there is no way from the UI to trigger the highlight generation? i.e. I need to login to the machine to trigger it and I can't share this page with alpha users to try out? Or that would work after the M12 completion? Or would that be T1-T11 ?

BTW an additional ask I have is to be able to localize the UI in Thai, Indonesian, Danish, Spanish, Chinese, Japanese and Korean languages. Can you make up a plan following the diligence of EN, KN and HI and write up a dedicated project proposal for the same so that I can approve and we can start that line of work in parallel.

I guess you can go ahead with B3, A1, A2, A4, A5, A8 and the doc hygiene work but I want to highlight that being able to trigger the tool from the UI is the goal, so feel free to help me prioritize it well after these are done.

450. assistant (2026-06-21T14:09:18.302Z)

Great questions ? let me address the conceptual one directly, then build the localization-expansion proposal with proper diligence. Kicking off the per-language i18n research and reading the template + font setup in parallel:

451. user (2026-06-21T14:09:33.852Z)

```
1      <!--
2      PROJECT DOC TEMPLATE ? copy this file to docs/<YOUR_PLAN>.md and fill it in.
3
4      A "tracked project doc" is how this repo carries substantial design/project work: it can
be
5      ITERATED on, its progress TRACKED, and it is MARKED DONE against explicit deliverables,
then
6      archived. Standardizes the emergent pattern in docs/RALLY_QUALITY_RESEARCH.md §7 and
7      docs/GOLDEN_REGRESSION_FIXTURES.md.
8
9      Rules of thumb:
10     - §7 (the progress tracker) is the SOURCE OF TRUTH ? one row per independently-shippable
PR.
11     - Keep it HONEST: tag each claim [verified] (checked against code), [researched] (needs
sign-off),
12     or [design] (proposed). The doc reflects real shipped state, never aspirational.
13     - It POINTS to detail (code anchors, research artifacts), it does not duplicate it.
14     - Sections marked OPTIONAL are included only when relevant (e.g. Foundation/Threat-model
for
15     research- or security/privacy-heavy work). Delete sections that do not apply.
16     - Move to docs/archives/ once Status = DONE (per the archive policy: only terminal-state
work).
17     -->
18
19     # <Project / Plan Title>
20
21     > Status: `DRAFT` ? `<owner-gated? | in progress>` . Owner: `<name>` .
Created: `<YYYY-MM-DD>` . Last updated: `<YYYY-MM-DD>`
22     > Lifecycle: `DRAFT ? IN PROGRESS ? DONE ? archived` (move to `docs/archives/` when
DONE)
23     > Tracking anchors: §7 progress tracker is the source of truth; indexed in
`docs/README.md`; pointer in memory `current-state` (and `docs/NEXT_STEPS.md` once work starts).
24     > Relation to existing docs: `<extends | supersedes | peer-of ?>`
```

```

25      > **Honesty note:** mark each claim `[verified]` / `[researched]` / `[design]`. Not
legal/financial advice where applicable.
26
27      ---
28
29      ## 0. TL;DR
30      <2?5 sentences: the goal, the approach, and the single most important caveat. Call out
any finding that is already actionable on its own.>
31
32      ## 1. Problem & goal
33      <Why now; the concrete user-facing outcome; what "good" looks like. Link the code/docs a
reader must read to get current.>
34
35      ## 2. Decisions locked
36      <Table ? capture answered questions so they are not relitigated.>
37
38      | # | Decision | Source / date | Implication |
39      |---|-----|-----|-----|
40      | D1 | ? | ? | ? |
41
42      ## 3. Foundation ? research / prior art *(OPTIONAL)*
43      <Include when the plan rests on external facts: cited research, legal/compliance,
benchmarks, prior-art survey. Mark researched-vs-verified; list any sign-off gates this
creates.>
44
45      ## 4. Design / architecture
46      <The actual plan. Data contracts, diagrams, and a REUSE MAP ? name the existing
modules/code this builds on so it is clear what is new vs reused.>
47
48      ## 5. Threat model / risk table *(OPTIONAL ? security / privacy / compliance work)*
49      <Who/what is protected vs NOT; residual risks stated plainly.>
50
51      ## 6. Honest limits ? what this does NOT do
52      <The false-confidence guardrail: what a green suite / a DONE status does NOT prove.>
53
54      ## 7. Deliverables & progress tracker   ? **source of truth**
55      Legend: ? Todo · ? In progress · ? Done · ? Blocked/gated. **One small PR per row**,
branched from `origin/master`, no stacking. Update the row (status + PR link) as work lands.
56
57      | ID | Deliverable | Depends on | Gated? | Status | PR |
58      |---|-----|-----|-----|-----|----|
59      | P0 | ? | ? | No | ? | ? |
60
61      ## 8. Open questions ? owner / external *(split blocking-gates from nice-to-knows; name
who decides)*
62      1. ?
63
64      ## 9. Definition of done
65      <Explicit acceptance checklist. When all ticked ? flip `Status ? DONE` and archive.>
66      - [ ] ?
67
68      ## 10. References
69      **Internal code anchors `[verified]`** `<paths>`. **Internal docs:** `<docs>`.
**External:** `<cited sources>`.
70
71      ---
72
73      ### Changelog
74      - `<YYYY-MM-DD>` ? `<what changed>`.
75

```

452. user (2026-06-21T14:09:34.027Z)

```

1      import { StrictMode } from "react";
2      import { createRoot } from "react-dom/client";

```



```

3      import App from "../App";
4      import "../i18n"; // initialize the i18next singleton before the app renders (tracker B1)
5      // Self-hosted OFL Noto fonts so Kannada/Hindi render crisply offline (tracker B4 /
decision D7) ?
6      // no Google Fonts runtime dependency. unicode-range-scoped, so only Indic codepoints
use them.
7      import "@fontsource-variable/noto-sans-kannada/index.css";
8      import "@fontsource-variable/noto-sans-devanagari/index.css";
9      import "../index.css";
10
11     const root = document.getElementById("root");
12     if (!root) throw new Error("#root not found");
13
14     createRoot(root).render(
15       <StrictMode>
16         <App />
17       </StrictMode>,
18     );
19

```

453. user (2026-06-21T14:12:22.347Z)

I now have all the authoritative data I need. Let me compile the report.

Localization Research: Adding 7 Languages (Thai, Indonesian, Danish, Spanish, Chinese)

****Scope:**** Extends the existing `en/kn/hi` i18n foundation (react-i18next + i18next-icu, ICU MessageFormat, self-hosted OFL Noto fonts, CI key-parity, native-speaker review) per `docs/I18N\_PLAN.md`. All facts below are sourced to authoritative references (CLDR v48 chart, SIL OFL, @fontsource, W3C/BCP-47, notofonts).

Key headline findings

- ****All 7 are LTR.**** No RTL layout work is needed (relevant because your current `hi`/`kn` are also LTR ? no new bidi handling).
- ****Spanish now has 3 cardinal plural forms**** in current CLDR: `{one, many, other}`. The `many` form is ****not**** a colloquial plural ? it triggers only on large/compact numbers (millions, scientific notation). This is the single most important authoring gotcha here.
- ****CJK + Thai need dedicated OFL Noto fonts; the 3 Latin-script languages (es, da, id) need no new font**** ? they're covered by your existing Latin UI font.
- ****CJK fonts are huge**** (~7?16 MB per weight unsubsetted). This is the dominant cost driver and the reason CJK should be phased last with lazy-loading.

Per-language table

Lang	BCP-47 code	CLDR **cardinal** plural categories	ICU authoring	Dir	Script	Dedicated font?	OFL?	@fontsource pkg	Notes
Thai	th	other (single-form)	one form only	LTR	Thai	No	No	No	(Latin) n/a (uses existing Latin font) Pure catalog add. ? OFL-1.1 @fontsource-variable/noto-sans-thai Small font (~Latin-ish, single script block). No-space word-wrap gotcha (below).
Indonesian	id	other (single-form)	one form only	LTR	Latin	No	No	No	(Latin) n/a (uses existing Latin font) Pure catalog add. **Danish** da one, other 2 forms LTR Latin No (Latin) n/a (uses existing Latin font) Same shape as English; `one` rule includes decimals (`n=1 or t!=0 and i=0,1`).
Spanish	es	one, many, other ?	3 forms	LTR	Latin	No	No	No	n/a (uses existing Latin font) `many` = millions/compact only. ~15?30% string expansion vs English. **Chinese** zh-Hans (decide vs zh-Hant) other (single-form) one form only LTR Han (Simplified) No Noto Sans SC ? OFL-1.1 @fontsource-variable/noto-sans-sc

****Script-variant decision required**** (below). Large font. |
| ****Japanese**** | `ja` | ****other**** (single-form) | one form only | LTR | Han+Kana | ****Noto Sans JP**** | ? OFL-1.1 | `@fontsource-variable/noto-sans-jp` | No region subtag needed. Large font. |
| ****Korean**** | `ko` | ****other**** (single-form) | one form only | LTR | Hangul | ****Noto Sans KR**** | ? OFL-1.1 | `@fontsource-variable/noto-sans-kr` | No region subtag needed. Large font (Hangul syllable blocks). |

1. BCP-47 codes + script-variant decisions

- ****`th`, `id`, `da`, `es`, `ja`, `ko`**** ? plain language subtag, ****no script or region subtag needed****. (Japanese and Korean each map to one script; no decision to make.)
- ****Chinese ? the one real decision.**** Per W3C BCP-47 guidance, when the distinction is ***script*** (Simplified vs Traditional), use the ****script subtag****, not a region code: prefer ****`zh-Hans`**** (Simplified) over **`zh-CN`**, and ****`zh-Hant`**** (Traditional) over **`zh-TW`**. This is the documented, increasingly-standard choice and is region-independent.
 - ****Recommendation:**** ship ****zh-Hans**** first (Simplified ? mainland + Singapore, the larger audience and your most likely market). Add ****zh-Hant**** later only if Taiwan/Hong Kong demand appears ? it's a ***separate catalog + a separate font (Noto Sans TC)***, not a free variant, because the character sets and many word choices differ. Do ****not**** label your catalog bare **`zh`**; key it **`zh-Hans`** so a future **`zh-Hant`** slots in cleanly under your existing supported-locales list.

2. CLDR cardinal plural categories (CLDR v48 chart ? authoritative)

- ****Single-form (`other` only):**** Thai, Indonesian, Chinese, Japanese, Korean. ICU authoring is trivial ? a single **`other`** arm; these languages do not grammatically inflect for number.
- ****Two-form (`one`, `other`):**** Danish ? same authoring shape as English. (CLDR **`one`** rule: **`n = 1 or t != 0 and i = 0,1`**.)
- ****Three-form (`one`, `many`, `other`):**** ****Spanish**** ?. The **`many`** arm exists in current CLDR (rule: triggers for integer millions / **`e`** (compact-exponent) cases ? **`i % 1000000 = 0 and v = 0`**, or compact exponent **`e ? 1.5`**). Practically: **`1 ? one`**, **`2 ? other`**, **`1000000 ? many`**. ****Every Spanish ICU plural message must include a **`many`** arm****, even if it usually mirrors **`other`**, or **i18next-icu/intl-messageformat** will fall back. This is recent enough that older translation tooling and many hand-written catalogs omit it ? flag it explicitly to your native-speaker reviewer and add it to the key-parity/lint check for **`es`**.

3. Text direction

****All 7 are LTR.**** Confirmed ? no RTL/bidi layout work, no **`dir="rtl"`** mirroring, no logical-property CSS audit needed for this batch. (If Arabic/Hebrew/Urdu ever come, that's a separate, much larger track.)

4. Fonts

****No new font (covered by your existing Latin UI font):**** Spanish, Danish, Indonesian ? these are pure catalog adds (Latin script, incl. Spanish **`ñ á í ó ú ü ¿`** and Danish **`æ ø å`**, all in any standard Latin font's Latin-1/Latin Extended coverage).

****Dedicated Noto fonts (all OFL-1.1 ? commercially bundleable under your §4.2(a) OFL-for-media exception; all on **`@fontsource`/`@fontsource-variable`**):****

Font	Script	OFL	Bundling	Size concern
--- --- --- --- ---				
Noto Sans Thai	Thai	? OFL-1.1	bundle freely	**Small** ? single Thai block (~few hundred glyphs); comparable to a Latin webfont. No subsetting headache.
Noto Sans SC	Simplified Chinese	? OFL-1.1	bundle freely	**Very large** ? full Noto Sans CJK SC is ~ **16 MB per weight** unsubsetted (OTF), near the 65,535-glyph OpenType ceiling.
Noto Sans JP	Japanese	? OFL-1.1	bundle freely	**Very large** ? same CJK family, comparable order of magnitude (~ **16 MB/weight depending on format).
Noto Sans KR	Korean	? OFL-1.1	bundle freely	**Very large** ? ~ 11,000+ Hangul syllable blocks plus Han; multi-MB/weight.

****Licensing:**** All Noto fonts (incl. the CJK family) are ****SIL Open Font License 1.1****. Google/SIL explicitly invite bundling into apps and devices; the only OFL restriction is you can't sell the fonts ***standalone***. This fits cleanly under the existing **`docs/I18N\_PLAN.md`** §4.2(a) "OFL-for-bundled-media" exception you already carved out for Noto Sans Kannada/Devanagari ? no new licensing decision required, just extend the same exception to these

4.

****The CJK subsetting concern (critical for an offline / air-gapped self-hosted bundle):****

- `@fontsource`` handles CJK by ****splitting each font into many small `unicode-range`-keyed `woff2` chunks**** (Latin fonts split into ~a handful; ****CJK fonts split into ~100+ chunks**** spanning the Han/Kana/Hangul ranges). The browser fetches only the chunks whose code points actually appear on screen ? so a typical UI loads a small fraction, not the whole 16 MB.
- ****But for an offline/air-gapped bundle this lazy-by-unicode-range model breaks down:**** you must ship ****the entire chunk set**** so any string renders without a network fetch. Plan for ****~5?16 MB per CJK language per weight on disk**** in the bundle. With 3 CJK languages (SC, JP, KR) × even 1?2 weights, that's ****tens of MB**** added to an otherwise small bundle ? this is the dominant size cost of this whole project and the reason CJK is phased last.
- ****Mitigations:**** (a) use the ****variable**** `@fontsource-variable`` packages (one file spans all weights, avoids Nx per-weight duplication); (b) consider **\*\*building a custom subset\*\*** restricted to the glyphs your reviewed `zh-Hans/ja/ko`` catalogs + dynamic data (rally counts, dates) actually use ? for a small fixed UI this can cut a CJK font from MBs to tens of KB, at the cost of a build step that must re-run whenever catalog text changes; (c) ****lazy-load CJK font CSS per active locale**** so non-CJK users never download it (works for online; for true air-gapped you still ship everything but only load on demand).

5. Rendering / layout gotchas

- ****Thai ? no spaces between words.**** Thai uses spaces as **phrase** delimiters, not word delimiters, so naive wrapping breaks mid-word. Browsers do dictionary-based line breaking but it's imperfect. Mitigations: `word-break: normal`` + `overflow-wrap: break-word`` as a safety net, optionally `line-break: loose/auto``, and avoid forcing breaks. Thai also has stacking combining marks (vowels/tones above/below) ? ensure line-height has headroom so diacritics aren't clipped.
- ****CJK line-breaking**** can break between (almost) any two characters, which is mostly fine, but watch ****kinsoku**** (forbidden line-start/end punctuation) ? `line-break: strict`` improves this. CJK glyphs are taller/denser; give generous `line-height``. CJK generally has ****no italic**** and reacts poorly to faux-bold ? rely on the variable-weight axis, not `font-style: italic``.
- ****Spanish string expansion**** ? expect ****~15?30% longer**** than English (and occasional longer single words). Buttons/labels/toasts must not be fixed-width or truncate; test the longest Spanish strings. (Your current `Rally Segment(s) Selected`` ? `t("compile.selected",{count})`` plural key is exactly the kind of element to stress-test.) Danish is similar but milder; Indonesian is roughly English-length.
- ****CJK/Thai are more *compact* horizontally**** than Latin, so the risk is the opposite (lots of whitespace), not overflow ? but ensure your layout doesn't assume a minimum string length either.

Recommended phasing

Order by **least new code/assets first**, then **one script-block font**, then **the heavy CJK + script-variant decision last**:

- ****Phase A ? Latin trio (catalog-only, zero new code/assets): `es`, `da`, `id`.****
No font, no lazy-loading, no bundle-size impact. The **only** engineering touch is making your ICU plural authoring + key-parity/lint check handle ****Spanish's 3-form `{one, many, other}`**** (Danish/Indonesian are 2-form/1-form and trivially fit existing tooling). Best first slice: highest ROI, proves the catalog pipeline scales beyond Indic without code changes. Spanish is the only one needing reviewer attention (the `many`` arm).
- ****Phase B ? Thai (`th`): one new small OFL font.****
Add ****Noto Sans Thai**** (`@fontsource-variable/noto-sans-thai``) ? small, single-script, no subsetting drama; reuses the exact font-bundling pattern you already established for Noto Kannada/Devanagari. Single-form plural (`other``). Add the Thai no-space word-wrap CSS safety net. Self-contained, low-risk.
- ****Phase C ? CJK (`zh-Hans`, `ja`, `ko`): the heavy slice ? needs new code/assets + a product decision.****
Requires: (1) the ****`zh-Hans` vs `zh-Hant` decision**** (recommend ship `zh-Hans`` only first); (2) **\*\*3 large OFL Noto fonts\*\*** (SC/JP/KR); (3) **\*\*per-locale lazy-loading of font CSS\*\*** so non-CJK users aren't penalized; (4) for the **\*\*offline/air-gapped bundle**, a CJK subsetting build step (or accept tens of MB). All single-form plural (`other``), so catalog authoring is the easy part ? the work is fonts + loading, not strings. Do this last and as its own PR(s); consider one PR per CJK language given the font weight.

Catalog-only vs new-code/asset summary

```
| Add | Catalog-only (code-free) | Needs new assets/code |
|---|---|---|
| `es` | catalog ? | minor: ICU `many` arm + lint/parity update for 3-form |
| `da` | catalog ? | none |
| `id` | catalog ? | none |
| `th` | catalog | **+1 OFL font** (Noto Sans Thai); Thai word-wrap CSS |
| `zh-Hans` | catalog | **+1 large OFL font** (Noto Sans SC) + lazy-load + script-variant
decision + offline subsetting |
| `ja` | catalog | **+1 large OFL font** (Noto Sans JP) + lazy-load + offline subsetting |
| `ko` | catalog | **+1 large OFL font** (Noto Sans KR) + lazy-load + offline subsetting |
```

Your existing **extensibility checklist** (``add locales/<code>/common.json ? i18n_draft ? human review ? add to supported-locales ? confirm font coverage ? CI key-parity green``) holds for all 7 ? the only checklist step that does real work is **"confirm font coverage"** for ``th`/`zh-Hans`/`ja`/`ko``, plus the **Spanish 3-form plural** wrinkle on the authoring side.

Sources

- Unicode CLDR v48 ? Language Plural Rules chart (cardinal categories per language): https://www.unicode.org/cldr/charts/48/supplemental/language_plural_rules.html
- Unicode CLDR ? Plural Rules spec: <https://cldr.unicode.org/index/cldr-spec/plural-rules>
- W3C i18n ? BCP-47 / Chinese script subtags (``zh-Hans`/`zh-Hant`` over ``zh-CN`/`zh-TW``): <https://www.w3.org/International/articles/bcp47/> and <https://www.w3.org/International/articles/language-tags/>
- IETF language tag (BCP-47) overview: https://en.wikipedia.org/wiki/IETF_language_tag
- SIL Open Font License (OFL) ? fonts list & terms: <https://openfontlicense.org/ofl-fonts/>
- Noto CJK license (SIL OFL 1.1) ? Google Open Source Blog + repo LICENSE: <https://opensource.googleblog.com/2017/04/noto-serif-cjk-is-here.html> and <https://github.com/notofonts/noto-cjk/blob/main/Sans/LICENSE>
- Noto CJK repo (SC/TC/HK/JP/KR variants, sizes): <https://github.com/notofonts/noto-cjk> and <https://github.com/notofonts/noto-cjk/blob/main/Sans/README.md>
- @fontsource variable packages (self-host, OFL): noto-sans-thai, noto-sans-sc, noto-sans-jp, noto-sans-kr ? e.g. <https://www.npmjs.com/package/@fontsource-variable/noto-sans-thai>, <https://fontsource.org/fonts/noto-sans-jp/install>, <https://fontsource.org/fonts/noto-sans-kr/install>
- W3C ? Thai line-breaking / no-space word wrap: <https://w3c.github.io/i18n-drafts/articles/typography/linebreak.en> and <https://www.w3.org/TR/thai-gap/>
- react-i18next + i18next-icu (ICU MessageFormat usage): <https://react.i18next.com/misc/using-with-icu-format> and <https://github.com/i18next/i18next-icu>

Verification notes (honest scope): CLDR categories, BCP-47 codes, OFL status, LTR, and @fontsource package availability are confirmed from the cited authoritative sources. **Exact self-hosted byte sizes** (the CJK ~7?16 MB/weight figures) are order-of-magnitude from notofonts/opensuse data ? confirm the precise per-package ``woff2`` chunk totals by running ``npm view @fontsource-variable/noto-sans-sc`` and inspecting ``node_modules`` before committing bundle-budget numbers to the project doc. **Font rendering** (Thai diacritic clipping, CJK kinsoku) needs an eyeball on the owner's machine, same caveat as the existing Kannada/Devanagari verification step in §6.

A note for the project doc: this batch fits your existing foundation with **zero changes** to the negotiation contract or catalog format ? the only foundation-level additions are (a) extending the OFL-for-media font exception to 4 more Noto fonts, (b) handling Spanish's 3-form plural in the ICU authoring + CI parity/lint check, and (c) a CJK lazy-load + offline-subset strategy. Everything else is "add a reviewed catalog," exactly as designed.

```
agentId: a636c498146ddc3fb (use SendMessage with to: 'a636c498146ddc3fb' to continue this agent)
<usage>subagent_tokens: 61803
tool_uses: 16
duration_ms: 169906</usage>
```

454. assistant (2026-06-21T14:13:27.893Z)

I have the verified i18n facts and the template. Let me answer your conceptual question first (below), and write the localization-expansion proposal. Branching off khelsutra master:

455. user (2026-06-21T14:13:31.045Z)

Switched to a new branch 'docs/multilingual-expansion-proposal'
on docs/multilingual-expansion-proposal @ 7750afc docs(dry-run): Path-B live e2e DONE ? in-UI
bucket?reel CUJ + screencast (DoD \$6 box 3 ?) (#57)

456. assistant (2026-06-21T14:13:37.888Z)

Now writing the proposal. On branch `docs/multilingual-expansion-proposal`:

457. assistant (2026-06-21T14:15:04.206Z)

Multilingual Expansion ? 7 new locales (th · id · da · es · zh-Hans · ja · ko)

```
> **Status:** `DRAFT` ? **owner-gated (approval needed to START this line)** . **Owner:** Avi  
Dullu . **Created:** 2026-06-21 . **Last updated:** 2026-06-21  
> **Lifecycle:** `DRAFT` ? IN PROGRESS ? DONE ? archived` (move to `docs/archives/` when DONE)  
> **Tracking anchors:** $7 progress tracker is the source of truth; index in `docs/README.md`;  
pointer in memory `current-state`; thread in `NEXT_STEPS.md` once work starts.  
> **Relation to existing docs:** **PEER-OF / extends**  
[`FRIENDLY_AND_MULTILINGUAL.md`](FRIENDLY_AND_MULTILINGUAL.md) (the `en`/`kn`/`hi` foundation)  
and the engine design `I18N_PLAN.md`. **Reuses that foundation verbatim** (react-i18next +  
i18next-icu + browser detector + key-parity/no-bare-strings CI + self-hosted OFL Noto via  
`@fontsource` + Gemini-draft?native-review). This is a **parallel** line: it adds locales, it  
does **not** re-architect i18n.  
> **Honesty note:** each claim tagged `[verified]` (checked vs the working tree this session),  
`[researched]` (sourced, needs sign-off), or `[design]` (proposed). Not legal advice ? the OFL-  
font carve-out is the owner's standing call (D7 in FRIENDLY).
```

0. TL;DR

Add ****Thai, Indonesian, Danish, Spanish, Chinese (Simplified), Japanese, Korean**** to the Studio SPA with the ****same rigor as en/kn/hi****. The good news ****[verified]****: the i18n foundation already scales ? ****zero changes to the negotiation contract or the catalog format****; adding a locale is "drop a reviewed `common.json` + (if non-Latin) bundle one OFL Noto font + keep CI key-parity green." The work splits cleanly into ****Phase A**** (es/da/id ? *catalog-only, no fonts*), ****Phase B**** (Thai ? *one small OFL font*), and ****Phase C**** (zh-Hans/ja/ko ? *large OFL CJK fonts + lazy-load + an offline-bundle subsetting decision + the zh script-variant call*). ****The single most important caveats:**** (1) ****native-speaker review per language**** is the real cost and the public-launch gate (Gemini drafts are *inference-only*, never the source of truth ? the standing guardrail); (2) ****CJK fonts add tens of MB**** to the otherwise-tiny air-gapped bundle; (3) ****Spanish now has a 3-form plural**** ({one, many, other}) ? the one authoring wrinkle.

1. Problem & goal

****Why now.**** en/kn/hi shipped and proved the foundation is locale-pluggable (a Kannada-default browser auto-renders Kannada; the switcher walks kn?hi?en). The owner wants to broaden reach beyond the India beachhead ? ****SE Asia (th, id), Europe (da, es), and East Asia (zh, ja, ko)****. ****Concrete outcome ("good")****: a user whose browser is in any of the 7 languages lands on the Studio (and, once `site/` is localized via FRIENDLY B3, the marketing pages too), and ? without help ? understands the product, runs the tool, and gets a reel, ****entirely in their language****, with the fonts rendering crisply offline.
****Read to get current:**** this doc \$3?\$4;
[`FRIENDLY\_AND\_MULTILINGUAL.md`](FRIENDLY_AND_MULTILINGUAL.md) (the foundation this rides on, esp. \$4 design + \$6.5 negotiation tests); `web/src/i18n/index.ts` (the detector/ICU/fallback config) + `web/src/main.tsx` (the `@fontsource` font-load pattern) +
`web/src/i18n/hygiene.test.ts` (key-parity + no-bare-strings + offline guards).

2. Decisions locked (inherited from FRIENDLY unless noted)

```
| # | Decision | Source / date | Implication |
|---|-----|-----|-----|
| ED1 | **Same framework, no re-architecture** ? react-i18next + i18next-icu + `i18next-browser-
  languagedetector`; one `en` source catalog + per-locale siblings; **always fall back to `en`** |
  FRIENDLY D-* / `i18n/index.ts` `[verified]` | Each new locale = add to `SUPPORTED_LOCALES` + a
  `common.json`; negotiation/format unchanged |
| ED2 | **Self-host OFL Noto via `@fontsource`** (the D7 OFL-for-media carve-out) ? **extend it
  to the 4 new Noto fonts** (Thai, SC, JP, KR) | FRIENDLY D7 (owner, 2026-06-21) | No new
  licensing decision; same `main.tsx` import + `unicode-range`-scoped pattern as kn/hi |
| ED3 | **Gemini-draft ? native-speaker review** (inference only, never training; every value
  flagged for review) | `CLAUDE.md` guardrail | Translation *quality* is human-gated + owner-
  paced; CI proves parity, not quality |
| ED4 | **CI key-parity + no-bare-strings stay merge gates** for every locale |
  `hygiene.test.ts` `[verified]` | A new catalog must match the `en` key set exactly or CI fails |
| ED5 | **All 7 are LTR** ? no RTL/bidi layout work in this batch | `[researched]` CLDR/W3C | No
  `dir="rtl"`, no logical-property audit; RTL (ar/he/ur) would be a separate, larger track |
| ED6 *(proposed)* | **Chinese = `zh-Hans` (Simplified) for v1**; `zh-Hant` (Traditional) is a
  *separate* catalog + font, added later only on demand | S8 Q2 `[design]` | Key the catalog `zh-
  Hans` (not bare `zh`) so `zh-Hant` slots in cleanly |
```

3. Foundation ? research / prior art `[researched]`

Per-language diligence (Unicode CLDR plural-rules chart; SIL OFL; `@fontsource`; W3C BCP-47 ? sources in §10):

```
| Lang | BCP-47 | CLDR **cardinal** plurals | Dir | Dedicated font (OFL) |
| @fontsource(-variable) `pkg` | Font size / notes |
|---|---|---|---|---|---|
| Thai | `th` | **`other`** (1 form) | LTR | **Noto Sans Thai** | `noto-sans-thai` | **small**
(single script block) · no-space word-wrap gotcha |
| Indonesian | `id` | **`other`** (1 form) | LTR | ? (Latin) | ? | pure catalog add |
| Danish | `da` | **`one, other`** (2) | LTR | ? (Latin) | ? | same shape as English (`æ ø å` in
Latin font) |
| Spanish | `es` | **`one, many, other`** (3) ? | LTR | ? (Latin) | ? | `many` =
millions/compact only; ~15?30% string expansion |
| Chinese (Simpl.) | **`zh-Hans`** | **`other`** (1 form) | LTR | **Noto Sans SC** | `noto-sans-
sc` | **very large** (~7?16 MB/weight unsubset) · script-variant decision |
| Japanese | `ja` | **`other`** (1 form) | LTR | **Noto Sans JP** | `noto-sans-jp` | **very
large** (CJK) |
| Korean | `ko` | **`other`** (1 form) | LTR | **Noto Sans KR** | `noto-sans-kr` | **very
large** (Hangul + Han) |
```

****Three findings that drive the design:****

1. ****Spanish's 3-form plural**** is the only authoring change. Current CLDR `es` cardinal = `{one, many, other}`; `many` fires only on large/compact numbers (e.g. millions), but every `es` ICU plural message must still carry a `many {?}` arm or `intl-messageformat` falls back. Flag it to the reviewer and extend the parity/lint check. (th/id/zh/ja/ko are ****single-form**** ? trivial; da is 2-form like en.)
2. ****Fonts split the batch.**** es/da/id need ****no new font**** (Latin). th needs one ****small**** OFL font. ****zh-Hans/ja/ko need large OFL CJK fonts**** ? all OFL-1.1 (bundleable under ED2), all on `@fontsource-variable`, but multi-MB each.
3. ****CJK + the offline guard collide.**** `@fontsource` serves CJK as ~100+ `unicode-range` `woff2` chunks, fetched on demand ? efficient **online**, but the SPA's ****air-gapped/offline guarantee**** (`hygiene.test.ts` "offline / self-contained") means we must ship ****all**** chunks so any glyph renders with no network ? ****tens of MB**** added with 3 CJK languages. Mitigations in §4.

4. Design / architecture

****Reuse map (what's reused vs new):****

- ***Reused unchanged:** the negotiation contract (persisted pref ? `?lang=` ? navigator ? `en`), the ICU catalog format, the `@fontsource` `unicode-range`-scoped self-host pattern (`main.tsx:7-8`), the key-parity + no-bare-strings + offline CI (`hygiene.test.ts`), the draft?review pipeline, the `` sync (`i18n/index.ts:59-65`).
- ***New:** 7 `common.json` catalogs; `SUPPORTED\_LOCALES` entries (`i18n/index.ts:18`); ****4 new Noto font imports****; an ****es 3-form plural**** allowance in the catalog + tooling; a ****CJK lazy-**

load + offline-subset** strategy (Phase C); Thai line-break CSS safety net.

- **? Catalog-home caveat `[verified]`:** catalogs today live at ``web/src/i18n/locales/{en,kn,hi}/common.json`` ? ****not**** a shared top-level ``locales/`` (the I18N_PLAN "D-CATALOG-HOME" diagram). This expansion adds siblings there. If FRIENDLY ****B3**** (the shared ``site/`` ``t()`` shim) hoists them to a shared dir first, these new catalogs move with them ? sequence-aware, not a blocker.

****Negotiation tests (per FRIENDLY \$6.5):**** every new locale gets an N1/N2-style test ? a browser set to that language ****auto-renders**** it (e.g. ``navigator.languages=["ja-JP"]`` ? ``ja``), region subtags normalize (``zh-Hans-CN`` ? ``zh-Hans``), and a missing key falls back to ``en``. ****Auto-selection is the promise; it's proven by a test, never assumed.****

****Phasing (least new code/assets first):****

- ****Phase A ? Latin trio (`es`, `da`, `id`): catalog-only, zero new assets.**** Highest ROI, proves the pipeline scales beyond Indic with no code. Only ``es`` needs the 3-form-plural touch.
- ****Phase B ? Thai (`th`): one small OFL font.**** Reuses the kn/hi font pattern exactly; add the no-space word-wrap CSS safety net. Self-contained, low-risk.
- ****Phase C ? CJK (`zh-Hans`, `ja`, `ko`): the heavy slice.**** Needs the zh-Hans/Hant decision, 3 large OFL fonts, ****per-locale lazy font-load**** (so non-CJK users never download CJK), and an ****offline-bundle subset strategy**** (ship-all-chunks vs a custom subset build step vs online-lazy-only). One PR per CJK language.

5. Privacy / scope note

i18n is ****presentation-only**** ? no new personal-data collection, no biometrics, minors-excluded posture unchanged (same as FRIENDLY \$5). Gemini-draft translation is inference, not training, with no user data in the loop. CJK/Thai font bundling is OFL media, not code/weights.

6. Honest limits ? what this does NOT do

- A green CI / DONE proves ****key-parity + render + no-bare-strings + auto-negotiation****, ****not**** translation **quality** (native-review-gated, ED3) or correct font rendering (owner-eyeball-gated, like kn/hi).
- ****`zh-Hant` (Traditional) is out of v1**** (separate catalog + font); only ``zh-Hans`` ships first.
- ****No RTL**** ? this batch is all LTR.
- The ****CJK offline bundle is large**** (tens of MB) unless a subset build is adopted ? a real size trade-off, not free.
- ``site/`` marketing localization for these 7 ****depends on FRIENDLY B3**** landing first (the shared shim); until then this is SPA-only.

7. Deliverables & progress tracker ? ****source of truth****

Legend: ? Todo · ? In progress · ? Done · ? Blocked/gated. ****One small PR per row****, branched from ``origin/master``, no stacking.

ID	Deliverable	Phase	Depends on	Gated?	Status	PR
E0	This proposal doc + inbound pointers (README/NEXT_STEPS/memory)				?	?
EA1	**Spanish (`es`)** catalog (Gemini-draft) + ICU **3-form `{one,many,other}`** handling + extend key-parity/no-bare-strings tooling for the <code>`many`</code> arm + negotiation test	A	E0	No	(after approval)	?
EA2	**Danish (`da`)** catalog + negotiation test	A	E0	No	?	?
EA3	**Indonesian (`id`)** catalog + negotiation test	A	E0	No	?	?
EB1	**Noto Sans Thai** bundle (OFL, <code>`@fontsource-variable`</code>) + Thai no-space word-wrap CSS safety net	B	E0	No (ED2 carve-out applies)	?	?
EB2	**Thai (`th`)** catalog ? render + negotiation test	B	EB1	No	?	?
EC0	**Decision + design:** <code>`zh-Hans`</code> vs <code>`zh-Hant`</code> (ED6) + CJK **lazy per-locale font-load** + **offline-bundle subset** strategy (ship-all-chunks vs custom subset vs online-lazy)	C	E0			
	Owner (Q2/Q3)	?	?	?		
EC1	**Noto Sans SC** + **`zh-Hans`** catalog + lazy-load + negotiation test	C	EC0	No	(after EC0)	?
EC2	**Noto Sans JP** + **`ja`** catalog + lazy-load + negotiation test	C	EC0	No	?	?
EC3	**Noto Sans KR** + **`ko`** catalog + lazy-load + negotiation test	C	EC0	No	?	?

EX1	****Native-speaker review**** of each AI-drafted catalog (one tracked row; per-language sign-off)	?	the catalog rows	****Owner (reviewer sourcing)****	?	?
EX2	****Owner-machine font eyeball**** per new script (th, zh, ja, ko) ? cloud CI can't render fonts	?	EB*/EC*	****Owner (owner machine)****	?	?
EX3	****`site/` marketing parity**** for the 7 (rides FRIENDLY ****B3**** shared `t()` shim)	?	FRIENDLY B3	Blocked on B3	?	?

8. Open questions ? owner

1. ****Approve starting this line**** (in parallel with the FRIENDLY UX/A-items), and the ****A ? B ? C phasing****? *(E0 gate.)*
2. ****Chinese:**** ship ****`zh-Hans` only**** for v1 (recommended), or ****both Hans + Hant****? *(EC0; Hant = a whole extra catalog + Noto Sans TC.)*
3. ****CJK offline strategy**** *(EC0; the real size trade-off):* (a) ship ****all**** font chunks ? simplest, ****+tens of MB**** to the air-gapped bundle; (b) a ****custom subset build**** restricted to glyphs the reviewed catalogs use ? KB-scale but adds a build step that re-runs on catalog change; (c) ****online lazy-load only**** ? small bundle but ****breaks the offline guarantee for CJK****. *Recommend (b) if offline-for-CJK matters, else (a).*
4. ****Native-reviewer sourcing**** for 7 languages ? who/how/budget? (Vendor like the Roora labeling pilot? community? paid freelancers?) This is the launch-gate cost.
5. ****Priority order**** ? which markets first? (Default A-then-B-then-C is by *engineering cost*; if a market is urgent, e.g. `id` or `ja`, it can jump within its phase.)
6. ****Does `site/` localize for these too**** for v1, or SPA-only until FRIENDLY B3 lands? *(EX3.)*

9. Definition of done

- [] All 7 catalogs (`es`, `da`, `id`, `th`, `zh-Hans`, `ja`, `ko`) live + ****rendering**** on the `web/` SPA; switcher + persistence + `en` fallback; ****no raw keys****.
- [] ****Auto-negotiation green**** ? a browser set to each language auto-renders it (N1/N2-style tests per locale); region subtags normalize; missing-key ? `en`.
- [] ****CI key-parity + no-bare-strings green**** (incl. the `es` 3-form plural) + the ****offline/self-contained guard**** still green after the CJK fonts land.
- [] ****Fonts render correctly**** ? owner eyeball per new script (th, zh-Hans, ja, ko).
- [] ****Native-speaker review**** complete + signed off per language (EX1).
- [] CJK ****offline/bundle-size**** strategy implemented per the EC0 decision; documented bundle-size impact.
- [] *(if in scope)* `site/` marketing renders the 7 via the shared shim (EX3, after FRIENDLY B3).
- [] S7 all rows ? (or explicitly deferred) ? flip ****Status ? DONE**** + archive.

10. References

****Internal `[verified]`:**** `web/src/i18n/{index.ts,hygiene.test.ts}`, `web/src/main.tsx` (font-load pattern), `web/src/i18n/locales/{en,kn,hi}/common.json`;
[`FRIENDLY\_AND\_MULTILINGUAL.md`](FRIENDLY_AND_MULTILINGUAL.md), engine `docs/I18N\_PLAN.md`, `docs/UI\_PROPOSAL.md`.
****External `[researched]`:**** Unicode CLDR v48 plural-rules chart (https://www.unicode.org/cldr/charts/48/supplemental/language_plural_rules.html) + plural-rules spec; W3C BCP-47 / Chinese script subtags (`zh-Hans`/`zh-Hant`); SIL Open Font License 1.1 + Noto CJK LICENSE (github.com/notofonts/noto-cjk); `@fontsource-variable/noto-sans-{thai,sc,jp,kr}`; W3C Thai line-breaking (`thai-gap`); react-i18next + i18next-icu.

Changelog

- `2026-06-21` ? created. Owner ask: add Thai/Indonesian/Danish/Spanish/Chinese/Japanese/Korean with en/kr/hi-level diligence, as a parallel line for approval. Grounded in a per-language i18n research pass (CLDR plurals, OFL Noto fonts, BCP-47, offline-bundle sizing) + the verified working-tree foundation. Phasing A (Latin) ? B (Thai) ? C (CJK); decisions ED1?ED6; 6 owner questions surfaced (E0 = approval gate).

458. user (2026-06-21T14:15:04.223Z)

File created successfully at: C:\Users\avidu\Projects\kheIsutra\docs\MULTILINGUAL_EXPANSION.md (file state is current in your context ? no need to Read it back)

459. assistant (2026-06-21T14:15:21.508Z)

Proposal written. Commit + push + open it as an approval PR:

460. user (2026-06-21T14:15:47.644Z)

```
[docs/multilingual-expansion-proposal 3297596] docs: propose Multilingual Expansion ? 7 new
locales (th/id/da/es/zh-Hans/ja/ko)
 1 file changed, 114 insertions(+)
 create mode 100644 docs/MULTILINGUAL_EXPANSION.md
3297596 docs: propose Multilingual Expansion ? 7 new locales (th/id/da/es/zh-Hans/ja/ko)
To https://github.com/avidullu/khelsutra.git
 * [new branch]      docs/multilingual-expansion-proposal -> docs/multilingual-expansion-
proposal
https://github.com/avidullu/khelsutra/pull/58
```

461. assistant (2026-06-21T14:16:59.803Z)

Proposal is up as **[khelsutra#58](https://github.com/avidullu/khelsutra/pull/58)** (left open for your approval ? it's a product/strategy doc, so owner-gated, not auto-merged). Now your conceptual question, which is the important one:

Can alpha users trigger generation from the UI today? Not yet ? but it's closer (and cheaper) than you think

You can trigger generation from the UI ? that's exactly what B-P2 shipped (paste a `gs://URI ? /api/process` ? poll ? rallies ? compile ? download; proven in the screencast). But you can't share the page with alpha users yet, for three specific reasons ? and none of them is M12, and none requires a GPU:`

1. **Nothing is hosted for them to reach.** Today the SPA runs via `npm run dev` on your machine, pointed at a local engine. There's no URL where the SPA and a reachable engine live together.`
2. **The engine has no auth.** It binds `127.0.0.1:8000`, CORS ```, single-tenant ? unsafe to expose without an edge-auth layer in front.`
3. **It needs a reachable engine with compute** ? and **Gemini is CPU-only**, so a plain CPU box suffices. GPU is only for WASB.

So the blocker is **not M12** (that's the Google-Drive read/write convenience ? a nice-to-have, not this) and **not the whole T1?T11** ? it's a small subset: **a hosted SPA + engine behind edge auth on a reachable box** = the appliance shell (`LOCAL_APPLIANCE` T2` + the reverse-proxy topology), plus the screens we already built.`

The cheap path you actually have: your **existing CPU droplet** (`168.144.126.176`) + Gemini can do this with no GPU subscription. You parked the droplet because the WASB CUJ needs a GPU ? but alpha sharing on Gemini doesn't. That reframes "trigger from the UI for real" from blocked-on-GPU to a handful of ungated steps.`

Recommended priority for the "alpha-shareable UI trigger" goal ? after B3/A1/A2/A4/A5/A8

1. **T4 ? in-UI local file upload** (the chunked-upload endpoints already exist; the SPA just needs to call them). **This is the one genuinely-missing UX piece:** alpha users have a **video file**, not a `gs://` bucket URI ? the current IngestPanel can't take their file yet.`
2. **T2 ? appliance shell on the droplet:** serve the SPA build + reverse-proxy `/api` ? the engine behind Cloudflare Access (you already use Cloudflare), engine on Gemini/CPU; rotate the chat-shared key.`
3. **T11 ? retention sweep** + basic hygiene before any non-owner exposure.

That's the alpha-shareable milestone ? no GPU, no M12. WASB/GPU and Cloud Run scale-to-zero (the serving session's M7) are **later** "better detection / zero idle cost" upgrades, not prerequisites. (One caveat: the engine processes **one job at a time** in-process ? fine for a few alpha users; it queues.)

So, the execution plan

- ? **This turn:** answered the above + shipped the **localization-expansion proposal**

[#58](https://github.com/avidullu/khelsutra/pull/58), awaiting your approval).
- **Next, in order (your go-ahead):** doc-hygiene cleanup ? B3 (site/ trilingual shim) ? A1 (plain-language copy) ? A2 (WhatsApp share) ? A4/A5/A8.
- **Then the goal you flagged:** the alpha-shareable UI trigger (T4 upload ? T2 droplet+edge-auth+Gemini ? T11 retention).

Want me to kick off the doc-hygiene + B3 now, or would you rather I bump T4 (in-UI upload) up front since it's the keystone of the "trigger from the UI" goal? Either way I'll hold the 7-language line until you approve #58.

462. user (2026-06-21T14:22:38.813Z)

Follow your original execution plan although I am not sure whether we should add WhatsApp share right now unless its a very scoped and limited work, we may leave the "sharing" aspect to a pure download and user can share themselves, but feel free to push back since I wanted to do all these "social media work" in a different scope. Apart from that your execution to finish A first and then make the UI runnable tool as the North Star makes sense. For the multilingua doc, please leave that as is and I will merge it, for others go ahead and merge on green and improved test coverage as usual.

463. assistant (2026-06-21T14:23:50.173Z)

Good steer ? and I agree on WhatsApp/A2. A *good* share isn't trivial: `wa.me` links can't attach a file (text-only), and real file-sharing needs the Web Share API with its own mobile-UX quirks ? that genuinely belongs in your separate "social media" scope. Dropping A2; the reel stays a pure download for now. Revised batch: doc-hygiene ? B3 ? A1 ? A4 ? A5 ? A8, then the runnable-tool North Star (T4 upload ? T2 droplet+edge-auth+Gemini ? T11 retention). I'll leave #58 for you to merge; everything else merges on green with rising coverage.

Starting the execution plan now ? first the doc-hygiene cleanup. Branching off master and reading the target docs:

464. user (2026-06-21T14:23:53.924Z)

Switched to a new branch 'docs/hygiene-refresh'
on docs/hygiene-refresh @ 7750afc docs(dry-run): Path-B live e2e DONE ? in-UI bucket?reel CUJ + screencast (DoD §6 box 3 ?) (#57)

465. user (2026-06-21T14:24:02.167Z)

```
1      # khelsutra docs index
2
3      ## Tracked work (this repo)
4      - **`DRY_RUN_CUJ.md`** ? **? active goal (owner directive 2026-06-21):** the UI drives
the tool on a
5          **GCS-bucket video** ? review/select ? compile ? **reel downloaded local**, verified
in Hindi. Grounds
6          the integration gap (no in-UI bucket-ingest **screen** yet ? the engine `/api/process`
already accepts
7          `gs://`, so Path B = SPA-only/B-P2) + a cheapest-path runbook + open decisions.
`DRAFT`
8          (owner-gated). Milestone of `LOCAL_APPLIANCE_ARCHITECTURE.md`.
9      - **`FRIENDLY_AND_MULTILINGUAL.md`** ? **? active focus (owner directive 2026-06-21):**
make the
10         product super user-friendly for non-tech-savvy coaches/parents **and** localize it in
Hindi +
11         Kannada, across both surfaces (`web/` SPA ? react-i18next, `site/` ? a shared `t()`)
shim). The
12         implementation project for the engine's `I18N_PLAN.md` (l10n) + the UX half of
`UI_PROPOSAL.md`.
13         `DRAFT` (owner-gated). Archived when en+kn+hi render on both surfaces.
14      - **`LOCAL_APPLIANCE_ARCHITECTURE.md`** ? the single-box, UI-powered rally indexer +
reel
15         generator (KhelSutra Studio): the `web/` SPA drives the full ingest ? index ? reel
```

```

loop on one
16     box behind an edge gate; the single-box deployment profile of the engine's decoupled
17     architecture. Umbrella for the console; per-rally selection is one screen of it.
`DRAFT`.
18     - **`PER_RALLY_SELECTION.md`** ? interactive pick-rallies ? reel builder (RallyPicker).
The alpha
19     SPA over the engine's rally index; hybrid grid + deterministic query bar on the
**existing**
20     `/api/query` + `/api/compile` contract. `IN PROGRESS` (decisions locked).
21     - **`REQUEST_ACCESS_FORM.md`** ? segmented lead-capture form + Worker/D1 backend
(shipped).
22     - **`PORTABILITY.md`** ? the Node+SQLite VM-host runbook (Cloudflare ? droplet parity).
23
24     ## Engine planning docs
25     Canonical planning docs currently live in the engine repo (`badminton-highlight-
indexer/docs`):
26     `UI_PROPOSAL.md` · `HOSTING_PLAN.md` · `UI_ALPHA_EXECUTION_PLAN.md`. They migrate here
when the
27     control plane lands (per ADR-009 in the engine repo's
`docs/archives/decisions/DECISIONS.md`).
28

```

466. user (2026-06-21T14:24:02.748Z)

```

1     # Single-box appliance ? UI-powered rally indexer + reel generator (KhelSutra Studio)
2
3     > **Status:** `DRAFT` ? owner-gated (auth choice, retention, cost ceiling, GPU host, P5
bridge) · **Owner:** Avi · **Created:** 2026-06-20 · **Last updated:** 2026-06-20
4     > **Lifecycle:** `DRAFT` ? IN PROGRESS ? DONE ? archived` (move to `docs/archives/` when
DONE)
5     > **Tracking anchors:** §9 progress tracker is the source of truth; indexed in
`docs/README.md`; pointer in `NEXT_STEPS.md` + memory `current-state` once work starts.
6     > **Relation:** peer-of `PER_RALLY_SELECTION.md` / `REQUEST_ACCESS_FORM.md` in this
repo; **consumes** the engine's decoupled `StorageBackend` + `python -m backend.worker` seams
and the FastAPI loop; **extends** the engine's `HOSTING_PLAN.md` gating + `UI_PROPOSAL.md` alpha
screens. Engine-side changes here are **proposals to coordinate** with the active
decoupled/worker sessions ? not written unilaterally.
7     > **Honesty note:** every engine-fact claim is tagged `[verified]` (read against engine
source this session), `[design]` (proposed here), or `[researched]` (needs sign-off). Not
legal/financial advice.
8
9     ---
10
11     ## 0. TL;DR
12     **KhelSutra Studio** turns the engine from a CLI into a **single-box, UI-powered rally
indexer + reel generator**: the khelsutra `web/` React SPA, served behind an **edge auth gate**,
drives the **existing** engine FastAPI loop (ingest ? index ? pick rallies ? reel ? download) on
**one machine**, with `StorageBackend=local` for the MVP. It is the **single-box deployment
profile of the just-shipped decoupled architecture** ? storage=local + compute=local is the
degenerate collapse of the same storage/worker split.

```

467. user (2026-06-21T14:24:12.796Z)

```

=== LOCAL_APPLIANCE T12 row ===
169:| T12 | Ingest from **bucket URI** (`s3://`/`gs://`) wiring ? **engine half DONE**
(`/api/process` accepts URIs, M2/[#280], `[verified]`); remaining = the SPA ingest call (=
**B-P2**) | T4 | coord | ? | engine done; SPA pending |
171:| T14 | *(reserved)* Bucket-worker bridge ? remote-GPU "compute as a setting" (P5) | T8,T12
| ? owner | ? | ? |
=== REQUEST_ACCESS_FORM P2 tracker row ===
84:| P2 | Worker + D1 + `wrangler.toml` + `schema.sql` + **unit tests (10 green)** | owner
deploy | ? | this PR (built+tested; owner runs `wrangler d1 create` + `deploy`) |
66:- **Anti-spam:** honeypot first; add **Cloudflare Turnstile** in P2 if spam appears.
67:- **Notify (P2):** optional email ping to `hello@` per submission (MailChannels / Cloudflare
Email).

```

```
84:| P2 | Worker + D1 + `wrangler.toml` + `schema.sql` + **unit tests (10 green)** | owner
deploy | ? | this PR (built+tested; owner runs `wrangler d1 create` + `deploy`) |
98:2. Email-notify per submission, or just query D1 weekly? (Default: notify in P2.)
=== PER_RALLY header ===
3:> **Status:** `IN PROGRESS` ? **MVP / M0 (P1?P5) COMPLETE 2026-06-20** (pick ? query ? select
? compile ? download, on existing engine endpoints; verified e2e against a stub engine, GCP-GPU
validation next). Design **APPROVED by owner 2026-06-20**. Downstream milestones M1?M5 (multi-
video auth, per-rally preview endpoint, corpus loop, localized NL) remain owner-gated. .
**Owner:** Avi . **Created:** 2026-06-20 . **Last updated:** 2026-06-20
```

468. assistant (2026-06-21T14:24:44.051Z)

Now the precise edits. README ? mark DRY_RUN delivered + add the multilingual proposal pointer:

469. user (2026-06-21T14:24:53.635Z)

The file C:\Users\avidu\Projects\khelsutra\docs\README.md has been updated successfully. (file state is current in your context ? no need to Read it back)

470. user (2026-06-21T14:25:00.727Z)

The file C:\Users\avidu\Projects\khelsutra\docs\LOCAL_APPLIANCE_ARCHITECTURE.md has been updated successfully. (file state is current in your context ? no need to Read it back)

471. user (2026-06-21T14:25:08.414Z)

The file C:\Users\avidu\Projects\khelsutra\docs\LOCAL_APPLIANCE_ARCHITECTURE.md has been updated successfully. (file state is current in your context ? no need to Read it back)

472. user (2026-06-21T14:25:19.588Z)

<tool_use_error>File has not been read yet. Read it first before writing to it.</tool_use_error>

473. user (2026-06-21T14:25:24.822Z)

```
80      | ID | Deliverable | Gated? | Status | PR |
81      |----|-----|-----|-----|----|
82      | P0 | This design doc + `NEXT_STEPS.md` | No | ? | #2 |
83      | P1 | Form UI (segments + GPU + storage + free text + consent) w/ `mailto` fallback |
No | ? | #2 |
84      | P2 | Worker + D1 + `wrangler.toml` + `schema.sql` + **unit tests (10 green)** | owner
deploy | ? | this PR (built+tested; owner runs `wrangler d1 create` + `deploy`) |
85      | P3 | Anti-spam (Turnstile) + per-submission email notify | No | ? | ? |
86      | P4 | Tiny admin/export view (auth'd) to read responses | owner | ? | ? |
87
```

474. user (2026-06-21T14:25:37.328Z)

The file C:\Users\avidu\Projects\khelsutra\docs\REQUEST_ACCESS_FORM.md has been updated successfully. (file state is current in your context ? no need to Read it back)

475. assistant (2026-06-21T14:25:47.672Z)

Now the PER_RALLY header (stale "stub engine, GCP-GPU next" ? the CUJ validated it live).
Reading then editing:

476. user (2026-06-21T14:25:47.837Z)

```
3      > **Status:** `IN PROGRESS` ? **MVP / M0 (P1?P5) COMPLETE 2026-06-20** (pick ? query ?
select ? compile ? download, on existing engine endpoints; verified e2e against a stub engine,
GCP-GPU validation next). Design **APPROVED by owner 2026-06-20**. Downstream milestones M1?M5
(multi-video auth, per-rally preview endpoint, corpus loop, localized NL) remain owner-gated. .
**Owner:** Avi . **Created:** 2026-06-20 . **Last updated:** 2026-06-20
```

477. user (2026-06-21T14:25:56.370Z)

The file C:\Users\avidu\Projects\khelsutra\docs\PER_RALLY_SELECTION.md has been updated successfully. (file state is current in your context ? no need to Read it back)

478. assistant (2026-06-21T14:26:02.651Z)

Doc-hygiene edits done across 4 khelsutra docs. Commit + push + PR:

479. user (2026-06-21T14:26:22.487Z)

warning: in the working copy of 'docs/REQUEST_ACCESS_FORM.md', CRLF will be replaced by LF the next time Git touches it

M docs/LOCAL_APPLIANCE_ARCHITECTURE.md

M docs/PER_RALLY_SELECTION.md

M docs/README.md

M docs/REQUEST_ACCESS_FORM.md

[docs/hygiene-refresh 6404f83] docs: hygiene refresh ? sync trackers to shipped reality
4 files changed, 14 insertions(+), 11 deletions(-)

6404f83 docs: hygiene refresh ? sync trackers to shipped reality

To https://github.com/avidullu/khelsutra.git

* [new branch] docs/hygiene-refresh -> docs/hygiene-refresh

https://github.com/avidullu/khelsutra/pull/59

480. assistant (2026-06-21T14:26:27.806Z)

Doc-hygiene PR up: [khelsutra#59](https://github.com/avidullu/khelsutra/pull/59). Watch CI ? merge on green:

481. user (2026-06-21T14:27:00.285Z)

web pass 21s

https://github.com/avidullu/khelsutra/actions/runs/27907355279/job/82578276165

Workers Builds: hidden-king-khelsutra pass 0

https://dash.cloudflare.com/213c12cf60de9041dd8a3521aa206472/workers/services/view/hidden-king-khelsutra/production/builds/183acd34-a0c0-4dab-b196-e17b97694938

site pass 12s

https://github.com/avidullu/khelsutra/actions/runs/27907355279/job/82578276185

web pass 21s

https://github.com/avidullu/khelsutra/actions/runs/27907355279/job/82578276165

Deleted branch docs/hygiene-refresh (was 6404f83).

khelsutra master: a0475bc docs: hygiene refresh ? sync trackers to shipped reality (#59)

482. assistant (2026-06-21T14:27:27.242Z)

Doc-hygiene **#59 merged** (khelsutra master `a0475bc`). Let me record this checkpoint in memory (carefully, in my own block), then give you the status.

483. user (2026-06-21T14:27:29.292Z)

19:B-P2 BUILT

19:ALL DONE + MERGED

484. user (2026-06-21T14:27:35.624Z)

19 - **? B-P2 BUILT (in-UI bucket ingest) ? khelsutra

[#56](https://github.com/avidullu/khelsutra/pull/56) OPEN, CI GREEN (web+site+CF build),

awaiting owner merge** (feature PR ? left for owner; precedent exists for me merging SPA PRs on owner's "merge as you go"). 100% `web/` + i18n, ZERO engine code (rides B-P1). New `web/src/{api/types,api/client(processVideo/getJob),lib/errors(parseDetail+classifyError),hooks/useIngestJob(idle?submitting?polling?done|error; recursive-setTimeout poll backoff+10min-ceiling+1 cid; success-shaped-failure?emptyResult),components/IngestPanel}` + `App.tsx` `videoId` useMemo?useState + IngestPanel-when-!videoId + history.replaceState; `ingest.\*` en/kn/hi (AI-

draft, native-review-pending). **123 web tests (+19), tsc/build green, browser-verified** (panel renders en+hi; live submit?friendly localized "is the engine running?"; cid logs). Live-fixed UX: bare 5xx/proxy-down ? friendly `networkError`, not "HTTP 500". **? ALL DONE + MERGED (owner: "merge green PRs, improve coverage, leave a CUJ screencast in the promo bucket; proof-and-validation-driven"):** **#56 MERGED** (`4525c5d`). **LIVE e2e CUJ recorded** ? real local engine (Gemini + `google-cloud-storage`+ADC for the gs:// fetch), Playwright-driven: SPA ingest `gs://mahadevpura\_smoke\_180s.mp4` ? **12 rallies** ? Build ? **reel downloaded (1080p, 76s)**. **? Recording it SURFACED A REAL BUG:** `api/compile` 500'd on a bucket-ingested video ? `videos.filepath` keeps the `gs://` source URI (M2 provenance), handed straight to the stitcher ? ffmpeg can't open `gs://`. **FIXED ? engine [#289](https://github.com/Khelsutra/badminton-highlight-indexer/pull/289) MERGED** (`e68b583`): compile resolves the source via `storage.acquire\_local\_input` first (identity local / fetch-to-scratch bucket) + regression test; full suite **1182?**. **Screencast ? `gs://khelsutra-rally-corpus/promo/khelsutra-bucket-cuj-screencast.mp4`** (42s, fast-forwarded Gemini wait + frozen reel-ready climax). **DoD S6 box3 ? ? khelsutra [#57] MERGED** (`7750afc`, +S9 run result). Engine+SPA+worktrees cleaned up. ? `google-cloud-storage` now pip-installed in the local engine env (enables local gs:// fetch). **? NEXT:** opt-3 droplet (deploy the fixed engine + GCS creds ? re-run the CUJ on Linux ? DoD box4); localization DoD tail (B3 site `t()` shim; native kn/hi review). [[gotcha-shared-checkout-branch-collisions]]

485. user (2026-06-21T14:28:00.550Z)

The file C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\memory\current-state.md has been updated successfully. (file state is current in your context ? no need to Read it back)

486. assistant (2026-06-21T14:28:19.754Z)

Done for this turn ? here's where we are.

Landed this turn

- **Decision: A2 (WhatsApp/social share) dropped** from the batch ? deferred to your separate "social media" scope. Reel stays a pure download; users share it themselves. (Confirmed the right call: `wa.me` can't attach files, and real file-share = Web Share API, which is its own mobile-UX scope.)
- **[khelsutra#58](https://github.com/avidullu/khelsutra/pull/58)** ? the **Multilingual Expansion proposal** (7 locales, en/kn/hi-rigor). **Left open for you to merge/approve** as requested. Nothing on that line starts until you approve E0.
- **[khelsutra#59](https://github.com/avidullu/khelsutra/pull/59)** ? **doc-hygiene refresh, merged on green** (`a0475bc`): DRY_RUN marked delivered + #58 pointer in README, LOCAL_APPLIANCE T12 ? ?, REQUEST_ACCESS P2 ? ?, PER_RALLY live-CUJ note. *(Engine-side staleness ? DOC_STATUS/NEXT_STEPS lagging serving M5?M8, RALLY_QUALITY n=6-vs-15 ? I left for the active serving session to avoid collision.)*

The confirmed plan (merge-on-green, rising coverage)

doc-hygiene ? ? B3 ? A1 ? A4 ? A5 ? A8 ? North Star (T4 ? T2 ? T11).

- **B3** *(next)* ? the shared `t()`/`data-i18n` shim so the **`site/` marketing pages go trilingual** (they're still English-only). It also has to resolve the catalog-home question (hoist `web/src/i18n/locales` to a shared dir vs import from there).
- **A1/A4/A5/A8** ? plain-language copy · mobile touch targets · friendly engine-error mapping · background-processing status chips.
- **North Star ? the runnable tool for alpha users:** **T4** (in-UI file *upload* ? the keystone, since alpha users have a video file, not a `gs://` URI) ? **T2** (host SPA + engine behind Cloudflare Access on the CPU droplet, Gemini/no-GPU) ? **T11** (retention). This is what makes the page shareable.

B3 is a real build (vanilla-HTML shim + `data-i18n` wiring + tests + the catalog-hoist) so I'll take it as the next focused PR. Want me to proceed straight into B3, or would you like to glance at / merge #58 first?

487. user (2026-06-21T14:28:21.770Z)

BTW you may take a pause before the runnable tool work because maybe your context is not enough by that point for that crucial work. Feel free to wrap up everything till that point.

488. assistant (2026-06-21T14:33:06.621Z)

Smart call ? I'll wrap the SPA UX line cleanly and hand off the runnable-tool work with full specs so a fresh session does that crucial part with clean context. Two parallel tracks: I'll (1) spin up a workflow to spec the remaining build items + the North-Star trio into ready-to-execute tickets for the handoff, and (2) ship **A1 (plain-language copy)** myself now.

Launching the spec workflow (it offloads the design to subagents, preserving my context for execution + the handoff):

489. user (2026-06-21T14:33:51.142Z)

Workflow launched in background. Task ID: w5x092blg
Summary: Produce ready-to-execute implementation tickets for the remaining SPA UX items (B3/A4/A5/A8) and the runnable-tool North Star (T4 upload, T2 droplet appliance shell, T11 retention), grounded in the actual code, for a fresh-context handoff
Transcript dir: C:\Users\avidu\claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\9f6357d8-0e35-491a-a830-052e094c7f37\subagents\workflows\wf_b7edcaa8-69d
Script file: C:\Users\avidu\claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\9f6357d8-0e35-491a-a830-052e094c7f37\workflows\scripts\wrapup-specs-wf_b7edcaa8-69d.js
(Edit this file with Write/Edit and re-invoke Workflow with {scriptPath: "C:\Users\avidu\claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\9f6357d8-0e35-491a-a830-052e094c7f37\workflows\scripts\wrapup-specs-wf_b7edcaa8-69d.js"} to iterate without resending the script.)
Run ID: wf_b7edcaa8-69d
To resume after editing the script: Workflow({scriptPath: "C:\Users\avidu\claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\9f6357d8-0e35-491a-a830-052e094c7f37\workflows\scripts\wrapup-specs-wf_b7edcaa8-69d.js", resumeFromRunId: "wf_b7edcaa8-69d"}) ? completed agents return cached results.

You will be notified when it completes. Use /workflows to watch live progress.

490. assistant (2026-06-21T14:34:57.932Z)

Got it ? **#58 merged** means the 7-language line is approved (E0 cleared); I'll fold it into the handoff as a greenlit parallel line (Phase A `es/da/id` is actionable; Phases B/C still need your answers on Q2 zh-Hans/Hant, Q3 CJK-offline, Q4 reviewer sourcing). I won't start that implementation now ? it's a parallel track and you flagged context ? but it's teed up.

Meanwhile the spec workflow runs; let me ship **A1 (plain-language copy)**. Branch + read the current en catalog:

491. user (2026-06-21T14:35:03.343Z)

Switched to a new branch 'feat/a1-plain-language-copy'
on feat/a1-plain-language-copy @ a20002f docs: propose Multilingual Expansion ? 7 new locales (th/id/da/es/zh-Hans/ja/ko) (#58)

492. user (2026-06-21T14:35:08.773Z)

```
1      {
2      "app": {
3        "badge": "per-rally selection · alpha",
4        "title": "Pick the rallies for your reel",
5        "intro": "Filter by <b>rally length</b> ? e.g. <i>?top 15 longest?</i> ? then fine-tune by tapping cards. Everything starts <b>selected</b>; just drop the ones you don't want."
6      },
7      "lang": {
8        "label": "Language",
9        "en": "English",
10       "kn": "?????",
```

```

11     "hi": "?????"
12 },
13 "bulk": {
14     "selectAll": "Select all",
15     "clear": "Clear",
16     "invert": "Invert"
17 },
18 "mode": {
19     "demo": "Demo data ? open with <code>?video=?video_id?</code> against a running
engine to load a real session.",
20     "loading": "Loading rallies for <code>{videoId}</code>?",
21     "error": "Couldn't load rallies for <code>{videoId}</code>: {error}",
22     "loaded": "Loaded <b>{count}</b> rallies from this session."
23 },
24 "query": {
25     "placeholder": "e.g. over 10s · top 15 longest · shortest first",
26     "ariaLabel": "Filter rallies by length, count or order",
27     "apply": "Apply",
28     "notAvailable": "not available yet",
29     "notAvailableTitle": "?{token}? needs {kind} detection ? not available yet",
30     "unreadable": "Couldn't read that ? try ?over 10s? or ?top 15 longest?.",
31     "hint": "Type over 10s for rallies longer than 10 seconds ? or tap a sample below.",
32     "try": "Sample queries:",
33     "support": "Filtering by <b>rally length</b> is fully supported. Shot-count is
<x>experimental</x> (under development).",
34 },
35 "grid": {
36     "empty": "No rallies to show yet.",
37     "rally": "Rally {n}",
38     "inReel": "Rally {n}, in reel",
39     "notInReel": "Rally {n}, not in reel",
40     "start": "start",
41     "long": "long",
42     "shots": "~{count} shots · experimental",
43     "shotsTitle": "Shot count is experimental (under development) and may be
inaccurate{conf}. Filter by rally length instead.",
44     "shotsConfidence": " ? confidence: {conf}"
45 },
46 "footer": {
47     "count": "{count, plural, one {# rally} other {# rallies}}",
48     "summaryRest": "selected · {total} total",
49     "reelName": "Reel name",
50     "reelNamePlaceholder": "my-highlights",
51     "build": "Build reel ({count})",
52     "building": "Building?",
53     "floorOne": "<b>Your only selected rally</b> has fewer than {min} shots ? the engine
would skip it, so the build would fail. Pick a longer rally.",
54     "floorAll": "<b>All {count} selected rallies</b> have fewer than {min} shots ? the
engine skips every one, so the build would fail. Pick longer rallies.",
55     "floorSomeLead": "{count, plural, one {# rally} other {# rallies}}",
56     "floorSomeRest": "with fewer than {min} shots will be skipped when building (the
engine's <code>min_shot_count</code> floor).",
57 },
58 "result": {
59     "ready": "Your reel is ready",
60     "failed": "Build failed",
61     "dismiss": "Dismiss",
62     "download": "Download {filename}"
63 },
64 "build": {
65     "noVideo": "Open with ?video=<video_id> against a running engine to build a reel ?
demo data can't be compiled."
66 },
67 "ingest": {
68     "title": "Process a video",

```



```

69         "intro": "Point KhelSutra at a video ? a bucket link or a file on the engine ? and
it finds the rallies. This can take a few minutes.",
70         "uriPlaceholder": "gs://your-bucket/match.mp4",
71         "uriAriaLabel": "Video location ? a gs:// bucket URL or a path on the engine host",
72         "hint": "Paste a gs:// bucket link, or a video path on the machine running the
engine.",
73         "submit": "Process video",
74         "submitting": "Submitting?",
75         "cancel": "Cancel",
76         "working": "Processing your video?",
77         "progress": "{progress}",
78         "retry": "Try again",
79         "failed": "Couldn't process the video: {error}",
80         "invalidUri": "Enter a video location first ? a gs:// link or a path on the
engine.",
81         "networkError": "Can't reach the engine. Is it running?",
82         "emptyResult": "Processing finished but found no rallies in this video. Try a
clearer match recording.",
83         "timeout": "This is taking longer than expected. The job may still be running ?
reload with the video link, or try again."
84     },
85     "selection": {
86         "count": "{count, plural, one {# rally selected} other {# rallies selected}}"
87     }
88 }
89

```

493. assistant (2026-06-21T14:35:47.623Z)

Now a focused de-jargon pass. First, let me check which tests assert the specific strings I'll change (so I update them in lockstep):

494. user (2026-06-21T14:35:50.120Z)

```

=== tests asserting strings I plan to change (mode/demo/loaded/sample/session/engine) ===
src/api/client.test.ts:74:   vi.stubGlobal("fetch", mockFetch(200, { stitching: {
min_shot_count: 4 } }));
src/components/IngestPanel.test.tsx:3:// The ingest screen end to end: type a URI ? submit ?
(mocked engine) ? onLoaded gets the video_id;
src/components/IngestPanel.test.tsx:49:  it("submits a URI, polls, and hands the video_id to
onLoaded", async () => {
src/components/IngestPanel.test.tsx:58:      const onLoaded = vi.fn();
src/components/IngestPanel.test.tsx:59:      render(<IngestPanel onLoaded={onLoaded} />);
src/components/IngestPanel.test.tsx:72:      expect(onLoaded).toHaveBeenCalledWith("vidZ");
src/components/IngestPanel.test.tsx:77:      render(<IngestPanel onLoaded={vi.fn()} />);
src/components/IngestPanel.test.tsx:87:      render(<IngestPanel onLoaded={vi.fn()} />);
src/components/IngestPanel.test.tsx:90:      expect(await screen.findByText(/can't reach the
engine/i)).toBeTruthy();
src/components/IngestPanel.test.tsx:96:      render(<IngestPanel onLoaded={vi.fn()} />);
src/hooks/useIngestJob.test.ts:74:   const onLoaded = vi.fn();
src/hooks/useIngestJob.test.ts:75:   const { result } = renderHook(() =>
useIngestJob(onLoaded));
src/hooks/useIngestJob.test.ts:80:   expect(onLoaded).toHaveBeenCalledWith("vid9");
src/hooks/useIngestJob.test.ts:92:   const onLoaded = vi.fn();
src/hooks/useIngestJob.test.ts:93:   const { result } = renderHook(() =>
useIngestJob(onLoaded));
src/hooks/useIngestJob.test.ts:98:   expect(onLoaded).not.toHaveBeenCalled();
src/hooks/useIngestJob.test.ts:157:   const onLoaded = vi.fn();
src/hooks/useIngestJob.test.ts:158:   const { result } = renderHook(() =>
useIngestJob(onLoaded));
src/hooks/useIngestJob.test.ts:166:   // Further time passes; nothing resumes, onLoaded never
fires.
src/hooks/useIngestJob.test.ts:171:   expect(onLoaded).not.toHaveBeenCalled();

```

495. assistant (2026-06-21T14:36:05.427Z)

Only `IngestPanel.test.tsx:90` asserts a string I'd touch (`ingest.networkError`). Let me confirm the App render test (`i18n.test.tsx`) doesn't assert the `mode.\*` strings before I change them:

496. user (2026-06-21T14:36:07.798Z)

```
=== i18n.test.tsx string assertions ===
27:   expect(screen.getByRole("heading", { level: 1 })).textContent().toBe("Pick the rallies for
your reel");
35:   expect(container.textContent).toContain("rally length"); // <Trans> hero intro + query
support
36:   expect(container.textContent).toMatch(/\d+ rallies selected · \d+:\d+ total/); // ICU
footer summary
37:   expect(container.textContent).toMatch(/Rally \d+/); // RallyGrid card
39:   expect(container.textContent).not.toMatch(/\b(app|grid|footer|query|mode|result|build)\.[
a-zA-Z]+\b/);
50:   expect(screen.getByRole("heading", { level: 1 })).textContent().toBe("????? ?????????
????????????? ?????");
56:   expect(screen.getByRole("heading", { level: 1 })).textContent().toBe("????? ?????????
????????????? ?????");
65:   expect(screen.getByRole("heading", { level: 1 })).textContent().toBe("????? ??? ?? ???
????????? ?????");
73:   expect(screen.getByRole("heading", { level: 1 })).textContent().toBe("Pick the rallies for
your reel");
94:   expect(detectBases()).toContain("kn"); // the kn-IN region subtag normalizes to base kn
100:  expect(detectBases()).toContain("hi");
120:  expect(screen.getByRole("heading", { level: 1 })).textContent().toBe("Pick the rallies for
your reel");
```

497. assistant (2026-06-21T14:36:51.959Z)

`i18n.test.tsx` only asserts strings I'm *not* changing (title, "rally length", the footer summary/selection ICU, "Rally N", and a no-raw-keys guard). The only test to update in lockstep is `IngestPanel.test.tsx:90`. Let me make the focused de-jargon edits to the en catalog:

498. user (2026-06-21T14:36:53.656Z)

The file C:\Users\avidu\Projects\khelsutra\web\src\i18n\locales\en\common.json has been updated successfully. (file state is current in your context ? no need to Read it back)

499. user (2026-06-21T14:37:00.596Z)

The file C:\Users\avidu\Projects\khelsutra\web\src\i18n\locales\en\common.json has been updated successfully. (file state is current in your context ? no need to Read it back)

500. user (2026-06-21T14:37:08.637Z)

The file C:\Users\avidu\Projects\khelsutra\web\src\i18n\locales\en\common.json has been updated successfully. (file state is current in your context ? no need to Read it back)

501. user (2026-06-21T14:37:15.856Z)

The file C:\Users\avidu\Projects\khelsutra\web\src\i18n\locales\en\common.json has been updated successfully. (file state is current in your context ? no need to Read it back)

502. user (2026-06-21T14:37:23.495Z)

The file C:\Users\avidu\Projects\khelsutra\web\src\i18n\locales\en\common.json has been updated successfully. (file state is current in your context ? no need to Read it back)

503. user (2026-06-21T14:37:29.356Z)

The file C:\Users\avidu\Projects\khelsutra\web\src\i18n\locales\en\common.json has been updated successfully. (file state is current in your context ? no need to Read it back)

504. assistant (2026-06-21T14:37:40.870Z)

Now update the one test that asserts that string:

505. user (2026-06-21T14:37:42.274Z)

The file C:\Users\avidu\Projects\khelsutra\web\src\components\IngestPanel.test.tsx has been updated successfully. (file state is current in your context ? no need to Read it back)

506. assistant (2026-06-21T14:37:49.558Z)

Let me run the full gate (typecheck + tests + build) and validate the JSON is well-formed:

507. user (2026-06-21T14:38:03.415Z)

en/common.json: valid JSON

=== typecheck ===

TSC OK

=== tests ===

Test Files 11 passed (11)
Tests 123 passed (123)
Start at 20:07:56
Duration 3.00s (transform 519ms, setup 145ms, collect 1.86s, tests 840ms,
environment 4.61s, prepare 2.21s)

=== build ===

dist/assets/index-yB_85rHa.js

346.60 kB ? gzip: 109.42 kB

? built in 1.81s

508. assistant (2026-06-21T14:38:14.776Z)

All green (123 tests). Let me quickly browser-verify the new copy renders (per the verification workflow):

509. user (2026-06-21T14:38:17.836Z)

```
{  
  "serverId": "23429133-14d2-40bc-a38f-a0cba378536f",  
  "port": 5174,  
  "name": "khelsutra-web",  
  "reused": false  
}
```

Server started successfully on port 5174.

510. user (2026-06-21T14:38:27.285Z)

"reloading en"

511. user (2026-06-21T14:38:34.208Z)

```
{  
  "badge": true,  
  "demoBanner": true,  
  "htmlLang": "en",  
  "ingestHint": true,  
  "ingestIntro": true,  
  "noRawKeys": true,  
  "oldJargonGone": true  
}
```

```
}
```

512. assistant (2026-06-21T14:38:41.896Z)

Verified live ? the plain-language copy renders (badge, demo banner, ingest intro/hint all updated), no raw keys, and the old jargon ("Demo data", "running engine", "video_id") is gone. Stopping the preview and shipping A1:

513. user (2026-06-21T14:38:42.230Z)

Server 23429133-14d2-40bc-a38f-a0cba378536f stopped

514. user (2026-06-21T14:39:07.446Z)

```
M web/src/components/IngestPanel.test.tsx
M web/src/i18n/locales/en/common.json
[feat/a1-plain-language-copy 600ddbc] feat(web): A1 ? plain-language copy pass on the en catalog (de-jargon)
 2 files changed, 14 insertions(+), 14 deletions(-)
600ddbc feat(web): A1 ? plain-language copy pass on the en catalog (de-jargon)
To https://github.com/avidullu/khelsutra.git
 * [new branch]      feat/a1-plain-language-copy -> feat/a1-plain-language-copy
https://github.com/avidullu/khelsutra/pull/60
```

515. assistant (2026-06-21T14:39:25.874Z)

A1 PR up: [khelsutra#60](https://github.com/avidullu/khelsutra/pull/60). Watch CI ? merge on green:

516. user (2026-06-21T14:39:58.289Z)

```
web      pass      25s
https://github.com/avidullu/khelsutra/actions/runs/27907677785/job/82579131561
Workers Builds: hidden-king-khelsutra pass 0
https://dash.cloudflare.com/213c12cf60de9041dd8a3521aa206472/workers/services/view/hidden-king-khelsutra/production/builds/f92d7522-f703-4fa9-b398-acb5fda07249
site     pass      13s
https://github.com/avidullu/khelsutra/actions/runs/27907677785/job/82579131579
web      pass      25s
https://github.com/avidullu/khelsutra/actions/runs/27907677785/job/82579131561
---merge---
Deleted branch feat/a1-plain-language-copy (was 600ddbc).
master: ccdcfcd feat(web): A1 ? plain-language copy pass on the en catalog (de-jargon) (#60)
```

517. assistant (2026-06-21T14:40:14.200Z)

A1 merged (master `ccdcfcd`). Now let me pull in the spec-workflow tickets for the handoff:

518. user (2026-06-21T14:40:19.024Z)

```
=== is the workflow process still running? ===
-rw-r--r-- 1 avidu 197609 0 Jun 21 20:03 C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-indexer\9f6357d8-0e35-491a-a830-052e094c7f37\tasks\w5x092blg.output
0 C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-indexer\9f6357d8-0e35-491a-a830-052e094c7f37\tasks\w5x092blg.output
```

519. assistant (2026-06-21T14:41:00.028Z)

Status check: the spec workflow is still running (it'll feed the handoff). A1 is merged. Here's where I'm landing this, given your context flag:

Done

- **A1 ? plain-language copy** ? [khelsutra#60](https://github.com/avidullu/khelsutra/pull/60)
merged (master `ccdcfcd`). De-jargoned the user-facing en strings (no more
`video\_id`/"engine"/"session"/`min\_shot\_count` shown to coaches/parents), honest caveats kept;
browser-verified; 123 tests green.
- **#58 acknowledged** ? merging it cleared **E0**, so the **7-language line** is approved.
Phase A (`es`/`da`/`id`, catalog-only) is actionable; Phases B/C still need your answers (Q2 zh-
Hans/Hant, Q3 CJK-offline strategy, Q4 native-reviewer sourcing). I'm **not** starting that
implementation now (parallel track + your context flag) ? it's folded into the handoff as a
greenlit line.

Decision on the rest (taking the pause you offered)

Rather than cram **A4/A5/A8 + B3** into the tail of an already-long session and risk a half-
finished state, I'm **pausing A-item implementation** after **A1** and converting the remainder into
a **fully-spec'd, ready-to-execute handoff** ? that's the clean stopping point before the
runnable-tool (North Star) work you flagged as needing fresh context. A fresh session will knock
out A4/A5/A8/B3 fast from the specs, then do T4?T2?T11.

The **spec workflow** (running now) is producing exactly those tickets ? for B3, A4, A5, A8, and
the North-Star trio (T4 upload, T2 droplet+Cloudflare-Access+Gemini, T11 retention), each with
files, change-spec, new i18n keys, tests, and cross-item coordination notes.

When it lands, I'll write the **session-handoff** (ramp-up kit + ordered tickets + open owner
decisions) and update the memory pointer, so the next session resumes cleanly. I'll pick that up
on the workflow's completion ? nothing else is in flight.

520. user (2026-06-21T14:42:05.924Z)

cool and also give me a great prompt to feed to the next session so we pick up wherer you are
leaving us! Appreciate the diligence and hard work!

521. assistant (2026-06-21T14:42:23.126Z)

Thank you ? it's been a genuinely fun, productive run. ? Let me grab the spec-workflow tickets
(it should be done now), then write the durable handoff and craft your next-session prompt.

522. user (2026-06-21T14:42:26.027Z)

STILL RUNNING (empty)

523. assistant (2026-06-21T14:43:22.660Z)

Workflow's still finishing, but I have full context on every item ? let me write the durable
handoff now (it doesn't need to block on the workflow). Writing the session-handoff:

524. assistant (2026-06-21T14:44:04.820Z)

```
---
name: session-handoff
description: "? READ FIRST for the next session. Where we are + ordered next steps (SPA UX
A-items ? runnable-tool North Star) + the approved 7-language line + open owner decisions +
ramp-up kit. Auto-refreshed at checkpoints."
metadata:
  node_type: memory
  type: project
---
```

Session handoff ? KhelSutra UX/localization + runnable-tool (refreshed 2026-06-21)

1. You are here

The **bucket?reel CUJ** is fully UI-driven and proven (B-P2 shipped + a live screencast in the
promo bucket), the **SPA** is trilingual (en+kn+hi), and the **plain-language copy pass (A1)** is
merged. The active line is **`FRIENDLY\_AND\_MULTILINGUAL.md`** (non-tech UX + localization).

The ****North Star** = make the tool runnable/shareable for alpha users from the UI** (a hosted SPA+engine behind edge auth ? ****NOT**** GPU-gated, ****NOT**** M12; cheapest = the existing CPU droplet + Gemini).

****Merged this session (all on green CI):**** engine [#284](https://github.com/Khelsutra/badminton-highlight-indexer/pull/284) (gs:// regression guard) .
[#289](https://github.com/Khelsutra/badminton-highlight-indexer/pull/289) (compile-from-bucket fix) .
khelsutra [#55](https://github.com/avidullu/khelsutra/pull/55) (stale-doc fix) .
[#56](https://github.com/avidullu/khelsutra/pull/56) (B-P2 in-UI ingest) .
[#57](https://github.com/avidullu/khelsutra/pull/57) (DoD \$6 box3 ?) .
[#59](https://github.com/avidullu/khelsutra/pull/59) (doc-hygiene) .
[#60](https://github.com/avidullu/khelsutra/pull/60) (A1 copy). ****Owner-merged:****
[#58](https://github.com/avidullu/khelsutra/pull/58) (MULTILINGUAL_EXPANSION proposal ? 7-lang line ****approved****). Screencast: `gs://khelsutra-rally-corpus/promo/khelsutra-bucket-cuj-screencast.mp4`.

2. Next steps ? ordered (merge each on green CI + rising coverage; one PR per item, off `origin/master`)

****A)** Finish the SPA UX A-items, then B3** *(active `FRIENDLY\_AND\_MULTILINGUAL.md` line; A1 ?, A2 DROPPED):*

- ****A4 ? mobile touch targets + visible tooltip text.**** Files: `web/src/components/{QueryBar,RallyGrid,SelectionFooter,IngestPanel,CompileResult,LanguageSwitcher}.tsx` + `index.css`. Bump interactive elements to 244px tap targets (`min-h-11`/py), enlarge the tiny query "parsed-pill" + sample chips, and surface `title` tooltips as visible helper text on phones. Effort M.
- ****A5 ? friendly engine-error mapping (build path).**** Reuse the existing `web/src/lib/errors.ts::classifyError` (built for ingest) on the ****build/compile**** path: `App.tsx::onBuild` currently shows raw `ApiError` text for 5xx/network. Add a ****shared `errors.\*` i18n namespace**** (recommend unifying IngestPanel onto it too), map onBuild's catch ? localized message in `CompileResult`. Effort S?M. New keys.
- ****A8 ? background-processing reassurance + status as icon+text chips.**** `IngestPanel` progress + the build state: reassurance copy ("can take a few minutes on slow connections") + render queued/running/done/failed as icon ****+ text**** chips (never icon-only). Effort S?M. New keys.
- ****B3 ? `site/` trilingual via a shared `t()` shim**** (carry the trilingual SPA win to the marketing site, which is still English-only). New ~30?50-line `t(key,vars)`/`data-i18n` shim over `site/public/{index,capture,own-model}.html`; `site/server.js` Accept-Language negotiation; extend `site/src/assets.test.js`. ****DECISION (D-CATALOG-HOME):**** hoist `web/src/i18n/locales/{en,kn,hi}/common.json` to a shared top-level `locales/` consumed by BOTH surfaces (recommended), vs the site importing/copying from `web/src/i18n/locales`. Effort M. *(Catalogs live under `web/src/i18n/locales/` today, NOT a shared dir ? B3 resolves this.)*

****B)** ? NORTH STAR ? runnable tool for alpha users (the crucial fresh-context work):**

- ****T4 ? in-UI local file UPLOAD (the keystone).**** Alpha users have a ****video file****, not a `gs://` URI. Wire the SPA to the ****existing**** engine chunked-upload endpoints (`backend/api/uploads.py`: `/api/upload/init|part|complete|DELETE`) + a file-picker UI in/beside `IngestPanel` ? on complete, hand the server path to the existing `/api/process` ? poll ? load ? compile ? download flow. Net-new: `web/src/api/client.ts` upload helpers + an upload mode; reused: the whole back-half. Effort M?L.
- ****T2 ? droplet appliance shell (hosted + shareable).**** Serve the built SPA + reverse-proxy `/api` ? engine `127.0.0.1:8000` behind ****Cloudflare Access**** (edge auth) on the ****CPU droplet `168.144.126.176`****, engine on ****Gemini/CPU (NO GPU)****. Deploy the #281/#289-fixed engine + `GEMINI\_API\_KEY` + `pip install google-cloud-storage`; ****rotate the chat-shared GEMINI_API_KEY****. Reuse `site/scripts/provision-vm.sh` + `docs/PORTABILITY.md`. Owner step: Cloudflare Access config. Effort M?L (+owner).
- ****T11 ? retention sweep**** for `output/` + `uploads/` originals before any non-owner exposure. Effort S. Owner: retention window.

****C)** 7-language line ? APPROVED (#58), parallel track, owner-paced:**

`MULTILINGUAL\_EXPANSION.md` §7. ****Phase A (`es`/`da`/`id`) is actionable now**** (catalog-only; ****es`** needs the 3-form ICU plural `{one, many, other}`\*\* + a parity/lint touch). **\*\*Phase B (Thai)\*\*** = 1 small OFL font. **\*\*Phase C (`zh-Hans`/`ja`/`ko`)**** = large OFL fonts + lazy-load + offline-subset ? ****blocked on owner Q2/Q3****. Native review (Q4) is the launch gate.

3. Open owner decisions (unblock the most)

- ****7-lang (in `MULTILINGUAL\_EXPANSION.md` §8):**** Q2 ship `zh-Hans` only or +`zh-Hant`? . Q3 CJK offline-bundle strategy (ship-all-chunks +tens-of-MB / custom-subset build / online-lazy-only) .

Q4 native-reviewer sourcing for 7 langs · Q5 priority order.
- **B3:** D-CATALOG-HOME (shared top-level `locales/` ? recommended).
- **T2:** Cloudflare Access setup + **GEMINI_API_KEY** rotation. **T11:** retention window.
Native kn/hi review + owner-machine font eyeball (the existing trilingual launch gates).

4. Ramp-up kit (read in this order)

1. This file ? memory `current-state.md` (today's checkpoints).
2. `khelesutra/docs/FRIENDLY\_AND\_MULTILINGUAL.md` §7 (the active UX/l10n tracker) + `DRY\_RUN\_CUJ.md` §9 (the delivered CUJ) + `LOCAL\_APPLIANCE\_ARCHITECTURE.md` §9 T2/T4/T11 + §4 topology (the North Star) + `MULTILINGUAL\_EXPANSION.md` (the approved 7-lang plan).
3. Code: `khelesutra/web/src/{App.tsx, components/IngestPanel.tsx, lib/errors.ts, api/client.ts, i18n/index.ts, i18n/hygiene.test.ts}`; `badminton-highlight-indexer/backend/api/uploads.py` (the upload endpoints for T4).
4. A `wf\_b7edcaa8` spec workflow produced detailed B3/A4/A5/A8/North-Star tickets (re-run `docs-next-steps`/`wrapup-specs` or check the workflow transcript if more file:line detail is wanted).

5. Key decisions ? don't relitigate

- **A2** (WhatsApp/social share) DROPPED ? owner defers ALL social/sharing to a separate scope; reel stays **pure download** (wa.me can't attach files; real file-share = Web Share API = own scope).
- **Runnable-tool** path = hosted SPA+engine behind edge auth; cheapest = droplet + Gemini/CPU. NOT GPU-gated, NOT M12. (The droplet was parked only for the WASB/GPU CUJ; Gemini alpha needs no GPU.)
- **B-P1** was already shipped (engine `api/process` accepts `gs://` since M2/#280) ? see `[[lesson-verify-engine-surface-before-scoping]]`; verify engine-surface claims at file:line.
- **DON'T** touch the serving track ? `COMPUTE_DECOUPLED_SERVING` / `backend/compute/` / Cloud-Run M5?M12` is a **concurrent session's** active line (M1?M8 merged, M9 #290 open). Engine doc-staleness (DOC\_STATUS/NEXT\_STEPS/RALLY\_QUALITY n) was left for that session. **Shared checkout** ? branch from `origin/master``, re-verify HEAD before commit, stage explicit paths (`[[gotcha-shared-checkout-branch-collisions]]`).
- **Merge authorization:** Claude merges its own green good-engineering/UX PRs on verified CI (`[[working-agreement-merging]]`); product/strategy/licensing + the 7-lang Phase-C/native-review stay owner-gated.

525. user (2026-06-21T14:44:04.823Z)

<tool_use_error>File has not been read yet. Read it first before writing to it.</tool_use_error>

526. assistant (2026-06-21T14:44:12.630Z)

Spec workflow finished with detailed tickets (B3 especially ? confirms the catalog-hoist recommendation). The handoff write needs the existing file read first. Let me read it, then write:

527. user (2026-06-21T14:44:13.565Z)

```
1      ---
2      name: session-handoff
3      description: "Living \"you are here\" handoff ? current state, next steps, open
decisions, ramp-up kit. AUTO-REFRESHED at meaningful checkpoints (PR merged, key decision, phase
shift, session wrap). \"update the handoff\" forces a full rewrite."
4      metadata:
5        node_type: memory
6        type: project
7        originSessionId: 45ea3a91-654d-4972-9cd3-65f35db54caf
8      ---
9
10     # Session handoff ? badminton-highlight-indexer (rename PENDING) + siblings: sports-
data-collector, rally-annotator, sports-obsreport
11
12     **Last updated:** 2026-06-21 (HUGE serving session ? M1?M9-auth shipped (#279?#291)
+ ops-agent codify + dry-run guide + the real M7 L4 run (22 rallies, $0); owner's top
NEXT priority = make it UI-DRIVEN + alpha-shareable ([[ui-driven-cloud-serving-gap]]); then
```

M11/M12. See the Icebreaker at the bottom of §"You are here"). **Repo is under the `Khelsutra` GitHub org**.

13

14 > **Read order:** this file ? [[current-state]] top (dense per-checkpoint detail) ?
`docs/NEXT\_STEPS.md` (the prioritized cross-track plan, refreshed 2026-06-21) ?
`docs/DOC\_STATUS.md` (doc lifecycle/hygiene).

15

528. assistant (2026-06-21T14:44:43.362Z)

The existing `session-handoff.md` is the **serving session's** engine handoff ? I must not clobber it. My UX/localization + runnable-tool handoff belongs in `next-session-ux-localization.md` (this track's start-here). Reading it (to refresh it), then writing the updated handoff:

529. user (2026-06-21T14:44:43.800Z)

```
1      ---
2      name: next-session-ux-localization
3      description: Consolidated handoff/plan for the NEXT session ? focus SOLELY on (A) making
the UI super user-friendly for non-tech-savvy users and (B) localizing in Hindi + Kannada.
Grounded in existing repo docs (owner directive 2026-06-21).
4      metadata:
5          node_type: memory
6          type: project
7          originSessionId: 577fd4df-ed3f-43d3-854d-d63d732e2e07
8      ---
9
10     **Owner directive (2026-06-21, session wrap):** the next session focuses **SOLELY on
making the product super user-friendly (non-tech-savvy) AND localized (Hindi + Kannada)**.
*Research first using the existing repo docs, THEN craft the plan* (don't reinvent ? there's
prior design). The end goal: **a non-tech-savvy coach/parent reads the page, confidently sets up
recording, runs the tool, and gets highlights ? in their own language.**
```

530. assistant (2026-06-21T14:44:50.170Z)

Now refreshing this track's handoff with the current state + ordered next steps:

531. assistant (2026-06-21T14:45:29.690Z)

```
---
name: next-session-ux-localization
description: "? START HERE for the khelsutra UX/localization + runnable-tool track. Where we are
+ ordered next steps (SPA UX A-items ? B3 ? runnable-tool North Star) + the approved 7-language
line + open owner decisions + ramp-up kit. Refreshed 2026-06-21. NOTE: the engine/serving track
has its OWN handoff in session-handoff.md ? don't conflate."
metadata:
  node_type: memory
  type: project
  originSessionId: 577fd4df-ed3f-43d3-854d-d63d732e2e07
---
```

Handoff ? khelsutra UX / localization / runnable-tool (refreshed 2026-06-21)

> This is the **khelsutra (web/ SPA + site/)** product track. The **engine/serving** track (Cloud-Run M5?M12, a CONCURRENT session) has its own `session-handoff.md` ? **don't touch its files** (`COMPUTE\_DECOUPLED\_SERVING`, `backend/compute/`, engine `DOC\_STATUS`/`NEXT\_STEPS`).

1. You are here

The **bucket?reel CUJ is fully UI-driven + proven** (B-P2 shipped + live screencast), the **SPA is trilingual (en+kn+hi)** , **plain-language copy (A1) is merged** , and the **7-language expansion is owner-approved** (#58). The **North Star = make the tool runnable/shareable for alpha users from the UI** ? a hosted SPA+engine behind edge auth. **It is NOT GPU-gated and NOT

M12** ? cheapest path = the existing **CPU droplet + Gemini**.

Merged this session (all green): engine [#284](https://github.com/Khelsutra/badminton-highlight-indexer/pull/284)(gs:// guard) · [#289](https://github.com/Khelsutra/badminton-highlight-indexer/pull/289)(compile-from-bucket fix) · khelsutra [#55](https://github.com/avidullu/khelsutra/pull/55)(stale-doc) · [#56](https://github.com/avidullu/khelsutra/pull/56)(B-P2 ingest) · [#57](https://github.com/avidullu/khelsutra/pull/57)(DoD box3) · [#59](https://github.com/avidullu/khelsutra/pull/59)(doc-hygiene) · [#60](https://github.com/avidullu/khelsutra/pull/60)(A1 copy). Owner-merged [#58](https://github.com/avidullu/khelsutra/pull/58)(MULTILINGUAL_EXPANSION). Screencast: `gs://khelsutra-rally-corpus/promo/khelsutra-bucket-cuj-screencast.mp4`.

2. Next steps ? ordered (one PR per item, off `origin/master`, merge on green + rising coverage)

A) SPA UX A-items, then B3 *(A1 ?; A2 DROPPED ? social/sharing is a separate scope):*

- **A4 ? mobile touch targets + visible tooltip text.** `web/src/components/{QueryBar,RallyGrid,SelectionFooter,IngestPanel,CompileResult,LanguageSwitcher}.tsx` + `index.css`: ?44px tap targets (`min-h-11`/py), enlarge the tiny query parsed-pill + sample chips, surface `title` tooltips as visible text on phones. (M)
- **A5 ? friendly engine-error mapping (build path).** Reuse `web/src/lib/errors.ts::classifyError` (built for ingest) on `App.tsx::onBuild` (currently shows raw ApiError for 5xx/network) ? localized message in `CompileResult`. Add a **shared `errors` namespace** (recommend unifying IngestPanel onto it). (S?M)
- **A8 ? background-processing reassurance + status as icon+text chips** (never icon-only) in IngestPanel progress + build state. (S?M)
- **B3 ? `site/` trilingual via a shared `t()` shim.** New ~70-LOC vanilla `t(key,vars)`/`data-i18n` shim (`site/public/i18n.js`) over `index/capture/own-model.html` + a switcher + `site/server.js` Accept-Language negotiation; extend `site/src/{server,assets}.test.js` + a shim unit test. **DECISION (recommended by the spec scan): hoist catalogs to repo-root `/locales/{en,kn,hi}/common.json` as the single source of truth, re-point the SPA's 3 imports (or a Vite `@locales` alias), and a tiny `site` prebuild copies them to `site/public/locales/` + a drift-guard test** (site is dependency-free/no-build today; this is the only new build step). Negotiation must match the SPA exactly (localStorage `khelsutra.lang` ? `?lang` ? navigator ? en; en-fallback never a raw key). (M)

B) ? NORTH STAR ? runnable tool for alpha users (the crucial fresh-context work the owner paused for):

- **T4 ? in-UI local file UPLOAD (keystone).** Alpha users have a **file**, not a `gs://` URI. Wire the SPA to the **existing** engine chunked-upload endpoints (`badminton-highlight-indexer/backend/api/uploads.py`: `/api/upload/init|part|complete|DELETE`) + a file-picker in/beside `IngestPanel` ? on complete feed the server path to `/api/process` ? existing poll?load?compile?download. New: `web/src/api/client.ts` upload helpers + upload mode; reused: the back-half. (M?L)
- **T2 ? droplet appliance shell (hosted + shareable).** Serve the built SPA + reverse-proxy `/api?engine` 127.0.0.1:8000 behind **Cloudflare Access** on the **CPU droplet** 168.144.126.176 **, engine on **Gemini/CPU (no GPU)**; deploy the #281/#289-fixed engine + `GEMINI\_API\_KEY` + `pip install google-cloud-storage`; **rotate the chat-shared key**; reuse `site/scripts/provision-vm.sh` + `docs/PORTABILITY.md`. (M?L, +owner Cloudflare Access)
- **T11 ? retention sweep** (`output/` + `uploads/`) before non-owner exposure. (S, owner: window)

C) 7-language line ? APPROVED (#58), parallel/owner-paced (`MULTILINGUAL\_EXPANSION.md` §7): **Phase A** (`es`/`da`/`id`) actionable now** (catalog-only; **`es` needs the 3-form ICU plural {one,many,other}**). **Phase B (Thai)** = 1 small OFL font. **Phase C (`zh-Hans`/`ja`/`ko`)** = large OFL fonts + lazy-load + offline-subset ? **blocked on owner Q2/Q3**. Native review (Q4) is the launch gate.

3. Open owner decisions

- **7-lang** (`MULTILINGUAL\_EXPANSION.md` §8):** Q2 `zh-Hans` only or +`zh-Hant`? · Q3 CJK offline strategy (ship-all-chunks / custom-subset / online-lazy) · Q4 native-reviewer sourcing · Q5 priority order.
- **B3:** confirm the catalog-hoist (repo-root `/locales/`) approach.
- **T2:** Cloudflare Access setup + **GEMINI_API_KEY rotation**. **T11:** retention window. Plus the standing launch gates: native kn/hi review + owner-machine font eyeball.

4. Ramp-up kit (read in order)

1. This file ? `[[current-state]]` top (today's checkpoints).
2. ``khelsutra/docs/`: `FRIENDLY_AND_MULTILINGUAL.md` §7 (active UX/l10n tracker) · `DRY_RUN_CUJ.md` §9 (delivered CUJ) · `LOCAL_APPLIANCE_ARCHITECTURE.md` §9 T2/T4/T11 + §4 topology (North Star) · `MULTILINGUAL_EXPANSION.md` (7-lang plan).`
3. Code: ``khelsutra/web/src/{App.tsx, components/IngestPanel.tsx, lib/errors.ts, api/client.ts, i18n/index.ts, i18n/hygiene.test.ts}`; `badminton-highlight-indexer/backend/api/uploads.py` (T4 endpoints); `khelsutra/site/{public/*.html, server.js, scripts/provision-vm.sh}` (B3 + T2).`
4. Detailed B3/A4/A5/A8/North-Star tickets: spec workflow ``wf_b7edcaa8`` (re-run skill or check the transcript for file:line specs).

5. Key decisions ? don't relitigate

- ****A2 (WhatsApp/social) DROPPED**** ? all social/sharing deferred to a separate scope; reel = ****pure download****.
- ****Runnable-tool = hosted SPA+engine + edge auth; cheapest = droplet + Gemini/CPU. NOT GPU, NOT M12.****
- ****DON'T touch the serving track**** (concurrent session):
``COMPUTE_DECOUPLED_SERVING`/`backend/compute/`/Cloud-Run M5?M12 (M1?M8 merged, M9 #290 open).`
****Shared checkout**** ? branch from ``origin/master``, re-verify HEAD before commit, stage explicit paths (`[[gotcha-shared-checkout-branch-collisions]]`).
- ****Verify engine-surface claims at file:line before scoping**** (`[[lesson-verify-engine-surface-before-scoping]]`) ? B-P1 was already shipped.
- ****Merge authorization:**** Claude merges its own green UX/good-engineering PRs (`[[working-agreement-merging]]`); product/strategy + 7-lang Phase-C/native-review stay owner-gated.

532. user (2026-06-21T14:45:29.737Z)

The file `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\memory\next-session-ux-localization.md` has been updated successfully. (file state is current in your context ? no need to Read it back)

533. assistant (2026-06-21T14:45:45.054Z)

Handoff refreshed. Let me update the `MEMORY.md` "? Start here" pointer to match, then give you the next-session prompt:

534. user (2026-06-21T14:45:51.051Z)

<tool_use_error>File has been modified since read, either by the user or by a linter. Read it again before attempting to write it.</tool_use_error>

535. assistant (2026-06-21T14:45:59.242Z)

`MEMORY.md` changed again (concurrent edits). Let me re-read and update the pointer:

536. user (2026-06-21T14:46:00.330Z)

```
1      # Memory Index
2
3      ## ? Start here
4      - [? NEXT SESSION: user-friendly + localized](next-session-ux-localization.md) ? **owner
directive (2026-06-21): next session focuses SOLELY on (A) non-tech-savvy UX + (B) Hindi/Kannada
localization.** Read the docs FIRST (`engine docs/I18N_PLAN.md` + `UI_PROPOSAL.md` + PMF/field
persona docs), then craft the plan. Clean-slate handoff + ramp-up kit + open owner decisions
inside.
5      - [Current state](current-state.md) ? **live micro-checkpoint** (updated every
checkpoint): open PRs, what just landed, next steps, open decisions. **Read this first.** Now
LIVE-only (handoff + today's checkpoints); kept read-first-short.
6      - [Current-state archive](current-state-archive.md) ? chronological checkpoint history
(2026-06-06 ? 06-18, 96 blocks) split out of current-state 2026-06-19. Deep history only; NOT
read each session.
7      - [Session handoff](session-handoff.md) ? current state · next steps · open decisions ·
ramp-up kit. AUTO-refreshed at meaningful checkpoints (PR merged, key decision, phase shift,
```

session wrap) ? see working-agreement-checkpoints staleness tripwire.

8 - [Code map artifact](code-map-artifact.md) ? for CODE/architecture ramp-up, read
`docs/CODE\_MAP.md` FIRST (dense anchor-rich map; replaces a file crawl). How to regenerate it.
9 - [File-placement conventions](file-placement-conventions.md) ? ****where new files go****
(scripts/ vs backend/eval/ vs tools/ vs scratch/ vs training/ vs artifacts/ ?); ****read before**
committing a new file.** Authoritative table = `docs/CODE\_MAP.md` §0 (PR #138); scratch/
gitignored (#136).

537. user (2026-06-21T14:46:10.018Z)

The file C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\memory\MEMORY.md has been updated successfully. (file state is current in your context ? no need to Read it back)

538. assistant (2026-06-21T14:46:58.500Z)

Everything's wrapped and the handoff is durable (in `next-session-ux-localization.md`, pointed to from `MEMORY.md` "? Start here"; the serving session's own handoff left untouched). Here's your next-session prompt ? paste it fresh:

? Picking up KhelSutra after the UX + runnable-tool session. State: the bucket?reel CUJ is fully UI-driven (B-P2, with a live screencast in the promo bucket), the SPA is trilingual (en+kn+hi), plain-language copy shipped (A1), and the 7-language expansion is owner-approved (#58).

? Read first (the "use the docs" rule): memory next-session-ux-localization.md (your full handoff ? you-are-here, ordered next steps, ramp-up kit, open owner decisions), then khelsutra/docs/FRIENDLY_AND_MULTILINGUAL.md §7, LOCAL_APPLIANCE_ARCHITECTURE.md §9 (T2/T4/T11) + §4, and MULTILINGUAL_EXPANSION.md.

? Clean slate: khelsutra master has #55?#60 merged; engine master has #284/#289; 0 of my PRs open.

? The engine/serving track (Cloud-Run M5?M12) is a CONCURRENT session ? do NOT touch COMPUTE_DECOUPLED_SERVING/, backend/compute/, or the engine DOC_STATUS/NEXT_STEPS. Shared checkout:
branch from origin/master, re-verify HEAD before commit, stage explicit paths.

Mission (in order; ONE PR per item, merge on green + rising coverage):

1. Finish the SPA UX A-items ? A4 (mobile ?44px touch targets + visible tooltips) ? A5 (friendly engine-error mapping on the BUILD path: reuse web/src/lib/errors.ts::classifyError, add a shared errors.* i18n namespace) ? A8 (background-processing reassurance + icon+text status chips).
2. B3 ? make the site/ marketing pages trilingual via a shared t()/data-i18n shim. Recommended: hoist catalogs to repo-root /locales/ + a tiny site prebuild copy (B3 ticket in the handoff); negotiation must match the SPA (localStorage khelsutra.lang ? ?lang= ? navigator ? en).
3. ? NORTH STAR (the goal ? runnable tool for alpha users): T4 ? in-UI file UPLOAD (wire the SPA to the EXISTING engine chunked-upload endpoints in backend/api/uploads.py; alpha users have a file, not a gs:// URI). T2 ? host SPA+engine behind Cloudflare Access on the CPU droplet (168.144.126.176) with Gemini/CPU (NOT GPU, NOT M12); rotate the GEMINI_API_KEY; reuse site/scripts/provision-vm.sh + PORTABILITY.md. T11 ? retention sweep.

Parallel line (owner-approved #58, owner-paced): MULTILINGUAL_EXPANSION ? add th/id/da/es/zh-Hans/

ja/ko. Phase A (es/da/id) is actionable now (es needs the 3-form ICU plural {one,many,other}). Phases B/C need owner answers (Q2 zh-Hans/Hant, Q3 CJK offline strategy, Q4 native reviewers).

Don't redo: the CUJ, B-P2, A1, the doc-hygiene, #58. A2 (WhatsApp/social) is DROPPED (separate scope) ? reel stays pure-download. Open owner decisions are in the handoff §3.

Ultracode's on ? fan out research/verification with workflows, verify against the docs, keep tight
one-item PRs, merge the good ones on green. Let's make the appliance point-and-go for alpha. ?
``

Session wrap-up

A clean stopping point, exactly before the crucial runnable-tool work:

- ****Shipped + merged (8 PRs, all green):**** the in-UI bucket?reel CUJ (B-P2) + a real ****live-recorded screencast**** in the promo bucket; ****two engine bugs the CUJ surfaced and I fixed**** (#281 queued-hang via the earlier session, #284 gs:// guard, #289 compile-from-bucket); the ****B-P1 staleness correction**** + a durable anti-staleness lesson; ****doc-hygiene****; and ****A1 plain-language copy****.
- ****Proposal delivered + you approved:**** the 7-language ****MULTILINGUAL_EXPANSION**** (#58) with real CLDR/font diligence.
- ****Two scans**** clarified the map: the Cloud-Run work is the concurrent session's active line (don't duplicate), and the runnable-tool goal is ****cheaper than feared**** ? droplet + Gemini, no GPU, not M12.
- ****Handoff + prompt**** so the next session resumes cleanly with fresh context for T4?T2?T11.

Truly enjoyed this one ? proof-and-validation-driven the whole way (the screencast literally caught a real bug). Good luck with the alpha launch; the path is clear. ??